

**GREEN
CLIMATE
FUND**

Meeting of the Board
17 - 20 July 2022
Incheon, Republic of Korea
Provisional Agenda Item 9

GCF/B.33/02/Add.04

24 June 2022

Consideration of funding proposals – Addendum IV

Funding proposal package for FP190

Summary

This addendum contains the following six parts:

- a) A funding proposal summary titled “Climate Investor Two” submitted by the Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden N.V. (FMO);
- b) No-objection letters issued by the national designated authorities or focal point(s);
- c) Environmental and Social report(s) disclosure;
- d) Independent Technical Advisory Panel’s assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel’s assessment; and
- f) Gender documentation of the funding proposal.

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled titled “Climate Investor Two” by the Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden N.V. (FMO) is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.

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Funding Proposal

Project/Programme title: Climate Investor Two

Country(ies): Bangladesh, Botswana, Brazil, Colombia, Côte d'Ivoire, Djibouti, Ecuador, India, Indonesia, Kenya, Madagascar, Maldives, Morocco, Namibia, Nigeria, Philippines, Sierra Leone, South Africa, and Uganda

Accredited Entity: FMO – Dutch Development Bank

Date of first submission: 2022/03/01

Date of current submission: 2022/06/21

Version number: V.05



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Note to Accredited Entities on the use of the funding proposal template

Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.

Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.

The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.

The recommended font is Arial, size 11.

Under the [GCF Information Disclosure Policy](#), project and Programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A. PROJECT/PROGRAMME SUMMARY				
A.1. Project or Programme	Programme		A.2. Public or private sector	Private
A.3. Request for Proposals (RFP)	Not applicable			
A.4. Result area(s)	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/Programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.			
			GCF contribution	Co-financers' contribution ¹
	Mitigation total		Enter number %	Enter number %
	☒ Energy generation and access		42,5% %	42,5 %
	☐ Low-emission transport		Enter number %	Enter number %
	☒ Buildings, cities, industries, and appliances		5 %	5 %
	☒ Forestry and land use		2,5% %	2,5% %
	Adaptation total		Enter number %	Enter number %
	☐ Most vulnerable people and communities		Enter number %	Enter number %
	☒ Health and well-being, and food and water security		23,75 %	23,75% %
	☒ Infrastructure and built environment		23,75 %	23,75 % ²
	☒ Ecosystems and ecosystem services		2,5%	2,5% ³
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed/sequestered)A	~44.65 million tons CO2e over the lifetime of the 20-year Programme ⁴	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	~ 11.18 million people provided with more climate-resilient water security and improved ecosystems over the lifetime of the 20 – year Programme	Indirect beneficiaries for ARA2, 3 and 4 are currently estimated at zero. ⁵
			0.42%	
A.7. Total financing (GCF + co-finance ⁶)	880,000,000 ⁷ USD		A.9. Project size	Large (Over USD 250 million)
A.8. Total GCF funding requested	145,000,000 USD			

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or Programme described in the funding proposal.

² ARA2 and ARA 3 are both selected as water infrastructure, which constitutes 47.5% of the expected results area, benefit both results area (an equal split was therefore chosen).

³ Forestry and land use was selected as some of the targeted ecosystems will have a significant impact on sequestration and avoidance of GHG emissions. The expected allocation to ecosystems (10%) is therefore split between ARA4 (50%) and MRA4 (50%).

⁴ The total lifespan of the project/Programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project/Programme implementation period.

⁵ The AE and EE commit to working with the Secretariat on developing a KPI for indirect beneficiaries, where relevant.

⁶ Refer to the Policy of Co-financing of the GCF.

⁷ The total target size of CI2 at Final Close is 1,100,000,000. 80% of this is estimated to be deployed within GCF NOL countries. Therefore, USD 880,000,000 represents total co-financing + GCF for the funded activity. The impact numbers are therefore calculated as 80% of the total CI2 impact.

A.10. Financial instrument(s) requested for the GCF funding⁸	☒ Reimbursable grant <u>145,000,000</u> ☐ Grant <u>Enter number</u> ☐ Loan <u>Enter number</u> ☐ Guarantee <u>Enter number</u> ☐ Equity <u>Enter number</u> ☐ Results-based payment <u>Enter number</u>		
A.11. Implementation period⁹	20 years (after the first close)	A.12. Total lifespan	20 years (after the first close)
A.13. Expected date of AE internal approval	TBD	A.14. ESS category	I-1
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work Programme?	Yes ☐ No ☒	A.18. Is this FP included in the country Programme?	Yes ☐ No ☒
A.19. Complementarity and coherence	Yes ☒ No ☐		
A.20. Executing Entity information	1. Climate Fund Managers B.V (hereafter: 'Climate Fund Managers' or 'CFM' or 'the Manager') Korte Vijverberg 4, 2513 AB The Hague, Netherlands 2. Stichting I Development Fund. (hereafter: 'the Development Fund' or 'the DF2') Korte Vijverberg 4, 2513 AB The Hague, Netherlands 3. Coöperatief Construction Equity Fund 2 U.A. (hereafter: 'the Construction Equity Fund' or 'the CEF2') Korte Vijverberg 4, 2513 AB The Hague, Netherlands 4. Financierings-Maatschappij voor Ontwikkelingslanden N.V (hereafter: 'the AE' or 'FMO') Anna van Saksenlaan 71 2593 HW The Hague, Netherlands		

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

The water cycle is one of the most important and often overlooked, parts of the global climate crisis. The feedback loop is complicated, as water can become a substantial carbon emitter when managed improperly, with methane emissions from untreated waste and wastewater accounting for 7% of global greenhouse gas emissions, while riverine, marine, and coastal ecosystems have the potential to act as a powerful ‘carbon sinks’ that sequester and avoid carbon emissions. The imbalance between the water cycle and the climate is threatening many livelihoods, as growing economies and populations are reducing freshwater availability, putting over USD 6 trillion of developing nations’ GDP at riskⁱ. The World Health Organization estimates that there already are 750 million people with no access to clean water, and 2.5 billion people lacking access to proper sanitation. Even the oceans, which cover 71% of the earth’s surface and hold 96.5% of all earth’s water, are under threat with unprecedented pollution rates, decreasing biodiversity, and rising sea levels. At current rates, by 2050 there will be a great greater volume of plastic in the oceans than all marine life combined.

There’s a clear need for climate-resilient infrastructure to mitigate the risks described above. This necessity is even further amplified in developing countries. Inadequate water and sanitation infrastructure are estimated to be related to 80% of all illnesses in the developing worldⁱⁱ. In total, the World Bank estimates that by 2050, water shortages caused by climate change could reduce the GDP of developing countries in Asia by USD 4.4 trillion and developing countries in Africa by USD 1.6 trillionⁱⁱⁱ. Climate finance is responding; however, investment requirements far exceed the current level of financing^{iv}.

Climate Investor Two (“CI2”) has the mission to develop and construct infrastructure projects in developing countries in the water, sanitation, and ocean sectors that reduce the effects and consequences of climate change by ameliorating GHG emissions and by increasing resilience. To do so, CI2 will deploy an innovative whole-of-life financing approach utilizing two independent but operationally inter-linked funds. The two funds focus on different stages of the project lifecycle; the Development Fund (“DF2”) provides development loans to early-stage projects and their ecosystems, whereas the Construction Equity Fund (“CEF2”) provides equity during the construction phase. With a Green Climate Fund (“GCF”) participation of USD 145 million, the prospective fund sizes are estimated at a USD 100 million DF2 and a USD 1 billion CEF2¹⁰ (out which 80% would be eligible for GCF proceeds – a USD 80 mln. DF2 and a USD 800 mln. CEF2). GCF participation would enable CI2 to scale significantly and provide scarce, highly additional donor capital to the structure which is required to crowd in further private sector participation.

CI2 will realize climate adaptation benefits in the water sector by building infrastructure that sources, transports, and treats the water necessary for both municipal and industrial users to be shielded from the consequences of climate change. In the sanitation sector, CI2 will realize mitigation and adaptation benefits through waste and wastewater treatment, reducing methane emissions and contributing to water re-use. Finally, in the ocean sector, CI2 will invest in physical infrastructures, such as environmental services for ports or green shipping initiatives that ameliorate GHG emissions. Additionally, CI2 will pursue investments in nature-based solutions to protect, preserve and restore riverine, wetland, coastland, and marine ecosystems (hereafter: “ecosystems”). Ecosystems are an important part of nature’s defenses against climate change and an important part of people’s livelihoods. They are currently

⁸ The USD 145 mln. will be subscribed to the Development Fund as a reimbursable grant. USD 100 mln. will then be subscribed to the Construction Equity Fund as equity (same structure as FP099). However, do note that the GCF FAA will therefore be for a reimbursable grant.

⁹ Investment period of the CEF is 15 years

¹⁰ The CEF includes three different tiers: (i) USD 200 mln. in junior-equity provided by Donors, (ii) USD 400 mln. of mezzanine provided by impact investors and DFIs and (iii) USD 400 mln. senior equity provided by commercial and institutional investors.

under stark threat facing ecological degradation at a rate ten to hundreds of times higher than the average over the past 10 million years, and with more than 50% of natural ecosystems at risk of being lost^{vi}. Through investments in nature-based solutions, CI2 targets cross-cutting climate change mitigation and adaptation outcomes that take advantage of the natural ability of restored ecosystems to sequester carbon and foster ecosystem-based adaptation.

By developing and constructing additional water supply, treatment, distribution, and waste-water treatment facilities, as well as physical and nature-based oceans infrastructure, CI2 will have a quantifiable and measurable social and environmental impact in developing countries. CI2 aims to construct up to USD 2.96 billion in infrastructure projects in developing countries (core indicator 3). The projects financed, developed, and constructed by CI2 will provide ~11.18 million people with year-round access to reliable and safe water supply despite climate shocks and stresses and protected and strengthened ecosystems (core indicator 2). In addition, the coverage of ecosystems protected and strengthened in response to climate variability and change will be ~1.7 million Ha (supplementary indicator 4.1). With regards to mitigation benefits, CI2 will reduce or avoid ~4.96 million tCO₂e per year for a total of ~44.65 million tCO₂e over the implementation period (core indicator 1). CI2 will create additional opportunities for scalability and replicability beyond its projects, triggering the paradigm shift described in more detail in section D2. The specific climate vulnerabilities within the countries eligible for GCF funding through CI2 are described in Annex 7.

FMO, the Dutch Development Bank, will act as Accredited Entity for this proposal and as an Executing Entity in respect of specific activities which will be elaborated in the FAA. Climate Fund Managers, a joint venture between FMO and Sanlam, will act as the Fund Manager of Climate Investor Two, and is the first Executing Entity. The DF2 and the CEF2, the two Funds within Climate Investor Two, will act as the second and third Executing Entities.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

The climate crisis

Globally, the seven hottest years in recorded history have been the six years from 2014 to 2020^{vi}, driven by increasing concentrations of greenhouse gases in the atmosphere. Simultaneously atmospheric carbon reached a corresponding record high in 2020, driven primarily by anthropogenic use of fossil fuels^{vii}. As of 2020, global average temperatures are 1.14°C higher than pre-industrial temperatures, the effects of this warming are already being felt, often disproportionately in developing nations^{viii}. For every 1°C increase in temperature, a developing nation can expect a decrease in the gross domestic product of 1.3%, whilst developed nations are far less affected by temperature variations^{ix}. The increase in global temperatures, combined with the vulnerability of developing nations to these temperatures highlights the critical need for climate finance to be invested in projects and programs which both decrease greenhouse gas emissions and increase the resilience of host nations to adapt to these changes. To date, climate finance has primarily invested in climate mitigation projects, receiving an average of USD 537 billion in 2017-18, compared to only USD 30 billion invested in climate adaptation projects^x.

The importance of the water cycle in the climate crisis

Water is an often-overlooked part of the global efforts to mitigate and respond to the climate crisis. Water is paramount to all life on earth, and any imbalance in its cycle or supply chain can quickly become detrimental to the global climate and the many livelihoods that depend upon it. Imbalances are created by a combination of growing economies and populations, changing weather patterns due to climate change, and a growing lack of resilient infrastructure. An example of how these imbalances have serious consequences for many is shown by the grim observation that more than 80% of illnesses in the developing world are linked to inadequate water and sanitation infrastructure.

CI2 aims to develop and construct the infrastructure that's required to future-proof parts of the water cycle. The investment thesis differentiates between three different sectors: (i) 'the water sector', (ii) 'the sanitation sector, and the (iii) 'the oceans and ecosystems sector'. The water sector is defined by projects that source, transport, and treat water to and from municipalities and sectors. The sanitation sector looks at treating waste and wastewater to avoid and sequester methane emissions while providing clean sanitation. The oceans and ecosystems sector aims to: (1) improve the ocean's healthy by providing environmental services to ports and harbors and (2) protect, strengthen, and restore important ecosystems.

The water sectors

Water is fundamental to human life and economic development but is also the medium through which the effects of climate change will be felt on earth, with a warming climate causing increasingly variable and unpredictable rainfall. Changing climate patterns, together with growing economies and populations could reduce urban water availability in 2050 by two-thirds compared to 2015 levels, putting over USD 6 trillion of developing nations' GDP at risk^{xi}. UNESCO estimates that over 2 bn. people already live in areas subject to water stress. A study done by the WHO and UNICEF in 2017 shows that globally more than 785 mln people did not have access to at least basic water services and more than 884 mln. people do not have immediate access to safe water to drink. Women are often more negatively impacted as they are responsible for collecting the water within the household. These responsibilities reduce the likelihood of women receiving education. One of the objectives of CI2 in the water sector is to realize

climate change adaptation impacts in countries undergoing, or expected to undergo water stress, through building the infrastructure which sources, transports, and treats the water necessary for both municipal and industrial users to be resilient to the localized impacts of climate change.

The sanitation sectors

Sanitation, defined as the sustainable collection, treatment, and disposal of waste and wastewater, is a major source of greenhouse gasses, environmental damage, and health hazards in developing countries. In developing countries, over 80% of all wastewaters are discharged into water sources without any treatment, resulting in severe and worsening pathogen pollution in one-third of all rivers. Pathogen pollution, caused by for instance seepage from landfills into water aquifers, affects surface water resources which supply 60% of the world's water consumption. As a result, water quality has significantly worsened in Asia, Africa, and Latin America since 1990^{xii}. More than 3 billion people worldwide lack adequate facilities to safely wash their hands at home, with stark concentrations of them living in Sub-Saharan Africa and Southern Asia^{xiii}. Similarly, over two billion people in developing countries have no access to solid waste collection, and with solid waste volumes projected to increase by 1 billion tons annually to reach 2.6 billion tons by 2050, serious environmental pollution will continue to occur without investment in sanitation (and waste) infrastructure. In improving groundwater quality and reducing pollution through its investments in sanitation infrastructure, CI2 increases the resilience of communities to climate change by improving the quality and availability of different water sources including rivers and groundwater. In turn, as in many cases livelihoods are dependent on these polluted water resources for sustenance or work, CI2 increases the resilience of communities to climate change through the preservation and restoration of existing and alternative sources. The lack of sanitation facilities is even more impactful for young women, as they are more likely to miss out on education due to insufficient sanitation at schools^{xiv}. The primary objective of CI2 in the sanitation sector is to deliver significant climate change mitigation impacts, through building the infrastructure which sustainably treats and manages solid wastes and wastewater, including, wherever possible, using waste and wastewater as a source for bio-energy generation by constructing waste to energy projects.

The oceans and their ecosystems

The oceans provide half of the Earth's oxygen, feed three billion people, and have absorbed an estimated 93% of the added heat arising from human-driven changes to the atmosphere^{xv}. In economic terms, the annual economic value-added output derived from the oceans is estimated at USD 2.6 trillion, making the oceans the world's seventh-largest economy, after the United Kingdom^{xvi}, showcasing its importance for many livelihoods. The climate crisis is imminent for the oceans, with climate change, causing warmer waters already being observed at depths of 1000m. Atmospheric warming is causing the melting of inland glaciers and ice, causing rising sea levels with significant impacts on shorelines (coastal erosion, saltwater intrusion, habitat destruction). The IPCC projects the global sea level to increase by 0.4 to 0.8 meters by the end of the century, putting shorelines, but also local water and sanitation infrastructure projects, further at risk^{xvii}. Oceans are under further serious pressure by uncontrolled pollution, often caused by the lack of infrastructure to properly treat (waste)water that's being dumped in the world's rivers and shores, and resource overexploitation.

In addition to reducing ocean pollution through its investments in pollution control through the sanitation sector, CI2 also aims to invest in infrastructure for ports, ships, and harbors that contributes to improved ocean health (by reducing greenhouse gas emissions). In addition, it will pursue investments and ecosystem-based approaches to protect, preserve and/or restore riverine, coastland, marine, and wetland ecosystems. Financing ecosystems has traditionally been seen as net-cost projects by

governments and businesses because external costs associated with pollution of the environment and GHG emissions are often not accounted for correctly. Through investments in ecosystem-based approaches, CI2 targets cross-cutting climate change mitigation outcomes through the natural ability of restored ecosystems to sequester carbon and adaptation outcomes, through ecosystem-based adaptation. CFM is a signatory to the United Nations Environment Program's Sustainable Blue Economy Finance Initiative and the investments of CI2 in the oceans' sector will be consistent with this framework.



While climate change resilient water and sanitation infrastructure is a global challenge, CFM recognizes that Adaptation requires tailored and localized solutions to have a legacy that endures. As a result, CFM will develop and construct interventions that fit within the host's country National Determined Contributions ("NDC's") and local context to maximize the impact they have on increasing resilience and reducing greenhouse gas emissions. Annex 7 of this Funding Proposal includes more detailed climate assessments about countries for which the AE has obtained, or is aiming to obtain, an NOL.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Climate Investor Two aims to create a paradigm shift by making investments which result in the scaling and replication of water, sanitation, and oceans infrastructure that is aligned to the adaptation and mitigation needs of developing countries. If it succeeds in this, low-emission, climate-resilient sustainable development will be catalyzed in the programme's host countries because successful climate investments in the target sectors, with a risk adjusted return, will be demonstrated to commercial

investors. In turn, this will foster significant contributions to blue and green climate finance in emerging markets, bringing countries closer to meeting their INDCs, and the global targets of the SDGs

To derive the above, CI2's theory of change relies on a number of reasonable assumptions. First, it is assumed that there will be sufficient expertise provided by project co-developers and co-investors for project development and implementation. This is an assumption further underpinned by CFM's inhouse technical, legal and ESG expertise, which is applied to projects accordingly in order to complement those specialties of co-developers or co-investors. Second, it is assumed that offtake and/or financial service contracts enable the financial sustainability of projects. This condition is considered and assessed prior to and during the development phase of a CI2 project. Third, it is assumed that targeted ecosystems are not further degraded by external variables. While this assumption is reliant on external factors, CFMs ESG and technical teams scrutinize different aspects of a potential project to evaluate and consider its long-term feasibility. Fourth, it is assumed that the projects will generate a material demonstration effect, leading to catalyzation of capital from external sources to similar projects. As in the case of assumption number one, this is supported by CFMs own in house expertise, which will drive knowledge transfer and skill sharing when working with local developers. Fifth, it is assumed that PPP regulation clearly defines a process for private sector participation. Sixth, it is assumed that wastewater quality standards are effectively enforced in the beneficiary countries. As the regulatory environments for water and sanitation in the emerging markets continue to improve in response to the threats posed by climate change, this assumption is likely to develop in CI2's favor over the 20-year implementation period. Similarly, the increased need to address infrastructure deficits support assumption seven, that country regulatory frameworks permit and facilitate private sector participation. Eighth, it is assumed there will be a stable political environment in operating geographies. While this too is dependent on externalities, CFM's regional offices employ the necessary expertise to take measures to minimize associated risks. This can include project structuring approaches, political risk insurance, and also the support of the CI2 investor community which contains a number of influential regional financial institutes. Finally it is assumed that institutional structures support commercial investment in CI2's target sectors. This assumption, like assumptions six and seven, is bolstered by the necessity for commercial investment to address the scale of the existing deficits within CI2's target sectors.

Furthermore, in order to generate a paradigm shift, CI2's theory of change identifies and aims to address or mitigate the following risks and barriers.

With respect to climate risk, the CI2 theory of change identifies that climate change, increasing population, and growing economies will reduce the availability of water if resilient infrastructure is not built. This will be exacerbated as untreated waste and wastewater will further increase GHG emissions while causing environmental pollution and health hazards. Simultaneously, the exploitation of the oceans will reduce biodiversity, increase environmental pollution, and negatively influence livelihoods for those most dependent on the oceans and their ecosystems. While this continues, the issue is compounded by increasing sea levels and water temperatures, which will jeopardize the health of the oceans and their ecosystems further. Finally, deterioration of the natural asset base of riverine, coastal, wetland, and marine ecosystems will reduce nature's ability to act as a carbon sink, reduce their regulatory abilities, threaten food stock, reduce biodiversity, and negatively influence local communities' livelihoods. When considered together, these five individual factors represent a network of related and interdependent drivers of climate risk.

Given CI2's planned interventions and the climate risks it intends to address, it is possible to identify a number of barriers and risks to the accomplishment of the paradigm shift it will trigger.

Barrier 1, is that the potential long-term benefits of investing in target sector projects may not outweigh the high perceived risks. This can be linked to Risk 1, that low volumes of private investments are mobilized due to a lack of demonstrating the commercial viability of investing in target sector projects.

The water, sanitation, and ocean sectors have primarily been within the public domain. Traditional private investors, therefore, perceive the sectors as high-risk. These (perceived) risks can be reduced by both (i) providing private sector investors with a blended finance solution that reduces their risks and (ii) by demonstrating the bankability of projects in the water, sanitation, and oceans sectors. A lack of long-term private sector buy-in will further jeopardize global temperatures and the resilience that developing countries have against it.

Barrier 2, is the limited know-how and capacity for the development, construction, and management of target sector projects in host countries.

There's often limited capacity for the development and construction of complicated and technology-abundant infrastructure projects. Adequately addressing this will require cooperation between all the developers and third parties.

Barrier 3, is the Increasingly limited public sector budgets and commitment for target sector projects in host countries.

The budgets that developing countries have available to invest in the water, sanitation, and ocean sectors are insufficient for the task at hand. Adequately addressing the lack of the water infrastructure that would be needed to stay within the 1.5-degree pathway while ensuring resilient water supply will require the mobilization of private sector finance.

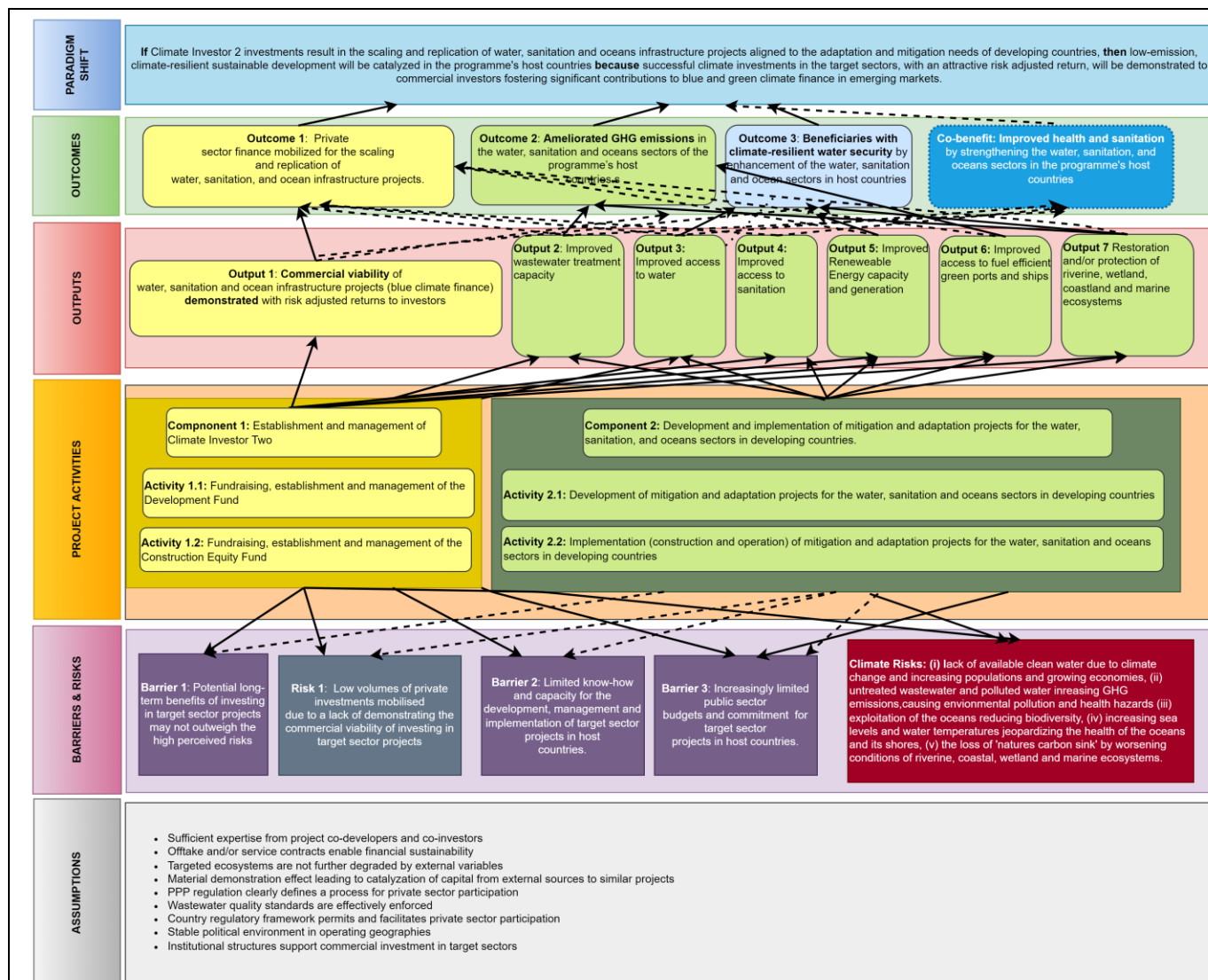
Climate Investor Two will aim to address these barriers and mitigate these risks by: (i) establishing and managing Climate Investor Two (including the Development – and the Construction Equity Fund) and (ii) development and implementation of mitigation and adaptation projects in developing countries.

The establishment and management of the two different funds, that together provide a whole-of-life solution to project financing, will help to showcase the commercial viability of water, sanitation, and oceans infrastructure projects by demonstrating the possibility of risk-adjusted returns to investors. This can help investors to see the long-term benefits of investing in such sectors and countries and mitigate some of the perceived high risks. Furthermore, it will showcase the effectiveness of blended finance vehicles such as Climate Investor Two in mobilizing (private sector) investors with different risk to return appetites. This will result in further mobilization of the private sector for the much-needed scaling and replication of water, sanitation, and ocean infrastructure projects in developing countries. Furthermore, showcasing the commercial viability of water, sanitation, and ocean infrastructure projects will contribute to providing know-how and capacity for the development and implementation of such projects in the host countries by providing a knowledge of transfer to local parties. CI2 aims to do this by including local partners and developers in both the development as well as the construction phases of its projects. Lastly, it will aim to include local financial intermediaries, including DAEs, to the refinancing of its projects.

Climate Investor Two will develop mitigation and adaptation projects by providing development loans from its DF2 to do the required feasibility studies, ESIA's, gender assessments, stakeholder engagements, potential pilot projects and everything else that's required before a project can go its construction phase. Once successfully developed, these projects graduate to the CEF2, which will implement (construct and operate) the projects by providing up to 100% of the equity (albeit it is

expected that most equity stakes would be below 75%). Successfully developing and implementing mitigation and adaptation projects for the water, sanitation, and ocean sectors in developing countries will: (i) improve wastewater treatment capacity, (ii) improve access to water, (iii) improve access to sanitation, (iv) improve renewable energy capacity and generation, (v) improve access to fuel efficient green ports and shipping, and (vi) restore and/or protect riverine, wetland, coastland, and marine ecosystems. Improving wastewater treatment and capacity, renewable energy capacity and generation, access to fuel-efficient green ports and shipping and restoration and/or protection of riverine wetland, coastal, and marine ecosystems will then result in ameliorating GHG emissions in the water sanitation and oceans sectors of the Programme's host countries. Improved access to water will and restoration and/or protection of riverine, wetland, coastland and marine ecosystems will result in beneficiaries with more climate-resilient water security in the Programme's host countries. As a co-benefit, improved access to sanitation, access to water and wastewater treatment capacity will result in improved health and sanitation by strengthening the water, sanitation, and oceans sectors in the Programme's host countries.

Altogether the above implies that **if** CI2's investments result in the scaling and replication of water, sanitation, and oceans infrastructure projects aligned to the adaptation and mitigation needs of developing countries, **then** low-emission, climate-resilient sustainable development will be catalyzed in the programme's host countries **because** successful climate investments in the target sectors, with an attractive risk-adjusted return, will be demonstrated to commercial investors fostering significant contributions to blue and green climate finance in emerging markets.



B.2 (b). Outcome mapping to GCF results in areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1: Private sector finance mobilized for the scaling and replication of water, sanitation, and ocean infrastructure projects.	X		X	X		X	X	X
Outcome 2:	X		X	X				

A,erlorated GHG emissionsin the water, sanitation, and ocean sectors of the Programme's host countries.								
Outcome 3: Beneficiaries with more climate- resilient water security by enhancement of the water, sanitation, and ocean sectors of the Programme host countries.						X	x	X

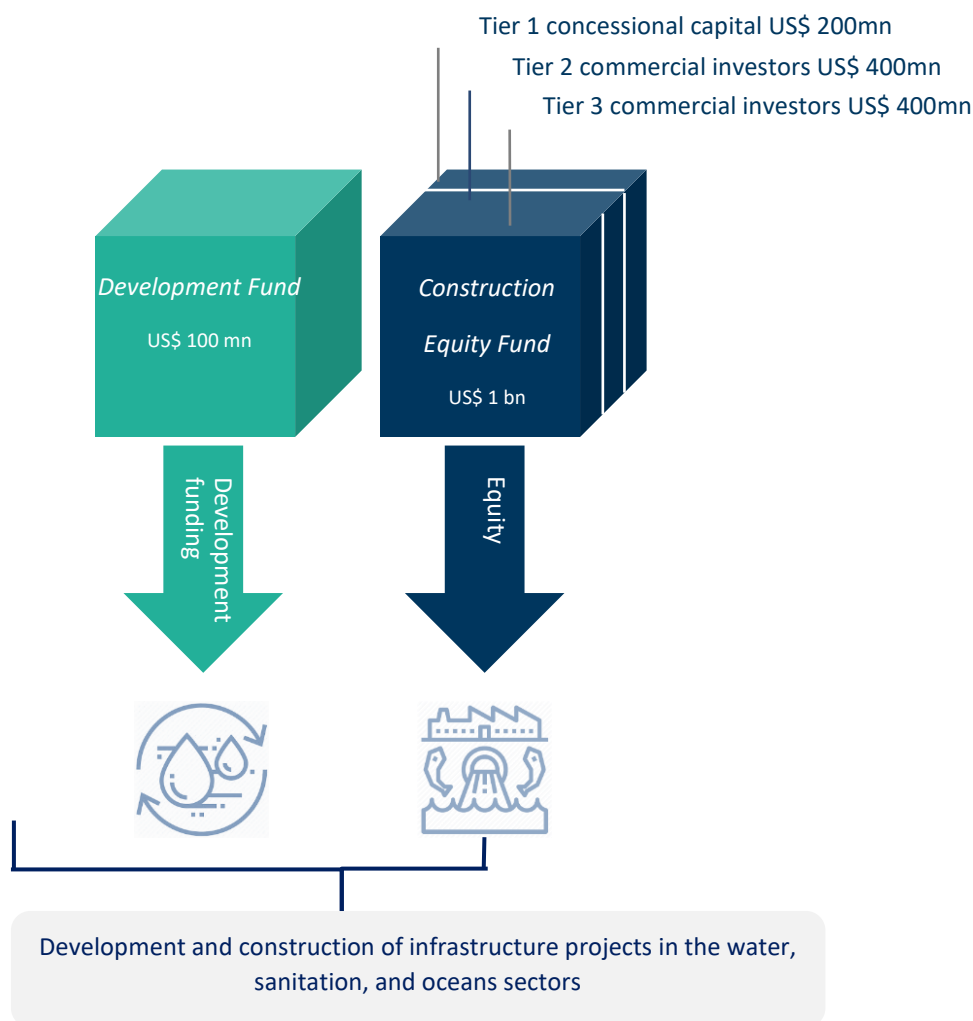
Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Improved health and sanitation by strengthening the water, sanitation, and oceans sectors of the Programme's host countries.	☐	☒	☒	☐	☐	☐

.3. Project/Programme description (max. 2500 words, approximately 5 pages)

Introduction to Climate Investor Two

Climate Investor Two is a blended finance facility mandated to develop and construct projects in the water, sanitation, and oceans sectors in developing countries to contribute to climate mitigation and adaptation benefits. CI2 duplicates the structure of the CI1 fund that launched in 2017 and received a GCF commitment in 2019 (FP099). CI2, like its predecessor CI1, was designed to address the lack of private financing options in developing countries by providing a whole-of-life project financing solution

that caters to investors with different risk to return profiles (ranging from Donors to commercial investors). CI2 constitutes two independent but interlinked funds, being two different legal entities, that operate at an arms' length: the Development Fund and the Construction Equity Fund.



Development Fund: The Development Fund ("DF" or "DF2") has been set up to provide development loans, accompanied by development support/assistance provided by CFM and 3rd parties, to project companies. The purpose of this is to fast-track qualifying development projects to financial close by reducing the development timeline and improving the ultimate bankability of infrastructure projects. The DF2 will increase the implementation rate of projects – while demonstrating the commercial viability of industries that have traditionally been part of a public mandate. The DF2's investor base mainly consists of Donors that want to get involved in the high-impact development phase. Once projects successfully complete their development phase by passing the assessments (including but not limited to gender assessment, ESIA's, and stakeholder engagements) they 'graduate' to the CEF2. The CEF2 then reimburses the grant to the DF2 and the project moves on to the construction phase. CFM ensures the DF2 remains 'revolvable' by recycling its capital. As it is expected that not all projects successfully complete the development phase (CFM expects 2/3rd of the projects to graduate), the CEF2 reimburses the grant with a premium that compensates for losses made on unsuccessful projects. The premium paid remains subject to the default rate within the DF2 (but would equal 50% if the default rate does end up

at 1/3rd.) This recycling character of the DF2 enables CI2 to complete multiple project cycles within its 20-year implementation period. CFM expects CI2 to complete two full project cycles (see D.6).

Construction Equity Fund: The CEF2 has been set up to provide to kick-start projects from their development phase, after the ‘proof of concept’, by providing up to 100 percent of the required equity (and repaying the development loan, including a premium, to replenish the DF2). The equity stake can include different sorts of equity (ordinary, preferred, shareholder loans) and will not always constitute a majority stake. It aims to further decrease the construction time and costs by: (i) reducing the complexity and costs at the project level through equity only funding strategy, (ii) reducing the construction timeline and thereby reducing the (costly) exposure period to investors, (iii) reducing costs by eliminating certain debt specific costs, (iv) recycling capital from completed projects to new project to reduce the time any funding ‘stands still’ (something that happens often in fundraising). The CEF2 uses a blended finance approach to mobilize three different tiers of investors: (i) Donors in the most junior tranche (representing 20% of the CEF2), (ii) DFI’s, and commercial/impact investors in the mezzanine tranche (representing 40% of the CEF2), and (iii) commercial and institutional investors in the senior tranche (representing 40% of the CEF2). The CEF2, like the DF2, recycles its capital over the 20-year implementation period (with the final deployment cycle occurring by year 15). The 20:40:40 ratios between the three different tiers are set and will always be maintained.

CFM, which will be responsible for the sourcing for and execution of, projects based within the following sectors:

Water, including the following sub-sectors:

- Bulk water supply and storage.
- Ocean-water desalination.
- Brackish groundwater desalination.
- Groundwater abstraction.
- Atmospheric water generation.
- Water treatment facilities.
- Water utilities and concessions.
- Pumping.
- Household water connections; and
- Decentralized water treatment and distribution.

Sanitation (waste and wastewater), including the following sub-sectors:

- Wastewater collection and sewerage networks.
- Wastewater treatment facilities, including wastewater to energy production and nutrient extraction and capture.
- Wastewater recycling facilities.
- Waste to value facilities, including waste sorting, recycling, composting, and waste to energy production; and
- Methane capture and destruction for waste facilities.

Oceans, including the following sub-sectors:

- Sustainable management, protection, and/or restoration of wetland, riverine, coastal, and marine ecosystems.
- Seaweed aquaculture.
- Climate-resilient port infrastructure.

- Sustainable solutions for ports, including onshore power, waste, and wastewater treatment.
- Port infrastructure for off-shore wind farm assembly, load-out, and servicing.
- De-carbonisation of port handling and rolling equipment; and
- De-carbonisation of coastal and inland shipping.

Other criteria

- Not a part of the Exclusion List.
- ESS Cat A, B, or C (or I-1, I-2, and I-3), with satisfactory safeguards as further detailed in the ESMS; and
- Standard and customary compliance requirements to be fulfilled for each investee company.

These projects will be assessed extensively on their contribution to either/or climate resilience and or GHG/reductions. Alongside that, the projects are screened on their commercial viability, ESG consideration, and governmental support. In addition, the partners (developers) are also screened on their ESG approach, reputation & local network, and track record.

Other investment restrictions / criteria for CI2 include:

Subject to the restrictions outlined below, the Fund will invest in the Mandate Countries listed below. The Fund will invest no more than:

- 15% of the Aggregate Fund Capital in any one Project Company / sub project, provided that such limit may be increased to a maximum of 20% of the Aggregate Fund Capital in a single Project Company / sub project with the consent of the CI Two Funds Advisory Board, which shall not be unreasonably withheld or delayed;
- 25% of its Aggregate Fund Capital into one single country;
- 50% of its Aggregate Fund Capital into Africa;
- 50% of its Aggregate Fund Capital into Asia;
- 50% of its Aggregate Fund Capital into Latin America;
- 75% of its Aggregate Fund Capital into the water sector (as defined above);
- 75% of its Aggregate Fund Capital into the sanitation sector (as defined above), and;
- 25% of its Aggregate Fund Capital into the oceans sector (as defined above).

Components, outcomes, and underlying (sub)activities Climate Investor Two.

Component 1: Establishment and management of Climate Investor Two

The first component aims at establishing and raising the funding for the two Funds that will form the basis for Climate Investor Two. Thereafter, it aims at managing the three legal entities (CFM, the 'Stichting' CI2 Development Fund (or: DF2), and the CI2 Coöperatief Construction Equity Fund (or: CEF2)'. This includes putting in place and maintaining, all the legal and governance arrangements (as described in B.4).

Activity 1.1: Fundraising, establishment, and management of the Development Fund.

CFM will continue to raise Donor funding, in the form of reimbursable grants for the DF2. CFM aims to raise an amount of USD 100mIn (USD 80 mln. for countries eligible for this proposal). It will do this by pro-actively reaching out to Donors, impact investors, and multilateral organizations that have a mandate that fits the Development Fund. Once the targeted amount has been reached, and the final

close achieved, CFM will oversee and manage all affairs of the legal entity for the entire implementation period of the DF2.

Activity 1.2: Fundraising, establishment, and management of the Construction Equity Fund

CFM will continue to raise funding for the three different tiers of the Construction Equity Fund (or: CEF2) up to the targeted amount of USD 1 bln (USD 800 mln. for countries eligible for this proposal). This will include funding from commercial investors (pension funds, asset managers, and other institutional investors, as well as impact-oriented investors (i.e. DFIs) and Donors (such as the European Commission, the Dutch government, and the GCF). The funding will be raised within the pre-determined ratio of 20:40:40 between the three different tiers. Once the targeted amount has been reached, and the final close achieved, CFM will oversee and manage all the affairs of the legal entity for the entire implementation period of the CEF2.

Outcome 1: Private sector finance mobilized for the scaling and replication of water, sanitation, and ocean infrastructure projects.

By successfully setting up the two different funds and raising USD 100 mln. Of funding for the DF2 (out of which USD 80 mln. is destined for GCF eligible countries) and USD 1 bln. For the CEF2 (out of which USD 800 mln. is destined for GCF eligible countries), the Programme will successfully engage the private sector in climate change adaptation and mitigation projects. It does this in three different manners: (i) it mobilizes private financing directly into CI2 by providing a blended finance solution that makes the return to risk profile acceptable for private investors and (ii) it demonstrates the potential that blended finance and whole-of-life all equity financing has for mobilizing private capital and project finance and (iii) it demonstrates the commercial viability/bankability of climate mitigation and adaptation infrastructure projects (linked to Activity 2.1 and Outcome 2, 3 and 4).

Component 2: Development and implementation of mitigation and adaptation projects for the water, sanitation, and oceans sectors in developing countries

Activity 2.1: Development of mitigation and adaptation projects for the water, sanitation, and ocean sectors in developing countries.

Activity 2.1.1: Identifying potential investment opportunities and initial screening by CFM IC:

The CFM deal teams will do the initial project sourcing. Once an opportunity has been found, and enough data have been gathered, an initial approval and guidance are requested by presenting the case to CFM's internal IC ("CFM IC"). The IC then decides whether: (i) the project fits within the investment mandate and (ii) whether additional information, restructuring, or due diligence is required before presenting it to the external IC. The CFM IC should be seen as a first 'quality control'.

Activity 2.1.2: (Re)structuring after CFM IC and initial due diligence

The CFM IC will provide the deal team with tasks to be completed before the actual Climate Investor Two Development Fund Investment Committee (DF2 IC). After that, the deal team will work to get all the information required to present the impact and business case of the project to the DF2 IC.

During this phase, the deal teams will gather all the information required to present the impact and business case towards the DF2 IC. The deal team will do initial due diligence on the project and the third party (co)-developers and sponsors.

Activity 2.1.3: Project approval by DF2 IC:

The deal team presents the case to the DF2 IC after doing the initial due diligence and completing the data information requested by the CFM IC. The DF2 IC will judge proposals, among other indicators, on the adaptation and mitigation impact they provide for the host countries. For adaptation, the deal teams will provide them with historic climate trends (trends (observed changes or variability in changes in the projects area), climate change projections in the area, and climate impact (an explanation on how these changes will hurt communities in the project area). Adaptation benefits will be further addressed by explaining how the project company's products, services, or technology will help the community adapt to these climate impacts, an explanation of how the proposed intervention will not lead to maladaptation, an explanation of the Rio Marker score for climate adaptation (to be achieved by the company) and an overview of the impact indicators that will be monitored to validate the above (including an identification of the baseline and mid-term (year 10) and final (year 20) projections). To assess mitigation benefits, CFM deal teams will explain how the project companies' products, services, or technologies will help reduce, avoid, or sequester GHG emissions, an explanation of the Rio Marker Score that's to be achieved by the project company and a description of the GHG gas accounting methodology to be used to validate the climate mitigation impact realized by the company's services, products, or technologies (including the baseline, mid-term, and final projections).

During the first DF2 IC, the DF2 IC will also approve terms of reference for an independent detailed adaptation and/or mitigation assessment based on the criteria set out above. This detailed independent assessment will be carried out as part of the more detailed due diligence that's conducted during the development phase (Activity 2.1.4.) put preceding the approval of the Construction Equity Fund 2 Investment Committee (CEF2 IC) (Activity 2.2.1). This independent assessment will then form part of the materials that the CEF2 IC receives and bases its approval on.

It is only after approval by the DF2 IC (including, for the avoidance of doubt, the satisfactory initial climate screening conducted by CFM) that a proposed investment would receive funding from the DF2 IC to move into the next stage of detailed due diligence and project development. GCF proceeds will only be used for projects in countries for which an NOL has been obtained and to projects which contribute to the envisioned climate impact. Likewise, the proceeds from divestments will run through the CEF2's waterfall on per project basis, ensuring that the GCF proceeds are not comingled with those from projects in countries for which no NOL has been obtained.

Activity 2.1.4: Project development and extensive due diligence (including but not limited to stakeholder engagement, gender assessment, and Environmental and Social Impact Assessment.)

The actual development phase starts after the approval of the DF2 IC. During this phase, CFM will conduct, supported by independent parties, extensive due diligence. This due diligence will include, but won't be limited to, stakeholder engagements, gender assessments, and Environmental and Social Impact Assessments (ESIA). The ESIA will then judge whether the projects have any resettlement risk (which is not expected to happen often) after which a Resettlement Action Plan (RAP) would be drafted. It's during this phase that all legal and regulatory permits are obtained and that the contractors are selected. This phase needs to be completed, on all important aspects, before the construction phase can start.

Activity 2.2: Implementation (construction and operation) of mitigation and adaptation projects for water, sanitation, and ocean sectors in developing countries.

Activity 2.2.1: Project approval by CEF2 IC

Upon successful completion of the development case, the deal team re-proposes the project to the CEF2 IC. This proposal will include all the findings off the due diligence and evaluations done in Activity 2.1.4. It is only after approval of the CEF2 IC (including, for the avoidance of doubt, the satisfactory independent screening conducted by a third party and satisfactory extensive due diligence) that a proposed investment would receive funding from the CEF2 IC. GCF proceeds will only be used for projects in countries for which an NOL has been obtained and to projects which contribute to the envisioned climate impact. Likewise, the proceeds from divestments will run through the CEF2's waterfall on per project basis, ensuring that the GCF proceeds are not comingled with those from projects in countries for which no NOL has been obtained.

Activity 2.2.2: Project construction:

The CEF2 will then repay the development loan from the DF2 (including a premium to ensure full revolvability of the DF2) and commit equity to start a project's construction phase after the successful approval of the CEF2 IC. During the construction phase, CFM will actively manage and oversee the project, in example by nominating directors or representatives on the boards or sub-committee of each investee company, to ensure optimal efficiency, (ESG) compliance, and maximum value creation.

Activity 2.2.3: Project refinancing exit and exit:

After successful construction, the project moves to the operational phase. Once a project is stable in terms of operations, impact, and cash flows, a process will commence procuring debt at the project level and, a few years later, the project's exit will be arranged following approval from the CEF2 IC. These two events will recapitalize CEF2 and enable it to commit funding to further projects.

Outcome 2: Ameliorated GHG emissions in the water, sanitation, and ocean sectors of the Programme's host countries (outputs 2, 5, 6, and 7).

Output 2: The Programme will avoid up to ~16.7 million tCO₂e in emissions by developing and constructing waste and wastewater treatment facilities,

Output 5: The Programme will develop and construct waste-to-energy projects that will produce ~314.3 GWh/yr in renewable energy (with a power capacity installed of (~111.5MW) providing up to ~ 314.3 thousand people with power. In addition, the Programme will develop and construct wastewater treatment, both municipal and industrial, that will produce 1,510.3 GWh/yr (with a power capacity installed of 147.0MW) contributing to the reduction of ~592.1 thousand tCO₂e annually and provide ~1,5 million people with power.

Output 6: The Programme will avoid up to ~955.8 thousand tCO₂e in emissions annually by developing and constructing energy-efficient infrastructure for green ports and/or shipping. Examples would be climate-resilient port infrastructure, onshore power, waste and wastewater treatment in ports, port infrastructure for off-shore wind farm assembly, load out and servicing, and de-carbonization of coastal and inland shipping.

Output 7: The Programme will restore and/or protect ~1.8 million hectares of riverine, marine, wetland, and coastal ecosystems, contributing up to ~2.2 million tCO₂e emissions avoided or sequestered annually. Furthermore, an estimate of ~2400 beneficiaries will benefit from restored and/or improved ecosystems.

In total, the Programme is expected to reduce up to ~ 44.65 million tCO₂e of emissions over its lifetime, with an annual impact of ~ 4.96 million tCO₂e in year 20.

The outcome will also contribute to outcome 1, further private sector involvement, by expanding the track record of private sector investments in a previously primarily public sector. Successful development and construction of waste-to-energy projects and protection and/or restoration of ecosystems will further enable private sector investments by reducing the (perceived) risks investors associate with the sector.

Outcome 3: Beneficiaries with climate-resilient water security by enhancement of the water, sanitation, and ocean sectors in host countries (outputs 3, 4 and 7).

Output 3: The Programme will provide up to ~ 11.18 million people with access to climate-resilient water infrastructure by developing and constructing infrastructure for bulk water supply and storage, ocean-water desalination, brackish groundwater desalination, groundwater abstractions, atmospheric water generation, water treatment facilities, water utilities, and concessions, pumping household water connections and decentralized water treatment.

Output 4: The Programme will provide up to ~1.97 million people with climate-resilient sanitation by developing and constructing infrastructure to collect wastewater, building sewerage networks, and treating and recycling waste.

Output 7: The Programme will improve the livelihoods of communities by constructing green infrastructure investments that restore and/or protect marine, riverine, coastal, or wetland ecosystems. Examples would be wetland restoration to improve water retention, flood control, and restoration of mangrove forests to provide coastal protection. The combined efforts will restore and/or strengthen up to ~1.8 million hectares of vulnerable ecosystems

Co-benefit 1: The Programme will contribute to health and sanitation. (output 2, 3, and 4)

Combined, outcomes 1, 2, and 3 are aimed at contributing to CI2's goal statement:

If Climate Investor 2 investments result in the scaling and replication of water, sanitation, and oceans infrastructure projects aligned to the adaptation and mitigation needs of developing countries, then low-emission, climate-resilient sustainable development will be catalyzed in the programme's host countries because successful climate investments in the target sectors, with an attractive risk-adjusted return, will be demonstrated to commercial investors fostering significant contributions to blue and green climate finance in emerging markets.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Flow of funds

Fund flows for investments

- The GCF and FMO will contractually agree that the GCF will commit USD 145 mln. In reimbursable grants, from which USD 100 mln. Is subscribed to a junior equity position (being the Tier 1 tranche of the CEF2), on terms to be agreed in the Funding Activity Agreement. For the avoidance of doubt, all the GCF funds for the purpose of CI2 will be kept on a separate ledger on FMO's GCF account.
- FMO will then make a USD 145 mln capital commitment. In reimbursable grants to the Development Fund. Subsequently, the Development Fund will make a USD 100 mln capital

commitment to the Tier 1 tranche of the CEF2 (which is its most junior capital tranche). The remaining USD 45 mln. Will remain within the DF2.

- FMO will request the GCF funds on a bi-(annual) basis based on CFM's estimates on the liquidity needs for the pipeline and portfolio.
- Based on estimates for capital requirements for approved DF2 or CEF2 investments as well as for costs and fees, the Manager will issue drawdowns to FMO.
- Upon receipt and after analyzing the compliance to its internal policies and procedures, FMO will remit the funds to the DF2 from its fiduciary GCF account. The DF2 can then remit the funds to the Construction Equity Fund, to the extent required.

Fund flows for distribution

- Upon partial or full divestment of a portfolio investment, a refinancing of equity by debt, or any other interim income received by the CCEF2, the funds will be remitted to the DF2 as per the distribution waterfall. At the end of the DF2 fund term and upon its dissolution, to the extent that funding remains, the DF2 will distribute funding received in accordance with the applicable law for Dutch trusts ("Stichting") to its contributors including to FMO in respect of any GCF funding (see section B.6 for more details).

Legal arrangements

FMO, in its capacity as Accredited Entity, will ensure that the following agreements are entered into for the implementation of the Funded Activity (each a "Subsidiary Agreement" and together the "Subsidiary Agreements"):

- A side letter between the Accredited Entity, the Manager, CEF and DF, respectively, to pass down to the DF and CEF all relevant obligations in accordance with the AMA and FAA (a "**Side Letter**"); the Side Letter shall also set out the mechanism pursuant to which the Manager shall seek the Accredited Entity's consent before presenting relevant matters (which may require GCF's approval) for approval to the relevant DF and CEF governing and decision-making bodies, and the Accredited Entity shall not provide such consent to the Manager unless it has sought the prior written approval of the GCF;
- The donors' agreement between the Manager, DF and donors to the DF, which agreement governs the rights and obligations of donors, as well as the governance and commercial arrangements between the Manager and DF ("**Donors' Agreement**");
- For the flow of the GCF Proceeds from FMO to the DF a subscription agreement entered into by FMO in relation to the DF, whereby FMO commits and the Manager accepts the relevant portion of the GCF Proceeds ("**DF Subscription Agreement**");

FMO, in its capacity as Accredited Entity of the Fund, will ensure that the following agreements are entered into for the implementation of the Funded Activity (each a "**Key Fund Document**" and together with the Subsidiary Agreements, the "**Key Fund Documents**"):

- Separate management agreements between CFM and the DF and CEF respectively pursuant to which CFM will agree to act as the manager of the DF and CEF (each a "**Management Agreement**");
- The Members' Agreement, between the Manager, CEF, investors to the CEF, which governs the rights and obligations of investors, as well as the governance and commercial arrangements between the Manager and CEF ("**Members' Agreement**");

- For the flow of the GCF Proceeds from the DF to Tier 1 of CEF a subscription agreement entered into by the DF in relation to the CEF, whereby the DF commits and CFM (as Manager of CEF) accepts the relevant portion of the GCF Proceeds ("**CEF Subscription Agreement**");
- A set of Articles of Association in respect of each of the DF and the CEF governing corporate legal matters (e.g. membership, representation, board composition, voluntary liquidation etc.) ("**DF Articles of Association**" and "**CEF Articles of Association**", respectively).

The terms of such Subsidiary agreements shall be consistent with the AMA and the Term Sheet (Annex 14), in particular in respect to the provisions set out in Schedules 1 & 2 to the term sheet.

Organizational structure & team experience

CFM is organized along regional and functional lines which combines; (i) a strong investment and operational focus with industry and local expertise and (ii) solid middle and back-office skills in operations, finance, and strategy (iii) sound risk management, coordinated legal / tax structuring, and robust compliance functionalities.

CFM will operate from its offices in the Hague, the Netherlands (head office), Cape Town, Singapore, and Bogota (regional offices) as follows:

Head office

CFM's head office is located in The Hague, Netherlands, and was established in 2015. At present the office has 24 full-time employees, with Andrew Johnstone, acting as CEO. The responsibilities of the Hague Office include (i) strategy & central oversight, (ii) tax & valuations, (iii) fund management, (iv) investor relations & capital raising, and (v) central environmental, social & governance management.

Regional Offices

CFM's regional offices are led by regional heads with strong networks and specific skills and experience in discrete geographies and sectors, thereby providing the regional teams with regional intelligence. Three regional offices have been established, in South Africa, Colombia, and Singapore.

CFM's regional office for Africa is located in Cape Town, South Africa. Established in 2017 the office has 19 full-time employees, with Sebastian Surie, Head of African Investments, acting as office head. The responsibilities of the Cape Town office include all investment activity in Africa, CFM's risk, compliance, and finance functions, as led by the CFO, and CFM's operations & human resources led by Ms. Caron Aitken.

CFM's regional office for Asia is located in Singapore, where it operates as Climate Fund Advisers (CFA). Established in 2017 the office has 14 full-time employees, with Nathan Schmidt, Head of Asian Investments, acting as office head. The Singapore office is responsible for all CFM's investment activity in Asia.

CFM's regional office for the LAC region was established towards the end of 2021 and is finalizing arrangements for its office in Bogota, Colombia. The team consists of 4 full-time employees and is led by Juan Paez, Head of Latin American Investments. The LAC office is responsible for all investment activity in the Latin American and Caribbean region.



Governance Arrangements

The Governance structure of CI2 has been designed to ensure appropriate supervision and investor representation to the Coöperatief CI2 Construction Equity Fund U.A. ("CEF2") and Stichting CI2 Development Fund ("DF2"), to guarantee adequate expertise is procured into each forum, as well as to create alignment of interest and control. The full governance structure of CFM and its managed vehicles can be seen below. It should be noted that any seat the GCF would acquire as per their rights will be filled by a representative of the AE at the request of the GCF.

Roles and responsibilities of AE and EE

Accredited Entity

The AE for the Programme is:

- FMO

FMO, in its role of AE, will oversee the Programme implementation and will ensure that relevant GCF policies are passed down to the Fund level and portfolio company-level. Specifically, it will dedicate full-time team members to monitor EE's compliance with FAA and AMA obligations, including retro-active policy compliance to the GCF's policies, to analyze quarterly and annual performance reports, to disburse GCF (equity) proceeds, and represent the GCF on CI2's Donors/Investors Meetings and CI2's Advisory Board. It will assess achievement of Programme, targeted outcomes, and results as per the log-frame, while commissioning independent interim and final evaluation reports as per GCF's requirements.

Executing Entities

The EEs for the programme are as follows:

- Climate Fund Managers B.V. (CFM)
- The Stichting CI2 Development Fund (DF2)
- The Coöperatief CI2 Construction Equity Fund U.A (CEF2)
- Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden N.V (FMO)

Roles and responsibilities of EEs

- Climate Fund Managers is the investment manager of Climate Investor Two and is responsible for investing the fund corpus and also managing the day-to-day activities of the Fund. The detailed role and responsibilities have been enumerated in the Fund documents.
- Stichting Development Fund is a ‘foundation’ set up in the Netherlands with the sole purpose of financing the development stage of projects with grant funding and making the Tier 1 Commitment of the Tier 1 tranche of the CEF2.
- The CEF2 is a cooperative established in the Netherlands designed to finance and expedite the construction phase of a project its Lifecycle with equity financing.
- FMO is the Accredited Entity for the Funding Proposal and will mainly focus on monitoring the EE’s (fiduciary) compliance on the Funded Activity, the FAA and the AMA, as well as discharging specific activities as an Executing Entity as will be further elaborated in the FAA.

Conflict of interest:

The AE, and the EE, are aware of potential conflicts of interest between FMO’s investments in CFM and the T2. The AE therefore always tests such constructs with its conflict-of-interest sounding board that sits in the compliance department. The AE, which has done many of such sub-delegation on behalf of Donors (e.g. the Ministry of Foreign Affairs, the European Commission, and the Green Climate Fund), has, therefore, set up Chinese walls, where appropriate, between the departments that do the investments from its balance sheet (in this case the Private Equity department) and the department that manages the investment on behalf of the Donor (the “Partnerships for Impact” department). The AE commits to delegating any positions within the governance of CI2, belonging to the GCF investments, to independent individuals in FMO. Furthermore, the AE will appoint a separate deal team to negotiate the FAA and other potentially sensitive documents.

Experience of the two parties:

FMO is the Dutch development bank and has invested in the private sector in developing countries and developing markets for more than 50 years. FMO invests in sectors with the highest long-term impact: financial institutions, energy, agribusiness, and infrastructure. FMO is a financial institution owned 51% by the Dutch State, 42% by Dutch banks, and 7% by employers’ associations, trade unions, and individual investors.

With an investment portfolio of ca. EUR 10 billion spanning over 85 countries, FMO is one of the larger bilateral private sector development banks in the world. FMO’s strategic goal is to be the preferred partner to invest in local prosperity.

FMO currently has two projects under implementation with the Green Climate Fund: FP099 (or Climate Investor One) and FP164 (or: the Green Growth Equity Fund). Alongside GCF funding, FMO also manages funding on behalf of the UK government (BEIS), USAID, the Dutch Ministry of Foreign Affairs, and the European Commission. FMO has been accredited with the Green Climate Fund since 2016, has signed its AMA in 2018, and its first FAA in 2019.

Climate Fund Managers is a B.V. incorporated under the laws of The Netherlands (hereafter “Climate Fund Managers” or “CFM”), incorporated on 30 October 2015, headquartered in the Hague, the Netherlands. CFM is a joint venture between the Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden N.V. (“FMO”) and Sanlam InfraWorks. Climate Fund Managers has successfully set up Climate Investor One (or: FP099) with a final close of nearly USD 1 bln. In 2019. Climate Investor

One focuses on renewable energy projects in Africa, Asia, and Latin America. CI1, in which the GCF is also invested, currently has over USD 400 mln. In commitments invested in over 7 different projects.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Worldwide 780 million people still do not have access to an improved water source. Demand shortages are more pronounced in Africa and Asia. Furthermore, in developing countries around 3.2 million children still die each year because of unsafe drinking water and poor sanitation. Waste is the fourth largest sector of all greenhouse gas emissions. Water and sanitation infrastructure are arguably the most important target sectors for climate adaptation interventions. However, adaptation finance faces a severe funding gap. The Economist estimates that USD 140bn to USD 300bn in climate adaptation finance is needed annually by 2030^{xxiii}. It is therefore of great importance to design water and sanitation infrastructure in a way that it i's: (i) robust in the face of climate-driven shocks – thereby delivering climate adaptation benefits and (ii) reducing GHG emissions by creating (bio)energy from the waste and wastewater that they treat.

These projects, whether designed by the public or the private sector, need to be implemented in a way that delivers on the additional climate requirements while keeping the costs of the services affordable to all those in society. This means that there's a need for both: (i) mobilizing the private sector to bridge the finance gap and (ii) making the services accessible and affordable for all.

CI2 has been designed to do the above by blending concessional funding with private funding in two structured funds. The CEF2 structure mobilizes private finance by reducing the risk with a junior-equity tranche (the T1) and reduces construction time of the required infrastructure by providing a life-cycle funding approach in conjunction with the DF2. The CEF2 has been modeled in a way that accommodates multiple investors' risk-return preferences while minimizing concessionality as much as possible. The three tranches are structured as follows:

- Tier 1 (T1) makes up [20%] of CEF2 and is comprised of junior equity.
- Tier 2 (T2) makes up [40%] of CEF2 and is comprised of ordinary equity.
- Tier 3 (T3) makes up [40%] of CEF2 and is comprised of senior equity.

The GCF funding will play a key role in enabling CI2 by providing a significant investment into Tier 1 for which funding is the scarcest. The GCF investment will therefore immediately mobilize the private sector in T2 and T3 by taking away some of the risks withholding commercial investors.). T1 ameliorates the risk associated with investing in climate-related infrastructure in developing countries to catalyze commercial investors. An important part of CI2's investment mandate is LCY denominated revenues. LCY risk is expected to be more profound in a larger part of CI2's portfolio than it was for CI1. Commercial investors are not keen on taking LCY risks in developing countries. The T1 tranche is therefore important to absorb and buffer any (perceptions) of LCY risks to mobilize commercial investors into T2.

In addition to the investments made by the GCF funding in water and sanitation infrastructure, there is also an urgent need for innovative financing solutions for the emerging sector of ecosystem-based approaches, or green infrastructure. In this regard, the GCF funding to the DF2 will enable CI2 to develop and finance ecosystem projects for which there is currently a limited track record, and practically no interest of the private sector, creating a demonstration effect that could open this sector for future private sector investment.

A GCF commitment to CI2 would unlock 31=4x total investment at the project level (see Investment Leverage Ratio below) for climate change adaptation and mitigation. CFM is applying for a total GCF contribution to CI2 of USD 145mn. An investment of this size provides GCF the opportunity to become the largest donor in the CI2 structure, potentially generating the highest additionality (and impact) of the CI2's members. Through the CI2 intervention, CFM has identified the following channels of GCF additionality through a CI2 intervention:

- Investment by GCF will support CI2's investment focus to a subset of countries, for which NOLs have been obtained, that have a greater need for intervention that fits within their NDCs
- A GCF intervention in the DF2 will catalyze a facility that offers scarce early-stage funding for project development including high-impact ecosystem restoration projects co-located to commercial projects. Access to development capital is one of the main unique selling points of CI2. By providing scale to the DF2 facility, GCF will enable CI2 to facilitate the development of a greater number of projects, all of which offer CEF2 a preferential financing opportunity.

B.6. Exit strategy (max. 500 words, approximately 1 page)

Sustainability of the paradigm shift

Through organic development of developing countries' financial markets and with a CI2 demonstration effect it is expected that other developers and equity funds will adopt a similar approach. Under these presuppositions, they are expected to fund a greater share of construction costs through an all-equity strategy where this is preferred over a traditional project finance approach. This may reduce costs across the market, both through lower equity risk premiums and by introducing lower-cost debt for projects. GCF funding would be instrumental to this market development through its catalytic impact in CI2 and the project companies in which CI2 invests.

The projects and companies financed through CI2 will focus on commercially viable assets with inherently long-lived technology and characteristics, in the range of 20-25 years. The commercial focus of CI2's investments ensures a long-term interest by owners and investors in the projects. In contrast to grant financed projects, satisfactory financial returns will only be realized if the projects continue to provide strong operational performance and goods and services that are valued by their users. By contrast, failure to ensure this – for example through poor maintenance – will have contractual or legal ramifications.

The sustainability of the paradigm shift will be further imprinted by involving local banks, potentially including Direct Access Entities ("DAE's), in the re-financing of the Programme's sub-projects after they have become operational.

Figure 2 – CI2 channels of long-term sustainability



Measurement and verification of the effectiveness of the projects in which CFM invests will be conducted in accordance with industry-standard protocols including the Greenhouse Gas Emissions Protocol, BS: EN 16247 (International Standard for Energy Auditing) and the International Performance Measurement and Verification Protocol (IMPVP). The performance will be reported to its various stakeholders, including investors, donors, the CI2 investment committees, and CFM's shareholders. CFM requires all projects in receipt of Construction Equity Fund financing to report in accordance with defined key performance indicators. In addition, the performance measurement framework indicators aforementioned will be incorporated into fund-level and project-level monitoring and evaluation activities that will be managed by CFM.

C. FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	145			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter</u> %		
(ii) Subordinated loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter</u> %		
(iii) Equity				Enter%		
(iv) Guarantees	<u>Enter amount</u>	<u>Enter years</u>				
(v) Reimbursable grants	145					
(vi) Grants	<u>Enter amount</u>					
(vii) Results-based payments	<u>Enter amount</u>					
(b) Co-financing information	Total amount			Currency		
	735			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Tier 3 investors	Equity	<u>320</u>	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>		<u>senior</u>
Tier 2 investors	<u>Equity</u>	<u>320</u>	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>	Enter	<u>senior</u>
Tier 1 investors (investments from Donors routed through and managed by FMO / the Accredited Entity)	<u>Equity</u>	<u>60</u>	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>		<u>junior</u>
The Development Fund (investments from Donors routed through and managed by FMO / the Accredited Entity)	<u>Reimbursable Grants</u>	<u>35</u>	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>		<u>junior</u>
©) Total finance (c) = (a)+(b)	Amount			Currency		
	<u>880</u>			<u>million USD (\$)</u>		

(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 pages)	The below refers to the entire Climate Investor Two program. The GCF Funding Proposal however only partakes in 80% and 'GCF's CI2' will therefore add up to USD 800 mln. for the CEF2 and USD 80 mln. for the DF2. CFM aims to raise up to 1,100 mln in funding for CI2. This consists of USD 100 mln. in reimbursable grants for the DF2 and USD 1,000 mln. in funding for the CEF2. CI2, being the DF2 and the CEF2, had its first close, for a total of USD 675 mln, in November 2021¹¹. This included investments from the European Commission, the Dutch Fund for Climate and Development (a climate adaptation fund from the Dutch government that's managed by FMO on behalf of CFM, WNF and SNV), IMAS Foundation, BNG bank, Sanlam, FMO, Nordic Development Fund, and KLP. FMO, the AE for this proposal, has invested USD 40 mln. directly into the T2. The DFCD and the European Commission investments, which have both invested in the DF2 and the T1, are also managed by FMO on behalf of either Donor. For the avoidance of doubt, all onboarded investors have successfully passed the AE's and EE's KYC policies. As explained in sections B.1, B.3, and B.5, the CEF2 is a structured fund with three different tiers with different investors that have different risk/return profiles. The first tier, or 'Tier 1', constitutes as soft-equity that has been provided by Donors and arguably the most additional. The second tier, or 'Tier 2', constitutes of impact investors and DFI's. The third most senior tier, or "Tier 3", constitutes of institutional and private investors. The ratio between the three tiers is set at 20/40/40. Whenever one or two out of the three tiers exceed that ratio that capital remains 'locked'. The total 'unlocked' commitments equal the value of the maximum value for which the 20/40/40 can be preserved. The initial close primarily consisted of T2 (USD 151 mln.) and T3 (USD 400 mln.) investments. This implicates that only USD 315 mln. of the CEF2's capital is currently 'unlocked' (as the 'unlocked' value is the maximum value for which 20/40/40 can be preserved). GCF participation in the T1 will therefore mobilize a large portion of private finance without requiring additional T3 investors coming in at later closes.
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C.2. Financing by component							
Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
<u>Establishment and management of Climate Investor Two</u>	<u>Further private sector investment in climate change adaptation and</u>	0	0	<u>Equity</u>	0	<u>Equity</u>	0

¹¹ The USD 675 mln. includes USD 19,2 mln. side pocket for 'frontier investments' that are specific to the investment of the Dutch Ministry of Foreign Affairs (via the Dutch Fund for Climate and Development). This USD 19.2 mln. is not counted in the above.

	<u>mitigation projects</u>						
<u>Development and implementation of climate mitigation and adaptation projects</u>	<u>Development of mitigation and adaptation projects and ecosystem building</u>	<u>80</u>	<u>45</u>	<u>Reimbursable grants</u>	<u>35</u>	<u>Reimbursable grants</u>	<u>Donors / T1 investors -</u>
	<u>Implementation of mitigation and adaptation projects</u>	<u>800</u>	<u>100</u>	<u>Reimbursable grants</u>	<u>700</u>	<u>Equity</u>	<u>Tier 1, tier 2, and tier 3 investors</u>

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 pages)

C.3.1 Does GCF funding finance capacity-building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

In some cases, the CI2 Fund will invest in companies that operate and have developed technologies in developed (or developing) countries, and CI2's investment proceeds will be used to deploy such technologies in the target countries. The technology and knowledge will be transferred by working together with local partners, developers, and third parties. Lastly, CFM will work together with local financial institutions, including DAEs, to refinance the projects after they have successfully graduated from the CEF2.

In terms of capacity building, the GCF proceeds in the DF2 can be used for the likes of feasibility studies, gender assessments, ESIAPs and other important capacity-building activities.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/Programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

- GHG emissions avoided: The Programme will reduce or avoid ~4.96 million tCO₂e per year, for a total of ~ 44.65 million tCO₂e over the lifetime of CI2. This consists of avoiding GHG emissions by (i) utilizing waste and wastewater as a source of renewable energy and waste to energy (constructing up to 258.74MW in Power Capacity and producing ~1824.65 GWh/yr) in total, (ii) reducing GHG emissions with fuel-efficient infrastructure for ports and green shipping and by (iii) reducing and/or sequestering GHG emissions by protecting and/or restoring ecosystems.
- Beneficiaries: The Programme will provide ~ 11.18 million people with year-round access to reliable and safe water supply despite climate shocks and stresses and restored and/or protected ecosystems. It will also provide ~1.97 million people with resilient infrastructure for sanitation. Lastly, it will provide ~1.82 million people with power.
- Hectares of ecosystems protected or restored: The Programme will protect and/or restore ~1.8 million Ha of ecosystems. This will improve the livelihoods of local communities and reduce GHG emissions by protecting or restoring nature's ability to act as a carbon sink.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

A paradigm shift for climate change occurs when there is a fundamental change in the way one, or many, perceive and respond to a climate change issue.

Climate Investor Two has the potential to cause such a paradigm shift at three levels: (i) the global development finance sector level, (ii) the investee country level, and (iii) at the sectoral level.

The first shift arises from the innovation of CI2's whole-of-life blended finance structure, further proving a new catalytic tool in the climate finance and sustainable development toolbox. Its application to a new thematic area sets a precedent for similar approaches to be implemented by other Development Finance Institutions, impact – and commercial investors. CI2's whole-of-life financing approach also eliminates a range of barriers to low carbon and climate-resilient development by ensuring that projects can progress smoothly and expeditiously from a concept or pilot project to a scaled, sustainable, and replicable operation with a clear function beyond CI2's exit. In addition, by demonstrating a replicable and adaptable approach to catalyzing private sector financing to climate-focused infrastructure, local project developers and international equity funds may adopt a similar approach building on CI2's precedent. CI2's potential to catalyze impact in this manner is amplified by the scarcity of infrastructure-themed climate adaptation funds, which despite the growing urgency for adaptation finance, can falter for a lack of established precedent. Bringing CI2 to market will provide recognition that climate adaptation-focused funds work in practice at scale, contributing to a re-balancing of the primarily mitigation themed funding available at present. Furthermore, as with CI1, the CI2 fund could further prove the scalability of blended finance.

At the investee country level, in some cases, CI2 will introduce the private sector into a space dominated by public funding and frequently facing shortfalls – in particular water supply & distribution, and ecosystem & oceans assets are typically publicly financed and/or operated. By being amongst the first to catalyze private climate finance to these sectors, CI2 will set a replicable precedent for the private sector across its thematic focus. In the long-term CI2 will also create opportunities for scalability and replicability beyond its

investments, through the transfer of knowledge and technical skills to local partners, and by contributing to improved regulatory and policy frameworks in the countries where it operates. Successfully demonstrating the commercial viability of ecosystems projects, that are traditionally public financed, will contribute to the development of proper accounting techniques for the actual economic value of ecosystems (i.e., by accounting for their ability to act as a 'carbon sink'). The lack of such accounting methodologies has been identified as one of the biggest potential paradigm shifts for ecosystems by the GCF^{xviii}.

Finally, at a sectoral level, CI2's projects will introduce innovative technologies and circular business models from source to sea, which will promote the preservation of water, and the climate-proofing of water infrastructure across the water cycle. CFM will work together with local partners, developers, and third parties to make sure CI2 contributes to technology and knowledge. Through the funding of sanitation projects, CI2 will also target reductions in ground and surface water pollution, removing a fundamental barrier to water reuse, and encouraging value creation and greenhouse gas (CH₄) capture from non-conventional waste and wastewater resources. Through the combination of these factors, CI2 targets long-term, systematic, and self-sustaining change in the sectors and regions where it is active.

Concluding, CI2 will provide a paradigm shift by fundamentally changing the way financial markets see adaptation-focused blended finance funds, demonstrating the commercial viability of sectors that were traditionally dominated by public funding and by transferring sectoral technology and knowledge to local communities.

D.3. Sustainable development (max. 500 words, approximately 1 page)

CI2 has identified the following SDGs as primary goals:

- Climate Action (SDG 13)
- Affordable and Clean Energy (SDG 7)
- Clean Water and Sanitation (SDG 6)
- Partnerships for the Goals (SDG 17)

By contributing to the above – CI2 sees potential co-benefits in the following SDGs:

- Good health and well-being (SDG 3).
- Gender Equality (SDG 5)
- Decent Work and Economic Growth (SDG 8)
- Industry, Innovation, and Infrastructure (SDG 9)
- Life below water (SDG 14)

Environmental co-benefits:

- Decontaminating the earth's water will increase biodiversity and improve the livelihoods and resilience of local communities.

Social co-benefits including health co-benefit:

- Safe access to water and sanitation will reduce the spread and severeness of diseases related to contaminated water and/or lack of sanitation.
- Development, construction, and operation of the infrastructure projects will leverage existing local capabilities and knowledge.
- Community Development programs planned and implemented alongside CI2 projects will tackle core issues identified by stakeholders in communities where CI2 is active.

Economic co-benefits:

- The construction and operation phases of all CI2's projects, combined with the required ESG compliance, will create fair jobs for local communities.
- Demonstration of the commercial viability of infrastructure projects will lead to a capital inflow/increase in GDP.
- Investing in innovative technology will lead to technological advancements.

Gender-sensitive development benefit:

- Gender assessments will be done for every project to ensure active inclusion of women in the entire sphere of influence of the project
- Access to water and sanitation will improve the livelihoods of women (as they are now often affected in an inequitable manner)
- Commitments in the Gender Action Plan will increase formal employment possibilities, improve working conditions, and create greater gender parity for women. This will reduce dependence on individual livelihood activities and enhance financial resilience for women.

D.4. Needs of the recipient (max. 500 words, approximately 1 page)

The Programme targets 18 different countries, spread over Latin America, Asia, and Africa, that all have significantly different scales and intensities of climate vulnerability (including water-stressed areas) . The Programme will therefore assess the needs, and the intervention required, on a case-by-case and per country basis.. The 18 countries have been identified on their National Designated Contributions (NDC) and the fit these have with the product offered by CI2. A detailed climate vulnerability assessment per country can be found in Annex 7.

D.5. Country ownership (max. 500 words, approximately 1 page)

FMO and CFM acknowledge that the point of departure for selecting a target/focus country is whether Country Ownership is encapsulated, specifically through the following three elements:

- Host nation climate plans, including Nationally Designated Contribution (NDC) commitments (and any subsequent updates) and the National Adaptation Plans, as well as sector plans from relevant ministries, including water, sanitation, energy, environment, and others as may be applicable. The countries selected by CFM recognize the importance of water, waste, and renewable energy, as highlighted in Annex 1. The initial short-listed countries cite specific targets to combat climate change (with a specific focus on adaptation) across various short, medium, and long-term periods. In addition, such short-listed countries align with CFM's regional office's investment opportunities, and therefore CI2 has identified potential projects that could qualify for funding.
- Positive and ongoing stakeholder engagement: CFM to date has built a reputable relationship with many of the identified country NDAs and other relevant stakeholders on the back end of receiving NOLs from such countries for the Climate Investor One (CI1) Programme. CI2, therefore, allows the CFM team to continue such engagements, with the intent to align such countries' climate change needs to the investment thesis of CI2. To further demonstrate Country Ownership at the project level, CFM is actively engaging with governments, local communities, and the private sector. This ongoing engagement process is fluid; hence it is conducted to attain broad continued support within these

target jurisdictions of CI2's investment activities. This process will be tailored per country as per the requests of the relevant NDA. CFM already has such processes in place with respective NDAs for CI1.

- **No Objection Letter (NOL) support:** In a similar mechanism to Climate Investor One, CFM and FMO will seek governmental support in the form of a No Objection Letters (NOL) and as per the Concept Note have approached twenty-five (25) countries to request NOLs. This process actively started in January 2021 and entails active engagement with the respective NDAs and other relevant stakeholders to ensure an alignment of interests. A summary overview of CI2 was shared with each NDA, along with the Concept Note. CFM has subsequently been in regular dialogue with NDAs and country representatives to address any clarifications that arise and will continue to do so in preparation for the GCF Board meeting, with the intent to try and secure all 25 NOLs. Should NOLs not be issued in time, additional countries might be added to the Programme after initial Board approval, provided that they submit an NOL and that such NOL is approved by the Board.

It is also the Managers, and AE's, ambition to include interested Directed Access Entities ("DAE's") to the pool of lenders that will constitute the Refinancing Fund after it has been set up. This will ensure further alignment of the Programme and the respective country ownership.

The D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

The GCF's investment has an expected mobilization ratio of 1:4 at the fund level (as every USD of T1 will mobilize 4 USD in the more senior tranches). A mobilization ratio of 1:4 is on the higher end of the benchmark and equal to the CI1 proposal^{xix}.

The below assumes an average 66% ownership of CI2 in the construction phase and two project life cycles during the implementation period.

GCF Leverage Ratios

Investment leverage ratio = value of investment (total project cost) divided by the total amount of GCF support relating to this investment.

The investment leverage ratio (referring to the total investments at project level catalyzed by the GCF contribution) = A / B which equals 10x whereby:

A = the DF2 (USD 100 million) + the CEF2 (USD 1 billion) + Partner Contributions (USD 363 million)

B = the GCF contribution (USD 145 million).

Including the 'recycled' funding from the CEF2, this equates to 20 whereby:

A = the DF2 (USD 180 million) + the CEF2 (USD 2 billion) + Partner Contributions (USD 727 million)

B = the GCF contribution (USD 145 million).

Private sector finance leverage ratio = amount of private sector financing in the investment project divided by the amount of GCF support. The private loans/equity leverage ratio (referring to the total investments made by the private sector catalyzed by the GCF contribution) = A / B which equals 16x whereby:

A = Private sector investors in CEF2 recycled (USD 800 million + USD 800 million) + private sector partners at project level recycled (USD 727mn).

B = the GCF contribution (USD 145mn).

CI2's primary efficiency and effectiveness benefits are derived from its whole of life financing approach:

- Funding made available by CI2 at all stages of the project including early-stage development and for pilot schemes, generates a pipeline of projects whilst building the capacity of local developers.
- The recycling character of CI2 enables more projects to get developed and constructed during the implementation period of the fund (the capital is expected to be recycled twice).
- The whole of life approach reduces the time project developers usually lose at fundraising at the different phases of project development.
- Flexibility to use all-equity financing during the construction phase of the project. This eliminates cash flow and financial distress from the project during construction, whilst also ensuring the long-term interest of the owners and investors in the project.
- Leaner capital expenditure of CI2 projects through exclusive technology partnership agreements with Original Equipment Manufacturers (OEMs), unlocking economies of scale for equipment sourcing, installation, and maintenance.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

- ☒ Reduced emissions (mitigation)
- ☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/Programme will contribute (Description)
	Description	Rating		
Scale (size of projects)	The water, sanitation, and ocean sectors have traditionally been primarily financed within the public domain. There's therefore often insufficient private finance available as private investors still perceive the sectors as too risky (i.e., because of a lack of projects that demonstrated bankability). This makes it difficult for projects, especially larger-scale projects, in the water, sanitation, and oceans sectors to receive enough funding. This is reflected in the average transaction size in the water and sanitation sector (an average of USD 76 mln. versus USD 301 mln. for all blended finance).	<u>Medium</u>	The potential target scenario would be a situation in which projects within the water, oceans, and sanitation infrastructure sectors in developing countries can reach enough scale as investors become more comfortable with the bankability and commercial viability of the (larger) projects.	The Programme will contribute to the potential target scenario by (i) mobilizing private finance at scale by offering a blended finance solution that caters to investors with different risk to return profiles and (ii) by reducing the perceived risks of investing in the water, sanitation, and oceans sectors by successfully developing and constructing bankable large-scale projects (demonstration effect)
Replicability (of CI2 / blended finance within the Sectors)	The water and sanitation sector has strong potential to strategically adopt innovative finance approaches like blended finance beyond CI2. Access to safe drinking water and sanitation is a basic human right and critical to ensuring human welfare and	<u>Medium</u>	The potential target scenario would be a situation in which a precedent is set whereby CI2 demonstrates the application of blended finance as a strategic option for the water and sanitation sector. In doing so, this could be further replicated by other entities, triggering the application of innovative financing	The Programme will aim to help achieve the potential target scenario by replicating the successful, and replicable, structure of Climate Investor One, in the Water, Sanitation, and Oceans Sectors. This could take form in different ways: (i) by establishing a precedent for blended finance at scale in the water, sanitation, and oceans sectors, (ii) by

	unlocking economic growth and productivity. Goal 6 (Clean Water & Sanitation) of the Sustainable Development Goals (SDGs) aims to ensure access to safe water sources and sanitation for all; however, the World Bank estimates that 114 USD bln per year would be required to meet this target. Despite this financing shortfall, the water and sanitation sector has not been as quick as other sectors to adopt alternative financing sources, including blended finance approaches to mobilize commercial capital.		solutions to address the 114 USD bln shortfall which currently exists.	replicating the all-equity full lifecycle approach of the Construction Equity Fund and (iii) by creating opportunities for co-investing alongside CI2.
Sustainability	The water, sanitation, and oceans infrastructure in developing countries are currently insufficiently equipped to use their potential for avoiding GHG emissions while dealing with the consequences of climate change.	<u>Medium</u>	The target scenario is a situation in which the infrastructure developed and built remains operational and resilient even after the implementation period of the Programme.	CI2 will focus on commercially viable assets with inherently long-lived technology and characteristics, in the range of 20-25 years. CI2 will ensure strong operational performance and goods and services that are valued and affordable by and for the users. Furthermore, the assets will cause a knowledge transfer and demonstration effect.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)^{12 13}

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ¹⁴	

¹²The stated [targets](#) are [the best preliminary estimates available at the FP design stage](#) and [are subject to change based](#) on both the number of NOLs and [sub-projects](#) achieved in the target countries.

¹³ The impact numbers are derived by taking 80% of the estimated impact of CI2 as a whole (as the GCF will only be an investor in 80% of the Projects due to the NOL requirement).

¹⁴ The final target means the target at the end of project/Programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

<u>MRA1 Energy generation and access</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Portfolio data (as provided in Annual Reports) and an independent verification / assurance before the interim and final evaluations.	0	~ 3.05 million tc2Oe	~ 16.7 million tCO2e	Amount of GHG avoided by waste to energy and wastewater treatment projects. This has been calculated by looking at existing projects (mainly from the GWI database). The implementation period of this Funding Proposal is equal to the lifespan and the outcome is therefore estimated at the end of the lifespan. Further calculations can be found in Annex 22.
<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.3: Installed renewable energy capacity</u>	Portfolio data (as provided in Annual Reports) and an independent verification / assurance before the interim and final evaluations.	0	~107MW	~ 259MW	Waste to energy and wastewater treatment projects installed. This has been calculated by looking at existing projects (mainly from the GWI database). The implementation period of this Funding Proposal is equal to the lifespan and the outcome is therefore estimated at the end of the lifespan. Further calculations can be found in Annex 22..
<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.4: Renewable energy generated</u>	Portfolio data (as provided in Annual Reports) and an independent verification / assurance before the interim and final evaluations.	0	~751GWh/yr)	~1,825 (GWh/yr)	Waste to energy and wastewater treatment projects RE generated. This has been calculated by looking at existing projects (mainly from the GWI database). The implementation period of this Funding Proposal is equal to

						the lifespan and the outcome is therefore estimated at the end of the lifespan. Further calculations can be found in Annex 22..
<u>MRA3 Buildings, cities, industries and appliances</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Portfolio data (as provided in Annual Reports) and an independent verification / assurance before the interim and final evaluations.	0	~1.6 million tCO ₂ e	~8.6 million tCO ₂ e	Amount of GHG avoided by green ports and shipping projects. This has been calculated by looking at existing projects (mainly from the GWI database). The implementation period of this Funding Proposal is equal to the lifespan and the outcome is therefore estimated at the end of the lifespan. Further calculations can be found in Annex 22..
<u>MRA4 Forestry and land use</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Portfolio data (as provided in Annual Reports) and an independent verification / assurance before the interim and final evaluations.	0	~ 3.5 million tCO ₂ e	~19.4 million tCO ₂ e	Amount of GHG avoided or sequestered by restoration and/or protection of ecosystems. This has been calculated by looking at existing projects (mainly from the GWI database). The implementation period of this Funding Proposal is equal to the lifespan and the outcome is therefore estimated at the end of the lifespan. Further calculations can be found in Annex 22..

<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Will depend on the sub-project but at a minimum by independent verification.	0	~720 (from which 450 females)	~1,920 from which 1200 females)	A correct KPI for direct beneficiaries will be formulated on a per (sub)-project basis. The numbers mentioned here are conservative estimates for one of the current pipeline projects.
	<u>Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice</u>	Will depend on project but at a minimum by independent verification.	0	~586 thousand Ha of marine ecosystems and ~ 147 thousand Ha of freshwater ecosystems	~1.4 million Ha of marine ecosystems and ~356 thousand Ha of freshwater ecosystems	The amount of hectares brought under protection or restoration has been estimated by multiplying the expected portfolio allocation with the average amount of Ha put under protection and/or restoration per USD of investment in reference projects
	<u>Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under resoration and/or improved ecosystems</u>	Will depend on project but at a minimum by independent verification	0	~586 thousand Ha of marine ecosystems and ~ 147 thousand Ha of freshwater ecosystems	~1.4 million Ha of marine ecosystems and ~356 thousand Ha of freshwater ecosystems	The ecosystems concern: coastal marine ecosystems (80%) and freshwater ecosystems (20%)

<u>MRA1 Energy generation and access</u>	<u>Core 3: Value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions</u>	Investment manager report	0	~USD 628.8 mln.	~USD 1.258 bln.	Sum of the expected project values at moment the CEF2 divests. Assumes an average 66% ownership of CI2 and two project life cycles. (total of ~ USD 2.96 billion and ~USD 1.48 bln.at the mid-term) ¹⁵
<u>MRA3 Buildings, cities, industries and appliances</u>				~USD 74 mln.	~USD 148 mln.	
<u>MRA4 Forestry and land use</u>				~USD 37 mln.	~USD 74 mln.	
<u>ARA2 Health, well-being, food and water security</u>				~USD 351.5 mln.	~USD 702.8 mln.	
<u>ARA3 Infrastructure and built environment</u>				~USD 351.5 mln.	~USD 702.8 mln.	
<u>ARA4 Ecosystems and ecosystem services</u>				~USD 37 mln.	~USD 74 mln.	

¹⁵ USD values are estimates of project value as derived from the CI2 fund model

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)						
Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage	
<u>Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation</u>	Asset development and construction in the water, sanitation, and ocean infrastructure sectors require sufficient knowledge and technical skills to succeed.	<u>medium</u>	Sufficient knowledge and technical skills at local partners and developers	The project will contribute to technology transfer by pro-actively cooperating (as stakeholder and Director of the Board) with local partners, financiers, and developers during the development as well as the construction of the assets.	<u>Multiple sub-national areas within a country</u>	
<u>Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u>	Projects in the water, sanitation, and oceans infrastructure sectors are currently often subject to shortcomings on Impact, Environmental, Social,, and Governance related issues.	<u>medium</u>	Better integration of IESG standards in the water, sanitation, and oceans sectors in the targeted countries.	The project will contribute by ensuring CFM's IESG standards, as reflected in the Responsible Investment Code (RIC) based on the IFC PS, and all other related policies, are complied with at a project and sub-project level. CFM will also screen all the partners, (co)-developers, and other third parties on their ESG standards and willingness to comply with these. CFM will do a full-fledged due diligence, including assessment done by independent third	<u>Multi-countries</u>	

				parties, during the development phase. None of the projects will receive construction funding if good practices, methodologies, and standards cannot be obtained (or maintained).	
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E.5. Project/Programme specific indicators (project outcomes and outputs)¹⁶¹⁷

Project/Programme results (outcomes/ outputs)	Project/Programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Output 2: Improved wastewater capacity treatment	Wastewater Safely Treated (m3/day) per day	Portfolio data (as provided in Annual Reports) and an independent verification / assurance before the interim and final evaluations.	0	~256 thousand (m3/day)	~623 thousand (m3/day)	Estimated based on reference projects (mainly from the GWI). Further calculations and underlying assumptions can be found in Annex 22.
Output 3: Improved access to water (m3/day)	Water supplied (m3/day)	Portfolio data (as provided in Annual Reports) and an independent verification / assurance before the interim and final evaluations.	0	~570 thousand (m3/day)	~1,4 million (m3/day)	Estimated based on reference projects (mainly from the GWI). Further calculations and underlying assumptions can be found in Annex 22.
Output 5: Improved renewable energy capacity and generation	People provided with power (# of people)	Portfolio data (as provided in Annual Reports) and an independent verification / assurance before the	0	~751 thousand	~ 1.8 million	Estimated based on reference projects (mainly from the GWI). Further calculations and underlying assumptions can be found in Annex 22.

¹⁶ The stated [targets](#) are [the best preliminary estimates available at the FP design stage](#) and [are subject to change based](#) on both the number of NOLs and [sub-projects](#) achieved in the target countries

¹⁷ The impact numbers are derived by taking 80% of the estimated impact of CI2 as a whole (as the GCF will only be an investor in 80% of the Projects due to the NOL requirement).

		interim and final evaluations.				
Project/programme co-benefit indicators						
Co-benefit 1: People Provided with Sanitation	People provided with Sanitation (# of people)	Portfolio data (as provided in Annual Reports) and an independent verification / assurance before the interim and final evaluations.	0	~800 million	~ 2 million	Wastewater safely treated per day divided by the average wastewater usage (as estimated by the GWI)

E.6. Project/Programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 1.1: Fundraising, establishment and management of the Development Fund	Fundraising for and management of the Development Fund		USD 100,000,000 raised for the DF2 from different Donors (out of which USD 80 mln. would be eligible for the GCF proceeds). and successful management of all affairs related to the DF2 over the entire implementation period of CI2.
Activity 1.2: Fundraising, establishment and management of the Construction Equity Fund	Fundraising for and management of the Construction Equity Fund		USD 1,000,000,000 raised for the CEF2 at final close (out of which USD 800 mln. would be eligible for the GCF proceeds). The USD 1,000,000 shall for 20% consist of Donor (T1 funding), 40% of DF1's and impact investors (T2), and 40% of commercial and institutional investors (T3) Successful management of all affairs related to the CEF2 over the entire implementation period of CI2.
Activity 2.1.: Development of mitigation and adaptation projects for the water,	Developing mitigation and adaptation projects for the water, oceans, and sanitation sectors in developing countries	Sub-activity 2.1.1: Identifying potential investment opportunities and initial approval of CFM IC	Identified opportunities and (successfully) presented to CFM's IC

sanitation, and ocean sectors in developing countries		Sub-activity 2.1.2: Restructuring after CFM IC and initial due diligence	Opportunities further screened after the guidance of CFM's IC.
		Sub-activity 2.1.3: Project approval by DF2 IC	Project approved by the DF2 IC to receive DF2 Funding.
		Sub-activity 2.1.4: Project development and extensive due diligence (including but not limited to stakeholder engagement, gender assessment, and Environmental and Social Impact Assessment.)	Projects successfully developed (including positive assessment on the stakeholder engagement, gender assessment, and ESIA)
Activity 2.2: Implementation (construction and operation) of mitigation and adaptation projects for water, sanitation, and ocean sectors in developing countries	Implementing (the construction and operation) of adaptation and mitigation projects for the water, sanitation, and ocean sectors in developing countries	Sub-activity 2.2.1. Project approved by the EF2 IC	Projects approved by the CEF2 IC to receive CEF2 Funding (including repayment of the Development Finance provided by the DF2).
		Sub-activity 2.2.3: Project construction	Construction of projects with climate adaptation and mitigation benefits in the water, sanitation, and ocean, sectors with CFM remaining an active investor to ensure efficiency, knowledge transfer, impact maximization, and (ESG) compliance.
		Sub-activity 2.2.3 Project refinancing and exit	Projects with climate adaptation and mitigation benefits in the water, sanitation, and ocean sectors successfully constructed and exited by the Construction Equity Fund.

E.7. Monitoring, reporting, and evaluation arrangements (max. 500 words, approximately 1 page)

Monitoring & Reporting

CFM reports on the investments done and financial performance of CI2's portfolio to all investors, including all Donors, on a quarterly basis. On an annual basis, a report will be provided that includes audited financial accounts and detailed information on investments, investment valuations, fees, and the impact indicators. In addition, the Accredited Entity, supported by CFM, will provide the GCF with an Annual Progress Report as mandated by the GCF.

Financial reporting:

Within sixty (60) calendar days after the end of each calendar quarter ended 31 March, 30 June, and 30 September, CFM shall prepare and provide all Investors with a quarterly report consisting of, inter alia:

- An unaudited profit and loss account, balance sheet, and cash flow statement of the Programme for the preceding quarter;
- Explanatory notes;
- An overview of the investment activity undertaken by the Programme within the reporting period;
- An overview of each project company, including:
 1. A description of the Project Company's business;
 2. The economic interest of the Programme in the Project Company;
 3. The percentage of the Project Company owned by the Fund;
 4. A discussion of recent key events;
 5. Selected financial information;
 6. If the Investment has been repaid or realized, the realized liquidation or disposal proceeds; and,
- Quarterly capital account statements.

Within one hundred and twenty (120) calendar days after the end of each financial year the Manager shall prepare and provide all Investors with an annual report (the "Annual Report") consisting of, inter alia:

- The annual accounts of the Programme, comprising at least a balance sheet, a profit, and loss account and explanatory notes thereto, as audited by the Auditor;
- A report of the Auditor on its findings resulting from its audit of the annual accounts;
- An overview of each Project Company acquired sold, or otherwise disposed of during the reporting period;
- A valuation of each Investment;
- A report on KPIs;
- A report on impact indicators
- A breakdown of fees received by CFM during the reporting period;
- A breakdown of the Programme's operating costs during the reporting period;
- A schedule of Drawdowns and distributions effectuated during the reporting period;
- The total amount of the Undrawn Commitments as per the end of the reporting period; and,
- A report detailing the activities of the Programme during the reporting period.

Environmental & Social Reporting

An important aspect of CFM's risk management process is the ongoing monitoring and review of each investment. Depending on which Fund that investment is drawing from (Development or Construction Equity), different monitoring requirements will be applied. Monitoring is commensurate with the risk categorization, with physical visits to the location(s) of higher-risk projects during construction.

All project companies drawing from the CEF2 are also required to monitor the performance of any contractors employed at the project site(s) and to participate in a Management Review held by CFM periodically during the lifecycle of the Fund to review overall E&S performance.

In addition, CFM will also ensure that any legal requirements for monitoring, for example, that may be included in permits and licenses for development, construction, and operation, are incorporated into the project monitoring and evaluation plan. CFM requires all projects in receipt of CEF2 are to report in accordance with defined key performance indicators (presented below). In addition, the performance measurement framework indicators presented in H1.1 and H1.2 will be incorporated into fund-level and project-level monitoring and evaluation activities that will be managed by CFM.

Evaluations

The Accredited Entity, together with CFM, will arrange for an interim and final evaluation to be outsourced to an independent evaluator after 10 and 20 years, respectively. The interim evaluation will focus on the extent to which CI2 is achieving its goals regarding the number of – mention GCF impact indicators -. The final evaluation will constitute an ex-post effectiveness study. The evaluation will focus on the DAC criteria, which focus on relevance, effectiveness, efficiency, impact, and sustainability and the GCF's evaluation criteria.

- Relevance: the extent to which the investments are suited to the priorities and policies of CI2
- Effectiveness: the extent to which the objectives of CI2 were achieved
- Efficiency: the extent to which activities were cost-efficient and timely
- Impact: determining the positive and negative changes produced by CI2, directly or indirectly, intended or unintended.
- Sustainability: the extent to which the investments of CI2 are likely to continue operating after CI2 has exited.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Selected Risk Factor 1		
Category	Probability	Impact
Technical and operational	Low	Medium
Description		
Investment Risk The risk is that the assets in the CFM managed assets' portfolio underperform investment expectations and decline in value.		
Mitigation Measure(s)		
- Robust asset selection processes in accordance with defined investment criteria, including IC approval - Comprehensive due diligence procedures and assessment of risk management framework in place at the asset - Oversight by way of representation on project company board and Involvement in asset strategy decisions - Monitoring of asset performance and revenue-drivers - Regular site visits to assets to remain updated on management and operational activities and progress		
Selected Risk Factor 2		
Category	Probability	Impact
Other	Low	Medium
Description		
Regulatory Risk The risk is that changes in the regulatory environment or regulator's attitude will have an adverse impact on CFM and the CFM managed assets' operations and/or strategy.		
Mitigation Measure(s)		
- Alignment with the needs of the recipient country, and NDA / NOL support - Continual dialogue with regulators/governments and continued evaluation, review of legislation - Strong contractual position in the project documents - Dedicated legal and compliance staff		
Selected Risk Factor 3		
Category	Probability	Impact
Technical and operational	Low	Low
Description		
Operational Risk Including Offshore Operations The risk that financial loss may occur through errors, omissions, or inadequacy of processing, people, infrastructure, and systems of CFM or relevant CFM managed assets.		
Mitigation Measure(s)		
- Adequate segregation of duties - Documented policies and procedures (updated as required) - Continuous training policy		
Selected Risk Factor 4		
Category	Probability	Impact
Other	Low	Medium
Description		

Compliance Risk The risk is that CFM and CFM-managed vehicles' compliance obligations (regulatory or internal) are not met, including reporting requirements.		
Mitigation Measure(s)		
- Dedicated legal and compliance staff - Quarterly compliance reporting and testing - Continuous training policy - Formal staff induction training - Documented policies and procedures (periodically updated as required)		
Selected Risk Factor 5		
Category	Probability	Impact
Technical and operational	Medium	Low
Description		
Conflicts of Interest Risk The risk that a conflict of interest occurs in an area of CFM's service provision to its Managed Vehicles, their investors, and its relationship with all of its stakeholders whereby CFM (or an investor or other stakeholder of CFM) may benefit whilst possible tangible loss may be caused to (another) investor to whom CFM provides a service.		
Mitigation Measure(s)		
- CFM has a comprehensive policy established to manage conflicts of interest - Following the identification of a potential conflict, the manager will implement measures to manage and mitigate the identified conflict will be implemented promptly through three basic mechanisms to manage a conflict, namely controlling, avoiding, and disclosing. - The appropriate mechanism to manage a conflict will depend on the circumstances and nature of any given conflict. - Documented policies and procedures (periodically updated as required)		
Selected Risk Factor 6		
Category	Probability	Impact
Credit	Medium	Low
Description		
Interest Rate Risk The risk that the cost of borrowing increases (and CEF2 Tier 2 returns decrease) due to changes in interest rates post refinancing event.		
Mitigation Measure(s)		
- Changes to debt facilities or hedging policy approved by the board. - Borrowing restrictions within the trust deed and prospectus or information memorandum, investment policy, regulatory requirements, considered when selecting debt facilities. - Fund board to be advised of the selection process and in the negotiation of terms and conditions of any debt package. - Where considered appropriate (and authorized by the fund agreements) derivatives used to manage interest rate exposure. - Calculations confirming adherence to borrowing limits, financial covenants under lending documentation, and hedging policies tabled at board meetings		
Selected Risk Factor 7		
Category	Probability	Impact
Forex	Medium	Medium
Description		

Currency Risk		
The risk is that future cash receipts/payments will decrease/increase due to adverse movement in exchange rates.		
Mitigation Measure(s)		
- Most off-take agreements contracted by CI2 at the asset level are directly or indirectly denominated in USD leading to an influx of USD (or equivalent) amounts. - It is therefore estimated that most off-take agreements will be indexed to USD currencies. If deemed desirable/necessary, CI2 has the option to explore hedging opportunities made available by local markets or by TCX (the Currency Exchange, a hedging mechanism that enjoys wide support from the DFI community including FMO, EBRD and KfW) - In principle, CFM may only enter into futures, options, or other derivative transactions for the purpose of, and only to the extent required for, hedging currency positions on non-USD Investments. CFM is not obliged to do this, however - A hedging policy may be adopted in respect of non-USD currency commitments for imminent acquisitions or divestments, considered on a project-by-project basis or hedging of short-term distributions - Distributions and other cash flows from assets are not hedged		
Selected Risk Factor 8		
Category	Probability	Impact
Technical and operational	Low	Low
Description		
Financial Close Risk		
Limited access to finance and protracted timelines to agree terms prevent projects from reaching financial close and subsequently refinancing.		
Mitigation Measure(s)		
- Providing a one-stop-shop for finance at development, construction, and refinance phases to avoid protracted negotiations with multiple parties. - Match-making projects with suitable local, international, or public finance.		
Selected Risk Factor 9		
Category	Probability	Impact
Technical and operational	Low	Low
Description		
Exit Risk		
The risk that CFM is not able to dispose of all of its assets in full.		
Mitigation Measure(s)		
- Development of high-quality projects - M&A activity in investment regions shows an appetite for water, sanitation, and oceans sector infrastructure. - Active divestment opportunity screening throughout the lifetime of a project		
Selected Risk Factor 10		
Category	Probability	Impact
Credit	Medium	Low
Description		
Counterparty / Credit Risk		
The risk that a counterparty (the other party with whom an investment or borrowing transaction is made) will fail to perform contractual obligations and default in either whole or part under a contract.		
Mitigation Measure(s)		

- Risk assessed as part of the due diligence process, preparing the underlying investment proposal and reviewed on an ongoing basis - International credit ratings checked where relevant		
Selected Risk Factor 11		
Category	Probability	Impact
Technical and operational	Medium	Low
Description		
Liquidity Risk The risk that CFM and a CFM-managed asset do not have sufficient cash to meet forward commitments. Meaning that revenues on the manager or project levels are below expenses and there is a lack of liquidity.		
Mitigation Measure(s)		
- Detailed 5-year cash forecast updated monthly "Forward commitments test" and liquidity limits under which available liquidity must not fall.		
Selected Risk Factor 12		
Category	Probability	Impact
Other	High	High
Description		
Political Risk The risk that sovereign rights exercised by governments are contrary to the rights of the Fund, or project company under the relevant agreements.		
Mitigation Measure(s)		
- Virtue of donors as Official Development Assistance (ODA) providers and DFI funding - Acquiring Political Risk Guarantee (World Bank, MIGA) when specific events heighten the political risk in a project company - Government Letter of Support - GCF funding would be complementary to already establish ODA providers (European Commission, Dutch Government) in the Development Fund and T1 Construction Equity Fund of CI2, in providing additional political risk mitigation capabilities on the back of being established under the framework of the UNFCCC.		
Selected Risk Factor 13		
Category	Probability	Impact
Reputational	High	Medium
Description		
Environmental and Social Risk Impact on natural resources, employees, investors, customers, and community by the business activities of CFM or a CFM managed asset fund and its investments.		
Mitigation Measure(s)		
- The application of FMO's developing countries' experience in the design of CFM's Environmental and Social Management System, which will apply to all CI2 investments - Adherence to international Environmental and Social standards as well as local legal requirements. - Disclosure of project-related information in accordance with local legal requirements as well as the IFC Performance Standards (PS). - Dedicated E&S specialist to implement CFM ESG due diligence processes at all stages. - Monitoring of implementation of the ESMS throughout the lifecycle of any project.		

Implementation of GIAP with particular focus on gender-responsive stakeholder engagement and consultation as well as gender-responsive grievance mechanisms		
Selected Risk Factor 14		
Category	Probability	Impact
Other	Low	Low
Description		
Technology Risk Risk of technology underperforming through inadequate natural resources, feedstock, and/or inadequate maintenance.		
Mitigation Measure(s)		
- Risk assessed as part of due diligence and preparing the underlying investment proposal - Investing in proven technologies from known/trusted manufacturers - Contracting independent experts to assess the viability of resource potential of project sites - Securing EPC contracts and OEM maintenance contracts		
Selected Risk Factor 15		
Category	Probability	Impact
Technical and operational	Low	Medium
Description		
Construction Risk The risk is that there are issues during project construction, triggering significant delays and construction overruns.		
Mitigation Measure(s)		
- Appropriate contractual arrangements and risk allocation mechanisms (including guarantees and liquidated damages) - High-quality planning and project management, and high-quality EPC and Balance of Plant contractors with an appropriate track record		
Selected Risk Factor 16		
Category	Probability	Impact
Other	Low	Low
Description		
Good Governance / Rule of Law / Corruption Risk of failed governance at the project company level; a counterparty or employee acting corruptly or fraudulently or in breach of law.		
Mitigation Measure(s)		
- A focus on good governance at both the fund and project level which will benefit local counterparties through strict adherence to the rule of law and in mitigating corruption risk - Existence of robust and clear compliance policies and procedures including KYC, Anti Money Laundering and Counter-Terrorist Financing and Anti-Corruption - Ensuring appropriate fraud insurance cover - Ongoing KYC, internal sanctions list, and anti-money laundering and counter-terrorist financing checks and verifications		

G. GCF POLICIES AND STANDARDS

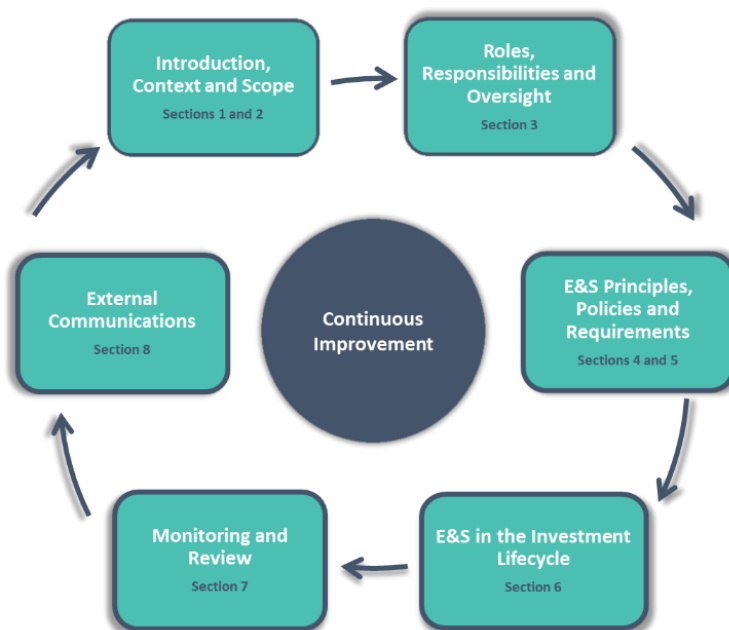
G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Environmental and Social Management System (ESMS)

CFM, in cooperation with, and as approved by, the AE, has developed an Environmental and Social Management System (“ESMS”) to manage impacts and risks associated with the Funds under its management. The ESMS manual has been disclosed in accordance with the GCF’s ex-ante disclosure requirements for category A/I-1 projects/Programmes. The document, provided in Annex 6, describes the requirements, processes and arrangements required to comply with the requirements of the Responsible Investment Code (“RIC”) and to uphold the commitments set out in CFM’s Responsible Investment Policy (“RIP”). It describes how the arrangements are embedded into CFM’s investment lifecycle (from development to operational). It also describes the environmental and social (“E&S”) management measures that need to be adopted by the Project Companies. The ESMS has been designed in accordance with IFC Performance Standard 1 (“PS 1”) and other international standards including ISO 14001 and ISO 45001 as well as the AE’s Sustainability Policy Universe.

The structure the ESMS is presented in the figure below and summarized as follows:

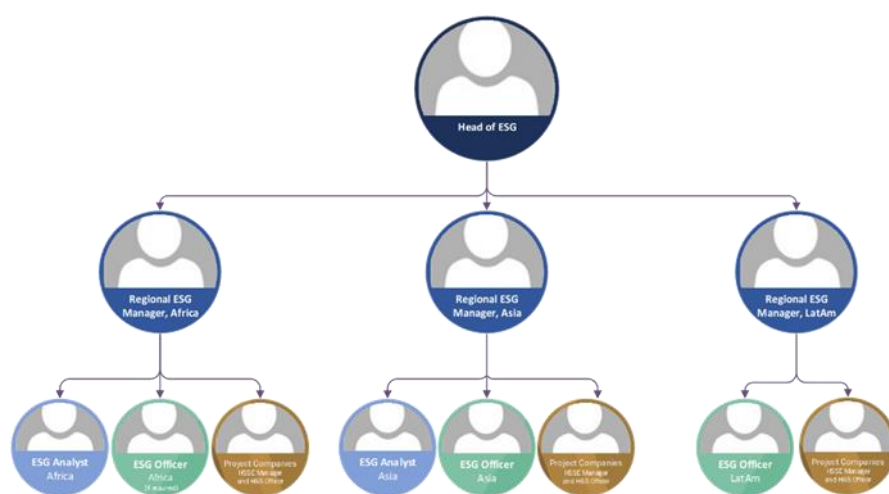
- Sections 1 and 2: Following an introductory section, section 2 of the ESMS outlines the context and scope, with a brief presentation of the CFM Managed Vehicles and the potential E&S impacts associated with them, a description of the key stakeholders with their needs and expectations and the scope of application of the ESMS.
- Section 3 presents the senior accountability in place for oversight of the ESMS, the resources assigned for its implementation with their roles and responsibilities and how the right level of expertise and capacity is ensured.
- Sections 4 and Chapter 5 describes CFM’s policies, principles, commitments, and E&S requirements that form the overall Responsible Investment Framework.
- Chapter 6 describes how CFM operationalizes the RIP across the investment cycle through risk management and delivering positive impacts.
- Chapter 7 presents the system in place for monitoring and periodically reviewing the effectiveness of the ESMS, through the definition of E&S performance indicators and impact indicators, audits, and management review.
- Chapter 8 outlines the arrangements in place for communication with external stakeholders, including reporting to investors, public disclosure, and responsiveness to grievances.
- The ESMS also includes a set of Annexures, including Annex 2 (List of Excluded Activities); Annex 4 (Indigenous Peoples Planning Framework), and Annex 5 (Resettlement Planning Framework).



E&S Governance and Management

FMO, as the AE, plays a central role in CFM's governance. Specifically in relation to E&S governance and management CFM has committed to FMO as AE to: (i) implement and maintain an ESMS to the satisfaction of FMO, (ii) appoint or acquire services of ESG specialists, (iii) submit annual reports including E&S performance and (iv) comply with all relevant GCF policies and procedures.

CFM, as the EE, has a designated Head of ESG that oversees and implements all E&S commitments and responsibilities. The Head of ESG directly reports to the MB and is part of the Executive Committee. Reporting to the Head of ESG is a broader team of ESG professionals. The team will be expanded as necessary to ensure adequate resourcing.



In addition, CFM will: (i) hire external E&S advisors to support with due diligence, specialist studies and ongoing monitoring; and (ii) partner with organizations (such as CSOs, development organizations, research organization and training institutions) to deliver specialist programmes including those focused on delivering community development interventions. Furthermore, all CFM Investment Staff have a responsibility for ensuring that all projects are

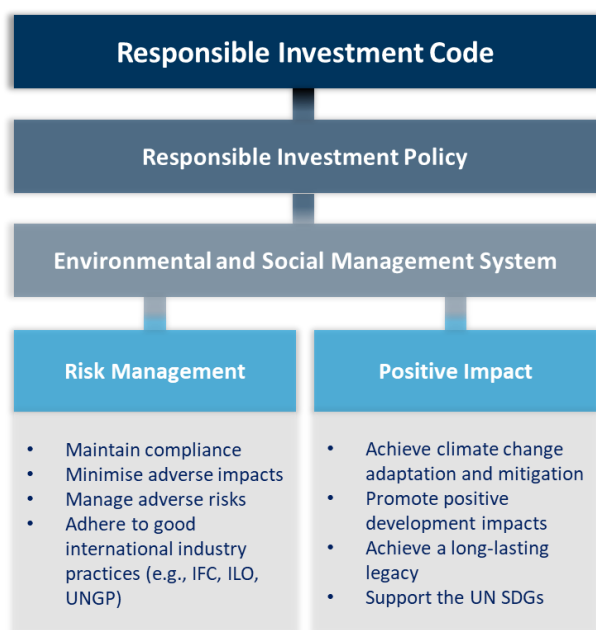
developed and implemented according to CFM's requirements as set out in the ESMS.

At the project level, CFM will require the project to manage E&S risks as set out in the ESMS. Each project will ensure they have a project level ESMS as well as adequate staffing and resources. CFM's ESG team will work closely together with the project company to ensure compliance.

E&S Risk Management

CFM's approach to responsible investment is enshrined in the RIC which guides all investment decisions and management activities relating to the CFM Managed Vehicles. The principles and undertakings of the RIC are reinforced by a set of commitments which are contained in the RIP. These commitments are implemented through the arrangements described in the ESMS Manual and supporting documentation. The underpinning ethos is of managing risk and delivering positive benefits in line with legal requirements, and good international industry practices. This is presented schematically below.

ESG considerations are fully integrated into the investment process, starting with providing inputs into the mandate fit stage of a potential investment, through screening and due diligence, to project development, construction, operation, and final exit. Each investment is assessed to confirm the potential for adverse impacts and risks, and this leads to a preliminary E&S risk categorization. As a general rule, Category A projects will not be pursued unless sufficient evidence is available to meet the conditions outlined in the ESMS manual. The risk category will inform the scope of the development activities that will be required, including the commissioning of an environmental and social impact assessment ("ESIA") for all Category A and B projects. Mitigation, management, and monitoring measures will be defined during the development phase for adoption by the Project Company from the start of the construction phase. Section 6 of the ESMS manual provides a description of the way in which E&S impacts and risks are assessed and integrated into the investment and decision-making process.



In addition to managing adverse impacts and risk, CFM is committed to maximizing opportunities for delivering environmental and social benefits, particularly to the local communities in the area of influence of its projects. These benefits are designed to be additional to the intrinsic positive impacts of the investments and are focused on local actions that can be taken to support progress towards achieving selected UN SDGs (refer to section D.3). Budget is committed to community development during the development, construction, and operational phases of an asset. A needs-based community development programme will be established during the development phase of the project, and the Project company will be required to implement it under the guidance of CFM. The programmes are typically delivered by development partners with the local network, local knowledge, and the expertise required to deliver successful outcomes.

Human Rights

CFM recognizes the centrality of human rights to sustainable development and ensuring fair distribution of development and opportunities and benefits. CFM is committed to operating in accordance with all legal requirements and agreements, including the ILO Core Conventions and the UN's Guiding Principles on Business and Human Rights. These principles have been implemented within CFM, eliminating all forms of unfair discrimination in the recruitment and selection of staff, and within all project companies, suppliers

and other companies CFM has control or influence over. Project companies will be required to provide information relating to their suppliers.

Gender Equality and Women's Empowerment

CFM's gender policy, as described in G.1 and annexed to the FP, details the commitments and arrangements for achieving gender equality and women's empowerment within the business and in relation to the projects. CFM recognizes that women have as much to gain from its projects (and particularly from increased access to energy and water and sanitation resources) as men but given their relative social position, will likely not profit as men from the access to energy, water, and sanitation resources. CFM considers women as employees, end-users and beneficiaries of its projects and requires risks and opportunities to be assessed and managed in accordance with a local gender action plan (see G.2) as well as in accordance with other E&S safeguards as described in the ESMS manual.

Of particular note is the issue of gender-based violence and harassment to which members of the Project workforce and those in the local communities can be exposed. CFM therefore requires all projects to implement measures to safeguard female members of its workforce as well as in the broader communities it serves. This includes consideration of the risks associated with gender-based violence at all stages of the investment lifecycle, adoption of controls to avoid cases and measures to respond to and address reported or suspected cases. The grievance mechanism adopted by every project will make specific provisions for a survivor-centered approach as well as providing the necessary support to witnesses and others such as whistleblowers, with those involved in managing the case receiving appropriate training in line with international guidance.

Indigenous Peoples

CFM recognizes the critical importance of respecting the rights of Indigenous Peoples ("IP") and of avoiding any adverse impacts on the IPs which may arise because of a Project. CFM will avoid proceeding with a Project beyond the Development Phase (activity 2.1.) where adverse impacts cannot be avoided. When avoidance is not feasible, CFM requires measures to be put in place to minimize and mitigate these impacts. Compensation shall only be considered as a last resort. Annex 4 of the ESMS manual describes CFM's Indigenous Peoples Planning Framework ("IPPF") which is designed to guide all projects in terms of the minimum requirements wherever IPs are identified in a project area of influence. The IPPF is informed by the requirements of IFC PS7 and the associated guidance note, and the Green Climate Fund IPP and associated Operational Guidelines.

Corporate Grievance Mechanism

CFM provides a fund-broad grievance mechanism for external stakeholders on its website. This grievance mechanism is intended for use by stakeholders such as NGOs and CSOs, government representatives, investors, and the wider public. Grievance mechanisms are also implemented by the Project Companies and are intended for use by the Project affected communities. All grievance mechanisms must be designed and operated in accordance with the Effectiveness Criteria detailed in Principle 31 of the United Nations Guiding Principles on Business and Human Rights. All grievance mechanisms will also consider gender-sensitivity and provide a survivor centered-approach for responding to reported cases of gender-based violence and sexual harassment.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

CFM recognizes that the projects can only be successful if they are gender sensitive. The use of the term ‘gender’ refers to the social relations between women and men; thereby the need to work with women and men to ensure both equitably benefit. It is further recognized that women have as much to gain from its projects (and particularly from increased access to water and sanitation) as men but given their relative social position, will likely not profit as men from the access to energy, water, and sanitation resources.

CFM, in cooperation with the AE, has developed a Gender Action Plan (“GAP”) to present an overview of relevant gender-related issues in the water, sanitation, and oceans sectors in the 25 targeted countries. The GAP has been developed based on country assessment and analysis, with close reference to the approach described in the GCF guidance document for mainstreaming gender in projects. The GAP will be supplemented by local analysis of gender issues and opportunities, and a local gender action plan that aims to address these challenges and opportunities. The Project company will be required to implement this action plan throughout the lifecycle of every asset (and will form part of the handover at the exit of an investment).

The GAP is intended to support the Manager in fulfilling its commitments to responsible investment in accordance with the Responsible Investment Framework and CI2’s Responsible Investment Policy. In particular, the GAP is based on the Framework’s commitment to “Do No Harm” and “Do Good” and the policy’s commitments. These include “avoiding and otherwise minimizing and mitigating the impact of its investments on local communities” and “maximizing opportunities for positive social benefits to be realized”. In this sense, the GAP provides a gender lens to the arrangements that already exist for addressing the social dimensions of CI2 financing. The GAP will include an (estimated) budget. The AE and the EE also confirm that the budget would be readjusted in case it would be deemed insufficient during the implementation of CI2.

The gender assessments and the full Gender Action Plan are to be provided as Annex 8.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

The financial management is done at two levels: (i) the AE (FMO) and (ii) the EE (CFM). On the AE’s level, the AE will enter into the Funding Activity Agreement (FAA) with the GCF. The AE will oversee the fiduciary management of the GCF’s funding in its GCF account and on a ledger specific to CI2. The AE will request the GCF funds, based on the funding needs as predicted by CFM, on a bi-annual basis. The AE will then disburse these funds to the DF2 whenever CFM issues a drawdown request. CFM will not manage any unused funds of the GCF. The drawdown requests and payments at FMO will follow FMO’s standard policies and procedures for PE fund investment.

Once received by CFM, the funds will be administered by CFM including drawdowns from and distributions to investors, as well as all fund-related accounting will be instigated, overseen, and performed by CFM (as asset manager). CFM has an Authorization and Payments Policy in place, which details the process of authorizing and making payments to external parties, including the various levels of internal authorizations required in order to ensure a sound control environment. CFM has controlled access to the CIO Fund bank accounts and is responsible for initiating and releasing payments according to the Authorization and Payments Policy and generally in accordance with the CI2 Funds’ governing documentation. All investor reporting is controlled and managed by CFM. A monthly review of the trial balance is performed by the CFM Finance Manager and quarterly and annual reports are prepared by CFM. The CI2 Funds’ and CFM’s accounts will be audited annually by PwC. Accounts will be prepared according to the International Financial Reporting Standards (IFRS) and will consist of, among others, a balance sheet, profit and loss account, explanatory notes, auditor’s report, an overview of project turnover activity, breakdown of

management fees paid, breakdown of fund operating costs, etc. CFM will require the project companies to prepare accounts in accordance with appropriately recognized financial reporting standards and each project company is required to appoint an Auditor. CFM's financial reports will be subject to FMO review and approval through FMO's participation on CFM's supervisory board & CFM's Audit, Risk, and Compliance Committee.

The AE's approved procurement policy will be applied, in addition to CFMs, throughout the programme including in relation to activities implemented by the EE. Please refer to Annex 10, Procurement Plan, for further details on procurement thresholds and requirements

G.4. Disclosure of funding proposal

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/Programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☒ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/Programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation, and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☒ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☒ Annex 17 Multi-country project/Programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence, or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)¹⁸
- ☐ Annex X Other references

¹⁸ Annex 22 is mandatory for mitigation and cross-cutting projects.

* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.

ⁱ World Bank. 2015. The Forgotten Factor of Climate Change - Water: Technical Papers. Washington, DC: Roson, Robert and Sartori, Martina.

ⁱⁱ https://www.international.gc.ca/world-monde/issues_development-enjeux_developpement/environmental_protection-protection_environnement/water-eau.aspx?lang=eng

ⁱⁱⁱ McKinsey. (2019). Unlocking private-sector financing in emerging-markets infrastructure. Retrieved from: <https://www.mckinsey.com/industries/capitalprojects-and-infrastructure/our-insights/unlocking-private-sector-financing-in-emerging-markets-infrastructure>

^{iv} International Monetary Fund. (2020). World Economic Outlook. Retrieved from <https://www.imf.org/en/Publications/WEO/Issues/2020/06/24/WEUpdateJune2020>

^v <https://www.theguardian.com/environment/2019/may/06/human-society-under-urgent-threat-loss-earth-natural-life-un-report>

^{vi} <https://grist.org/climate/flatten-the-curve-coronavirus-climate-emissions/>

^{vii} <https://www.globalcarbonproject.org/carbonbudget/archive.htm>

^{viii} <https://www.ncdc.noaa.gov/sotc/global/202001>

^{ix} McKinsey. (2019). Unlocking private-sector financing in emerging-markets infrastructure. Retrieved from: <https://www.mckinsey.com/industries/capital-projects-and-infrastructure/our-insights/unlocking-private-sector-financing-in-emerging-markets-infrastructure>

^x International Monetary Fund. (2020). World Economic Outlook. Retrieved from <https://www.imf.org/en/Publications/WEO/Issues/2020/06/24/WEUpdateJune2020>

^{xi} <https://blogs.scientificamerican.com/eye-of-the-storm/earth-had-its-second-warmest-year-in-recorded-history-in-2019/>

^{xii} xxvii UNEP 2016. A Snapshot of the World's Water Quality: Towards a Global Assessment. United Nations Environment Program, Nairobi, Kenya. 162 pp.

^{xiii} [https://www.cdc.gov/healthywater/global/wash_statistics.html#:~:text=Worldwide%2C%20780%20million%20people%20do,world%E2%80%99s%20population\)%201%2C%203](https://www.cdc.gov/healthywater/global/wash_statistics.html#:~:text=Worldwide%2C%20780%20million%20people%20do,world%E2%80%99s%20population)%201%2C%203)

^{xiv} <https://washdata.org/sites/default/files/2020-08/jmp-2020-wash-schools.pdf>

^{xv} Hoegh-Guldberg, O. et al. 2015. Reviving the Ocean Economy: the case for action - 2015. WWF International, Gland, Switzerland., Geneva, 60 pp.

^{xvi} <https://www.nature.com/articles/s41467-021-23168-y>

^{xvii} https://www.ipcc.ch/site/assets/uploads/2018/02/WG1AR5_Chapter13_FINAL.pdf

^{xviii} <https://www.greenclimate.fund/sites/default/files/document/sap-technical-guidelines-ecosystems-and-ecosystem-services.pdf>

^{xix} <https://www.convergence.finance/resource/35t8IVft5uYMOGOaQ42qgS/view>



Economic Relations Division
Ministry of Finance
Sher-e-Bangla Nagar
Dhaka-1207

D.O. No. 09.00.0000.105.05.011.17-

Dated 09 September 2021

Re: **Funding proposal for the GCF by FMO (being the Netherlands Development Bank) regarding the programme "Climate Investor Two (CI2)"**

Dear Sir,

We refer to the multicounty programme **"Climate Investor Two (CI2)"** in Bangladesh as included in the funding proposal submitted by FMO to us on 18 January 2021.

The undersigned Fatima Yasmin is the duly authorized representative of the National Designated Authority of Bangladesh.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our preliminary no-objection to the programme **"Climate Investor Two (CI2)"** as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Bangladesh has no-objection to the programme **"Climate Investor Two (CI2)"** as included in the funding proposal;
- (b) The programme **"Climate Investor Two (CI2)"** as included in the funding proposal is in conformity with Bangladesh's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme **"Climate Investor Two (CI2)"** as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme **"Climate Investor Two (CI2)"** as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme, including the water, sanitation, oceans and related infrastructure sectors projects and activities.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Fatima Yasmin

Executive Director
Green Climate Fund
Songdo Business District, 175 Art Center-daero
Yeonsu-gu, Incheon 22004, Republic of Korea

TELEPHONE: 3950100
FAX: 3956086
Toll free: 0800600773



REPUBLIC OF BOTSWANA

MINISTRY OF FINANCE AND ECONOMIC
DEVELOPMENT
PRIVATE BAG 008
GABORONE

REF: MFED170/1/4A Vol.5 (34)

Date: 04th April, 2022

The Executive Director,
The Green Climate Fund
Songdo International Business Centre
175, Art Center-Daero Yeonsu-gu, Incheon 22004
Republic of Korea

Dear Sir,

**RE: FUNDING PROPOSAL FOR THE GCF BY NEDERLANDSE
FINANCIERINGS-MAATSCHAPPIJ VOOR ONTWIKKELINGSLANDEN
(FMO) REGARDING CLIMATE INVESTOR TWO PROGRAMME**

We refer to the programme titled Climate Investor Two as included in the funding proposal submitted by Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden N.V. (FMO) to us on 1st March, 2022.

The undersigned is the duly authorized representative of the Ministry of Finance and Economic Development, the National Designated Authority of Botswana. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

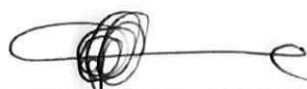
- a) The Government of Botswana has no-objection to the programme as included in the funding proposal;
- b) The programme as included in the funding proposal is in conformity with Botswana's national priorities, strategies and plans;
- c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours sincerely,



Keineetse Lepekoane

GCF Focal Point for Botswana
Ministry of Finance and Economic Development



MINISTÉRIO DA ECONOMIA
Secretaria Especial de Comércio Exterior e Assuntos Internacionais
Secretaria de Assuntos Econômicos Internacionais
Subsecretaria de Instituições Internacionais de Desenvolvimento
Coordenação-Geral de Instituições Globais de Desenvolvimento

OFÍCIO SEI Nº 117330/2022/ME

Brasília, 20 de abril de 2022.

Mr. Yannick Glemarec

Deputy Executive Director
Green Climate Fund
G-Tower, 24-4 Songdo-dong, Yeonsu-gu
Incheon City, Republic of Korea

Re: Funding proposal for the GCF by fmo regarding the program "Climate Investor Two (CI2)"

Dear Mr. Glemarec,

1. We refer to the program titled "**Climate Investor Two (CI2)**" as included in the funding proposal submitted by FMO to us on March 1 2022.
2. The undersigned is the duly authorized representative of the Secretariat for International Economic Affairs, Ministry of Economy, the National Designated Authority of Brazil.
3. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.
4. By communicating our no-objection, it is implied that:
 - a) The government of Brazil has no-objection to the programme as included in the funding proposal;
 - b) The program as included in the funding proposal is in conformity with Brazil's national priorities, strategies and plans;
 - c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.
5. We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.
6. We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

7. We acknowledge that this letter will be made publicly available on the GCF website.⁶⁸

Best regards,

Documento assinado eletronicamente

ERIVALDO ALFREDO GOMES

Secretary for International Economic Affairs

National Designated Authority for Brazil



Documento assinado eletronicamente por **Erivaldo Alfredo Gomes, Secretário(a) de Assuntos Econômicos Internacionais**, em 20/04/2022, às 12:36, conforme horário oficial de Brasília, com fundamento no § 3º do art. 4º do [Decreto nº 10.543, de 13 de novembro de 2020](#).



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Esplanada dos Ministérios, Bloco K, 8º andar, sala 874 - Bairro Zona Cívico-Administrativa

CEP 70048-900 - Brasília/DF

(61) 2020-5644 - e-mail cgpin.sain.df@fazenda.gov.br - gov.br/economia

Processo nº 12120.100144/2021-11.

SEI nº 24169775



El futuro
es de todos

DNP
Departamento
Nacional de Planeación

69

Bogotá D.C., viernes, 13 de mayo de 2022

DADS



Al responder cite este Nro.
20224500389531

The Green Climate Fund – GCF
Att. executive-office@gcfund.org

Subject: Funding proposal for the GCF by Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden N.V. (FMO) – regarding Climate Investor Two.

Dear Madam, Sir,

We refer to the programme titled “Climate Investor Two” in Colombia, as included in the funding proposal submitted by FMO to us on March the 1st 2022.

The undersigned is the duly authorized representative of the National Planning Department of Colombia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Colombia has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Colombia;
- (c) In accordance with the GCF’s environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.





El futuro
es de todos

DNP
Departamento
Nacional de Planeación

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We acknowledge that this letter will be made publicly available on the GCF website.

Best Regards,

CAROLINA DIAZ GIRALDO

Director of Environment and Sustainable Development
National Planning Department

CC o.arbelaez@climatefundmanagers.com; k.anderson@climatefundmanagers.com;
j.paez@climatefundmanagers.com; vvilloria@gcfund.org; mrestrepo@dnpgov.co

Prepared: Lukas Socarras
Revised: Maria del Pilar Restrepo





To: The Executive Director
The Green Climate Fund ("GCF")
Songdo International Business Centre
175, Art Center-Daero Yeong-gu
Incheon 22004
Republic of Korea

**Re: Concept Note for the GCF by Nederlandse Financierings-Maatschappij Voor
Ontwikkelingslanden N.V. (FMO) regarding Climate Investor Two (CI2) programme.**

Dear Madam, Sir,

We refer to the programme title Climate Investor Two (CI2) in Djibouti as included in the concept note submitted by Nederlandse Financierings-Maatschappij Voor Ontwikkelingslanden N.V. (FMO) to us on 14 January 2021.

The undersigned is the duly authorized representative of the Ministry of Environment and Sustainable Development the National Designated Authority / Focal Point of Djibouti.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Djibouti has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Djibouti;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mr. Dini Abdallah Omar

Ministry of Environment and Sustainable Development





Ministerio del Ambiente, Agua
y Transición Ecológica

Oficio Nro. MAATE-SCC-2022-0229-0

Quito, D.M., 23 de junio de 2022

Asunto: Funding proposal for the GCF by FMO regarding the initiative "Climate Investor two CI2

Señor
Yannick Glemarec
Executive Director
GREEN CLIMATE FUND
En su Despacho

We refer to the programme titled **Climate Investor two “CI2”** in Ecuador as included in the funding proposal submitted by FMO.

The undersigned is the duly authorized representative of Ministry of Environment, Water and Ecological Transition, which acts as the National Designated Authority of Ecuador.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

1. The government of Ecuador has no-objection to the programme as included in the funding proposal;
2. The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Ecuador
3. In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Ministerio del Ambiente, Agua
y Transición Ecológica

Oficio Nro. MAATE-SCC-2022-0229-0

Quito, D.M., 23 de junio de 2022

Atentamente,

Mgs. Karina Maribel Barrera Moncayo
SUBSECRETARIA DE CAMBIO CLIMÁTICO

Copia:

Señora Secretaria
Teresa Del Pilar Pérez Toscano
Secretaria de la Subsecretaria



No. CC-08008/3/2022-CC
20th May, 2022

To

The Green Climate Fund ("GCF")
Songdo Business District
175 Art Center-daero
Yeonsu-gu, Incheon 22004 (Republic of Korea)

Subject: Funding Proposal for the GCF by FMO regarding "Climate Investor Two"

Dear Madam/ Sir,

We refer to the programme, "**Climate Investor Two**" in India as included in the funding proposal submitted by FMO to us on 4th March 2022.

2. The undersigned is the duly authorized representative of Ministry of Environment, Forest and Climate Change, the National Designated Authority/Focal Point of India.

3. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

4. By communicating our no-objection, it is implied that:

- The Government of India has no-objection to the programme as included in the funding proposal;
- The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of India;
- In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

5. We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

6. We also confirm that our no-objection applies to all activities/projects to be implemented within the scope of the programme.

7. We acknowledge that this letter will be made publicly available on the GCF website.



Yours sincerely,

Rajasree

(Rajasree Ray)

Economic Adviser and GCF Focal Point,
National Designated Authority, India



ए-650, छठा तल, अग्नि विंग, इंदिरा पर्यावरण भवन, जोर बाग रोड, नई दिल्ली-110 003

फोन: (011) 20819197, टेलीफैक्स: (011) 20819329

A-650, 6th Floor, Agni Wing, Indira Paryavaran Bhawan, Jor Bagh Road, New Delhi-110 003

E-mail : rajasree.r@nic.in Tel.: (011) 20819197, Telefax: (011) 20819329





75

MINISTRY OF FINANCE OF THE REPUBLIC OF INDONESIA
FISCAL POLICY AGENCY

R.M. NOTOHAMIPRODJO BUILDING 2ND FLOOR JALAN DR. WAHIDIN RAYA NOMOR 1 JAKARTA 10710
TELEPHONE (+62 21) 3441484; FACSIMILE (+62 21) 3848049; WEBSITE www.fiskal.kemenkeu.go.id

Ref. : S-53/KF/2022

28 April 2022

Mr. Yannick Glemarec
Executive Director
Secretariat of the Green Climate Fund
175, Art center-daero
Yeonsu-gu, Incheon 406-840
Republic of Korea

Subject : Funding Proposal for the GCF by FMO regarding Climate Investor Two (CI2)

Dear Mr. Glemarec,

We refer to the program of Climate Investor Two (CI2) as included in the funding proposal submitted by FMO to us on 17 March 2022.

The undersigned is the Chairman of Fiscal Policy Agency, Ministry of Finance as the National Designated Authority of Indonesia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the program as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Indonesia has no-objection to the program as included in the funding proposal;
- (b) The program as included in the funding proposal is in conformity with Indonesia's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the program as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to program as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the program.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours faithfully,



Ditandatangani secara elektronik
Febrio Nathan Kacaribu
Chairman



No-objection letter template for programmes



GREEN
CLIMATE
FUND

**MINISTERE DE L'ENVIRONNEMENT
ET DU DEVELOPPEMENT DURABLE**

DIRECTION GENERAL

**DIRECTION DE LA LUTTE
CONTRE LES CHANGEMENTS
CLIMATIQUES**

**Autorités Nationale Désignée
(AND)**

REPUBLIQUE DE CÔTE D'IVOIRE
Union – Discipline – Travail



The Green Climate Fund
The Executive Director
Songdo International Business Centre
175, Art Center-Daero
Yeonsu-gu, Incheon 22004
Republic of Korea

22 January 2021

Re: Funding proposal for the GCF by the Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden N.V. (FMO) regarding Climate Investor Two.

Dear Madam, Sir,

We refer to the programme titled "Climate Investor Two" as included in the funding proposal submitted by FMO to us on 18 January 2021.

The undersigned is the duly authorized representative by Mr. Kouadio Kumassi Philippe, the National Designated Authority / Focal Point of Cote d'Ivoire.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Cote d'Ivoire has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Cote d'Ivoire;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mr. Kouadio Kumassi Philippe
National Designated Authority
Cote d'Ivoire





REPUBLIC OF KENYA

THE NATIONAL TREASURY AND PLANNING

Telegraphic Address: 22921
FAX NO. 310833
Telephone: 2252299

THE NATIONAL TREASURY
P O BOX 30007 – 00100
NAIROBI

When Replying Please Quote

Ref: CONF/MOF/36/021/D/ (56)

Date: 17th September, 2021

Yannick Glemarec
Executive Director
Green Climate Fund
Songdo Business District
Yeonsu-gu, Incheon 22004
Republic of Korea

Dear

Yannick

Re: Funding proposal for the GCF by Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden N.V. ("FMO") regarding Climate Investor Two (CI2)

We refer to the programme '*Climate Investor Two*' in Kenya as submitted by Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden N.V. ("FMO") to us in February 2021.

The undersigned is the duly authorized representative of The National Treasury and Planning, the National Designated Authority of the Republic of Kenya.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The Government of Kenya has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Republic of Kenya;

- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed. We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours

Sincerely



JULIUS MUIA, PhD, CBS

PRINCIPAL SECRETARY/NATIONAL TREASURY

Copy to: David Kuijper
Manager, Public Investments and Blended Finance
FMO

Kevin Anderson
Head: Strategic Initiatives
Climate Fund Managers



SECRETARIAT GENERAL

**BUREAU NATIONAL DES CHANGEMENTS
CLIMATIQUES ET DE LA REDD+**

Antananarivo, 27 May 2022

**AUTORITE NATIONALE DESIGNEE
DU FONDS VERT POUR LE CLIMAT**

N° 01 -22/MEDD/SG/BNCCREDD+/AND FVC

**To: The Green Climate Fund ("GCF") Secretariat
Song Do, South Korea**

**Re: Funding proposal for the GCF by FMO – Dutch Development Bank regarding Climate
Investor Two**

Dear Madam, Sir,

We refer to the programme titled *Climate Investor Two* in Madagascar as included in the funding proposal submitted by FMO – Dutch Development Bank to us on 1 March 2022.

The undersigned is the duly authorized representative of the Ministry of Environment and Sustainable Development, the Focal Point of Madagascar.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Madagascar has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Madagascar;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Lovakanto RAVELOMANANA
Coordinator of the National Office
Of Climate Change and REDD+ (BNCCREDD+)
Focal Point of GCF
MADAGASCAR

Copy to:
The Minister of Environment and Sustainable Development
The General Secretary of MEDD

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



Ministry of Environment, Climate Change and Technology
Male', Republic of Maldives



Date: 10 May 2022

No: 438-CCD/PRIV/2022/161

Mr. Yannick Glemarec,
Executive Director,
Green Climate Fund ("GCF"),
Songdo International Business Centre
175, Art Center-Daero Yeonsu-gu
Republic of Korea.

Sub: Funding proposal for the GCF by Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden (FMO) regarding the Climate Investor Two programme.

Dear Sir,

We refer to the programme titled Climate Investor Two as included in the funding proposal submitted by Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden to us on 16 November 2021.

The undersigned is the duly authorized representative of the Ministry of Environment, Climate Change and Technology, Ms. Aishath Aileen Niyaz, the National Designated Authority of Maldives.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Maldives has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Maldives;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.



We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely,

A handwritten signature in black ink, appearing to be 'Aishath Aileen Niyaz'.

Aishath Aileen Niyaz,
Director

MINISTÈRE DE L'ÉNERGIE, DES MINES ET DE L'ENVIRONNEMENT
DÉPARTEMENT DE L'ENVIRONNEMENT



وزارة الطاقة والمعادن والبيئة - قطاع البيئة

To: The Executive Director
The Green Climate Fund
Songdo International Business Centre
175, Art Center-Daero
Yeonsu-gu, Incheon 22004
Republic of Korea

03043

Rabat, March 17th 2021

Re: The proposal for the GCF by Nederlandse Financierings-Maatschappij Voor Ontwikkelingslanden N.V. (FMO) regarding Climate Investor Two.

Dear Madam, Sir,

We refer to the programme “Climate Investor Two” in Morocco as included in the funding proposal submitted by Nederlandse Financierings-Maatschappij Voor Ontwikkelingslanden N.V. (FMO) to us on January 2021.

The undersigned is the duly authorized representative of M. Bouzekri RAZI, the focal point of Morocco.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the proposed programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Morocco has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Morocco's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal, is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal, has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

M. Bouzekri RAZI

Director of Climate Change, Biodiversity and Green Economy
Ministry of Energy Mining and Environment
Kingdom of Morocco
GCF National Focal Point

ny

Directeur des Changements
Climatiques, de la Diversité Biologique
et de l'Economie Verte

Signé : Bouzekri RAZI



REPUBLIC OF NAMIBIA

MINISTRY OF ENVIRONMENT, FORESTRY AND TOURISM

Tel. No. 061 – 2842111
Fax. No. 061 - 229936
E-mail: Paulus . Ashili@met.gov.na
Enquiries: Mr. P. Ashili

Cnr of Dr. Kenneth David Kaunda Street
 & Robert Mugabe Avenue
 Private Bag 13306
Windhoek

Date: 02 December 2021

The Executive Director
 The Green Climate Fund
 Songdo International Business Centre
 175, Art Center-Daero Yeonsu-gu, Incheon 22004
 Republic of Korea

RE: Concept note for the GCF by Nederlandse Financierings-Maatschappij Voor Ontwikkelingslanden N.V. (FMO) regarding Climate Investor Two programme

Dear Sir/Madam,

We refer to the programme "*Climate Investor Two*" as included in the concept note submitted by the Nederlandse Financierings-Maatschappij Voor Ontwikkelingslanden N.V. (FMO) to us on 12.02. 2021.

The undersigned is the duly authorized representative of the Ministry of Environment, Forestry and Tourism, the National Designated Authority of Namibia. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the concept note.

By communicating our no-objection, it is implied that:

- (a) The government of the Republic of Namibia has no objection to the programme as included in the concept note;
- (b) The programme as included in the concept note is in conformity with the national priorities, strategies and plans of Namibia;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the concept note is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no objection to the programme as included in the concept note has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

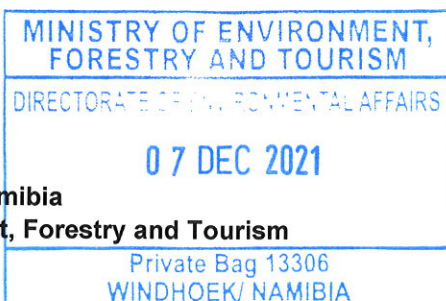
We acknowledge that this letter will be made publicly available on the GCF website.

Yours Sincerely,

Petrus Muteyauli

GCF Focal Point for Namibia

Ministry of Environment, Forestry and Tourism





FEDERAL MINISTRY OF ENVIRONMENT

HEADQUARTERS, MABUSHI, ABUJA.

FMENV/DCC/GCF/035/33

9 April, 2021

Ref. No.....

Date.....

The Green Climate Fund ("GCF")
Songdo International Business District,
175, Art Centre- Daero,
Yeonsu-gu, Incheon 406-840
Republic of Korea

Dear Sir,

**Re: Funding proposal for the GCF by Climate Fund Managers regarding the Nederlandse
Financierings- Maatschappij Voor Ontwikkelingslanden FMO "Climate Investor 2 Programme"**

We refer to the programme "Climate Investor 2" in the Federal Republic of Nigeria as included in the Concept Note submitted by Climate Fund Managers to us on 21st January 2021.

The undersigned is the duly authorized representative of Federal Ministry of Environment, the National Designated Authority/focal point of the Federal Republic of Nigeria.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of the Federal Republic of Nigeria has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the Federal Republic of Nigeria's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Halima Bawa-Bwari
GCF Focal Person Nigeria,



Republic of the Philippines
DEPARTMENT OF FINANCE

Roxas Boulevard Corner Pablo Ocampo, Sr. Street
Manila 1004

Mr. YANNICK GLEMAREC

Executive Director
Green Climate Fund (GCF)
175 Art Center-daero
Yeonsu-gu, Incheon
Republic of Korea

SUBJECT : No Objection Letter (NOL) for the Funding Proposal "Climate Investor Two Programme" (CI2) by the FMO - Dutch Entrepreneurial Development Bank

Dear **Mr. GLEMAREC**:

We refer to the concept note and funding proposal titled "Climate Investor Two Programme" (CI2) by FMO - Dutch Entrepreneurial Development Bank.

The undersigned is the duly authorized representative of the Department of Finance, the National Designated Authority of the Republic of the Philippines.

Pursuant to GCF decisions B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the concept note and funding proposal.

By communicating our no-objection, it is implied that:

- (a) The Government of the Republic of the Philippines has no-objection to the programme as included in the concept note and funding proposal;
- (b) The programme as included in the concept note and funding proposal is in conformity with the national priorities, strategies and plans of the Government of the Republic of the Philippines; and
- (c) In accordance with the GCF's environmental and social safeguards, the project included in the concept note and funding proposal is in conformity

A handwritten signature in black ink, appearing to be a stylized 'G' or similar character.

with relevant national laws and regulations.

We also confirm that our national process for ascertaining no objection to the programme as included in the concept note and funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Very truly yours,


CARLOS G. DOMINGUEZ

Secretary of Finance

MAY 16 2022





GOVERNMENT OF SIERRA LEONE
Environment Protection Agency
Ministry of the Environment
92, Dundas Street, Freetown



Ref: EPA-SL/GCF/NDA/FMO/CN/4/22

To: The Green Climate Fund ("GCF")

4th May 2022.

Re: Funding proposal for the GCF by Nederlandse Financierings-Maatschappij Voor Ontwikkelingslanden N.V. (FMO) regarding Climate Investor Two

Dear Madam, Sir,

We refer to the programme titled the climate investor two, a multi-country intervention, including Sierra Leone in as included in the funding proposal submitted by submitted by FMO to us on 22nd November 2021

The undersigned is the duly authorized representative of Environment Protection Agency, the National Designated Authority/focal point of Sierra Leone.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Sierra Leone has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Sierra Leone;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

.....
Dr. Bondi Gevao
National Designated Authority
Executive Chairman, EPA-SL



forestry, fisheries & the environment

Department:
Forestry, Fisheries and the Environment
REPUBLIC OF SOUTH AFRICA

Private Bag X447, Pretoria, 0001, Environment House, 473 Steve Biko Road, Pretoria, 0002 Tel: +27 12 399 9000, Fax: +86 625 1042

Ref: EDMS 217568

Enquiries: Shahkira Parker

Tel: 012 399 9240 **Email:** sparker@environment.gov.za

The Green Climate Fund
Songdo Business District
175 Art center-daero
Yeonsu-gu
INCHEON 22004
Republic of Korea

Email: fundingproposal@gcfund.org

FUNDING PROPOSAL FOR THE GREEN CLIMATE FUND BY NEDERLANDSE FINANCIERINGS- MAATSCHAPPIJ VOOR ONTWIKKELINGSLANDEN (FMO) REGARDING THE CLIMATE INVESTOR TWO PROGRAMME

Dear Sir/Madam

We refer to the programme titled Climate Investor Two as included in the funding proposal submitted by the Nederlandse Financierings-Maatschappij Voor Ontwikkelingslanden (FMO) to us on 19 March 2022.

The undersigned is the duly authorized representative of the Department of Forestry, Fisheries, and the Environment, the National Designated Authority/focal point for South Africa.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the Programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of South Africa has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of South Africa ;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.



Batho pele- putting people first

The processing of personal information by the Department of Forestry, Fisheries and the Environment is done lawfully and not excessive to the purpose of processing in compliance with the POPI Act, any codes of conduct issued by the Information Regulator in terms of the POPI Act and / or relevant legislation providing appropriate security safeguards for the processing of personal information of others.

**FUNDING PROPOSAL FOR THE GREEN CLIMATE FUND BY NEDERLANDSE FINANCIERINGS-⁹²
MAATSCHAPPIJ VOOR ONTWIKKELINGSLANDEN (FMO) REGARDING THE CLIMATE
INVESTOR TWO PROGRAMME**

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours sincerely

A handwritten signature in black ink, appearing to read 'Nomfundo Tshabalala', with a stylized flourish at the end.

Ms Nomfundo Tshabalala
DIRECTOR-GENERAL
DATE: 18/05/2022

Telephone : 256 41 4232095/341286
 Fax : 256 41 4233524
 Email : finance@finance.go.ug
 : treasury@finance.go.ug
 Website : www.finance.go.ug
 Plot No. 2-8 Apollo Kaggwa Road
 In any correspondence on
 This subject please quote No. ALD 79/251/03



Ministry of Finance, Planning &
 Economic Development,
 P.O Box 8147
 Kampala, Uganda

21st February 2022

The Executive Director,
 Green Climate Fund,
 Songdo Business District,
REPUBLIC OF KOREA.

**RE : LETTER OF NO OBJECTION FOR THE CLIMATE
 INVESTOR TWO (CI2)**

I refer to the above programme, which has been prepared and submitted to you by the FMO Climate Fund Managers for funding by the Green Climate Fund.

The undersigned is the duly authorized representative of the Ministry of Finance, Planning and Economic Development, the National Designated Authority focal point of Uganda.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the above programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

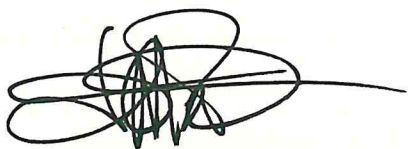
- (a) The Government of the Republic of Uganda has no-objection to the funding proposal;
- (b) The programme is in conformity with Uganda's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the program has been duly followed and that it applies to all projects or activities to be implemented within the scope of the programme.

Mission

"To formulate sound economic policies, maximize revenue mobilization, ensure efficient allocation and accountability for public resources so as to achieve the most rapid and sustainable economic growth and development"

We acknowledge that this letter will be made publicly available on the GCF website.



Ramathan Ggoobi

**PERMANENT SECRETARY/ SECRETARY TO THE
TREASURY/NATIONAL DESIGNATED AUTHORITY FOCAL POINT
FOR THE GREEN CLIMATE FUND**

Mission

"To formulate sound economic policies, maximize revenue mobilization, ensure efficient allocation and accountability for public resources so as to achieve the most rapid and sustainable economic growth and development"

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Climate Investor Two (CI2)
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Private
Accredited entity	Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden N.V. (FMO)
Environmental and social safeguards (ESS) category	Category I-1
Location – specific location(s) of project or target country or location(s) of programme	Bangladesh, Botswana, Brazil, Colombia, Djibouti, Ecuador, India, Indonesia, Ivory Coast, Kenya, Madagascar, Maldives, Morocco, Namibia, Nigeria, Philippines, Sierra Leone, South Africa, and Uganda*
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	Friday, February 25, 2022
Language(s) of disclosure	English, Arabic, Bengali, French, Hindi, Indonesian, Portuguese, and Spanish*
Explanation on language	English is the relevant language of Botswana, Kenya, Maldives, Namibia, Nigeria, the Philippines, Sierra Leone, South Africa, and Uganda. Arabic is the relevant language of Morocco. Bengali is the official language of Bangladesh. French is the relevant language of Djibouti, Ivory Coast and Madagascar. Hindi is the relevant language of India. Indonesian is the relevant language of Indonesia. Portuguese is the relevant language of Brazil. Spanish is the relevant language of Colombia and Ecuador*
Link to disclosure	https://www.fmo.nl/disclosure-esmf-ci2

Other link(s)	Indonesian https://climatefundmanagers.com/wp-content/uploads/2022/06/CFM-ESMS-Indonesian.pdf Bengali https://climatefundmanagers.com/wp-content/uploads/2022/06/CFM-ESMS-Bengali-.pdf Hindi https://climatefundmanagers.com/wp-content/uploads/2022/06/CFM-ESMS-Hindi.pdf Arabic https://climatefundmanagers.com/wp-content/uploads/2022/06/CFM-ESMS-Arabic.pdf French https://climatefundmanagers.com/wp-content/uploads/2022/06/CFM-ESMS-French.pdf Portuguese https://climatefundmanagers.com/wp-content/uploads/2022/06/CFM-ESMS-Portuguese.pdf Spanish https://climatefundmanagers.com/wp-content/uploads/2022/06/CFM-ESMS-Spanish.pdf English https://climatefundmanagers.com/wp-content/uploads/2022/06/CFM-ESMS-English.pdf
Remarks	An Environmental and Social Management System (ESMS) consistent with the requirement for a Category I-1 programme is contained in the “Environmental and Social Management System (ESMS) Manual”.* Climate Fund Managers (CFM) is the fund manager for the Climate Investor Two (CI2) fund. As co-founder and initiator of CI2, FMO has also taken a significant ownership stake in CFM. Considering (i) the status of FMO being a 50 percent shareholder of CFM, as well as (ii) the fact that all previous E&S safeguards-related information on CI2 has been published on the CI2 and CFM websites, the present ESMS disclosure is being done via the CI2 and CFM websites instead of the FMO website. Links to the CI2 and CFM websites may be found on FMO’s CI2 web page. For further details on the ownership role of FMO in relation to CFM, as well as on the overall involvement of FMO in CI2 as an initiator and investor, please refer to the below link to the FMO website: https://www.fmo.nl/governance-related-to-cio-and-cfm

Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity's website	IPP Framework and RPF are part of the ESMS Manual*
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, February 25, 2022
Place	Online and in CFM offices in: Cape Town (4th Floor, The Citadel 15 Cavendish Street, Claremont, Cape Town 7700, South Africa), and The Hague (Korte Vijversberg 4 2531BA – the Hague) and Singapore
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	TBD
Date of GCF's Board meeting	Sunday, July 17, 2022*

Note: This form was prepared by the accredited entity stated above.

*Subsequent to the disclosure of this form to the Board and active observers on 25 February 2022, the following updates have been made: Egypt, Mexico, Peru, Senegal, Thailand and Viet Nam have been deleted as they have been removed from the programme. Moreover, after further due diligence on the funding proposal, the Secretariat has confirmed that the "Environmental and Social Management System (ESMS) Manual" is consistent with the requirements for a Category I-1 programme. The accredited entity has also updated the ESMS Manual to clarify the procedures and requirements for managing environmental and social risks and impacts pursuant to the ESS standards and comply with the revised Environmental and Social Policy adopted by decision B.BM-2021/18 as regards SEAH with respect to projects and programmes approved at or after the thirty-second meeting of the Board (B.32). Lastly, the date of the GCF Board meeting in which the funding proposal is to be considered has been updated.

Independent Technical Advisory Panel's assessment of FP190

Proposal name:	Climate Investor Two
Accredited entity:	FMO – Dutch Entrepreneurial Development Bank
Project/programme size:	Large

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential *Scale: Medium*

1. Overview: Climate Investor Two (CI2) intends to develop and construct infrastructure projects in developing countries in the water, sanitation and ocean sectors that reduce the effects and consequences of climate change by mitigating greenhouse gas (GHG) emissions and by increasing resilience. The CI2 structure builds on and repurposes the multi-tranche blended finance approach utilized by Climate Investor One (FP099) (approved by the Board at its twenty-first meeting (B.21)) while targeting new sectors. CI2 seeks to accomplish the programme's objectives by mobilizing public investments and unlocking private capital across the following sectors:

- **Water:** The water sector is defined by projects that source, transport and treat water from municipalities and other sectors. Climate adaptation benefits include building infrastructure that sources, transports and treats the water necessary for both municipal and industrial users in countries/subnational regions undergoing or expected to undergo water stress because of a changing climate;
- **Sanitation:** The sanitation sector looks at treating waste and wastewater to avoid and sequester methane emissions and increase water availability in areas experiencing increasing water stress as a result of climate change;
- **Ocean:** The oceans and ecosystems sector aims to (a) reduce GHG emissions from and enhance the resilience of ports and shipping; and (b) protect, strengthen and restore important ecosystems to safeguard their services in buffering droughts, floods and storm surges intensified by climate change.

2. CI2 intends to deploy a whole-of-life financing approach utilizing two independent but operationally interlinked funds. The two funds focus on different stages of the project lifecycle: a Development Fund (DF) provides reimbursable grants to early-stage climate projects, whereas the Construction Equity Fund (CEF) provides follow-on finance during the construction phase. The DF has been set up to provide development loans (in the form of reimbursable grants), accompanied by technical assistance provided by Climate Fund Managers (CFM) and third parties, to project companies. The purpose of this is to fast-track qualifying development projects to financial close by reducing the development timeline and improving the ultimate bankability of infrastructure projects. Once projects successfully complete their development phase, CEF reimburses the grant to DF, and the project moves on to the construction phase. In this way, CEF ensures that DF remains capable of recycling its capital.

3. CEF has been set up to finance the construction phase by providing up to 75 per cent of the required project equity. The CEF equity approach is designed to address a

structural barrier of many local banks and lending institutions being unwilling to assume construction risk. It further aims to decrease the construction time and costs by: (i) reducing the complexity and costs at the project level through an equity-only funding strategy; (ii) reducing the construction timeline and thereby reducing the exposure period to investors; (iii) reducing costs by eliminating certain debt-specific costs; and (iv) recycling capital from completed projects to new projects to reduce the time any funding 'stands still'.

4. CEF itself is structured as a blended finance vehicle to better mobilize commercial capital at scale into CEF. It is made up of a multi-tiered structure to accommodate multiple investors' risk/return preferences. The three tranches are structured as follows:

Tier 1 makes up 20 per cent of CEF and is comprised of junior equity;

Tier 2 makes up 40 per cent of CEF and is comprised of ordinary equity; and

Tier 3 makes up 40 per cent of CEF and is comprised of senior equity.

5. The CI2 funding proposal requests a total GCF contribution of USD 145 million, of which USD 45 million would be directed to DF and USD 100 million to Tier 1 of CEF. The expected capital size of the prospective funds is estimated at USD 90 million for DF and USD 1 billion for CEF. GCF funding would play a key catalytic role in enabling CI2 by providing junior Tier 1 capital at concessional rates (0 per cent + inflation Consumer Price Index (CPI)). GCF participation in Tier 1 is intended to reduce investment risks for the larger Tier 2 and Tier 3 tranches and thus mobilize private sector finance into the facility. For instance, the junior status of Tier 1 capital ameliorates risks associated with investing in climate-related infrastructure in developing countries such as local currency fluctuation risk. This facilitates CI2's ability to invest in projects where revenues are denominated in local currency while at the same time mitigate real and perceived local currency risk on the part of commercial investors.

6. This programme would be GCF's first at-scale private sector programme in the water and sanitation sector. The programme aims to mobilize capital at scale. The USD 145 million investment by GCF will potentially catalyse USD 945 million at the fund level, and with further leverage assumed at the project level, the total capital expected to be mobilized will be higher.

7. The Dutch Entrepreneurial Development Bank (FMO) will act as the accredited entity (AE) for this proposal. CFM, a joint venture between FMO and Sanlam, will act as the fund manager of CI2 and is the first executing entity (EE). DF and CEF, the two funds within CI2, will act as the second and third EEs.

8. Impacts: CI2 aims to make investments which result in the scaling and replication of water, sanitation and ocean infrastructure that is aligned with the adaptation and mitigation needs of developing countries. The investments themselves are expected to generate measurable climate impacts and are also intended to have a market demonstration effect. CI2 contends that the financial and impact success of low-emission, climate-resilient sustainable development in host countries, with a risk-adjusted return, will catalyse commercial investors through demonstration.

9. The water cycle is a key component of the global climate crisis. The World Bank estimates that by 2050, water shortages caused by climate change could reduce the gross domestic product (GDP) of developing countries in Asia by USD 4.4 trillion and developing countries in Africa by USD 1.6 trillion, compounding existing health and development challenges. Water can also be a substantial carbon emitter when managed improperly, with methane emissions from untreated waste and wastewater accounting for 7 per cent of global GHG emissions. Riverine, wetland, marine and coastal ecosystems have the potential to act as 'carbon sinks' that sequester and avoid carbon emissions, and restoring them can enhance their services in buffering against climate change-intensified disasters.

10. The financing of the water, sanitation and ocean sectors has primarily been within the public domain. However, the budgets that developing countries have available to invest in these sectors are insufficient for the task at hand. Adequately addressing the lack of low-emission water infrastructure that would contribute to staying within the 1.5-degree pathway while ensuring resilient water supply in areas experiencing reduced water availability because of climate change will require the mobilization of private sector finance. Traditional private investors, however, perceive the sectors as high-risk, particularly in developing countries.

11. CI2's two-fund blended approach is intended to address these risks (real and perceived) by both (i) providing private sector investors with a blended finance solution that reduces their risks; and (ii) demonstrating the commercial viability of water, sanitation and ocean infrastructure projects by demonstrating the possibility of risk-adjusted returns to investors. This can help investors to see the long-term benefits of investing in such sectors and countries and mitigate some of the perceived high risks.

12. Furthermore, highlighting the commercial viability of water, sanitation and ocean infrastructure projects will help provide know-how and capacity for the development and implementation of such projects in the host countries by providing a transfer of knowledge to local parties. CI2 aims to do this by including local partners and developers in both the development as well as the construction phases of its projects. Lastly, it will aim to include local financial intermediaries, including direct access entities, in the refinancing of its projects.

13. Successfully developing and implementing mitigation and adaptation projects for the water, sanitation and ocean sectors in developing countries will involve: (i) improving wastewater treatment capacity; (ii) improving access to water; (iii) improving access to sanitation; (iv) improving renewable energy capacity and generation; (v) improving access to fuel-efficient green ports and shipping; and (vi) restoring and/or protecting riverine, wetland, coastal and marine ecosystems. In terms of mitigation potential, improving wastewater treatment and capacity, renewable energy capacity and generation, and access to fuel-efficient green ports and shipping, and restoring and/or protecting riverine, wetland, coastal and marine ecosystems will then result in reduced GHG emissions in the water, sanitation and oceans sectors of host countries. With regard to mitigation benefits, CI2 is expected to reduce or avoid approx. 6 million tonnes of carbon dioxide equivalent (tCO₂eq) per year for a total of approx. 56million tCO₂eq over the implementation period.

14. In terms of adaptation potential, developing and constructing additional water supply, treatment, distribution and wastewater treatment facilities, as well as enhancing ecological infrastructure, will enable CI2 to reduce the vulnerability of developing countries to climate hazards. CI2 aims to construct up to USD 3.7 billion in infrastructure projects in developing countries. The projects financed, developed and constructed by CI2 are expected to provide approx. 14 million people with year-round access to reliable and safe water supplies despite climate shocks and stresses. They will also protect and strengthen ecosystems covering approx. 900,000 hectares to help them adapt to climate variability and change.

15. CI2 is also expected to contribute to technology transfer by proactively cooperating with local partners and developers during project development as well as during the the construction of the infrastructure assets.

16. It is important that all investment projects in which GCF funds are used to take an equity stake in CI2 do ultimately deliver on the specific climate change adaptation and mitigation impacts sought in the funding proposal. All potential investment projects claiming climate change adaptation impacts must go through a robust climate adaptation assessment at the level of investee, activity and/or site prior to any decision on investment through DF

and CEF. With regard to the broad way in which the funding proposal discusses interwoven development and climate challenges, the independent Technical Advisory Panel (TAP) identified a risk that a shortage of viable deals could lead to low standards being applied by CFM in the selection of investment projects aimed at adaptation, with CI2 ultimately funding business-as-usual water and sanitation investment projects that lack any clearly motivated, specifically contextualized climate adaptation logic. In response to questions from the independent TAP on this, the AE highlighted the screening process the Climate Investor Two Development Fund Investment Committee will use to judge proposals with intended adaptation benefits: “The deal teams will provide [the Climate Investor Two Development Fund Investment Committee] with historic climate trends (observed changes or variability in changes in the project areas), climate change projections in the areas, and climate impact (an explanation on how these changes will hurt communities in the project area). Adaptation benefits will be further addressed by explaining how the project company’s products, services, or technology will help the community adapt to these climate impacts, an explanation of how the proposed intervention will not lead to maladaptation, an explanation of the Rio Marker score for climate adaptation (to be achieved by the company) and an overview of the impact indicators that will be monitored to validate the above (including an identification of the baseline and mid-term (year 10) and final (year 20) projections).”

17. Overall impact potential is assessed as Medium. The Medium rating is conservative and reflects pipeline risk (Do these project opportunities turn into concrete investments?) and execution risk. CFM demonstrated its capacity to manage a similar vehicle in Climate Investor One, but migrating to new and arguably more challenging sectors will require new capacity and expertise, leading us to hedge our impact rating somewhat.

1.2 Paradigm shift potential

Scale: Medium

18. The focus of the funding proposal is an innovative investment scheme enabling new investors with no or minimal track record in developing markets to enter new sectors previously considered as high risk (water, sanitation, oceans), accelerating the development and implementation of projects and reducing costs across the entire process. CI2 provides different types of technical support to the private sector at the development stage as well as ready investment for the construction and operational stages, all of which are fundamental elements of the investment scheme and might be considered innovative for the targeted host countries.

19. CI2 aims to create a paradigm shift by making investments which result in the scaling and replication of water, sanitation and ocean infrastructure that is aligned with the adaptation and mitigation needs of developing countries. These are sectors that are perceived as risky or better suited for public finance, especially in developing countries. CI2 has the paradigm shifting potential of introducing the private sector into a space dominated by public funding, although the success of these efforts will depend on participating countries’ policy and regulatory environments. Additionally, CI2 has the potential to demonstrate the actual versus perceived risk of climate-beneficial investments in the water sector and blue infrastructure generally. Data and learning from CI2 has the potential to change the way financial markets see these sectors by demonstrating the commercial viability and transferring sectoral technology and knowledge to local communities.

20. Moreover, CI2’s whole-of-life financing approach itself is paradigm shifting by eliminating a series of barriers to low-carbon and climate-resilient development by ensuring that projects can progress smoothly and expeditiously from a concept or pilot project to a scaled, sustainable and replicable operation. CI2’s structure is a useful catalytic tool, as demonstrated by Climate Investor One’s success and now by the willingness and appetite of CFM and its investors to shift into more commercially difficult sectors with CI2. Its

application to a new thematic area sets a precedent for similar approaches to be implemented by other institutions and in other sectors. By demonstrating a replicable and adaptable approach to catalysing private sector financing of climate-focused infrastructure, local project developers and international equity funds may adopt a similar approach, building on CI2's precedent.

21. CI2's potential to catalyse impact in this manner is amplified by the scarcity of infrastructure-themed climate adaptation funds, which, despite the growing urgency for adaptation finance, can falter for a lack of established precedent. Bringing CI2 to market will provide recognition that climate adaptation-focused funds can work in practice at scale, contributing to a rebalancing of the primarily mitigation-themed funding available at present.

22. In the longer term, CI2 will also create opportunities for scalability and replicability beyond its investments, through the transfer of knowledge and technical skills to local partners.

23. Overall paradigm shift potential is assessed as Medium. Replicability of the investment scheme will depend on its success and results.

1.3 Sustainable development potential

Scale: High

24. Alignment with Sustainable Development Goals (SDGs): CI2 contributes to the achievement of several SDGs, primarily Climate Action (SDG 13), Affordable and Clean Energy (SDG 7), Clean Water and Sanitation (SDG 6), and Partnerships for the Goals (SDG 17). By contributing to the above, CI2 can provide potential co-benefits relating to the following SDGs: Good Health and Well-Being (SDG 3), Gender Equality (SDG 5), Decent Work and Economic Growth (SDG 8), Industry, Innovation, and Infrastructure (SDG 9) and Life Below Water (SDG 14).

25. Environmental co-benefits: CI2 can improve water quality and in turn increase biodiversity and improve the livelihoods and resilience of local communities. Ecosystems and ecosystem management will also protect communities from floods and increase biodiversity and food stock.

26. Social and health benefits: Changing climate patterns, together with growing economies and populations, could reduce urban water availability in 2050 by two-thirds compared to 2015 levels, putting over USD 6 trillion of developing nations' GDP at risk. A study done by the World Health Organization in 2017 shows that globally, more than 785 million people did not have access to at least basic water services, and more than 884 million people do not have immediate access to safe drinking water. One of the objectives of CI2 in the water sector is to realize climate change adaptation impacts in countries undergoing, or expected to undergo, water stress, by building the infrastructure which sources, transports and treats the water necessary for both municipal and industrial users to be resilient to the localized impacts of climate change.

27. Gender-sensitive development impact: The lack of sanitation facilities is even more impactful for young women, as they are more likely to miss out on education due to insufficient sanitation at schools.¹ Additionally, women are often more negatively impacted as they are frequently responsible for collecting the water for use in the household. This and other household responsibilities reduce the likelihood of women receiving education. By

¹ UNICEF. 2020. Progress on drinking water, sanitation, and hygiene in schools. Available at <<https://washdata.org/sites/default/files/2020-08/jmp-2020-wash-schools.pdf>>.

providing and improving sanitation infrastructures, CI2 will enhance the livelihoods of women.

28. Economic co-benefits: In economic terms, the annual economic value-added output derived from the oceans is estimated at USD 2.6 trillion, making the oceans the world's seventh-largest economy (after the United Kingdom),² showcasing the ocean's importance for many livelihoods. Regarding the water and wastewater treatment sectors, equipment market size was valued at USD 61.6 billion in 2020. The market value is expected to reach USD 84 billion by 2028, growing at a Compound Annual Growth Rate of four per cent between 2021 and 2028.³ This growth will be fuelled by a rising demand for clean water amid shrinking water access worldwide.

29. The independent TAP assesses this programme as having High sustainable development potential with significant social, economic and health co-benefits.

1.4 Needs of the recipients

Scale: Medium–High

30. The United Nations Educational, Scientific and Cultural Organization estimates that over 2 billion people currently live in areas subject to water stress, with water availability in many areas worsening because of changes in precipitation patterns. Additionally, as noted earlier, up to 80 per cent of illnesses in the developing world are linked to inadequate water and sanitation, as well as pollution combined with climate change-induced rising sea levels, which contaminates trusted water sources globally. These factors can detrimentally impact local health, safety and economic opportunity. CI2 will provide much needed infrastructure and nature-based solutions for local communities of targeted host countries.

31. The CI2 programme targets 25 different countries, spread over Latin America, Asia and Africa, that all have significantly different scales and intensities of climate vulnerability (including water-stressed areas). CI2 is designed to assess the needs and determine the intervention required on a case-by-case and per country basis. The 25 countries have been selected based on their nationally determined contributions (NDCs) and whether CI2 products are suitably aligned.

32. To attract capital into host countries, CI2 requires investment from GCF as catalytic capital to help countries attract capital from private investors who have not been active in the water, sanitation and ocean sectors.

33. The independent TAP assesses this programme as Medium–High in terms of needs of recipients.

1.5 Country ownership

Scale: Medium–High

34. CFM has a demonstrated track record from Climate Investor One in building a positive relationship with many of the identified country national designated authorities (NDAs) and in procuring no-objection letters (NOLs) from such countries. CFM and the AE have already sought NOLs from 25 countries for CI2. A summary overview of CI2 was shared with each

² Sumaila, U.R., Walsh, M., Hoareau, K. et al. Financing a sustainable ocean economy. Available at <<https://www.nature.com/articles/s41467-021-23168-y#citeas>>.

³ Statista 2022. Available at <<https://www.statista.com/statistics/1258838/water-and-wastewater-treatment-equipment-market-size/>>.

NDA, along with the CI2 concept note. As of May 2022, CI2 has received 14 NOLs and were expecting approximately 18–20 NOLs to be in place by B.32.

35. CFM maintains regional offices in the relevant CI2 regions. CFM's regional office for Africa is in Cape Town, South Africa. Established in 2017, the office has 19 full-time employees. CFM's regional office for Asia is in Singapore. Also established in 2017, the office has 14 full-time employees. CFM's regional office for the LAC region was established towards the end of 2021, and it is finalizing arrangements for its office in Bogotá, Colombia. The team consists of four full-time employees. To strengthen this local presence, the AE (FMO) also has headquarters in Kenya, Singapore and South Africa.

36. The country selection for CI2 was based on national climate plans, including NDCs and national adaptation plans, as well as sector plans from relevant ministries, including water, sanitation, energy and environment, and other ministries as may be applicable. The countries selected recognize the importance of water, wastewater management and renewable energy.

37. We assess the country ownership of the funding proposal as Medium–High.

1.6 Efficiency and effectiveness

Scale: Medium–High

38. The GCF's investment in CI2 is expected to mobilize a ratio of 1:4 at the fund level (as every USD 1 of Tier 1 will mobilize USD 4 in the more senior tranches). The total leverage amount of private sector financing resulting from GCF funding is estimated at 1:16 (fund-level plus project-level mobilization). The impacts of the programme are dependent on the ability of FMO and CFM to raise additional capital of USD 945 million from investors. CI2 has already achieved its first close with total commitments of USD 675 million.

39. CI2's primary efficiency and effectiveness benefits are derived from its whole-of-life financing approach. Funding made available by CI2 at all stages of the project, including early-stage development and for pilot schemes, generates a pipeline of projects while building the capacity of local developers. The recycling character of CI2 enables the development and construction of more projects during the implementation period of the fund (the capital is expected to be recycled twice).

40. The whole-of-life approach also reduces the time project developers usually lose to fundraising at the different phases of project development. CI2 also provides financing flexibility by preferring equity financing during the construction phase of the project. This eliminates cash flow and financial distress from projects during construction, while also ensuring the long-term interest of the owners and investors in the project.

Table. Project funding and costs

Total framework size	USD 1,090.0 million
Requested GCF amount	USD 145.0 million
Total greenhouse gas emission reduction	about 50.0 MtCO ₂ eq
Estimated overall cost	USD 2.9/tCO ₂ eq
Estimated GCF cost	USD 21.8/tCO ₂ eq

41. We note that although the CFM is experienced in fund management and in making equity investments in climate-focused sectors and projects, it does not have investment experience/a track record in the water sector and ocean-related investments. Hence, the success of this CI2 hinges upon the ability of CFM to establish a team of investment and climate professionals who have sufficient experience and a track record in investing in water, sanitation and ocean-related investments.

42. This includes the need for expertise in the relatively untested and challenging area of "green infrastructure" investments (grouped under the Oceans theme on the grounds of a

“catchment-to-coast” approach), for example wetland restoration for water retention or flood control, catchment restoration for water infiltration and erosion prevention, or mangrove restoration for coastal protection. The origination of such investment projects may rely on complex deals involving voluntary carbon market funding or other “payments for ecosystem services” such as income streams, and the establishment of innovative public-private partnerships. The existence of DF and the expertise of CFM in a range of adaptation strategies, including ecosystem-based adaptation, give the independent TAP some confidence that these challenges can be met. Although ecological infrastructure investments are likely to form only a small portion of the overall CI2 portfolio, this kind of innovation would be welcome.

43. The funding proposal is assessed with a rating of Medium–High in terms of efficiency and effectiveness.

II. Overall remarks from the independent Technical Advisory Panel

44. Based on the assessment above, the independent TAP endorses this funding proposal.

45. We are pleased to see earlier GCF-approved blended finance models being expanded and applied to harder-to-solve financial barriers: mobilizing private finance for mitigation and adaptation in the water sectors.

46. The one area that gave us some pause was the level of concessionality requested (0 per cent return + CPI for GCF funding), albeit the request was in line with the level previously provided to Climate Investor One. In our view, the appropriate level of concessionality required to leverage private sector finance is an evolving target, as more firms become experienced in climate investments and are engaged in developing countries generally. What might have been the appropriate amount of concessionality needed to mobilize private funding in 2017 may be greater than what is needed in 2022.

47. However, the information provided in the funding proposal and discussions with the AE made clear that the perceived risks associated with the water, sanitation and ocean sectors, and the lack of familiarity of private sector investors with this area, supported the requested levels of concessionality. Ultimately, we were persuaded that the risk of over-subsidization is low in this case.

48. Nevertheless, we would prefer to see future blended finance instruments with flexible structures that allow the GCF to make a return if the expected risks and barriers to fund mobilization and investment return prove to be overly conservative. We want to make sure that GCF concessionality strives to be priced (i) at the lowest necessary level to achieve the desired mobilization and climate impacts; (ii) to avoid crowding out commercial initiatives that do not have access to concessionality; and (iii) to support the financial sustainability of GCF itself.

49. We also note approvingly that CFM is providing a General Partner contribution of 1 per cent (up to USD 10 million) into CI2. Investors in private sector funds typically prefer to see managers allocate some of their fees into the fund itself or otherwise contribute. This is a sign of confidence in the expected returns and aligns investor and manager interests. We would prefer to see this become more commonplace in fund vehicles where GCF concessional funds are being requested. Additionally, we take note that CFM’s shareholders (which include the AE) are investing a combined total of USD 70 million into CI2.

50. Finally, we observe that true paradigm shift will be achieved only if the CI2 funds are able to build a track record of performing impactful and successful investments, which increases the confidence of asset owners and encourages new investment managers to

structure new vehicles in these sectors. For that reason, it is important that CI2's underlying investments ultimately deliver on the specific climate change adaptation and mitigation impacts sought in the funding proposal. All potential investment projects claiming climate change adaptation impacts are dependent upon a robust climate adaptation assessment being conducted at investee/activity/site-level prior to making a decision on investment. Owing to the programmatic nature of the proposal, such assessments would need to be carried out before project-specific investment decisions are made by CI2. This introduces a level of uncertainty as to the exact projects that will ultimately be financed. This is the nature of fund models where we must rely to a great extent on the capacity of AEs and their selection processes. In this case, we were ultimately able to become comfortable with the approach to be taken by CI2, including the use of third-party assessments to validate scoring at least 2 or higher in the Rio markers for climate mitigation and/or adaptation, as well as robust measurement, reporting and verification processes.

Reply to the Independent Technical Advisory Panel assessment findings (FP190)

Proposal name:	Climate Investor Two
Accredited entity:	Nederlandse Financierings-Maatschappij voor Ontwikkelingslanden N.V. (FMO)

Impact potential
The AE agrees with the iTAP's observation that Climate Investor Two has a challenging mandate for a private sector Programme for the reasons mentioned. However, the AE believes the mandate to be highly impactful, with a significant part of its mandate focusing on much-needed adaptation and resilience projects, and that pipeline risks are being mitigated as much as possible. The AE would therefore argue that the Impact Potential would warrant a high score.
Paradigm shift potential
The AE would like to highlight that the investment scheme has already been successfully piloted by Climate Investor One (FP099) since its first close in 2017.
Sustainable development potential
This is well noted
Needs of the recipient
This is well noted
Country ownership
This is well noted
Efficiency and effectiveness
The AE agrees that the Ecosystem part of the proposal has high impact potential but remains somewhat untested and challenging due to the lack of clear revenue streams. The AE does however already see a promising pipeline and is confident in its capabilities to source and execute deals for an anticipated 10% of the GCF's proceeds.
<i>Overall remarks from the independent Technical Advisory Panel:</i>
The AE agrees with most of the iTAP remarks. However, the AE would note that:

- It is of the opinion that the + CPI return for the Tier 1 providers is warranted due to the perceived investment risks of the water, sanitation, and oceans sectors (i.e., because of the lack of 'clear' revenue streams). Mobilizing private sector finance in these sectors still requires such levels of concessionality (which is also reflected by the status of the Tier 2 fundraising)
- The Climate Adaptation Assessment, and more detailed climate studies, are typically done during the Development Phase (pre-approval of the Construction Equity Fund)

Climate Investor 2: A Gender Assessment

Prepared for CFM

Date: 24 June 2022

Authors: Claire Hughes, Sue Cavill, Martha Keega, Georgia Plank, George Murray, Daan Schuttenbelt, and Victoria Tully

Submitted by Itad



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Executive Summary

Overview of Scope and Methodology

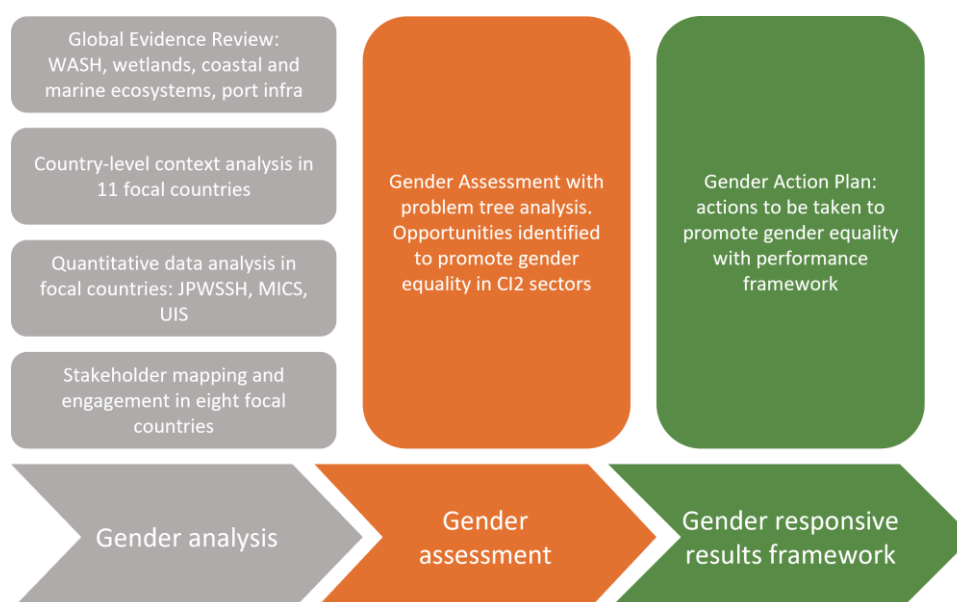
This Gender Assessment and Gender Action Plan forms part of Climate Fund Managers (CFM) funding proposal to the Green Climate Fund (GCF) for financial support to the Climate Investor 2 (CI2) to invest in infrastructure for water and sanitation services, the use and protection of wetlands, coastal and marine ecosystems as well as climate resilient ports in up to 25 countries.

The Gender Assessment explores how, in the context of climate change women and men are affected by problems in three main areas: i) access to and management of water and sanitation (WASH) services; ii) the protection and management of wetland, coastal and marine ecosystems; and iii) the development of climate resilient port infrastructure.

The preparation of the Gender Assessment comprised four main components (Figure 1):

- A global evidence review, of 85 studies in the areas of WASH, wetlands, coastal and marine protection and port infrastructure;
- Analysis of national WASH, climate change and gender policies in 11 priority countries, Bangladesh, Botswana, Brazil, Djibouti, Indonesia, Ivory Coast, Kenya, Morocco, Namibia, Nigeria, Uganda;
- Analysis of selected SDG related quantitative datasets available for access to WASH, the protection of wetland, coastal and marine ecosystems and girls' education.
- Stakeholder analysis in 8 out of the 11 priority countries to map actors in the WASH sector and engage stakeholders in the Gender Assessment process.

Figure 1: Methodology for the Gender Assessment and Action Plan



Findings from each of these components were used to develop a problem tree analysis for each of the three areas covered by the Gender Assessment and identify opportunities for promoting gender equality through CI2 investments.

The opportunities identified in the Gender Assessment are expanded upon in the linked Gender Action Plan which provides further details on how CFM and its partners will take advantage of the opportunities available to promote gender equality internally within Project Companies and in communities where CI2 supports investment. The Gender Action Plan includes a performance management framework with which CFM can be held accountable.

Overview of Global Evidence Review

WASH

WASH infrastructure and services for households, in health care facilities and in schools are inadequate in most developing country contexts. Their supply is further undermined by climate change, which affects weather patterns, reducing water availability and damaging WASH infrastructure.

Women and girls are disproportionately affected by inadequate WASH infrastructure and services. Their domestic and care-giving roles mean they are responsible for collecting water, for WASH practices within the household and for caring for family members when they are sick, sickness which often is a result of low-quality WASH services available to them. Women and girls can often travel long distances to collect water, limiting the time they can spend on other, more enriching activities such as attending school, or engaging in productive activities. It exposes them to the risk of violence whilst travelling to and from water points, and to a heightened risk of disease through greater exposure to poor quality WASH.

Women and girls' physiology means they have greater need for WASH facilities, for example, during menstruation or pregnancy, needs which often cannot be met. Cultural norms sometimes prevent women from sharing latrines with men, especially during menstruation, thereby limiting women and girl's access to WASH and forcing them into situations where they are exposed to the threat of violence. Girls often miss school when menstruating as toilet facilities are inadequate. As primary users of health services, women and girls face increased risk of infection and disease as a result of poor WASH facilities in health centres.

Although most affected by limitations in the WASH system, women and girls have limited influence over the planning, design and management of WASH infrastructure and services. Social norms often exclude women from public consultation spaces or silence their voices when they are invited in and so their needs and priorities do not inform the development of the WASH system. Female representation in the workforce is a major challenge in most countries in both government and NGO WASH services. Where women are found in the WASH sector workforce, they are more likely to be in junior or unpaid roles, although pioneering work in Asia is starting to break down traditional barriers.

Wetlands, Coastal and Marine Ecosystems

Relevant evidence identified through literature searches for the gender assessment speaks to: i) women and men's use of and interaction with wetland, coastal and marine areas; ii) how climate change is affecting these different uses and interactions, and the impact on the populations concerned; and, lastly, iii) the different ways in which men and women are, are not and could be involved in the protection and management of wetland, coastal and marine areas.

Key findings include that, as regards their participation in oceans-based industries (of which most of the relevant literature relates to fishing) women are disproportionately represented in informal and unregulated economies and concentrated in less profitable parts of value chains. Women's participation is curtailed by gendered norms which restrict their mobility, constrain their access to finance and discourage other community members from viewing them as "real fishers". Women's participation in the fishery industry is often overlooked and undervalued, resulting in the marginalisation of women who do participate in the management of wetland, coastal, and marine areas.

At the same time, and whilst the evidence base on the gendered impacts of unsustainable fishing practices is underdeveloped, it is women who are likely to be disproportionately affected by rising sea temperatures, levels and acidity.

As regards gender mainstreaming in sustainability and management initiatives, women's participation is currently very limited – a hindrance for the improvement of, for example, fishing sustainability because women tend to be more willing to engage in progressive sustainability practices. Policy and decision making remains mostly gender blind, with data collected by governments tending to focus on formal economies and evidence on gender-differentiated effects of innovation and technological advancements lacking. Unsurprisingly, then, existing women's empowerment interventions, tend to take a segregated, 'gap filling' approach which fails to address underlying structural barriers, perpetuating discriminatory norms and gendered inequalities.

Ports

The evidence base on the intersect between gender, climate change and port infrastructure is particularly limited, primarily focusing on i) the different ways in which men and women use and benefit from available port infrastructure, as well as ii) those in which men and women are, are not and could be involved in the development of climate resilient port infrastructure.

Women are not able to fully reap the benefits of economic growth from ports development due to their exclusion from related employment and, particularly, from operational and managerial positions. At the same time, there is a lack of evidence on the gender-differentiated effects of ports' negative impacts on ecosystems.

The development and expansion of new climate-resilient ports creates jobs and therefore opens up opportunities to train and hire women. Women's recruitment, integration and progression is starting to be supported through training, mentorship and initiatives to improve working environments. More broadly, women have an important role to play in promoting climate resilient port development for the good of wider communities – but continue to be marginalized in discussions around new approaches. In the same vein, data on gender-differentiated benefits of optimised infrastructure and energy efficiency remains scant.

Context Analysis in Focal Countries¹

As part of this gender assessment, 11 in-depth country analyses were completed to understand the national policy context for WASH in programming countries, as well as to identify relevant issues, opportunities and challenges at the national level to drive more equal outcomes. These analyses present a top-line picture of access to WASH facilities and look at the WASH context to understand how gender is addressed throughout. Summaries of these analyses can be found in Section 6 of this report, and the full country analyses are included in Annex 5, including a stakeholder mapping of the WASH sector.

Overall, most countries have access to improved drinking water for around 90% of their population, with a few outliers. Drawing on WHO/UNICEF's classification for WASH services, analysis suggests that the rate of access to different quality levels of service (e.g. either safely managed or basic drinking water services) varies more significantly. In terms of sanitation, access to improved facilities is more varied between countries, with some countries having eradicated open defecation while others are still working to get half of their population access to improved sanitation facilities.

¹ This section is based on detailed gender and energy profiles for each programming country. Key insights as they relate to the renewable energy sector are included in [Annex A].

The policy frameworks across different focal countries address a range of aspects related to water, sanitation and hygiene. All of them place a strong emphasis on the impact of climate change in exacerbating already existing issues in water availability, health problems, and inequality. For instance, it is widely recognised that climate change exacerbates the already challenging WASH landscape for women. At the same time, the degree to which gender is mainstreamed throughout policy and practice varies significantly. For instance, while all policies in Kenya concretely identify opportunities for women's participation and job opportunities at different levels; Ivory Coast's are relatively gender blind and do not propose how to create gender-inclusive environments. Many countries, such as Namibia, often reference gender as a cross-cutting issues in water and sanitation but fail to explore in much detail what role they can play or what opportunities exist for them.

In terms of policy priorities, policies commonly identify a need to create more equal participation and outcomes for women. The notion of inclusion is echoed in national development plans, as well as more specific gender, water, health, and climate change plans. At the same time, many countries do not identify concrete actions to achieve these objectives. Nevertheless, this policy analysis identifies a number of common entry points across focal countries to improve the position for women, including greater participation of women in the workforce (including the WASH sector), education and training opportunities in specific areas (such as Science, Technology, Engineering and Mathematics (STEM)) and ensuring that women and young people are consulted and have sufficient voice in policy and investment decisions.

Overview of Gender Action Areas for CI2

This Gender Assessment brings together analysis of the position, opportunities and challenges faced by women and men across CI2 sectors (drawing on the review of literature and the deep-dives into the analysis of country contexts). Building on this assessment, the Gender Action Plan annexed to this Assessment in Annex 6 (and summarised in Section 8) identifies action areas and entry points CFM and its Project Companies can leverage to enhance participation and equal outcomes for women in CI2 investments in these sectors.

Improving women and girls' access

CFM and its Project Companies will use its investment projects and community development projects to address the physical, financial, attitudinal and safety barriers that may limit women's access to WASH and Oceans Infrastructure.

Using project development processes (such as Environment and Social Impact Assessment, community needs assessment and targeted Gender Assessments), CFM will explicitly consider the barriers which affect women and girls' access, specifically focusing on indigenous households and women and girls with disabilities. It will incorporate, where possible, appropriate actions to address these barriers into the design of investment projects (and will monitor progress in this area). CFM will seek to use community development projects to increase the direct benefits for women and girls resulting from investment by addressing identified barriers which cannot be addressed through the main investment. Examples of potential projects include working with families and communities to build their understanding of and support for good WASH practices and modifying water points to ensure access. Additional projects are outlined in the Gender Action Plan.

Giving women voice in the planning, design and management of WASH facilities

As a priority beneficiary group for CI2 investments, it is essential that stakeholder engagement and consultation processes service the needs and experiences of women and girls. This is to ensure that investment projects and community development projects can be responsive, as far as possible, to the WASH needs of women and girls. This is particularly pertinent given that social norms in CI2 focal countries often devalue and silence women's voices, even when public consultation mechanisms are in place.

CFM and its Project Companies will adopt an inclusive stakeholder engagement approach that provides conducive spaces for women and girls to voice their needs and experiences, including identifying women and girls, as well as other identified marginalized social groups in the targeted countries, as a distinct stakeholder group with whom to directly engage, identifying women's leaders to work with to convene appropriate fora to hear a range of women's and girls' voices and working with community and men's leaders to help them understand the rationale for engaging women in the planning of WASH facilities.

In some projects, community-led WASH management committees may play a role in the management of WASH facilities. Where this is the case, CFM and its Project Companies will support committees to operate in a way that provides space for women's voices to be heard, including by identifying and recruiting to the committee local women who are perceived (by men and other women) to have some legitimacy in speaking out on behalf of the community. Planned actions to provide a safe space for women's voices in the planning, design and management of WASH facilities will be set out in the investment's stakeholder engagement plan. Additional action points to ensure women and girls have an equal voice are outlined in the Gender Action Plan.

Providing women with formal employment opportunities in the WASH sector

Low levels of secondary education completion amongst girls in several of the CI2 focal countries combined with gender norms around acceptable productive roles for women and men can make the formal employment of women in the WASH sector challenging.

CFM Project Companies will be proactive in hiring a gender-balanced team, making adjustments to working practices where this could level the playing field between male and female candidates. It is, however, recognised that in several focal countries, there are systemic issues beyond CFM's control which are likely to limit the Companies' ability to recruit senior female project staff. In such contexts, CFM will scope out opportunities to encourage more women into target sectors, including through scholarships for women embarking on tertiary education in relevant Science, Technology, Engineering and Mathematics (STEM) fields.

At the project level, Project Companies will identify employment opportunities for women in WASH enterprises and put in place appropriate mechanisms to recruit, train and support women in these roles. Opportunities might include roles as sanitation sale agents, in meter reading, water tank construction or cleaning, among others. Additional action points are outlined in the Gender Action Plan.

Creating an enabling environment for gender mainstreaming in the CI2 Fund and the Project Companies

CFM recognises the need to draw on specialist skills and advice to guide the company's leadership on how to operationalise its gender commitments across the business and within the investments it manages. This is expected to be achieved through a network of gender and inclusion specialists who can be hired on an as-needed basis to provide fund and project specific support.

CFM will support Project Companies to put in place essential internal systems which are gender-responsive and gender-transformative, and capacities that will facilitate gender mainstreaming through the project and its operations, thereby operationalising this CI2 Fund Gender Action Plan. Arrangements will include, developing a local (Project-specific) gender action plan, agreeing and implementing a Project-specific Gender Equality Policy which sets out the company's commitments to gender equality and Project Companies also take steps to require their principal contractors to give due attention to gender issues in their own operations (such as developing an operational Gender Equality Policy and recruiting a gender-balanced team). Further commitments are set out in the Gender Action Plan.

List of acronyms

BCCSAP	Bangladesh Climate Change and Action Plan
CFM	Climate Fund Managers
CI2	Climate Investor 2
GCF	Green Climate Fund
HCF	Health care facilities
JPWSSH	Joint Programme for Water Supply, Sanitation and Hygiene
KESHP	Kenya Environmental Sanitation & Hygiene Policy
KRA	Key Result Areas
MH	Menstrual hygiene
MOWS	Ministry of Water and Sanitation
NDP5	Namibia's 5th National Development Plan
NCCSAP	National Climate Change Strategy Action Plan
NDP III	National Development Plan III
NHPSWSS	National Hygiene Promotion Strategy for Water Supply and Sanitation Sector
NPCC	National Policy on Climate Change
NPGD	National Policy on Gender and Development
NSWSS	National Strategy for Water Supply and Sanitation
NSDP	National Sustainable Development Policy
OD	Open defecation
STEM	Science, Technology, Engineering and Mathematics
ToR	Terms of Reference
MTP III	Third Medium Term Plan
UNFCCC	United Nations Framework Convention on Climate Change

Glossary of Selected Gender Terms

Empowerment	The process through which people take control and action in order to overcome obstacles of structural inequality which have previously put them in a disadvantaged position.
Gender	Sex refers to the biological differences between male and female bodies. Gender, on the other hand, refers to the socially constructed attitudes, values, roles and responsibilities of women and men, in a given culture and location. These attitudes, values and roles are influenced by perceptions and expectations arising from cultural, political, economic, social and religious factors, as well as from custom, law, class, ethnicity and individual or institutional bias. Gender attitudes and behaviors are learnt and change over time.
Gender analysis	The close examination of a problem or situation in order to identify to differences in the gender roles, activities, needs, and available opportunities of men and women. Gender analysis of a development program involves identifying the gender issues within the problem which is being addressed and in the obstacles to progress, so that these issues can be addressed in all aspects of the program - in project objectives, in the choice of intervention strategy and in the methods of program implementation.
Gender-aware policy	A policy which takes into account the social relationships of women and men as well as the differences in their needs, as opposed to a policy that is gender neutral and implicitly assumes that women and men have the same needs.
Gender blindness	The failure to recognize that the needs of men and women are different. A gender-blind approach assumes that gender is not an influencing factor in projects, programs or policies.
Gender division of labour	Is an overall societal pattern where women are allotted one set of gender roles and men another. An unequal gender division of labor refers to a division of labor in which there is an unequal gender division of reward. Discrimination against women in this sense occurs when women get most of the burden of labor, and most of the unpaid labor, but men collect most of the income and rewards resulting from labor. In many countries, the most obvious pattern in the gender division of labor is that women are mostly confined to unpaid domestic work and unpaid food production, whereas men dominate in cash crop production and waged employment.
Gender equality	Means that there is no discrimination on grounds of a person's sex in the allocation of resources or benefits, or in the access to services. Equality exists when both men and women are attributed equal social value, equal rights and equal responsibilities, and have equal access to the means (resources, opportunities) to exercise them. Gender equality may be measured in terms of whether there is equality of opportunity, or equality of results.
Gender equity	Means fairness and justice in the distribution of benefits and responsibilities. Gender equity is the process of being fair to women and men. To ensure fairness, measures must often be available to compensate for historical and social disadvantages that prevent men and women from otherwise operating on a level playing field. Equity leads to equality.
Gender gap	A gender gap is an observable and sometimes measurable gap between men and women in terms of socioeconomic indicators, such as ownership of land, attendance at school or participation in the labor force, which is understood to be unjust and provides evidence of a gender issue to be addressed.
Gender mainstreaming	Is the systematic process of ensuring that women and men have equal access and control over resources, development benefits and decision-making at all stages of the development process through ensuring that the respective needs, interests and priorities of women and men are integrated into in all policies, programs and

	projects. It requires: (a) continuous assessment of the implications and effects of energy-related projects and policies on women and men; (b) designing and implementing strategies and actions that aim to better address the needs of women and men, improve their well-being, and facilitate their participation in the development process; and (c) a mechanism for ensuring this is done and for reporting to managers.
Gender neutral and gender blind	Gender-neutral policies are not specifically aimed at either men or women and are assumed to affect both sexes equally. However, they may actually be gender-blind. Gender-blindness means ignoring the different roles, responsibilities, capabilities, needs and priorities of women and men. Gender-blind policies are based on information derived from men's activities and/or assume those affected by the policy have the same (male) needs and interests.
Gender relations	The social relationships and power distribution between men and women in both the private (personal) and public spheres.
Gender roles	Gender roles are sets of behaviours assigned to men and women respectively, according to their cultural norms and traditions that determine which activities, tasks and responsibilities are perceived as male and which female. Gender roles are affected by age, class, race, ethnicity and religion, and by the geographical, economic and political environment. These roles shape identity, determining how we are perceived by others, how we are expected to think and act as women and men. Changes in gender roles often occur in response to changing economic, natural or political circumstances, including development efforts.

1 Background to the Assessment

Climate Fund Managers (CFM) is preparing a funding proposal for the Green Climate Fund (GCF) to be presented to the GCF Board in June 2022. If successful, CFM will use its financing facility Climate Investor 2 (CI2) to invest in infrastructure for water and sanitation services, the use and protection of wetlands, coastal and marine ecosystems as well as climate resilient ports in up to 25 countries.

Both GCF and CFM are committed to promoting gender equality and the empowerment of women through their investments. By adopting a gender mainstreaming approach in investments, GCF and CFM expect not only to mitigate the risks for women and men associated with funded projects, but also to benefit women and men equally, reducing any gender differentials associated with vulnerabilities exacerbated by climate change and promoting sustainable and equitable development.

This assessment has been commissioned to inform CFM's funding proposal to GCF. In line with the assignment's Terms of Reference (ToR), it aims to understand, within the context of climate action, how women and men are affected by problems in three main areas: i) access to and management of water and sanitation (WASH) services;² ii) the protection and management of wetland, coastal and marine ecosystems; and iii) the development of climate resilient port infrastructure. The specific issues covered under each area are summarised in Table 1 below.

Table 1: Issues covered in the gender assessment, by area

Sector	Issues covered
Water and Sanitation	• Women and men's, girls' and boys' access to and use of WASH services, the barriers they encounter, including gender-based violence, and how climate change is affecting these • How women and men, girls and boys are affected my challenges in accessing quality WASH • The roles that men and women, boys and girls play in the WASH sector, including formal employment opportunities • Women and girls' participation and leadership in the design, planning and management of WASH infrastructure and services • National policy in relation to gender, water and sanitation • Stakeholders involved in WASH • WASH problem tree analysis
Wetland, coastal and marine ecosystems	• Women and men's use of, and interaction with, wetland, coastal and marine areas • How climate change is affecting these different uses and interactions, and the impact on the populations concerned • The different ways in which men and women are, are not, and could be involved in the protection and management of wetland, coastal and marine areas • Problem tree analysis
Port infrastructure	• The different ways in which men and women use and benefit from available port infrastructure • The different ways in which men and women are, are not and could be involved in and can benefit from the development of climate resilient port infrastructure • Problem tree analysis

² Although the assignment ToR refer to 'water and sanitation' only, the gender assessment also covers hygiene, including, to some extent, menstrual hygiene. This is because improvements in hygiene practices can often be achieved through improvements in access to water and sanitation services, and the benefits of these are considerable for women and girls. Given this slightly broader focus, we tend to refer to water, sanitation and hygiene and use the acronym 'WASH'.

Findings from the assessment have informed the development of a Gender Action Plan, which sets out how CFM will mainstream gender and inclusive approaches in order to promote gender equality through the CI2 investment and supported projects.

This report presents a gender assessment of the three areas of investment interest. It has been prepared over two iterations. The first iteration provided a high-level assessment of gender issues in the water and sanitation sectors, as well as country level context analysis for 8 priority countries: Bangladesh, Djibouti, Ivory Coast, Kenya, Morocco, Namibia, Nigeria and Uganda. Responding to GCF feedback on the first iteration, this second iteration has an expanded scope. It now includes a high-level global gender assessment of wetlands, coastal and marine ecosystems and port infrastructure, stakeholder mapping and engagement for the original 8 priority countries, as well as country level context analysis for an additional 3 priority countries: Botswana, Brazil and Indonesia.³

1.1 This report

The findings from the gender assessment are presented in this report in the following eight sections.

We start by briefly outlining the methodology used in preparing the assessment (Section 2).

In Section 3, we describe, at a global level, the main gender issues identified in the water and sanitation sectors, and how these gender issues are affected by climate change, drawing on evidence from academic papers and practitioner literature.

In Sections 4 and 5, we conduct a similar exercise in the areas of wetlands, coastal and marine ecosystems and climate resilient port infrastructure.

In Section 6 we analyse the policy context and data in 11 focal countries in which we describe the policy direction in relation to WASH, gender and climate change, and identify entry points at the national level to improve women's participation and equal outcomes. For eight of these countries, those designated priority countries 1 and 2, the context analysis also includes a mapping of key stakeholders in the WASH sector from national to community levels.

In Section 7 we draw together the literature reviews and country-level analyses into problem tree analyses which unpack the causes and effects of inadequate infrastructure and services for women and girls across the CI2 sectors.

In Section 8, we use the problem tree analyses and the evidence reviewed to identify four opportunities for CI2 to promote gender equality through large scale investment in WASH, wetland, coastal and marine ecosystem protection and climate resilient port infrastructure development.

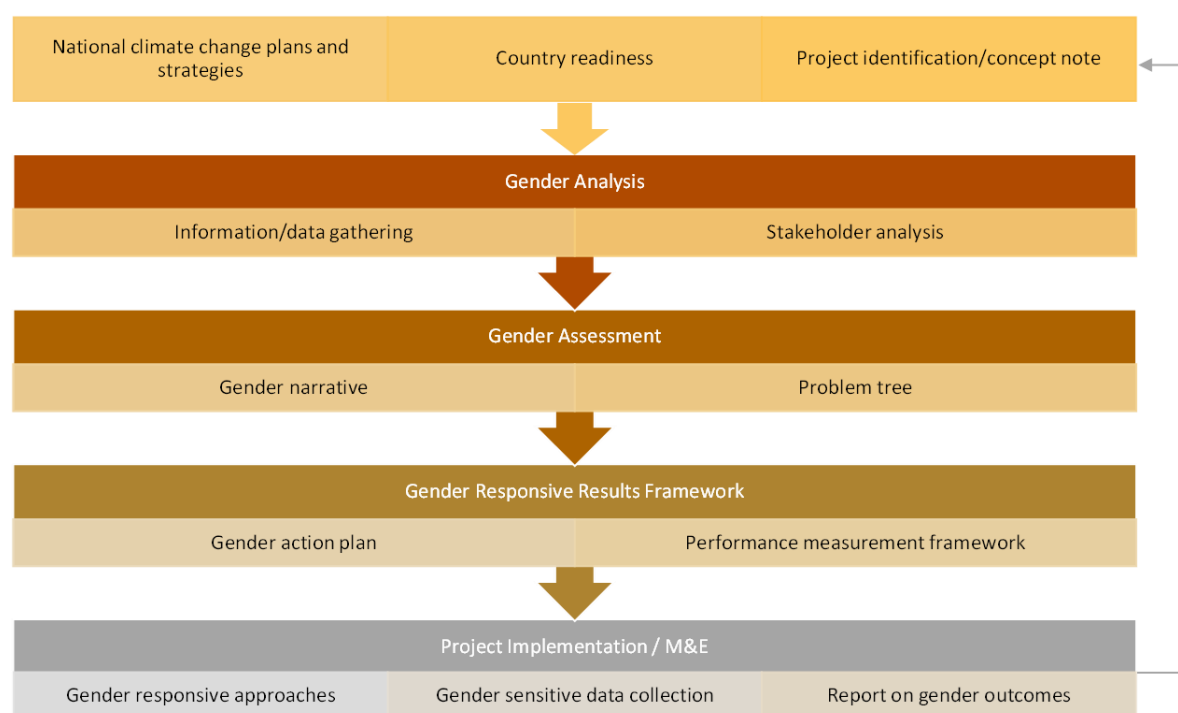
In the final section (Section 9), we provide an overview of key indicators to monitor the results and impact of gender mainstreaming activities.

³ It was not possible to conduct stakeholder mapping and interviews for these 3 additional countries as No Objection Letters were received too late in the process for this Gender Assessment.

2 Gender Assessment methodology

The methodology for this Gender Assessment and the subsequent Gender Action Plan has been guided by GCF's approach to gender mainstreaming (Figure 2). As part of it, we have conducted gender analysis of global evidence on gender issues in WASH, the protection of wetlands, coastal and marine ecosystems and port infrastructure, national policies for priority countries and available quantitative datasets; mapped country level stakeholders; and used this evidence to prepare the narrative and problem tree analysis that are presented in this report. Subsequently, the Gender Assessment has been used to inform a Gender Action Plan, which will guide CFM's gender mainstreaming efforts in CI2. This Section describes in more detail the methodology used in preparing the Gender Assessment.

Figure 2: GCF's gender mainstreaming approach⁴



This is a second draft of the Gender Assessment and Action Plan, which, following feedback from GCF, has been developed further. The second iteration responds to this feedback, especially focusing on the inclusion of Oceans Infrastructure in addition to the water and sanitation sector and reflect on findings from stakeholder analysis and engagement and will include a performance measurement framework and be relevant to a wider set of countries where CI2 will provide support.

2.1 Information and Data Gathering

The information/data gathering element of the Gender Assessment focused on:

1. **A review of global evidence in WASH, the protection of wetland, coastal and marine ecosystems, and the development of port infrastructure in developing country contexts.** This included men's and women's, girls' and boys' access to and use of available resources, and associated constraints, including gender-based violence; their participation in the

⁴ Source: [guidelines-gcf-toolkit-mainstreaming-gender_0.pdf \(greenclimate.fund\)](#)

planning and design of initiatives to improve facilities in the three thematic areas, as well as employment opportunities.

2. **National policies in WASH, gender and climate change** in 11 focal countries: Bangladesh, Botswana, Brazil, Djibouti, Indonesia, Ivory Coast, Kenya, Morocco, Namibia, Nigeria and Uganda.⁵
3. **Quantitative datasets** available for access to WASH, the protection and management of wetland, coastal and marine ecosystems and girls' education, data primarily used for reporting against relevant SDGs.

Review of global evidence in WASH, the protection of wetland, coastal and marine ecosystems and port infrastructure

For the review of global evidence, we identified relevant search terms/keywords for the three thematic areas. These included: "gender", "women", "boys", "girls", "water", "sanitation", "hygiene", "menstrual hygiene", "handwashing", "WASH", "blue economy", "ocean economy", "sustainable ports", "climate resilient ports" and "climate change". These terms were used to manually search (through trialling different combinations of search terms) search engines and the websites of the organisations known to have a role in gender, WASH, oceans, port infrastructure and climate change. Through this, we identified relevant online resources from academic and practice-focused sources, which we then used to snowball additional sources by examining reference lists within the literature found.

A total of 85 studies were identified and reviewed, including 71 focused on WASH, nine on wetland, coastal and marine ecosystems and eight (two being used for both) on port infrastructure (See Annex 1 for bibliography). A selection of academic papers was included as well as multilateral agency reports and selected practitioner literature. To narrow down the extensive literature on WASH, the review focused primarily on what the Joint Programme Water Supply, Sanitation and Hygiene (JPWSSH) refer to as higher levels of service, in line with CI2's focus on infrastructure. Although hygiene is not specified in the assignment ToR, improved water and sanitation infrastructure will facilitate better hygiene, and so sources on hygiene were also reviewed.

The global evidence review found that, with one notable exception (the Water for Women Fund), research with a comprehensive and combined focus on gender, WASH and climate, is relatively sparse. More studies were identified on the effects of climate change on WASH infrastructure than on its gendered impacts on WASH services. The exceptions here are the Water for Women studies which focused on rural settings and community managed WASH systems in the East Asia and Pacific region. In WASH overall, there is limited qualitative research and a lack of longitudinal research assessing the effects over time. For urban settings, utility level reports, complaint registers and WASH programme evaluations could provide additional sources of information.

The global evidence review also found that most literature on wetland, coastal and marine ecosystems is largely gender blind. Most reports primarily examined benefits and impacts from an economic and biodiversity perspective, with limited discussion of benefits and impacts for people. When people were discussed, this was mostly associated with economic benefits, such as those relating to tourism and fishing, with little consideration of social impacts.

⁵ The context analysis concentrated on WASH only, and did not include wetland, coastal and marine ecosystems, or port infrastructure, areas that were covered through global level analysis only.

Policy Analysis in focal countries

This analysis focused on national policies, strategies and action plans in the 11 focal countries in the following areas: overarching national development, water, sanitation, hygiene, gender, climate change, health and education. Our review covered a total of 65 policies currently operational, including seven for Bangladesh, five for Botswana, nine for Brazil, six for Djibouti, five for Indonesia, four for Ivory Coast, seven for Kenya, five for Morocco, five for Namibia, six for Nigeria and six for Uganda (see full set of policies reviewed in Annex 2). All of them were sourced online. Most were in English, although those for Djibouti, Ivory Coast and Morocco were largely in French and those for Brazil were largely in Portuguese.

Once collated, key data on core areas of interest (Box 1) was extracted from the policies into a spreadsheet and then analysed drawing out the overarching WASH policy direction, the extent to which gender issues are considered and tackled in WASH policy, and how climate change is affecting all of this.

Box 1. Areas of interest in the policy analysis

1. What are the main water and sanitation and hygiene challenges identified?
2. How is climate change affecting these challenges?
3. How are men, women, boys and girls affected differently by these challenges?
4. What are the main health consequences associated with the water and sanitation and hygiene challenges faced? Who is primarily affected by these?
5. What are the main policy objectives in water and sanitation and hygiene?
6. How do these policy objectives seek to address the impact of climate change?
7. What formal and informal roles does the policy foresee men, women, boys and girls playing in the water and sanitation and hygiene sectors, and in adapting water and sanitation and hygiene management practices to minimise the effects of climate change?
8. How does the policy framework encourage and support women and girls to pursue careers in areas relevant to water and sanitation systems development and management?
9. Which agencies are responsible for overseeing and implementing the policy objectives outlined and what are their roles?

Quantitative data analysis

Quantitative data on population access to WASH facilities and access to WASH facilities in schools and health centres has largely been drawn from the WHO and the UNICEF JPWSSH.⁶ This data has been complemented with data from UNICEF's Multiple Indicator Cluster Surveys (MICS)⁷, where available, and, for data on women's education levels, from UNESCO's Institute of Statistics.⁸ The latter is to understand the extent to which women may be equipped to access formal employment within the WASH sector. For the protection of wetlands, coastal and marine protection, we drew on SDG14 reporting, but very little of it tracked men's and women's engagement.

2.2 Stakeholder analysis

⁶ [Data | JMP \(washdata.org\)](https://washdata.org/)

⁷ [Surveys - UNICEF MICS](https://data.unicef.org/mics/)

⁸ [UIS Statistics \(unesco.org\)](https://uis.unesco.org/)

The stakeholder mapping was done through a series of interviews with key sector players in eight of the priority countries.⁹ Relevant actors were identified using the team's personal contacts and through online sources. From this longlist, a selection of key informants were identified from national ministries in charge of water, sanitation, hygiene, gender and climate change, CSOs working on gender and/or WASH issues and other country-based WASH experts. We contacted these individuals to request a short interview. A total of 50 individuals were contacted in the eight priority countries, 29 of whom responded and were interviewed (Table 3).

Table 2: Stakeholders interviewed, by country

Country	Interviews	Type of Stakeholder Interviewed
Bangladesh	3	CSO (2), National WASH Expert (1)
Djibouti	4	CSO (3), National Government (1)
Ivory Coast	2	National Government (1), CSO (1)
Kenya	7	National Government (2) CSO (5)
Morocco	0	-
Namibia	2	Multilateral (1), National WASH Expert (1)
Nigeria	3	CSO (2), Multilateral (1)
Uganda	8	National Government (3) Multilateral (2), CSO (3)
TOTAL	29	

Stakeholder interviews were used to explore three main issues:

- Current efforts to promote gender equality in water and sanitation infrastructure development;
- The main actors involved in promoting gender equality in water and sanitation infrastructure development and their roles; and
- Stakeholder recommendations for CI2 investments in their country and feedback on the opportunities identified in the first iteration of the Gender Assessment.

The interview guide is attached in Annex 3, with the list of stakeholders involved and the main points emerging provided in Annexes 4 and 5 respectively.

The interview guide is attached in Annex 3, with the list of stakeholders involved and the main points emerging provided in Annexes 4 and 5 respectively.

2.3 Gender Assessment Narrative and Problem Tree Analysis

Findings from the global evidence review of WASH and protection of wetlands, coastal and marine ecosystems and port infrastructure, as well as the deep dive analysis of focal countries in the WASH sector, are presented separately in this Gender Assessment, in Sections 3, 4 and 5 respectively. We have used these findings to prepare problem tree analyses for each of the three thematic areas, all of which can be found in section 7.

The opportunities for promoting gender equality through CI2 investments presented in section 8 of this report draw on the full dataset collated through the global literature review, the stakeholder mapping, the national policy review in 11 countries and problem tree analysis in the three areas. The

⁹ It was not possible to conduct stakeholder interviews with stakeholders in the remaining three countries as No Objection Letters were received too late in the process.

analysis process to identify these opportunities involved two steps. Firstly, as part of the first iteration of this Gender Assessment, we used the findings from the literature review of gender issues in WASH to identify potential opportunities for promoting gender equality through WASH. We then validated these using the country analysis in an initial set of 8 focal countries. Overall, there was a good alignment between the two. Secondly, in the second iteration of this Gender Assessment, we used findings from the literature review of gender issues in the protection and management of wetland, coastal and marine ecosystems, and in the development of port infrastructure, to identify opportunities for promoting gender equality in these areas. The opportunities aligned well with those already identified for WASH and the two have been combined. This integrated set of opportunities form the basis of the material presented in section 8 of the report.

2.4 Limitations

Availability of global evidence on the protection of wetlands, coastal and marine ecosystems, and climate resilient port infrastructure

The existing evidence base on gender as it relates to oceans and, particularly, port infrastructure is extremely small. Many qualitative reports reviewed were gender-blind and did not explore how benefits and impacts may differ for women and men. No reports explored differentiated impacts for boys and girls, with children in general referred to only in relation to food security.

More broadly, most reports primarily examined benefits and impacts from an economic and biodiversity perspective with limited discussion of benefits and impacts for humans. When people were discussed, this was mostly associated with economic benefits, such as those relating to tourism and fishing, with little consideration of social impacts. With the majority of ocean-related reports authored by INGOs and other issue-focused organisations who collaborate through groups such as Ocean Panel and WorldFISH, many reports are sector specific, particularly fisheries and, to a lesser extent, tourism.

The literature on good practices for involving women in, and maximising benefits for them from, the protection and management of wetland, coastal and marine areas or the development of climate resilient port infrastructure is also scant.

Quantitative data is similarly scarce. Most data identified focused on developed countries or regions (e.g., the EU). Available data also tends to focus on the formal sector with little or no consideration of impacts in the informal sector, where women tend to be concentrated. The most recent unpublished statistics for SDG 14 are from 2014 and, as above, focus on physical indicators such as marine and coastal protected areas and fish stocks, rather than related human benefits or impacts.

Availability of national policy documents

The research team has encountered some challenges in accessing required data. For example, it has not been possible to include in the review policies that are not accessible online. This has led to some significant omissions, for example, we have not been able to access Water and Sanitation policies for Nigeria and Uganda, and we have only been able to review three policies for Ivory Coast. Similarly, there are gaps in the quantitative data available which are noted in the country summaries in Section 6. Furthermore, the gender analysis in policies can be quite scant, suggesting that a full consideration of the gender issues in WASH has not been done at the policy level.



Time pressures for conducting the stakeholder mapping

With only three weeks to prepare the stakeholder mapping, the scope for engaging stakeholders has been extremely tight. It is positive that 29 stakeholders have been engaged but there are gaps: we have not been able to engage any stakeholders in Morocco, nor have we been able to conduct a stakeholder mapping for the three countries added in the second iteration of the Gender Assessment, namely Brazil, Botswana and Indonesia, for which No Objection Letters came through too late in the process.

3 Global Evidence Review: WASH & Climate Change

In this Section, we present findings from the global evidence review exploring the main gender issues in WASH. It considers the following questions, identified as the more pertinent issues based on our review of the relevant literature:

- Women and men's, girls' and boys' access to and use of WASH services, the barriers they encounter, including gender-based violence, and how climate change is affecting these;
- How women and men, girls and boys are affected by challenges in accessing quality WASH;
- The roles that men and women, boys and girls play in the WASH sector, including formal employment opportunities;
- Women and girls' participation and leadership in the design, planning and management of WASH infrastructure and services.

3.1 Differences in WASH access and consequences

Women and girls are largely responsible for household water collection and latrine cleaning, yet there are often differences in their ability to access and use WASH facilities. In 2017, WaterAid estimated that WASH-related diseases were the fifth biggest killer of women worldwide, causing more deaths than AIDS, diabetes or breast cancer. This section outlines the differences in access to and use of water supply, sanitation and hygiene in households, health facilities and schools. Some of the key barriers which affect access to WASH facilities are summarised in Table 3 below.

Access to water in household settings

Fetching water is a physically challenging task and is typically carried out by women and children. It has musculoskeletal health risks and other potential injuries, including spinal damage, hernia and prolapse, as well as an increased risk of miscarriage¹⁰. Distance to reaching water and sanitation facilities makes people vulnerable to physical or verbal abuse, rape and assault¹¹. Other risks include rough terrain (particularly when carrying water) and attack from dangerous animals¹². Water collection can affect people's quality of life in a number of ways: time devoted to water-related tasks limits women's time for other household chores, rest and leisure and income-generating activities¹³.

Increasing access to sufficient quantities of water can also contribute to better hygiene (such as handwashing with soap or face washing) and thus improve health (in terms of diarrhoea and respiratory infections). Women have an increased need for water for hydration, sanitation and hygiene during menstruation, pregnancy, the postnatal period and caring for sick family members or young children¹⁴. Limited supplies of water, as well as competing water needs in the household, can trigger stress and violence between spouses and parents and children¹⁵. For this reason, women might put partner/family water needs above their own to avoid abuse. There is some evidence that household piped water helps reduce intra-household conflict¹⁶. Examples of sextortion (the practice

¹⁰ Fisher (2008)

¹¹ Geere et al. (2018)

¹² Caruso et al. (2021)

¹³ Sorenson et al. (2011)

¹⁴ Kayser (2019)

¹⁵ House et al. (2014)

¹⁶ Devoto et al. (2012)

of extorting sexual favours) for public water supply in low-income settlements have been documented in Kenya¹⁷.

Access to sanitation facilities in household settings

Many communities have strict cultural norms around men and women sharing household or public toilets. Where norms prevent women from sharing latrines with men during menstruation, this often compels open defecation. Social expectations of modesty mean that women should not be seen urinating and defecating. In order to avoid using the toilet during the day, women report reduced intake of food/drink or taking anti-diarrhoea tablets¹⁸. Poor sanitation practices (such as open defecation) and WASH-related diseases can induce stress during pregnancy and may also contribute to miscarriage and still birth¹⁹. Pay-per-use latrines are often inequitable for women as there are often fewer cubicles for women (leading to longer queues), and women use toilets more frequently than men, making it more expensive²⁰. Women report becoming practised at going for several hours without urinating due to the lack of facilities in a variety of settings (including in the workplace, in public buildings and spaces or when travelling). Those needing more frequent toilet use (e.g. pregnant women) or incontinence (older women) are limited in their ability to travel far from home²¹. Dehydration from not drinking due to poor latrine access (especially for women and girls) can affect personal hygiene and enhances the risk of reproductive and urinary tract infections.

Men tend not always to use a toilet for defecation, either when they are at home or away. A range of reasons are cited for this, including: to slow the rate of the pit filling, men having more time available for open defecation (OD), not having a toilet where they work, the greater mobility of men and reverting to OD when household toilets are in high demand²². Open defecation is considered less risky for men regarding security/safety concerns and perhaps, in some cases, less shame about OD. Male open urination is a commonly accepted practice, although it can offend public decency, and is often prohibited through by-laws and punishable with fines²³.

Table 3: Examples of barriers inhibiting use

Physical	• Women have more limited land rights, which constrains their ability to decide whether or not to construct a household latrine or improved water point. • Use may also be limited by rough terrain (particularly when carrying water). • Men migrating for paid work means less support for physical labour. Women may need to pay men for construction of infrastructure²⁴ • Dependence on others for water assistance or visits to the latrine could result in stress and anxiety for people with disabilities, sick or older relatives and others.
Financial	• Paying for water has gender implications: women earn lower wages, give up paid work to collect water and have less control over assets within the household. Water was a major expense for many women. Constrained finances limited

¹⁷ KEWASNET (2020)

¹⁸ Kulkarni et al. (2017)

¹⁹ Bijaya et al., (2015)

²⁰ WaterAid et al. (2019)

²¹ Scott, Sohail and Cavill, (2017)

²² Chambers and Myers (2016)

²³ Scott, Sohail and Cavill (2017)

²⁴ Megaw, Kohlitz and Chong, (2020).

	women's access to preferred water sources, desired water quantity and water treatment methods. • Wealthy women report spending money to access water sources with shorter wait times or higher quality, or to compensate male neighbours, laborers or drivers for water collection. • Using public pay-per-use toilets mean women incur higher costs than men due to more frequent need.
Safety	• General safety concerns include feeling unsafe collecting water or feeling unsafe where they bathe and wash clothes. • Long absences from home for water collection could lead to conflicts because of husbands suspecting wives of infidelity. • Inadequate access to sanitation (when toilets lack doors, locks, lighting or are far away) is also associated with a heightened vulnerability to various forms of violence by men²⁵. This includes taunting, teasing, name calling and throwing things at women or girls when going to latrines and open defecation sites, especially after dark. • Women frequently accompanied one another or asked men to accompany them for safety when open defecating. • Women might only let their daughters go out for open defecation at night or go with them²⁶. • Women use make-shift arrangements (like potties) within the household to avoid using public toilets²⁷. • Quarrels over contributions toward maintenance of shared toilets affects ability to use such facilities.
Attitudinal	• Women are often unable to use sanitation facilities or touch water when menstruating due to perceptions that menstruation is polluting. Taboos around menstruation can also affect participation in community life²⁸. Menstrual hygiene is linked to emotional and psychosocial stress, resulting in feelings of shame, fear, anxiety and distraction. • Participation in a water project can lead to intrahousehold conflict and violence or shunning, for instance, if women spoke in meetings. • Husbands became angry or physically abusive when women took too long to collect water, when there was not enough water available for bathing and other purposes or when they were asked to assist with water collection. • Women and girls report shame related to sanitation and menstruation behaviours and experiences.

Access to hygiene in household settings

Globally, men are reportedly less likely to consistently wash their hands with soap at critical times. In households, women are typically the ones targeted by hand hygiene promotion through WASH programmes (often in their role as mothers). Women play a key role in supporting the hand hygiene of family members, for instance, by ensuring soap and water is available for hand hygiene or reminding family members to wash their hands before eating or after using the toilet. Women, in their role as

²⁵ Heller, 2014; House et al., (2014)

²⁶ Kulkarni et al. (2017)

²⁷ Lennon (2011)

²⁸ Megaw, Kohlitz, Gero and Chong (2020)

community health promoters, also play a role in supporting this wider practice within the community. The Global Scaling Up Handwashing Project²⁹ in Senegal included men in handwashing promotion discussions after recognition that: (1) men control their household's access and provide funds for soap, (2) men can follow up the outreach session to ensure that household members wash their hands with soap and (3) men can enable, reinforce and sustain behavioural change by washing their hands with soap while encouraging others to do the same.

Adequate menstrual hygiene (MH) refers to using a clean menstrual management material to absorb or collect menstrual blood. It can be changed in privacy as often as necessary for the duration of a menstrual period, using soap and water for washing the body as required. It also encompasses having access to safe and convenient facilities to dispose of used menstrual management materials (Joint Programme for Water Supply, Sanitation and Hygiene, 2020). Clean water is a fundamental aspect of safe and dignified menstruation. Water is needed for (1) personal washing during menstruation and (2) washing of reusable menstrual products like menstrual cloths. MH is limited by inadequate access to water and sanitation infrastructure, limited availability of menstrual products (absorbents), lack of a private place to wash or change products, poor hygiene awareness and problems arising from stigma and taboo.

Access to WASH in health facilities and schools

Health facilities

According to the JPWSSH (2020), a quarter of health care facilities (HCF) lacked basic water services in 2019. HCF may rely on handpumps on their premises for their water supply. If the handpump stops working, staff have to collect water from nearby communities. Smaller facilities may not have the systems needed to provide buffer capacity by storing water on-site, meaning that where there is no caretaker, medical staff must collect water daily (or more often) for use in the HCF. In times of shortage, family members, likely to be women, may be expected to bring drinking water for patients as well as water for bathing. Water quality may also not be sufficient, and handpumps may not always be chlorinated (or the quality of water tested) on a regular basis so health care facilities may not always have the quality of water necessary for use.

Integrating handwashing/basic hygiene is fundamental to primary health care and the prevention of Hospital Acquired Infections. Although clinic staff may have knowledge of the importance of handwashing, HCFs do not always have functional handwashing stations and soap. In 2019, one in three HCFs did not have hand hygiene facilities at the point of care. During the COVID-19 pandemic, handwashing stations have been increasingly deployed at the entrance of HCFs to ensure patients and staff, including all visitors, adhere to COVID-19 measures. Health facilities may lack sufficient numbers of functional sanitation facilities, separate for staff and patients: in 2019, 10% of health care facilities had no sanitation service. Female friendly public toilets would: be safe and private, cater for hygiene requirements, be accessible for users, be affordable and available when needed, be well maintained and managed, and meet requirements of caregivers and parents. There would also be sufficient numbers of cubicles for men and women to reduce queues³⁰.

Increasingly, development partners are supporting improved WASH in HCF with improved WASH infrastructure and training on proper handwashing and handwashing stations. In particular, WASH can also support dignity in childbirth: a survey by the White Ribbon Alliance (2019) found that women

²⁹ WSP (2010)

³⁰ WaterAid et al. (2019)

want WASH in health care facilities for a dignified birth experience: expectations include running water, indoor toilets, clean beds and sheets.

Schools

Globally, 69% of schools had a basic drinking water service in 2019 while 63% had a basic sanitation service³¹. Whilst there are global and national standard ratios for pupils' latrines, many schools are unable to provide enough toilets for pupils, or their limited budget means that materials are not always available to regularly clean toilets or to ensure anal cleansing materials and soap are always available. Soap should be included in the school budgets, but parents may also be encouraged to contribute funds.

WASH in schools is intended to have a positive impact on child health, thereby reducing absences from school. WASH services in schools can be an incentive for parents to send their children to school, particularly girls in the transition from primary to secondary schools³². Adequate consumption of water at school can also lead to better hydration, which has a number of health benefits as well as benefits for learning (Hunter et al., 2015).

A lack of WASH at home can impact girls' education, for instance, when they are enlisted to help with water or sanitation-related tasks at home. Where there is no caretaker, students and teachers have a role to ensure the WASH facilities are maintained and that clean water is available. When girls attend school, they may also be asked to collect water for the school or clean toilets instead of being in class. There is some evidence that adolescent boys are impacted by poor sanitation in secondary schools; however, they are less impacted than girls. Overcrowding in schools has been associated with both boys and girls avoiding toilets. Students avoid using the toilet due to the anxiety of using a filthy latrine, queuing during the break or a lack of privacy when toilets do not have doors or locks.

Globally, 57% of schools had a basic hand hygiene service in 2019³³. Handwashing in schools is promoted through a number of WASH in-school programmes and by school Health Clubs that promote WASH among the school community. Female teachers are often the main champions of handwashing with soap in the school environment.

Self-reporting by girls suggests that inadequate toilet facilities in schools (lack of toilets, filthy toilets, insufficient water and limited access to menstrual materials) are a contributory factor for truancy, failing classes, absenteeism and drop out. Similarly, the lack of facilities for female teachers is a factor in staff retention/turnover. WASH programmes are increasingly tackling taboos and stigmas about menstruation through education and increasing as well as availability of WASH facilities at school by, for example, increasing girl's access to sanitary absorbent materials.

3.2 Effects of climate change on access to and use of WASH services

At its most basic level, changing weather patterns affect WASH facilities by reducing water availability and damaging WASH infrastructure (Figure 3). Low-income households are more likely to be situated in areas affected by water shortages, floods, landslides and sewage overflow during floods. They are also more likely to use WASH facilities that can be easily destroyed or disrupted by climate hazards.

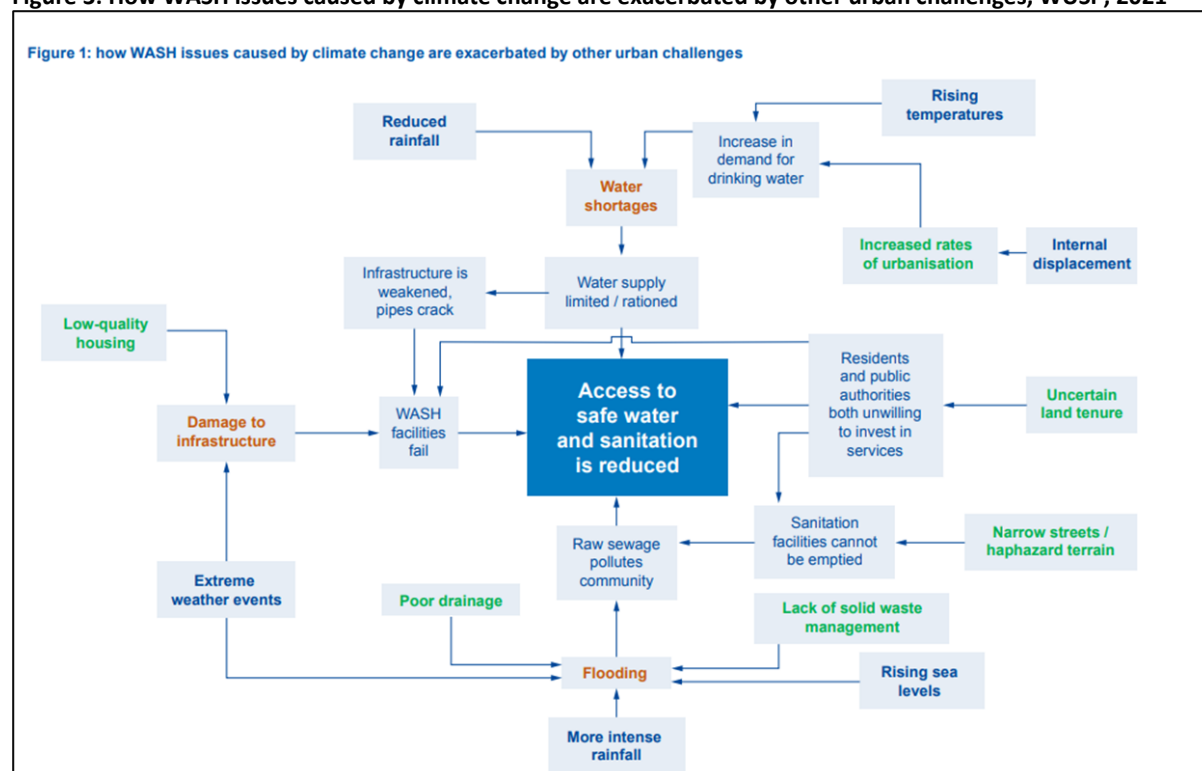
³¹ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

³² Chard et al., (2019)

³³ JMP, (2022)

The poorest typically have the least capacity to cope with and adapt to shocks³⁴. This section reviews the general impacts of climate change on access to and use of WASH services, before reviewing the specific effects on water, sanitation and hygiene services.

Figure 3: How WASH issues caused by climate change are exacerbated by other urban challenges, WUSP, 2021



The literature review suggests that climate change affects access to and use of WASH facilities in five main ways:

- Nuisance and disruption:** Disruption to residents is an immediate implication of climate-related damage to WASH infrastructure. For instance, flooding could lead to sewage back-up in homes connected to the sewerage network, overflowing pit latrines, as well as flooding of homes. As household managers or home-based workers, women are more likely to be affected by the nuisance caused by the maintenance and repair of WASH systems and disruption to WASH services caused by climate change, for instance, in ensuring the cleanliness of the home or dealing with nuisance such as odours. Clean-up activities are likely to cause temporary closure of markets, shops and roads leading to disruption for women in particular.
- WASH health outcomes:** Waterborne diseases are predicted to become more prevalent due to an increase in climate-related hazards³⁵. In addition, exposure of residents to untreated faecal sludge or wastewater poses an immediate threat to public health, including faecal-oral disease and vector-borne diseases. Increased faecal sludge contamination is associated with health risks meaning that temporary evacuations of homes may be required. Women are more likely to be caring for sick with WASH-related diseases and elderly relatives within households.

³⁴ Mills, Willetts, Evans, Carrard and Kohlitz (2020)

³⁵ Ibid

- **Elevated levels of depression and other mental health outcomes:** Water insecurity and scarcity, as well as open defecation, are considered significant stressors for individuals³⁶. Water insecure communities experience higher levels of anxiety and depression, and this is particularly the case for older women³⁷. High stress levels are a result of perceived unfairness in water distribution³⁸. Loss of WASH services to residents is also highly likely to cause stress and reduced mental health for residents, and this may be due to a decrease in the functionality of water-based toilets, toilets not flushing, unusable taps or a lack water for bathing etc. Water stressed communities are **more prone to arguments, disagreements and violence**³⁹.
- **Increased WASH costs:** Inadequate WASH can also increase financial strains⁴⁰. Where men are the decision makers over household's budgets, this can increase stress for women. Climate related damage to WASH infrastructure will require increased maintenance, repair and running costs. For those with networked services, increased preventive maintenance and water quality management costs are likely to be passed on to the household through tariffs. A rise in water fees or the cost of buying water from private providers can make safe water unaffordable for poorer households.
- **Loss of productive time/income:** Homes, schools, HCFs and businesses may incur clean-up costs and a loss of income due to climate related events. Repair works are likely to cause temporary closure of businesses. Businesses that require water for operations e.g. catering, tea stalls, brewing, hairdressing or laundry, often run by women, are more likely to be affected by loss of income.

Access to water in household settings

Following extreme weather events, water scarcity can result from resource limitations (i.e. physical hydrological constraints) or accessibility/affordability constraints.

Damage to piped water networks can mean that water is supplied through mobile tankers instead. Women and girls might have to queue to collect the water which is then stored in the household and its use carefully rationed. Women typically have higher water needs for hygiene and are thus impacted more. This brings a stress on families as a result of this water scarcity. Disputes are reported when men use too much water for bathing or when children spill water⁴¹. Households without many storage containers must make more trips to collect water instead. Warmer temperatures lead to increased demand for drinking water, with the poorest city residents particularly at risk due to the phenomenon of urban heat island⁴². On extremely hot days, water-carrying women and girls (especially elderly women) can be affected by heat stress/exhaustion. Women with limited physical abilities (e.g., pregnant women or the elderly) are more likely to struggle to access waterpoints and toilets when land conditions become waterlogged, muddy or slippery.

The poorest people in cities, already reliant on intermittent services, are most affected by water shortages⁴³. Augmentation strategies include the use of multiple water sources to meet household

³⁶ Young et al. (2021)

³⁷ Simiyu et al. (2021)

³⁸ Brewis et al. (2021)

³⁹ Adams et al. (2021)

⁴⁰ Javidi and Pierce (2018)

⁴¹ Caruso et al. (2021)

⁴² WSUP (2021)

⁴³ Ibid

water needs, and this has implications for water quality, cost and the time required to collect water. Residents might have to pay private water providers for additional water deliveries or find other private water solutions such as kiosks and bottled/sachet water during periods of water scarcity. To save costs, the poorest may use unsafe sources such as surface water or hand-dug/shallow wells. Increasing rainwater storage capacity could help to address water shortages, but poor households are less likely to have resources to store water (e.g., safe storage tanks). Water borrowing from neighbours is a strategy to overcome access constraints, particularly for those with longer round-trips to fetch water⁴⁴, but this is associated with feelings of shame, fear and indebtedness⁴⁵.

Access to sanitation facilities in household settings

Climate change will accentuate existing weaknesses in sanitation services. In many cities, little of a household's sewage reaches the treatment plant due to poor design or poor management, or as a result of broken sewer pipes, pump failures or power supply outages or a non-functional treatment plant. Instead, faecal sludge is dumped or discharged into open drains, bodies of water and the surrounding environment. The public health risk this poses will increase in the context of climate change e.g., increased faecal contamination of water sources and residential areas. Bad practices in rainy season include taking advantage of flooding to flush out pit latrine contents (instead of paying for pit emptying). Men often provide pit latrine emptying services and experience increased health risks from working with inadequate sanitation services without properly health and safety equipment.

Where toilets are no longer usable because they've been destroyed or damaged by climate hazards, poor households are more likely to struggle to repair toilets and may be more likely to be driven to revert to open defecation. During periods of water scarcity in a peri-urban community in Botswana, residents with toilets connected to a sewer reverted to using pit latrines⁴⁶. The poorest live on land at risk from flooding and human-driven deforestation, which can create flash floods and landslides. Rebuilding damaged facilities or purchasing new devices after extreme weather events can be difficult, particularly for those living in remote areas and who may be cut off from access to markets that provide products and services. Gendered norms, such as men being responsible for decision making on household purchases or re-constructing damaged facilities, can result in delays rebuilding if male figures are absent (e.g. because of out-migration for work) or if men consider sanitation a low household priority⁴⁷. Continual damage or destruction of a facility has an impact on WASH behaviour, and it can result in fatigue/apathy such that households must be motivated to re-build a toilet and handwashing facility, or it may result in reversion to open defecation (permanently or on a seasonal basis). Women and girls are likely to face issues with privacy or harassment while defecating or managing menstrual hygiene.

Adaptations for climate change include the scheduled emptying of latrines in advance of flood seasons, low water-use toilets and improved construction quality to reduce the infiltration of water into septic tanks or sewers⁴⁸.

Access to hygiene in household settings

Hygiene improvements create demand for water. Where water supplies are limited, households typically prioritize consumption over hygiene, and this has implications for women and girls in

⁴⁴ Rosinger et al. (2020)

⁴⁵ Wutich et al. (2008)

⁴⁶ McGill et al. (2019)

⁴⁷ Kohlitz et al. (2020)

⁴⁸ Oates et al. (2014)

particular. This section reviews hand and menstrual hygiene in the context of increased high-intensity rainfall, decreased rainfall, different types of flooding and tropical cyclones and other weather events.

Extreme weather events (such as heavy rain, landslides, floods and cyclones) can destroy or damage hand hygiene facilities, disrupt the functionality of the facility, or make it difficult for people with physical limitations to access facilities. For example, where households have their handwashing station outdoors or next to their toilet, storms could cause damage to the station. Handmade devices are more prone to break down; however, variable rainfall also undermines the operations of handwashing facilities served by a piped water supply. Climate change also affects people's journeys to wash their hands. This can be a particular problem for children, people with disabilities or older people⁴⁹.

Handwashing with unsafe water may increase the risk of diarrhoea⁵⁰. For instance, handwashing with water contaminated with high levels of *E. coli* is likely to transfer more *E. coli* onto hands than it removes through washing⁵¹. Intensive handwashing with contaminated water can cause skin lesions that can lead to infections entering the bloodstream⁵².

Climate hazards can mean that access to urban centres or other places to purchase soap or commercial menstrual products may not be accessible all year-round. Climate-related events can disrupt transport networks, affecting the final link of a supply chain i.e. preventing products available in the district town from reaching the village shop or household in sparsely populated rural areas.

Water scarcity can force choices about how water is used at the household level. With menstrual hygiene already insufficiently prioritized in many cases, this challenge could easily become even more problematic for women and girls in the face of climate-induced water shortages.

Water shortages at home and at school make personal hygiene more difficult to maintain during menstruation. Hotter days, more rain and higher humidity can increase the risk of reproductive tract infection, making personal hygiene that prevents disease even more critical. Washing intimate areas with unsafe water may plausibly increase the risk of infections. Saline intrusion in coastal areas can make water unsuitable for bathing or washing menstrual cloths e.g. causing fabrics to become rough and chaffing skin, which necessitates the usage of single-use, disposable products instead. Hand washing after changing a sanitary product or handling menstrual waste is often promoted as a critical time for hand hygiene, again highlighting the importance of access to water.

Prolonged shortage of water affects how reusable menstrual product are used, the frequency and manner in which they are washed and whether fuel is available to make it possible to sterilise them with boiling water. It also affects whether women and girls are able to wash their menstrual cloths effectively. Prolonged rainfall or very windy weather can make it impossible to dry reusable cloths in direct sunlight as recommended for hygiene reasons since drying and storing cloths in a dark or shady place (or using damp cloths) can lead to infections.

Climate change may make conventional disposal options for single-use products irrelevant, both those that are safe and hygienic (disposal in solid waste systems, incineration, composting and deep burial)

⁴⁹ Kohlitz et al. (2020)

⁵⁰ McCormick et al. (2016)

⁵¹ Verbyla et al. (2019)

⁵² Lemery et al. (2015)

or improvised (flushing products down the toilet, shallow burying of menstrual waste and throwing used products into drainage channels or open spaces). High-water table, heavy rainfall and overflowing sewers and flooding mean burial is not a viable option. Incorrect disposal of single-use menstrual products can exacerbate climate change if they block sewers or drainage systems, cause flooding and pollute water. Inadequate disposal options can result in the further stigmatisation and marginalisation of menstruating women and girls. During climate related disasters, social norms and stigma that impact MH often become more entrenched. This could make it difficult to discuss menstruation, hygiene and more sustainable ways for women to manage their menstruation.

Access to WASH in health facilities and schools

Staff, students and patients in schools and health care facilities require sufficient WASH facilities. Increasingly deep boreholes are being installed with overhead tanks to provide a higher-level of climate resilience to services in schools and health care facilities⁵³. The use of renewable energy sources (such as solar energy) for pumping systems can further optimize their resilience. Storage containers provide water supply in drought and can provide a gravity-fed water supply to handwashing/drinking water tap stations. Simple adaptations to the water pipe system include water-efficient fixtures and leakage management.

Moreover, the greater rooftop area and storage capacity can mean that institutions provide greater efficiency in rainwater collection. The water collected can then be used for toilet flushing, cleaning, washing hands and also drinking, with adequate filtration and proper treatment. In coastal areas affected by salinity, water storage facilities based on the rainwater harvesting can provide continuity of safe water to communities. Greywater can also be captured and recycled for non-potable uses such as irrigation of trees.

Repair works are likely to cause temporary closure of schools. Additionally, increased domestic WASH work leaves less time for education, and this is likely to affect girls in particular. During weather events, schools are often used as shelters for temporarily displaced people, and this can place strain on WASH services. When schools re-open for students, latrines may need to be emptied and cleaned and supplies of water treatment products and soap restocked. If it is too dangerous to go to school or if schools are closed, girls might not be able to collect their entitlement of sanitary pads (if the government has a pad distribution programme). All of this has consequences for girls and women's ability to access their preferred menstrual product, or to change the product as often as needed.

3.3 Agency in accessing and managing WASH facilities

Women, men, boys and girls have different roles in WASH services in household, community and schools and health care facilities. Where women have relatively lower social and legal status and hold less political power within a community, this limits their influence on decisions about how families and communities maintain WASH in accessing, using, developing and managing services against climate change impacts⁵⁴.

Household

Women are very often given responsibility for household WASH, for example, fetching water for the household, storing water safely, washing plates and cleaning toilets. Women and girls are usually

⁵³ Charles et al. (2009)

⁵⁴ Kohlitz et al. (2020)

tasked with filtering or boiling water before drinking⁵⁵. Women also take care of the toileting needs of children and older people. They also spend considerable time caring for family members ill from water-related disease. Despite being primary managers of water and charged with the WASH related care for their families and households, women are not always consulted in decisions on household WASH services. Routray et al. (2017) found that in rural Odisha (India), the majority of construction of household level sanitation facilities was done exclusively by the male head, attributed to women's lack of education and limited exposure to the world beyond their home and inability to influence the household's financial decisions.

Care and the domestic WASH-related work burden falls disproportionately on women in all cultures, but the degree of this inequality varies. A shift in gender roles, with men taking some responsibility for cleaning the house, toilets, and the environment can help women and girls take on other roles and options within and outside the home, including work and schooling. Men more often take on household roles related to both building and financing WASH facilities.

Community

Women's perspectives or voices are not always reflected in WASH decision making. Particularly in indigenous communities, women often hold important traditional knowledge about how best to manage and use both water and land resources, making them valuable experts⁵⁶. Community WASH committees may have a requirement for female representation, for instance, the Government of Nepal's 2011 Sanitation Master Plan requires 33% of WASH committee members to be women; however, women might not be empowered to contribute fully to WASH decision making. Women's groups are rarely identified as stakeholders in the process of developing City Sanitation Plans, and so have limited input to guide strategic planning and investment decisions. Unless men are engaged in facilitating opportunities for women's decision-making in WASH processes, a WASH programme can inadvertently run the risk of increasing burdens for women. Time spent by women on community WASH activities can be a source of conflict within the household, particularly if this is viewed as impacting on women's time for household and family responsibilities⁵⁷.

When the household is invited to attend village meetings, typically the male head-of-household will go and women will stay in the home to take care of domestic responsibilities⁵⁸. Facilitating men's allyship to support women's active participation in WASH decision-making is an emerging area in the WASH sector.

Workforce

UN Global Analysis and Assessment of Sanitation and Drinking Water (GLAAS) data indicates that a large majority of countries lack sufficient human resources to implement national WASH plans. There are insufficient workers with the appropriate competencies in the right positions, including in rural or hard-to-reach areas or communities.

Men hold the majority of professional positions in the WASH sector. Studies in Vietnam, Indonesia and Timor and elsewhere indicate that men hold the majority of professional positions in water and sanitation companies and utilities⁵⁹.

⁵⁵ Megaw, Kohlitz, Gero and Chong (2020)

⁵⁶ USAID (2020)

⁵⁷ IWDA (2015)

⁵⁸ Megaw, Kohlitz and Chong (2020)

⁵⁹ Leahy et al., 2017; Grant et al. (2017)

Female representation in the workforce is a major challenge in most countries in both government and non-governmental WASH services. While men take on more paid roles, women are more likely to be offered unpaid roles as community health volunteers or health promoters. In many countries, women are missing in technical fields or management roles in the WASH sector. For instance, the HR Capacity assessment in Bangladesh found that the HR composition of government and WASH sector agencies is dominated by engineers and technicians, most of whom are men. The number of women employed in the sector is very low, and the majority tend to work as health extension workers and on community level hygiene promotion activities. Data from the IBNET database⁶⁰ reveals that Bangladesh had around 6% female staff in utilities (1% female engineers) in 2017. In Sri Lanka⁶¹, the number of female engineers in the WASH sector has grown steadily over the past three decades. Sri Lanka's free education policy has had enormous impacts on the sector, improving the ratio of women on engineering and technical water and sanitation courses in higher education. In Sri Lanka, international agencies prefer female candidates for WASH activities, and there are numerous rural water supply schemes run by women-led organisations.

SACI Waters pioneering work in 2010 documented the experience of female water professionals in the region and shared their personal stories⁶². Since then, participation of women in WASH enterprises is increasing⁶³. In Nepal⁶⁴, having female sanitation sale agents is found to drive higher overall latrine use, and additionally, women are more effective at selling water and sanitation products/services to marginalized households⁶⁵. WaterShed (2019) has worked in rural Cambodia to support women's engagement in private WASH markets, including latrine sales and WASH product retail. Women's ability to succeed often depends on their having self-belief, technical knowledge and safe workplaces free of harassment. It is also important they have the ability to work from home and simultaneously complete housework or supervise children as well as having family buy-in and permission to be away from home. Elsewhere, women have engaged in water vending, selling water from kiosks, meter reading, water tank cleaning, water filter selling and rainwater harvesting tank construction. Women also work in water-dependent, income-generating activities like catering, brewing and hairdressing. Additionally, women can work in sanitation related occupations such as toilet pan selling, toilet cleaning, latrine construction, serving as toilet attendants, doing manual scavenging work/pit emptying, managing public toilets or running faecal sludge management enterprises. In India, it is common for Dalit woman sanitation worker to clean and remove faecal waste from dry toilets, without the proper safety gear, risking infections⁶⁶: *'A woman sanitation worker, working as a sweeper on the road, has no facilities such as toilets, or a resting shed to take a break, especially when working under the hot sun These women do not raise their voice against it, because they feel powerless and replaceable.'* Improvement in household water and sanitation services have freed up time for other income-generating activities.

Schools and health care facilities

Gendered power issues are at play, not only in how WASH is managed within households, but also in community interventions. For instance, latrine cleaning in schools often remains a female teacher or

⁶⁰ <https://database.ib-net.org/DefaultNew.aspx>

⁶¹ IWA (2013)

⁶² SaciWaters (2010)

⁶³ Indarti et al., 2018; Indarti et al. (2019)

⁶⁴ USAID and IDE (2020)

⁶⁵ Ibid

⁶⁶ Participatory Research in Asia (2019)

girls' role. School-based WASH behaviour change programmes focus on the practice of WASH behaviours in all students. Integrating household WASH issues, gender and climate awareness within 'life skills' curricula could support change. Play-based WASH activities have been developed to help girls and boys learn WASH behaviour, but have also been used to teach boys about the importance of taking on WASH related work in the home and thereby challenge their preconceptions of which tasks are 'appropriate' for their gender.

3.4 Effects of climate change on agency

In some cases, climate change is entrenching existing gender norms and roles, but in other cases, it is contributing to a (temporary) shift in norms and expectations around WASH related work in the household, community and workforce⁶⁷.

In extreme weather events, due to the dangerous conditions, men and women make more decisions together about when and how to do WASH related tasks, as opposed to in normal conditions when women mostly make these decisions. Men and boys may take over water collection (usually women's duty) due to the dangerous conditions that can arise during extreme rain and wind, for example, slippery surfaces and falling trees. Men may assist women and girls more with washing clothes due to the heavy loads and security issues associated with walking longer distances to water sources. Clothes are heavy, and therefore, they will help with this heavy work⁶⁸. Men might also support in the bathing of children. However, when this happens it appears more on a temporary basis during a crisis and is done to 'help' women rather than signifying a more permanent shift in gender roles. Men and boys may still be embarrassed to be seen collecting water and are more likely to use bicycles, wheelbarrows or motorcycles to collect water. Anecdotally, some husbands help their wives in collecting water from far away (with donkey carts). They may also help when wives are sick or pregnant or when they are in the postnatal period⁶⁹.

At the community level, men may take on more responsibility and work load for public clean up and for repairing damage to infrastructure after a climate related weather event such as strong winds and heavy rain⁷⁰. Dangerous conditions may also lead to men fixing water systems and building toilets alone rather than with help from boys⁷¹.

Men often have better access to education and information about climate change and water management than women (such as access to climate forecasts, early warning information and emergency and post-disaster services), and this can reinforce their role as WASH decision-makers⁷². Men often take more responsibility making decisions for restoring WASH access following a weather event⁷³. Men may attend more community activities and meetings than women in dangerous conditions.

Women's work inside the home generally increases with water scarcity due to gender roles and norms that govern house work and decision-making. Women often take on responsibility for managing scarce

⁶⁷ Dery, Bisung, Dickin and Dyer (2020)

⁶⁸ Kohlitz et al. (2020)

⁶⁹ Assefa et al. (2021)

⁷⁰ Leahy, Kohlitz and Chong (2020)

⁷¹ Ibid

⁷² Megaw, Kohlitz, Gero and Chong (2020)

⁷³ Ibid

water resources in the household⁷⁴. Climate change can also lead to the migration of men to find alternative jobs outside of the village, which increases the burden on women in terms of caring for their household, including taking sole responsibility for their household's WASH needs. Water sources become more time consuming to access due to climate-related shortages⁷⁵; accessing them requires waking up early to queue for water, walking long distances and includes long waiting times. It also leads to quarrels and conflicts at water points and the need to identify multiple water sources to diversify risk⁷⁶, all of which are stressful for women⁷⁷. Women may also expect that men will assist more with collecting water if they have to travel longer distances to reach it, which suggests some potential for conflict⁷⁸. Young girls and boys may also be asked to help mothers with increased workloads. Deteriorating water quality (or use of surface water instead) means that women and girls will usually be tasked with treating contaminated drinking water to make it safe to drink⁷⁹. Constraints on women's time can limit their ability to access the fuel needed to treat or boil water. When people with disabilities or older people in the household are no longer able to access sanitation facilities, either because they cannot reach the facilities or the facilities are not functional, they will depend on others (usually women) to help them take care of their toileting and hygiene needs⁸⁰. Women may be dependent on men to generate and spend income for WASH climate change adaptations such as storm-resistant toilets or water storage tanks for the dry season⁸¹.

Access to information influences how people are impacted by, and respond to, climate change. Women are less likely to have access to information about climate events (e.g., early warning systems), and therefore are likely to be less able to prepare (e.g. to store clean water in their home before a storm hits). Women often acquire information from other women about water issues and are often excluded from decision-making processes on the maintenance of water resources and assets that could make services more resilient⁸². Megaw, Kohlitz and Chong (2020) report how this inequality in decision-making is linked to the perception of the man's role in physical response, such as building back infrastructure, as being more vital to decisions than the woman's role in caring for the house, family possessions and the wellbeing of household members. WASH clubs in schools can help raise awareness and foster individual behaviour change around climate and WASH at the household level.

⁷⁴ Leahy, Kohlitz and Chong (2020)

⁷⁵ Caruso et al. (2021)

⁷⁶ Daly et al., 2021; Elliott et al. (2019)

⁷⁷ Megaw, Kohlitz and Chong (2020)

⁷⁸ Leahy, Kohlitz and Chong (2020)

⁷⁹ Megaw, Kohlitz and Chong, (2020)

⁸⁰ Leahy, Kohlitz and Chong (2020)

⁸¹ Megaw, Kohlitz, Gero and Chong (2020)

⁸² Kohlitz, Chong and Willetts, (2019)

4 Global Evidence Review: Gender & wetland, coastal and marine ecosystems

In this Section, we present the findings of the global evidence review in the protection of wetland, coastal and marine ecosystems. It considers the following areas of discussion, identified as the more pertinent issues based on our review of the relevant literature:

- Women and men's use of, and interaction with, wetland, coastal and marine areas
- How climate change is affecting these different uses and interactions, and the impact on the populations concerned
- The different ways in which men and women are, are not, and could be involved in the protection and management of wetland, coastal and marine areas.

4.1 Different interactions with wetland, coastal and marine areas

Gendered roles in the fishery industry

Fishing at sea with specialised equipment is usually carried out by men, while women generally fish on shallow reefs, tidal flats, and coastal estuaries. Formal and paid fishing activities are usually dominated by men, with women only representing 2% of maritime workers.⁸³ In most places, women only have access to the less profitable parts of the fish for selling, and have limited trade connections, as well as less access to boats, fishing gear, and limited financial capital due to barriers to taking out loans and accessing credit⁸⁴. This has contributed to a gender pay gap, with women earning 64% of men's wages in the aquaculture industry for carrying out the same work⁸⁵. Within value chains, women predominate in the post-harvest processing, marketing, and trading of fish, while most fishery agencies focus on the production aspect; 15% of harvesting jobs and 90% of fish processing jobs are carried out by women globally.⁸⁶ Moreover, women are disproportionately employed in the informal and unregulated fish economies, including shellfish gathering and food processing. This can lead to exploitation, as there is little oversight and labour trafficking and sanitary issues can be easily masked.⁸⁷

Gender norms as a barrier to women's participation in formal activities

Women's participation in the fishery industry is often overlooked and undervalued, resulting in the marginalisation of women who do participate in the management of wetland, coastal, and marine areas. Studies show that fishing is seen as a "man's job", and women are not expected to fish due to the arduous labour and long hours that keep them outside of the home for long periods of time.⁸⁸ For example, gender norms in the Philippines prevent women from straying far from home to go fishing.⁸⁹ In Zambia, fishing gear is often only owned by men.⁹⁰ As mentioned before, in many countries, women

⁸³ Stuchtey et al, 2020 & Österblom et al, 2020.

⁸⁴ Österblom et al, 2020.

⁸⁵ Friends of Ocean Action, 2020.

⁸⁶ OECD 2020.

⁸⁷ Stuchtey et al, 2020.

⁸⁸ Stuchtey et al, 2020

⁸⁹ Österblom et al, 2020

⁹⁰ Ibid.

are denied bank loans or do not have access to credit, which prevents them from buying large fishing boats and other equipment needed to engage in formal fishing. Women may not be viewed as constituting part of the market, and tend to lack access to bank accounts or their own personal financial capital.⁹¹

Maternal health concerns related to consuming large amounts of fish

High reliance on marine organisms for nutrition leads to higher exposure to methylmercury and organic pollutants. Pregnant women are impacted by high levels of mercury from consuming seafood and must avoid certain fish species.⁹² Infectious diseases from fishing can impact food security, and industrial activity near oceans can lead to contaminants ending up in food webs and human food supply. *No further gender specific impacts pertaining to these health concerns have been found in the literature.*

Lack of evidence on the gender-differentiated effects of unsustainable fishing practices

Food insecurity leads to reliance on seafood, which causes overexploitation of fish stock. This causes fishers to spend more resources locating and catching stock, as well as moving to shallower waters which leads to more overfishing.⁹³ Tourism, mining, and energy industries disrupt natural habitats, displace marine species, and the spread of contaminants to food webs and human food sources.⁹⁴ *Gender-specific impacts of overexploitation are not mentioned in the literature reviewed.*

Seafood makes up for 16% of all animal protein consumed globally. Fishing from oceans provides jobs, nourishment and livelihoods for coastal communities. Apart from fishing, sources of income include ports, shipping, tourism, and energy. The fishing sector continues to grow, with fishery productions expected to expand by 20% over the next ten years worldwide (OECD 2016). However, we have not yet seen any wholesale shift from extractive fishing practices to biologically sustainable production practices; this is likely to prove particularly difficult given the great (and increasing) pressure on the fishing industry to provide livelihoods to entire communities and societies. *The literature does not explore gender-specific impacts in relation to the supply of food or employment.*

4.2 Effects of climate change on different uses and interactions

Rising sea temperatures are likely to disproportionately affect women's livelihoods and food security

Rising sea temperatures due to climate change causes changes in the behaviour of fish stock, permanent changes to ecosystems, and extinction of some species. The elimination of important ocean habitats will have serious effects on the food security of vulnerable communities. Whilst not explicit in the oceans literature, we know that women tend to bear the responsibility for managing crises within households and often reduce their own food consumption relative to that of other family members in contexts of food insecurity. Job opportunities will decrease in a context where there are already limited opportunities for employment, especially for women who are excluded from much of the fishery industry. Climate change disproportionately affects coastal communities who do not have

⁹¹ Ibid.

⁹² UN Environment, 2019.

⁹³ High level panel for a sustainable ocean economy, 2020 & UN Environment, 2019.

⁹⁴ UN Environment, 2019

alternatives to certain fish supplies and do not have the resources to respond to changing fish behaviours.⁹⁵ Because women are often barred from taking out loans or accessing credit, they are less likely to be able to access resources that would enable them to adapt to these changing behaviours.

Rising sea levels are likely to disproportionately affect women's livelihoods and food security

Heat-induced expansions of water lead to a rise in sea levels which increases the risk of storm surges for islands and coastal communities. Exposed infrastructure is also at risk.⁹⁶ Rising sea levels leads to submerged lands, salinised soil and groundwater, and erosion of coastlines.⁹⁷ Tropical storms put coastal regions at risk of significant economic damage, especially due to damage to ports and harbours. Women and children are especially vulnerable in these instances as it is more difficult for the former to relocate and find work outside their communities.

Rising acidity in oceans are likely to disproportionately affect women's livelihoods and food security

Greenhouse gas (GHG) emissions have increased acidity in oceans by 26% since the industrial revolution.⁹⁸ This harms shell-forming animals such as corals, phytoplankton, bivalves, and zooplankton, who make up the basis of many marine food chains. Poor water quality and rising acidity causes shifts in fish stock behaviour and movements, which leads to food insecurity for many coastal and island communities (OECD 2016). Food supplies will shift locations, and coral reefs that serve as tourist attractions will die. Vulnerable communities, especially women and young workers, who already face difficulty in finding jobs, are unlikely to have the resources necessary (or for women, sufficient mobility) to follow the movement of fish stocks.

4.3 Involvement in the protection and management of wetland, coastal and marine areas

Women's (potential) roles in community-based management

Provision of clear boundaries for access and rights to fisheries are associated with a community commitment to usage rights. When this occurs, management is often more sophisticated with better registration records for boats and fishers.⁹⁹ Managing specific areas that communities understand as "theirs", leads to the elimination of outsiders skimming off the benefits of good management behaviours.

The role of women in sustainability and management initiatives for the ocean economy is very limited. Because women are rarely viewed as members of the formal fishing sector (or as "real" fishers), they are often excluded from discussion around the rights and needs of communities. They may even face negative social repercussions for participating in harvesting processes, despite the nutritional needs of their families.¹⁰⁰ As a result, recorded engagement of women in new management initiatives is very

⁹⁵ Stuchtey et al, 2020.

⁹⁶ UN environment, 2019.

⁹⁷ Stuchtey et al, 2020.

⁹⁸ Stuchtey et al, 2020.

⁹⁹ Österblom et al, 2020.

¹⁰⁰ Österblom et al, 2020.

low,¹⁰¹ causing their ideas to go unheard in community-based management and protection of marine and coastal life.

Women's exclusion is a hindrance for the improvement of fishing sustainability because they are often more cognisant of the need to make conditions better for future generations, and are more willing to engage in progressive sustainability practices.¹⁰² For example, when First Nation Heiltsuk women in British Columbia took on leadership roles in the management of wetlands, they helped transform the governance of herring in their coastal region (Ibid).

However, as developments in the ocean economy progress, there have been more pushes to include local voices and developing communities in decision making processes – with some attempts at inclusion involving women¹⁰³ (Österblom et al, 2020). With this new call for bottom-up processes, women are more likely to be included in discussions, but empowerment of civil society and localised movements (still new developments in the ocean economy) is necessary for women to exercise agency and decision making power (UN Environment, 2019).

Gender-blind policy and decision making

Institutions need to be agile, adaptable and follow 'top-down, bottom-up' practices that are informed by science. Women face barriers to accessing governing institutions, with their voices often left out of decision making practices. In Senegal, for example, women constitute only 5% of fishing management bodies, causing decisions to be made that only benefit some groups (Österblom et al, 2020), making it difficult to understand how women are affected by industry developments.

Data collection by governments often only includes information on men's engagement in the paid harvesting sector of the fish production chain and excludes women's engagement (Adam et al, 2021). Such conventional gender-blind approaches to research and policy development for the fishery industry which only take into account formal and paid fishing activities mean that women's activities in the sector are largely unaccounted for, with the resultant policies, including those concerning livelihoods, not designed with women in mind.¹⁰⁴ For example, spatial policies that designate certain areas as 'no-take' lands, disproportionately impact women who do not have the resources to fish elsewhere. In these cases, women have had to resort to fishing illegally, which removes them still further from the formal fishing industry.¹⁰⁵

Limited evidence on the gender-differentiated effects of technological advancements

Innovations in marine resource management have often been assumed to be "gender neutral", meaning that approaches "leave gender out of it".¹⁰⁶ In reality, such initiatives' "gender neutrality" has contributed to male bias in their design.¹⁰⁷

Climate Smart Aquaculture (CSA) is a new technology that is used to ensure sustainable fishing and the continuity of the fish supply chain. Limited research has been conducted to see how women

¹⁰¹ Adam et al, 2021.

¹⁰² Österblom et al, 2020.

¹⁰³ Österblom et al, 2020.

¹⁰⁴ Österblom et al, 2020.

¹⁰⁵ Adam et al, 2021.

¹⁰⁶ Adam et al, 2021

¹⁰⁷ Ibid.

engage with and benefit from this new technology, despite the fact that climate change affects rural, low-income, communities of women particularly adversely.¹⁰⁸ Innovation in feed and seed breeding technologies are also lacking in gender-related analysis, resulting in a lack of evidence as to how new technological advancements are affecting women.¹⁰⁹

However, there have been some studies on gender differences in CSA initiatives. When a study was conducted in Bangladesh to investigate the potential role of women in CSA, several conclusions were made about women's empowerment as a result of training, including over 75% of women reporting that they had experienced positive changes in their decision-making power.¹¹⁰ In Bangladesh and India, women were found to relate fish trait preferences to household food and nutrition security.¹¹¹

Prevalence of gender-segregated interventions which fail to address underlying barriers

Capacity development for local communities to engage in good fishing practices aims to maximise profits whilst reducing overfishing.¹¹² Effective fishing practices that include women would not only generate wealth but also allow for better distribution of resources, and an ocean economy where all voices have decision making roles.¹¹³ Building institutions that support women's employment in the fishing industry will protect against inequity, and prevent inefficiencies and unidimensional decision making. Including women in all segments of the fishery industry requires transparency, and monitoring of the supply chain for those who participate in illegal activity and exploitative practices such as labour trafficking.¹¹⁴

However, existing training for women in the fishing sector largely follows a "gap filling" approach which seeks to accommodate (but, critically, not address) the normative barriers that women face.¹¹⁵ Training tends to be held close to women's homes and often the technology or strategies introduced relate to jobs that can be performed from home. These interventions target women separately from men, and have limited impact as they fail to address the underlying power dynamics and barriers that perpetuate gendered inequities at the foundation of the fishing industry. These interventions often do not recognise that women are not viewed as "true" fishers, themselves treating women separately and as different from men.

In select instances, gender transformative measures which address these underlying normative issues have been introduced in natural resource management, as seen in a study on the implementation of women-targeted training in Bangladesh.¹¹⁶ However, implementation in developing countries remains difficult. Barriers to gender-transformative approaches include low political and social will, poor messaging about gender norms and a lack of assistance from governments and NGOs.¹¹⁷

¹⁰⁸ Adam et al, 2021.

¹⁰⁹ Ibid.

¹¹⁰ Ibid.

¹¹¹ Ibid.

¹¹² UN Environment, 2019.

¹¹³ Stuchtey et al, 2020.

¹¹⁴ Ibid.

¹¹⁵ Österblom et al, 2020.

¹¹⁶ Ibid.

¹¹⁷ Ibid.

5 Global Evidence Review: Gender & climate resilient port infrastructure

In this section, we present the findings from the global evidence review on climate resilient port infrastructure. It considers the following areas of discussion, identified as the more pertinent issues based on our review of the relevant literature:

- The different ways in which men and women use and benefit from available port infrastructure; and
- The different ways in which men and women are, are not, and could be, involved in and can benefit from the development of climate resilient port infrastructure.

5.1 Different uses and benefits from available port infrastructure

Women do not benefit equally from the employment benefits of port infrastructure

Ports are a major hub for trade, and are vital for national economies as they provide job creation and employment opportunities. In Eastern and Southern Africa, seaborne trade accounted for 80% of total trade volume in 2016.¹¹⁸ Seaports have contributed to the expansion of urban populations and growth of local markets.¹¹⁹ Women, however, are not able to fully reap the benefits of this economic growth due to their exclusion from operational and managerial positions.¹²⁰

Lack of evidence on the gender-differentiated effects of ports' negative impacts on ecosystems

Activities related to port infrastructures can damage city port, delta and river ecosystems, through land reclamation, dredging and construction. As ports grow in number, it is difficult to find areas that have the space for sustainable expansion.¹²¹ There is a need for port infrastructure development that respects the environmental constraints of a particular area, and can adapt to climate and technological change¹²². Ports were originally built with the prioritisation of trade and economic growth in mind, but little consideration of environmental and health impacts. *No gender-disaggregated impact analysis could be found in this specific literature.*

5.2 Different involvement and benefits from the development of climate resilient port infrastructure

Women's (potential) role in employment and leadership roles in climate resilient ports

The expansion and development of new climate-resilient ports creates jobs and therefore opens up opportunities to train and hire women. However, according to reports from seafaring institutions, female trainees only made up 6% of training participants in 2021.¹²³

¹¹⁸ Humphreys et al, 2019.

¹¹⁹ Ibid.

¹²⁰ Patrick Verhoeven, 2020.

¹²¹ Shipper et al., 2015.

¹²² Ibid.

¹²³ BIMCO & International Chamber of Shipping, 2021.

Women's recruitment, integration and progression is starting to be supported through training, mentorship and working environment initiatives. Training institutions for maritime trade management are looking to include women in their training programmes. One step institutions have taken is to offer a 10% discount to female students.¹²⁴

Mentorship programmes have been implemented to connect women to those with experience in senior level roles, including roles that work with digital technology and autonomous vehicle operations.¹²⁵ These initiatives focus on the challenges women encounter in the maritime industry, finding ways of advancing and retaining female talent. With women rarely involved in the operational aspects of ports, initiatives may have a particular focus on teaching women how to use certain vehicles and technologies to build operational knowledge, skills and confidence.

Through family-friendly human resources (HR) policies and adequate sanitation facilities for women (e.g. separate toilets), the development of new climate-resilient ports can encourage women to work in the port environment.¹²⁶

Women's (potential) roles in promoting the importance of ecosystems

More broadly, women have an important role to play in promoting climate resilient port development for the benefit of wider communities. Through the utilisation of ecosystem-based management approaches, the health, function and resilience of ecosystems can be restored and protected. No-impact port development involves the development of ports which have no negative impact on the environment, and recognises the complexity of ecological systems, as well as the intricate ways in which elements of oceans and coasts interact with each other. One aspect of this is reducing noise pollution in order to reduce physiological stress and change on the environment. To limit disturbances to current ecosystems, no-impact ports are preferably created in natural locations that do not require dredging.¹²⁷ Women are often excluded from discussions around new approaches, but tend, as noted above, to be more willing to engage in sustainable or environmentally friendly practices, and should be enabled to participate meaningfully in consultation and planning processes.

Lack of evidence on gender-differentiated benefits of optimised infrastructure and energy efficiency

Increased trade capacity and employment opportunities can stimulate local economies. The expansion of ports will create many jobs in the areas of operations and building, as well as increase business to the ports. This has the potential to grow under-developed economies in an environmentally friendly way. Moreover, better ports will lead to a decrease in the cost of sea and inland transport and operational expenses, increasing profits for port-operators and decreasing prices for port-users.¹²⁸ The optimisation of infrastructure and the use of digital technologies to combat climate change can benefit port users while also minimising the impact of port activity on the environment. *However, the literature does not explore whether or how the optimisation of port infrastructure can benefit men and women differently.*

¹²⁴ Ibid.

¹²⁵ Patrick Verhoeven, 2020.

¹²⁶ Patrick Verhoeven, 2020.

¹²⁷ Schipper et al. 2015.

¹²⁸ WWF & Deltares 2015.

Through the production and sourcing of local renewable energy, encouragement of clean energy initiatives and increasing energy-efficient port equipment, negative environmental externalities of ports can be reduced. These energy initiatives also improve the health and safety awareness of employees and local communities. *However, the literature does not explore whether or how these changes benefit women and men differently.*

GHG emissions from ships can be reduced through onshore power supplying, incentivising best-performing vehicles and investing in energy efficient infrastructure.¹²⁹ Operations can be made more energy efficient through innovative processes, production technologies and the implementation of clean and renewable energy. An incentive scheme can help to promote these actions, such as an Environmental Ship Index that incentivises best-performing vessel initiatives. Concession agreements and lease clauses could be used to encourage clean energy initiatives from third parties.¹³⁰ However, as it stands, the introduction of new technologies is likely to further marginalise women, who are often excluded from technical training and leadership positions. In order to limit inequities in port infrastructure development, it could be beneficial to engage in mentorship and training programmes for women first, before enacting any technical improvements.

¹²⁹ Patrick Verhoeven, 2020.

¹³⁰ Patrick Verhoeven, 2020.

6 National WASH policy and context in focal countries

In this Section, our focus shifts to the country level to present a summary of national WASH policy and context in the 11 focal countries: Bangladesh, Botswana, Brazil, Djibouti, Indonesia, Ivory Coast, Kenya, Namibia, Nigeria, Morocco and Uganda. For each of these countries, we present a summary of country analyses that were completed and which can be found in full in Annex 5. These country analyses provide a top-line picture of access to WASH facilities, drawing on data from the WHO/UNICEF Joint Programme for Water Supply, Sanitation and Hygiene (JPWSSH)¹³¹ and making reference to JPWSSH's ladders for drinking water and sanitation service (Table 4) as a means of differentiating the level of water and sanitation services the population has access to. Each country summary also appraises the WASH policy context to understand the extent to which, and how, current policy frameworks identify and seek to address gender issues in WASH, whilst building climate resilience. In 8 out of the 11 country summaries, we also include a stakeholder mapping.¹³²

Table 4: Joint Programme for Water Supply, Sanitation and Hygiene ladders for drinking water and sanitation service

Drinking water ladder		Sanitation service	
Safely managed	Drinking water from an improved water source that is accessible on premises, available when needed and free from faecal and priority chemical contamination	Safely managed	Use of improved facilities that are not shared with other households and where excreta are safely disposed of in situ or removed and treated offsite
Basic	Drinking water from an improved source, provided collection time is not more than 30 mins for a roundtrip including queuing	Basic	Use of improved facilities which are not shared with other households
Limited	Drinking water from an improved source for which collection time exceeds 30 mins for a roundtrip including queuing	Limited	Use of improved facilities shared between two or more households
Unimproved	Drinking water from an unprotected dug well or unprotected spring	Unimproved	Use of pit latrines, without a slab or platform, hanging latrines or bucket latrines
Surface water	Drinking water directly from a river, dam, lake, pond, stream, canal or irrigation canal	Open Defecation	Disposal of human faeces in fields, forests, bushes, open bodies of water, beaches and other open spaces or with solid waste

¹³¹ A joint WHO/UNICEF initiative to improve global monitoring of drinking water, sanitation and hygiene in the context of the new 2030 Agenda for Sustainable Development, [How we work | JMP \(washdata.org\)](https://washdata.org)

¹³² A stakeholder mapping could not be conducted for Brazil, Botswana and Indonesia as No Objection Letters were received late in the Gender Assessment process.

6.1 Bangladesh

Key Messages

Bangladesh generally has high levels of access to improved drinking water sources and relatively high levels of access to some form of sanitation facility. Improving water, sanitation and hygiene is a key part of Bangladesh's national policy framework, with the Bangladesh Climate Change and Action Plan (BCCSAP) (2008) identifying connections between WASH and negative impacts caused by increased weather extremities. The plan identifies the need to take specific approaches for vulnerable people, to mainstream gender in improving hygiene promotion, including by shifting organisational culture and ways of thinking and by developing procedures relating to gender responsibility, accountability, coordination and personnel policy. While most WASH-related policies do not explicitly consider advancing women's formal roles in the WASH sector, the National Women Development Policy, identifies a need to advance career development opportunities for women employees and equal participation of women in decision-making and governance at all levels. In general, women are limited in their careers prospects within the WASH sector by low rates of secondary education and policies which focus on opportunities in low-skilled jobs.

Headline data on access to WASH services

Access to Drinking Water: 98.9% of the population has access to an improved source for drinking water, of which the majority, 58.5%, is safely managed drinking water and women are the main collectors of water¹³³.

Access to Sanitation: 78% of the population has access to an improved facility, of which 38.7% has access to safely managed sanitation services. None of the population is reported to practice open defecation.

Policy framework

Improving water, sanitation and hygiene is a key part of Bangladesh's national policy framework, as identified in their overall sustainable development strategy, as well as further, more specific water strategies. The policies recognise the increasing impacts of climate change, rapid urbanisation and water contamination, and inequitable access, identifying these as challenges for future development in water, sanitation and hygiene.

Bangladesh's location makes it highly susceptible to the adverse effects of climate change, hindering its development, **including access to safe and economic opportunities for women and other vulnerable groups**. The Bangladesh Climate Change and Action Plan (BCCSAP) of 2008 looks further into the negative impacts caused by increased weather extremities. Strategic objectives to combat identified challenges include ensuring safe drinking water, **undertaking specific approaches for vulnerable people, gender mainstreaming**, improving hygiene promotion, adapting to climate change and undertaking integrated and accountable development approaches. These objectives seek to include **gender mainstreaming in all the service planning, implementation and monitoring** by shifting organisational culture and ways of thinking and by developing procedures relating to **gender responsibility, accountability, coordination and personnel policy** and the consideration of women's role in hygiene promotion, both as a token role and also as an empowered advocate.

¹³³ MICS (2019)

Most of the policies reviewed did not explicitly consider advancing women's formal roles in the WASH sector, but they still discussed the need for **community involvement** in the planning, implementation, operation and management of water and sanitation facilities. Some policies present opportunities to consider for the advancement of women's involvement in, for example, entrepreneurial activities. Within the National Women Development Policy, there is a broader emphasis, within all sectors, on **career development opportunities for women employees** and equal participation of women in decision-making and governance at all levels. Additionally, due to only 21.1% of women completing upper secondary schooling¹³⁴ and 19.8% enrolment in tertiary education,¹³⁵ **women are limited in the careers prospects** within the WASH sector, and a focus on policies is placed on **opportunities in low-skilled jobs**.

Stakeholder Mapping

In Bangladesh, the Policy Support branch of the Ministry of Local Government, Rural Development and Cooperatives has primary responsibility for Water and Sanitation Supply and Strategy. Other national Ministries with some WASH responsibility include the Ministry of Health and Family Welfare, for hygiene, and Ministry of Water Resources (MoWR), covering irrigation, flooding, and climate-related water protection.

6.2 Botswana

Key Messages

Botswana has high levels of access to at least basic sources of drinking water and basic sanitation services. In the policy environment, key challenges identified in the WASH sphere include water shortages and droughts in the water sector, and the inadequate capacity of sanitation infrastructure, and pressure on sanitation and health services because of increasing levels of urbanisation, facilitate access to water of suitable quality adopting 'smart agriculture' as an opportunity and working at the interplay between climate change, the water sector and public health. No specific policies related to women and WASH are identified, but the National Development Plan calls for equal participation and representation of all women and men in all development processes. Key challenges and opportunities identified in the policy landscape include the currently low participation of females in STEM subjects, a need for gender-sensitive policies for climate change and including gender and climate change into academic curriculum.

Headline data on access to WASH services

Access to drinking water: 92.2% of the population has access to a basic level of drinking water and 4.7% of the population has access to a limited level of drinking water. 1.7% of the population relies on unimproved drinking water whilst 1.3% of the population has access only to surface drinking water. Th

Access to sanitation: 80% of the population has access to basic sanitation services and 5.6% of the population has access to a limited level of sanitation services. 9.8% of the population practices open defecation in sanitation services.

¹³⁴ UNIS (2019)

¹³⁵ UNIS (2020)

Policy framework

Botswana's WASH policy landscape is focused around its 11th National Development Plan, including Botswana's Vision 2036. One of the key development challenges facing Botswana's population includes water resources, and related sections on migration, population distribution and urbanisation. Challenges identified here, and further elaborated on in the National Water Policy include **water shortages and droughts in the water sector**, and the inadequate capacity of sanitation infrastructure, and pressure on sanitation and health services because of **increasing levels of urbanisation**, as well as further WASH-related health consequences.

Future objectives is set around a framework that will **facilitate access to water of suitable quality** and standards for the citizenry and provide the foundations for sustainable development of water resources in support of economic growth, diversification and poverty eradication. Climate is a key consideration throughout all documents, including the Climate change policy. They identify water shortages and droughts and water supply for agriculture and industry as key issues, and expanding on adopting **'smart agriculture' as an opportunity**. Between documents, **interplay between climate change, the water sector and public health** is pointed out. The National Health Policy elaborates on the impact of water shortages and their potential for causing malnutrition, and also identifies the negative public health consequences of a lack of access to adequate sanitation and safe drinking water.

The National Development Plan largely ignores specific gender considerations in the water and sanitation sectors. However, it does state that the government recognizes the need for **equal participation and representation of all women and men in all development processes**. In further policies, an opportunity is identified in working towards gender and social equity in accessing water resources, in particular, women shall be empowered to participate fully in issues and decisions relating to sustainable development and management of water resources. Although it is identified often as an important consideration, throughout all policies it is **not further elaborated how this will be operationalised**.

There are some key gender challenges and opportunities identified around over the policy landscape. First, the **low participation of females in Science, Technology, Engineering and Mathematics (STEM)** subjects is identified as a key challenge facing the education sector, despite women having relatively high school completion rates. However, they fail to articulate how this challenge impacts women's employment in the water and sanitation sectors. Second, Botswana's climate policies discuss that measures should be gender sensitive, by **empowering women and youth to actively participate** in the implementation of climate response measures, adopting strategies that are targeted at increasing women and youth's resilience to climate change, and **including gender and climate change into academic curriculum at all levels**.

6.3 Brazil

Key Messages

Brazil has a high rate of access both to improved water sources for drinking water and to improved sanitation services. Brazil's National Policies for Water and Sanitation identify the importance of WASH in daily life and lay out clear objectives to ensure the impact on human livelihoods and activities. These policies present a number of opportunities within the sector, for instance working to increase environmental education and capacity building in technology and science, and both seek to incorporate gender equality in all actions and to develop ongoing training that promotes female

leadership. In addition, at the Federal level, policies emphasise the need to achieve equality, including from a gender perspective. There are few specific recommendations with regard to WASH and gender at either national or Federal level, however.

Headline data on access to WASH services

Access to drinking water: 99.3% of the population has access to an improved water source for drinking water, of which the majority, 85.8%, has access to safely managed drinking water, and only 0.1% has access to a limited level of drinking water.

Access to sanitation: 90.1% of the population has access to improved sanitation services, of which 48.7% has access to safely managed sanitation service, while 9.8% of the population uses unimproved sanitation services, including 0.3% who practice open defecation².

Policy framework

Brazil's National Policies for Water and Sanitation both discuss in-depth the importance of WASH in daily life and lay out clear objectives to ensure the impact on human livelihoods and activities, such as business, is limited. They examine the negative impacts that water scarcity creates in vulnerable communities, including increase in conflict, siltation of water, water pollution, and changes in natural characteristics from high degradation. They add the impact of climate change leaders to further alterations to available water resources and impacts both the energy and agricultural sector. They present innovative opportunities within the sector, all working to increase environmental education and capacity building in technology and science for water resource management. They also seeks to incorporate gender equality in all actions and develop ongoing training that promotes female leadership and empowerment, but does not provide any detail on practical implications of this intention.

In terms of sanitation, Brazil's policies identifies issues with parts of the population continuing to lack access to basic sanitation facilities and/or proper disposal of solid waste and associated health consequences. There are clear objectives and programmes to overcome identified issues, such as increasing and improving infrastructure, management, and monitoring of sanitation facilities, and the suggestion of creating a circular economy for solid waste disposal. It does not explicitly mention gender in the policy objectives. The cross between health and climate change is brought up the country's climate change policies, discussing the impacts extreme weather has on the health and livelihoods of people, which includes the effect it has on water reserves in the country.

As a federal state, Brazil's lower tiers of government play an important role in the provision of sustainable WASH services. The current National Policy for Regional Development II (2019) emphasises achieving equality, within regions and between regions. It seeks to ensure equal access to economic opportunities within all the regions, both from a socio-economic and gender perspective. However, there is no specific discussion of WASH-related challenges and objectives in the policy, nor are there are gender specific strategies within outlined climate action measures. Brazil's policies further explore the importance of including a gender lens in the sustainable use of water, and emphasises the importance of valuing women's contribution to farm work and equal participation in leadership and decision making. Moreover, the Education and Health Plans aim to reduce inequality in their respective systems.

6.4 Djibouti

Key messages

Djibouti generally has high levels of access to improved sources of drinking water and relatively high levels of access to improved sanitation services. It's policy framework identifies a number of areas of connection between WASH and climate change and although connections between WASH and gender aren't typically made explicitly, it's Vision 2035 acknowledges women's role in enforcing new behaviours and practices as well as the importance of engaging men to change their views to women's role. Policies also acknowledge that various stakeholders, including women and youth, should have a larger role in the policy-making process. Finally, policies recognise opportunities to incorporate women into the workforce due to their higher levels of education.

Headline data on access to WASH services

Access to drinking water: 90.8% of the population has access to an improved source of drinking water, although for the majority (76%), this is basic access^[1] and women are the main collectors of water.^[2]

Access to sanitation: 73.8% of the population has access to improved sanitation services, of which 37.1% has access to safely managed sanitation services.

Policy framework

There are a number of key policies which relate to WASH and gender in the Djibouti policy environment and which offer opportunities for engagement. For instance, the Djibouti 2035 Vision (2010–2035) presents a long-term developmental vision for sustained growth and with respect to WASH, **the vision focuses primarily on water supply throughout the country** and its implications for other areas of the primary sector, with a lesser focus on sanitation and hygiene. This is complemented by the climate section of the vision, which includes sustainable water management as an area of priority, amongst other strategy objectives unrelated to WASH. Furthermore, **Vision 2035 identifies the link between climate change and WASH**, specifically in terms of how they will experience increased instances of droughts and flooding which will impact food production and supplies throughout the country.

Vision 2035 includes a specific section on gender which acknowledges **womens' role in enforcing new behaviours and practices**; however, it emphasises how 'in order to improve women's situation, men should be included in the project. It is important to work with men to push them into changing their approaches towards women and their approach with themselves' (Djibouti 2035 Vision, 2010, p. 98).

The 2010 Development Plan for the Primary Sector and the 2015 Strategy of Accelerated Growth and Promotion of Employment echo the focus on access to water for agriculture and livestock, as well as the hygiene risks that a lack of sanitation infrastructure poses in cities. Although the latter two policies touch on gender and the need to address gender inequity throughout the primary sector and for growth, they do not directly link WASH issues to gender. The problems identified, such as health consequences and hard to reach water points, are known to generally affect women more, but this is not specified in the policies. They do acknowledge that various stakeholders, including women and youth, should have a larger role in the policy-making process.

Education is a strong focus of Djibouti's 2011–2021 Gender Policy, the policy that most identifies how water and sanitation challenges are an extra burden on women. The policy covers accessibility to water and sanitation facilities in their communities, remoteness of water points and sanitation deficits in their immediate environments. Currently, the completion rate for upper secondary is higher for women than for men; **this presents a great opportunity to incorporate women into the workforce due to their higher levels of education.** To this end, the 2018–2022 National Strategy for Social Protection identifies a role for women in the country's goal to increase primary sector employment, specifically in fishing and agriculture.

Stakeholder Mapping

In Djibouti, the Ministry of Agriculture, Water, Fisheries, Livestock and Fisheries Resources and the National Office of Water and Sanitation have primary responsibility for WASH. Other national Ministries with some WASH responsibility include the Ministry of Health, the leading ministry in implementing sanitation and hygiene programmes at national level, and the Ministry of Housing, Urbanism, Environment and Territorial Development, who monitor WASH service delivery.

6.5 Indonesia

Key Messages

Indonesia has both high levels of access to improved water sources for drinking water and access to improved sanitation services. Faced by the challenges of climate change and rapid urbanisation, in terms of the policy environment it sees water supply and increasing sanitation facilities as a key component to ensure self-reliance and development for all people. National climate policies highlight that climate-related challenges disproportionately impact women due to issues such as limited access to water supplies for domestic and economic use and lower rates of access to safe sanitation facilities. WASH and health policies recognise the need for gender mainstreaming to reach the country's long-term goals but entry points in WASH are not explored, such as providing career opportunities in the WASH sector. However, the National Plan identifies more broadly the need to increase women's role in development and decision-making, especially within health, education and the labour market.

Headline data on access to WASH services

Access to drinking water: 93.3% of the population has access to an improved water source for drinking water, of which almost all, 92.4%, have access to a basic level of drinking water.

Access to sanitation: 92.6% of the population has access to improved sanitation services, of which 86.5% has access to a basic level of sanitation service, and 6.1% who practice open defecation¹³⁶.

Policy framework

As a country with over 200 million spread out over 6,000 islands, and a major reliance on raw water supply to provide for their water resource, Indonesia has placed natural resource promotion and recovery as a priority. With the growing effects of climate change impacting the water cycle, rapid urbanization, and contamination of water supply it sees water supply and increasing sanitation facilities as a key component to ensuring self-reliance and development for all people. Whilst it

¹³⁶ JMP 2020

recognises the need for gender mainstreaming in policies to reach the country's long-term goals, the specific gender dimensions of WASH are not explored.

The National Plan talks about using the 3R approach - Reduce, Reuse, Recycle - and the circular economy to increase effective waste disposal, to address cross-cutting issues of climate and WASH. They aim for 90% of the population to have access to safe and proper sanitation. As for water resources, they seek to improve infrastructure to meet the goal of 100% of the population having access to safe drinking water. Policies see community involvement as a main way to ensure implementation and management of safe and adequate water and sanitation facilities and practices. This includes **increasing women's role in development and decision-making, especially within health, education, labour market, economy, and public sectors**. National climate policies highlight that climate-related challenges are something that disproportionately impacts women due to limited access to water supplies for domestic and economic use, limited access to safe sanitation facilities, and an imbalance in capacity for understanding the importance of safe water and sanitation practices.

Health concerns due to poor handwashing and water contamination persist in the country, and this is attributed to the lack of progress by local governments. The national Health Strategy presents opportunities for governments to partner with NGOs and the private sector, subsidise communal sanitation facilities, and encourage community engagement. The Health Strategy does not consider gender-related issues in community-based sanitation and there is no consideration to the specific sanitation needs of women. However, all the policies discuss increasing the capacity of communities and workers to lead to increased community engagement and awareness of these issues, but there is limited or no discussion about providing women specifically with career opportunities in the WASH sector.

6.6 Ivory Coast

Key Messages

Ivory Coast has average levels of access to improved sources of drinking water as well as improved sanitation services, although they perform stronger on drinking water access. A high proportion of their population still practice open defecation. It's policy framework has a strong focus on structural economic reforms, balanced with an understanding that these go hand in hand with the development of human capital. The two largest problems that are brought out are very low water availability, worsening due to climate change, as well as high levels of disease due to poor water and sanitation facilities. Policies argue for diversified representation and improved governance from better consideration of gender in public policies. However, all policies are quite gender blind and not propose how to create gender-inclusive WASH environments, other than brief mention of increasing inclusion of women in the employment sector.

Headline data on access to WASH services:

Access to drinking water: 79.8% of the population has access to an improved water source, of which only 35.2% has access to safely managed drinking water. 20.2% of the population access unimproved

drinking water, including 6% who only access surface water.¹³⁷ Women are the main collectors of water.¹³⁸

Access to sanitation: 57.1% of the population has access to improved sanitation services, of whom 34.6% has access to basic sanitation service, while 25.3% of the population still practices open defecation.

Policy framework

Ivory Coast's national development plan has a strong focus on **structural economic reforms**, which is balanced with an understanding that these go hand in hand with the **development of human capital**, one of which is water and sanitation. The challenges that Ivory Coast faces in water and sanitation range from governance of resources, quality of health services and the population's health, and lastly, lacking infrastructure. Beyond the general lack of access to water and sustainable sanitation systems, it is acknowledged there is a heavier burden on populations in cities that are rapidly urbanizing. **Consequences**, such as the rapid spread of diseases like cholera, and food shortages due to water scarcity often **impacts women more severely**. In terms of solutions, they aim to **increase accessibility to drinking water**, build new water pumps and transfer the responsibility and ownership of adopting laws and regulations to local communities. Local communities should also **include women's associations and further diversified representation**.

Climate Change programmes further discuss health consequences of water scarcity and cleanliness, in changing circumstances. Suggested strategies focus on more than just the construction of new facilities and assure uptake of these new facilities to ensure their effectiveness. They have demonstrated that urban populations are more vulnerable, but there is **no mention of the differentiating impact between women and men**. The National Development Plan does, however, identify how improved governance will result from **better consideration of gender and the environment in public policies**. With regard to gender, the strengthening of the institutional and regulatory framework will be necessary, as will the development of the skills of actors who apply it in plans, programs and projects. Furthermore, the plan proposes **increased vocational training for women** to include them in the employment sector, boosted by the improvements in water and sanitation infrastructure.

Stakeholder Mapping

In Ivory Coast, the Ministry of Hydraulics, Sanitation and Hygiene has primary responsibility for WASH. Other national Ministries with some WASH responsibility include the Ministry of the Environment of Urban Sanitation and Sustainable Development, who have a key role in the adaptation to climate change, and the Ministry of Health and Public Hygiene, leading on hygiene policy and programmes

¹³⁷ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹³⁸ MICS (2016)

6.7 Kenya

Key Messages

Kenya has average levels of access to improved sources of drinking water as well as improved sanitation services, although they perform stronger on drinking water access. It's policy framework focuses on climate change, lacking infrastructure and lack of strategy as the main causes of lacking water and sanitation access. Approaches for improvement focus on capacity building, public participation, adoption of innovative technical approaches, and multi-sectoral collaboration. Women's more disadvantaged position in society, and the negative consequences poor WASH facilities have on health and education are recognised throughout and a big emphasis is put on job opportunities and the inclusion of women in leadership. The impacts of climate are discussed but do not relate directly to gender.

Headline data on access to WASH services

Access to water: 71.1% of the population has access to an improved water source, of which 61.6% has access to basic drinking water and 9.5% has access to drinking water categorised as 'limited.'

Access to sanitation: 58.2% of the population has access to improved sanitation services while 41.8% of the population uses unimproved sanitation services, with 8.5% practicing open defecation.¹³⁹

Policy framework

The drive to advance livelihoods in Kenya is laid out through the four key areas, dubbed the Big Four Agenda, within the National Development plan. Water, sanitation and hygiene continues to be a key sub-focus area. Within the plan it sees environmental degradation, climate change, inadequate infrastructure and lack of water conservation strategies as hindrances to the **development of a clean, safe environment and access to water and sanitation services**. Approaches identified to combat these challenges are capacity building, public participation, adoption of innovative technical approaches, multi-sectoral collaboration and adequate water and sanitation facilities.

There is mention of the **under-valuation of women's contribution to the economy**, under-employment and skills mismatch. In relation to health issues, key issues are identified such as the **mortality and morbidity burden** associated with unsafe water and sanitation and the risk of rape and domestic violence for women and girls due to inadequate sanitation facilities. Moreover, within schools, girls tend to be more affected by inadequate WASH facilities, and this impacts their attendance and can **expose them to increased risks of gender-based violence**. To limit the impact on children's schooling, the National Education Sector Strategic Plan includes, among its objectives, the construction of facilities for safe and clean drinking water in schools and **separate sanitation facilities for boys and girls**.

To address these issues, they intend to scale up access to improved rural and urban sanitation and **build governance and leadership capacity for sanitation**. The policy discusses an increase in education, training and skill development for sanitation, including increasing the availability of skilled personnel in the planning and management of sanitation facilities, though it **does not suggest these efforts are focused on any one gender/sex**. Interwoven throughout these policies is the continual discussion on the inclusion of equal opportunities for stakeholder participation throughout the planning, implementation and management of water and sanitation projects/programmes and the

¹³⁹ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

push for participation of female leaders on community-led committees through gender mainstreaming. While most policies include opportunities for job creation, **many do not discuss career opportunities for women specifically**, with a specific focus on seeking to encourage women's participation in the technologies and sciences and STEM based courses.

With climate change being a major discussion in most of the policies, the National Climate Change Policy examines how increases in weather extremities, including droughts, floods and water salinisation greatly impacts the water, sanitation and hygiene sector, with the biggest focus being on inadequate sanitation systems which lead to clogged drains and water contamination.

Stakeholder Mapping

In Kenya, the Ministry of Water and Irrigation has primary responsibility for WASH. Other national Ministries with some WASH responsibility include the Ministry of Health, responsible for sanitation and Hygiene provision and policy, and the Ministry of Environments and Forestry, who set environmental standards, and the Ministry of Housing, who handle WASH infrastructure.

6.8 Namibia

Key Messages

Namibia has a high level of access to improved sources of drinking water, with over 90% of the population, while less than half of the population has access to improved sanitation services. It's policy framework looks at water and sanitation as two separate sectors, with the former focusing on water availability and infrastructure and the latter focuses on health consequences and more on delivering through advocacy efforts. While all policies reference gender as a cross-cutting issue in the water and sanitation sectors, none explore in any great detail how women, youths and marginalised groups are impacted nor the roles that these groups can play in these sectors. The policy landscape recognizes that women continue to be under-represented, and highlights the importance of education.

Headline data on access to WASH services

Access to drinking water: 91.4% of the population has access to an improved source of drinking water, of which 84.3% has access to basic drinking water and 7.1% has access to limited drinking water.

Access to sanitation: 48.1% of the population has access to improved sanitation services, of which 35.3% has access to basic sanitation services. 47.2% of the population continues to practice open defecation.

Policy framework

Namibia's National Development Plan looks at **water and sanitation as two separate sectors**, rather than looking at water and sanitation through a combined WASH lens. As such, analysis of the challenges faced in the water and sanitation sectors fall within two separate pillars. The first is **economic progress**, focusing on water availability and demand in urban areas and for agriculture and infrastructure. The second is **social transformation**, focused on improved sanitation systems and infrastructure as well strengthening advocacy.

Namibia's Water Supply and Sanitation Policy (2008) goes into detail on the challenges themselves, the context, the health consequences and Namibia's policy objectives in response to these challenges.

They continue to handle water availability and infrastructure separately from sanitation challenges. Namibia's 2011 National Policy on Climate Change also analyses the issues of water supply and provision of sanitation services, but focuses on the impact of climate change on both sectors, as well as the potential health consequences and Namibia's vulnerability of water supply to climate change is recognized.

While **all policies reference gender as a cross-cutting issue** in the water and sanitation sectors, none explore in any great detail how women, youths and marginalised groups are impacted nor the roles that these groups can play in these sectors. Namibia's National Gender Policy explicitly tackles the gender issues affecting the water and sanitation sectors. The policy argues that **women continue to be under-represented** in policy and decision-making forums on the utilisation of natural resources, environmental development and water management. The policy also explores the roles that women play within the water and sanitation sectors, as well as exploring the health inequalities related to water and sanitation. A lack of these basic amenities has a detrimental effect on the health of rural communities, and this disproportionately affects women and female-headed households.

The policy states that the government will include gender and climate change, including issues such as water supply, in the curriculum of education and training programmes. However, all policies are very limited in their consideration of women's employment levels in the water and sanitation sectors and in terms of opportunities for women to access employment in these sectors. It is put forward that, as a strategy for gender mainstreaming, women should be involved in interventions to achieve the objectives of environmental sustainability, including in the water and sanitation sectors.

Stakeholder Mapping

In Namibia, the Ministry of Agriculture, Water and Land Reform has primary responsibility for WASH. Other national Ministries with some WASH responsibility include Ministry of Urban and Rural Development, ensuring implementation of national WASH strategy for urban settlements, and the Ministry of Health and Social Services, implementing and enforcing hygiene practices.

6.9 Nigeria

Key Messages

Nigeria has relatively high levels of access to improved water sources, although much of this is still at a basic level, but less than half of their population has access to improved sanitation services. It's policy framework presents a balance of WASH-related issues between economic impacts, governance and coordination issues, and health and culture. The importance of including women in the planning, implementation and management of programmes is recognised throughout, seeking to integrate a gender perspective in all their solutions. In all policies reviewed, there was a clear theme around capacity building and job creation. Their policies do not go much further than recognising the importance of inclusion into suggesting gender specific policies and opportunities.

Headline data on access to WASH services

Access to drinking water: 78% of the population has access to an improved water source for drinking water, of which the majority, 55.9%, is access to a basic level of drinking water. 40% of the population are women who are the main collectors of water.

Access to sanitation: 43% of the population has access to improved sanitation services, of which 30.5% has access to safely managed sanitation service. 18.7% practice open defecation.¹⁴⁰

Policy Framework

As Nigeria continues to experience rapid urbanisation and growth, they also aim to socially develop in areas of water resources and sanitation, water and gender equity and employment, and job creation. Challenges identified include ensuring economic diversification, private sector investment in WASH services, lack of coordination and communal disputes, negative impacts of climate change, contamination of water sources and its impacts on health and poor operation and maintenance culture. Some of the main ways they seek to overcome these challenges is by continuing to improve sanitation services, extending water supply access and adopting an integrated planning and policy formulation approach, while also increasing training, developing the bioeconomy and circular economy, and increasing access to finance and technical support.

One recurring theme in their policies is the increase in negative impacts on Nigeria's development brought about by climate change. Their climate change policy examines the **impact that extreme weather places on an already strained water, sanitation and hygiene sector**. Nigeria seeks to overcome issues related to climate change and inadequate water and waste infrastructure through the **implementation of a circular economy**, as well as increasing infrastructure and reforms in the establishment, management, operation and maintenance of water supply and sanitation schemes and services. They recognise the importance of **including women in the planning, implementation and management of programmes**, and seek to integrate a gender perspective in the adoption of new technologies and infrastructure.

In all the policies reviewed, there was a clear theme around **capacity building and job creation**. Within the Climate Change Policy, there is discussion on **collaboration with the Federal Ministry of Women Affairs** to ensure equitable policies and programmes. This discussion seeks to encourage women's participation in the planning and management of climate change programmes. This includes investment to promote water security and climate change resilient growth and development to sustain jobs, employment, economic growth and social stability. The National Gender Policy of 2006 looks at the importance of **including women in the work force and actively participating in decision making**, calling for equal inclusion throughout.

Stakeholder Mapping

In Nigeria, the Ministry of Water Resources has primary responsibility for WASH. Other national Ministries with some WASH responsibility include The National Council on Water Resources, a water resources policy formulating body, and the Ministry of Health and Ministry of Education, who have roles formulating community sanitation and hygiene.

¹⁴⁰ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

6.10 Morocco

Key Messages

Morocco has a very high level of access to improved water sources and relatively high levels of access to improved sanitation services. Its policy framework is centred around creating a sustainable water and sanitation system as it identified the important crossover between climate change and WASH. Throughout its policies, there is an emphasis on combatting social inequalities; however, they often remind quite gender blind and don't present concrete action points on how to create gender-sensitive programmes and systems. There is recognition that women are more vulnerable to WASH-related health and climate impacts, and places importance on equal participation, as well as capacity building and training for all communities.

Headline data on access to WASH services

Access to drinking water: 90% of the population has access to an improved water source for drinking water, of which the majority, 79.9%, has access to safely managed drinking water.

Access to sanitation: 87% of the population has access to improved sanitation services, while 9.5% of the population uses unimproved sanitation services, including 1.7% who practice open defecation.¹⁴¹

Policy Framework

Due to continued threats to available water resources throughout the country, the National Sustainable Development Policy 2017 (NSDP) recognises the importance of water, sanitation and hygiene improvements for the development of the country. Not only does it seek to improve facilities, access and programmes around water, sanitation and hygiene, it identifies the opportunities available for transitioning to a circular economy through recycling water usage. It also examines the need for human development and **combat social inequalities and promotes equality**, however, it remains quite **gender blind with regard to the incorporation of gender in policies**.

Moroccan Climate Change Policy 2014 highlights the impact of climate change on health and water access in the country, which impact women and girls more than men as **women are typically responsible for managing water**. It also points to the importance of increasing financing and programmes focused on combating climate change, including **education and awareness on alternative solutions** and the re-use of water resources. Morocco's water policies also sets out the need to protect communities against floods, especially for sanitation facilities and agricultural drainage systems.

Within all these policies, there is **no clear examination of how to increase job opportunities for women**. However, they discusses increasing employment opportunities more generally. The Gender Strategy sees the value of ensuring there is **equal participation of women and men in communities**, especially around decision-making, and identifies the impact that inadequate water and sanitation facilities place on women, such as leaving them more vulnerable to illness, it also discusses the importance of **equal access to education**. The Climate Change Policy further discusses the importance of **capacity building and training for individuals, communities, interest groups** and public and private enterprises.

¹⁴¹ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

Stakeholder Mapping

In Morocco, the Ministry of Interior has primary responsibility for WASH. Other national Ministries with some WASH responsibility include The General Directorate of Hydraulics, the lead agency for water resource planning, and The Ministry of Health, responsible for drinking water quality control and water-related laws.

6.11 Uganda

Key Messages

Uganda has a relatively high access to improved sources of drinking water, but has low access to improved sanitation services. It's policy framework strikes a balance between a good quality of life for all and economic productivity, therefore focusing their water programmes on investing in the effective management of the entire WASH value chain segments. The country's policies are built around the worsening impacts of climate change. Connections between climate change, WASH, and gender are quite explicit throughout, with emphasis on the disproportionate burden that women and girls suffer. There is a commitment to mainstream gender issues throughout policy and practice, and a strong focus on including women in employment or income-generating opportunities. Although there is a lack of practical points to implement gender-sensitive policies, there is a commitment to stakeholder participation throughout all phases of a programme.

Headline data on access to WASH services:

Access to drinking water: 83.1% of the population has access to an improved water source for drinking water, of which a large part, 39.2%, has access only to a basic level of drinking water.

Access to sanitation: 37.4% of the population has access to improved sanitation services, while 62.6% of the population uses unimproved sanitation services, including 4.7% who practice open defecation.¹⁴²

Policy Framework

There has been a recent decreased attention to water, sanitation and hygiene in national policy in Uganda. Nevertheless, water, sanitation and hygiene remain a core area of Uganda's Human Development Programme, one which seeks to **improve labour productivity for increased competitiveness and better quality of life for all**. It also foresees investing in the effective management of the entire WASH value chain segments, including containment, emptying, transportation, treatment, safe reuse and disposal.

It is recognised that all development will be **more challenging due to the effects of climate change**, with more extreme weather patterns leading to increased water stress, impacting water availability and quality for both domestic and commercial uses as well as the water and sanitation infrastructure already in place. The Uganda National Climate Change Policy (2015—2020) articulates the challenges further and highlights that, given their domestic role, **women are likely to experience the effects of climate change on water and sanitation facilities most keenly**. It commits to **mainstream gender issues across climate change adaptation** and mitigation actions including ensuring climate change related policies and activities are gender sensitive.

¹⁴² Joint Programme for Water Supply, Sanitation and Hygiene (2020)

The National Water and Sanitation Gender Strategy seeks to ‘empower men, women, boys, girls and vulnerable groups through **ensuring equity in access to and control of resources** in the water and sanitation sub-sector’. The Strategy **highlights the burden and risks for women and girls** that come with poor access to water, sanitation and hygiene. In Uganda’s Water Policies, the most direct benefits to women and girls are likely to come through the rehabilitation and expansion of water and sanitation facilities, as well as support so women and girls can **take advantage of employment or income-generating opportunities**. The Water and Sanitation Gender Strategy recognises the importance of building the capacity of Ministry of Water and Education staff on gender and of **encouraging a gender-sensitive work environment**, but stops short from promoting actions to increase women’s participation in formal roles in the planning, development, management and maintenance of water and sanitation facilities. Interestingly, the National Development Plan III outlines clear plans for **expanding education provision** in Science, Technology, Engineering and Mathematics and increasing the number of graduates in these areas, but places **no emphasis on encouraging women** specifically into education and careers in these areas.

There is a **commitment to stakeholder participation** in the planning, implementation and maintenance of water and sanitation projects. Whilst this commitment provides opportunities for women and girls to influence the development of water and sanitation infrastructure in their communities, they bring an **additional time burden** which women and girls may struggle to take on. Social traditions which encourage women and girls to defer to men may inhibit women and girls’ abilities to participate further. The Second National Health Policy and the National Education and Sports Strategic Plan also highlights that, as the main users of primary health centres, **women and girls can be severely affected by inadequate water, sanitation and hygiene practices** in these centres. Similarly, inadequate toilet facilities are often a barrier to girls’ attendance at school. The Education and Sports Sector Plan includes, amongst its objectives, the expansion and improvement of safe water supply infrastructure to government supported primary schools as well as the construction of sanitation facilities primarily in primary schools.

Stakeholder Mapping

In Uganda, the Ministry of Water and Environment has primary responsibility for WASH. Other national Ministries with some WASH responsibility include the Ministry of Health, responsible for sanitation and hygiene, as well as WASH in health care facilities, and the Ministry of Education and Sports, which has responsibility for WASH in schools.

7 Problem Tree Analyses

The global evidence reviews for WASH, Wetland, coastal and marine ecosystems, and port infrastructure, have identified the most pertinent issues related to gender and climate change in these sectors. To pull together our findings and present the main causes and effects of issues from a gender perspective per sector, problem tree analyses have been created per sector. They have been prepared in line with GCF guidance and draw on and distil the main issues emerging from the global evidence reviews, and the national policy reviews for WASH. By unpacking the causes and effects, it identifies some of the central issues that CI2 investments need to address to promote gender equality, issues, which have directly informed the opportunities presented in the section 8 of this report.

7.1 WASH

The WASH global evidence review and national policy review have both highlighted the multiple ways in which women and girls are affected by inadequate WASH infrastructures and services. They have also pointed to some of the causes of this situation. Whilst national policies in focal countries commit to tackling this core problem, in addition to tackling climate change and mainstreaming gender through policy action, few countries appear to have a considered and comprehensive approach promoting gender equality through investments in the WASH sector. The problem tree analysis presented here seeks to unpack the causes and effects of inadequate WASH infrastructure and services from a gender perspective.

Figure 4: WASH Problem Tree Analysis

ROOT CAUSES	Governments have been slow to take action to mitigate effects of climate change and build resilience	A weak state-citizen social contract inhibits policy implementation and investment in essential public services	Gender norms relegate women and girls to the domestic space, undermine investment in girls' education and exclude women and girls from formal and informal decision-making fora		Women and girls' physiology	Weak public health systems
CAUSES	Climate change depletes available WASH infrastructure and resources	Insufficient investment into WASH infrastructure, maintenance and service management	Women and girls are responsible for household chores, including water collection, and caring for family members	Women and girls do not influence the design, planning and management of WASH services	Women and girls' have increased and specific WASH needs	Men and women, boys and girls have insufficient knowledge about the need to prioritise good hygiene practices
PURPOSE STATEMENT INADEQUATE WASH INFRASTRUCTURE AND SERVICES PLACE A HIGH BURDEN ON WOMEN AND GIRLS						

IMMEDIATE EFFECTS	Women and girls spend more time on household chores and family care giving Women and girls face increased risk of sexual harassment and gender-based violence Families, especially men, are unwilling to spend scarce resources on WASH services Girls, especially adolescent girls, do not attend school consistently Families, but especially women and girls, face an increased risk of ill-health and low levels of well-being, including mental well-being Available WASH infrastructure and services do not respond to women and girls' needs or priorities Women and girls are not able to attend to their physical needs
EFFECTS	Women and girls are not able to invest in empowering activities such as going to school, income generating activities, community leadership People, especially women and girls, have a poorer physical and mental health status Girls are not able to complete basic levels of education Few women have the education and skills to work in the WASH sector Gender norms and inequalities are sustained

7.2 Wetland, coastal and marine ecosystems

As regards wetland, coastal and marine ecosystems, the Gender Assessment found that women are mostly excluded from the formal sector and most profitable parts of value chains in oceans-based economies, as well as from dialogue and decision making around community-based protection and management initiatives. At the same time, and in spite of gaps in the evidence base on climate-related impacts, they are likely to be disproportionately affected by damage to ecosystems. The causes and effects of this situation are summarized in the Problem Tree Analysis below (Figure 5).

Figure 5: Wetland, coastal, and marine ecosystem Problem Tree Analysis

ROOT CAUSES	Governments have been slow to take action to mitigate effects of climate change and build resilience	Gender norms relegate women to the domestic space: limiting the amount of time they can spend outside, and distance they can stray from, the home; and constraining women's access to credit and specialised equipment		Design of community protection and management initiatives does not take into account restrictions on women's participation
CAUSES	Climate change damages wetland, coastal and marine ecosystems	Women are mostly excluded from the formal sector in oceans-based industries, and are disproportionately represented in informal and unregulated economies	Women are concentrated in less profitable parts of value chains	Low female participation in community protection and management initiatives across relevant sectors
PURPOSE STATEMENT				
WOMEN ARE MARGINALISED IN THE USE, PROTECTION AND MANAGEMENT OF WETLAND, COASTAL AND MARINE ECOSYSTEMS				
IMMEDIATE EFFECTS	There is limited evidence on the gender-differentiated effects of damage to ecosystems At the same time, rising sea temperatures, levels and acidity are all expected to disproportionately affect women's livelihoods, with women facing disproportionate and additional barriers to adaptation Women working in oceans-based industries earn less than men for carrying out the same work Women working in informal or unregulated economies are vulnerable to exploitation, including unsanitary work environments			
EFFECTS	Policy- and decision-making are gender-blind, overlooking women's needs, interests and potential roles in protection and management Women and their families have poorer economic status and food security Gender norms and inequalities are sustained or exacerbated			

7.3 Resilient Port Infrastructure

In spite of particularly limited evidence on the intersect between gender, climate and port infrastructure, the Gender Assessment likewise identified widespread exclusion of women from the development and operation of port infrastructure, particularly from employment in operations and management. The Problem Tree Analysis below summarises the causes and effects of this situation (Figure 6).

Figure 6: Resilient Port Infrastructure Problem Tree Analysis

ROOT CAUSES	Governments have been slow to take action to mitigate effects of climate change and build resilience		Gender norms dictate that port employment is overwhelmingly dominated by men		
	Operation of traditional port infrastructure damages ecosystems	Women comprise only a fraction of recruits and trainees to the port industry	Female employees are particularly underrepresented in port operations and management	There is a lack of support to the integration and progression of female employees (for example, through mentorship programmes)	Working environments are insufficient to enable retention of female recruitments (e.g. by providing separate sanitation facilities and family-friendly HR policies)
PURPOSE STATEMENT WOMEN ARE EXCLUDED FROM THE DEVELOPMENT AND OPERATION OF (CLIMATE-RESILIENT) PORT INFRASTRUCTURE					
IMMEDIATE EFFECTS	There is a lack of evidence on the gender-differentiated effects of ports' negative impacts on ecosystems, as well as of optimised infrastructure and energy efficiency Women do not benefit equally from the employment benefits of port infrastructure Women are excluded from discussions around new approaches such as no-impact port development, despite tending to be more willing to engage in sustainable or environmentally friendly practices				
EFFECTS	Policy- and decision-making are gender-blind, overlooking women's needs, interests and potential roles in the development of climate resilient port infrastructure Women have poorer economic status Gender norms and inequalities are sustained				

8 Opportunities for promoting greater gender equality

Although global evidence with a combined focus on gender, WASH and climate remains relatively sparse, our review has highlighted the significant burden placed on women and girls by inadequate access to WASH, access that volatilities associated with climate change are exacerbating. Our national policy and context analysis echoes this, pointing to the opportunity costs associated with the time women and girls spend collecting water and their increased risk to disease and gender-based violence. Distilling this evidence in a problem tree analysis highlights the need for more investment in climate resilient WASH, investment which women and girls have the opportunity to influence so that it responds to their needs and from which they should expect to benefit.

The existing evidence base on the intersect between gender, climate and, each of, the protection and management of wetlands, coastal and marine ecosystems and climate resilient port infrastructure is more scant still, but likewise suggests a lack of understanding of the gender-differentiated impacts of climate change in these sectors. The literature also points to widespread exclusion of women from participation in formal economies, as well as dialogue and decision making around relevant initiatives.

Despite this sombre picture, our policy analysis shows that national governments in the focal countries have made important commitments to continue improving access to WASH for households, as well as in health and education facilities, and to do this in a climate-smart way. Generally speaking, there is an intention to do this in a gender-responsive way, with opportunities for women to shape service design, delivery and management.

Drawing on our problem analyses, and the evidence we have reviewed, we can identify four areas of opportunity for CFM to promote gender equality through CI2 investments. These areas of opportunity are discussed in this section 8 of the gender assessment. The gender action plan in Annex 6 summarises the actions to be taken by CFM, and at Fund level, in order to mainstream gender issues in the investment lifecycle of the CI2 fund.

8.1 Opportunity Area 1

Making better quality, climate-smart WASH facilities more accessible in urban and rural communities and in schools and health facilities

Building on the progress already made to improve the population's access to WASH facilities has the potential to bring multiple direct benefits to women and girls. Such benefits include:

- Reducing the time women and girls spend collecting and cleaning water, freeing up time for other activities, such as attending school, or engaging in income generating activities.
- Reducing the incidence of illness in the family and the associated care burden that women and girls carry.
- Reducing women and girls' risk of being subjected to GBV as a result of travelling long distances to fetch water, to inadequate sanitation services, and to conflict in the household associated with expenditure on WASH services.
- Increasing women and girls' physical and mental well-being for being better able to care for themselves, including during menstruation, and for an improved sense of personal dignity.

- Development of WASH facilities in primary and secondary schools to contribute to improving girls' school attendance and their completion of basic education. This brings benefits to the individual girls concerned, but as evidence suggests, also brings it to their families and to society more generally.
- Development of WASH facilities in health centres and hospitals, not only to reduce the incidence of disease but also to help lower mortality rates.

JPWSSH data suggests that in many of the focal countries, good progress is being made in providing the population with access to an improved water source for drinking water, although there is clearly scope for improving the level of service by making drinking water available on premises and improving water quality. However, there is an urgent need to tackle the much lower levels of access to improved sanitation services and to ending the practice of open defecation.

Whilst improving physical access to WASH services is essential to benefit women and girls, maximising the benefits of this also requires financial, attitudinal and safety barriers to be addressed. This calls for infrastructure development initiatives to complement other initiatives addressing these barriers. Examples of complementary initiatives which could be considered include:

- Building family and community understanding of the health benefits of good WASH practices: this appreciation is critical if families, especially men, are to accept the need to use scarce financial resources to cover costs associated with WASH services. Such interventions could also usefully cover the importance of good hygiene practices in an effort to erode the taboos often associated with menstruation. As men often control financial decision-making in households, involving men in these conversations and garnering their support is critical.
- Cross-subsidisation approaches to make sustainable WASH facilities accessible to poorer sections of the community: in such scenarios, wealthier sections of the community would pay higher rates so that user rates for the poorest households could be reduced or eliminated.
- Adapting WASH facilities so they are physically accessible to women and girls with disabilities.

It should be noted that working with households and communities on good hygiene practices, including during menstruation, can help increase demand for WASH facilities and therefore the return on investment.

8.2 Opportunity Area 2

Giving women voice in the planning, design and management of facilities and initiatives

If WASH facilities, wetland, coastal and marine ecosystem protection and management initiatives, and climate-resilient port infrastructure are to meet the needs of women and girls, then their planning, design and management need to be done with an awareness of those needs and a sensitivity to them. As we have seen in the WASH policy analysis, most countries foresee community participation in the sector. There are, however, strong social norms which devalue and often silence women's voices, even when formal public consultation spaces are available.

A different approach to public consultation and participation is needed to truly provide space for women and girls to inform the planning, design and management of WASH facilities, wetland, coastal and marine ecosystem protection and management initiatives, and climate-resilient port infrastructure. This approach needs to first and foremost identify women and girls as a specific

stakeholder group (rather than part of ‘the community’) and put in place an engagement strategy which counters the power dynamics which often silence women and girls to allow them to be heard and understood. Such an approach might involve:

- Recognising that within the broad category of ‘women’ and ‘girls’, there are intersecting sub-groups with different needs e.g. indigenous women, women with disabilities, chronically poor women and adolescent girls, and that it is important for each of these sub-groups to be given the opportunity to be heard. Ways to reach out to and convene representatives of these sub-groups will need to be identified.
- Identifying women’s leaders to work with to convene appropriate fora to hear a range of women’s and girls’ voices about their needs and challenges. Leaders within critical sub-groups of women and girls could also be identified as a conduit for engaging the wider group.
- Working with community leaders and men’s leaders to help them understand the rationale for engaging women in the planning of facilities and initiatives, and to build their support for that, thereby avoiding any backlash.
- Holding consultation meetings at times and at venues which are convenient to the women and girls being targeted, considering how the direct and indirect costs of attending can be off-set, particularly for the chronically poor and for women and girls with disabilities.

Taking this approach to engaging with women and girls has the potential to bring real insights into their needs and experiences, but these insights will only lead to gender responsive facilities and initiatives if those employed in designing them are sensitive to the importance of benefitting women and girls, and most crucially, decision-making gives due attention to such issues. Enabling this will require those leading the design to set the tone in this regard, clearly communicating expectations and taking action to ensure those expectations are met.

If women are to influence the management of WASH facilities and oceans ecosystem protection initiatives involving community led committees, as is sometimes intended, strategic consideration of how to create the space for women’s voices to be heard will similarly be needed. The challenge is the same: how to counter social norms and power differentials which tend to silence women. The approach is likely to be context specific, but could involve:

- Identifying and recruiting to the committee women in the community who are perceived by men and other women to have legitimacy in speaking out on behalf of other women, including women from different sub-groups.
- Building the skills of the women representatives on the committee to consult with a diverse range of women and girls in the community to get their feedback on the WASH facilities or oceans ecosystem protection initiative and to represent these views in committee meetings and decision-making.
- Building male committee members’ understanding of the need for management to be informed by women and girls’ experiences and developing the skills to ensure women’s voices inform committee decision-making.

8.3 Opportunity Area 3

Providing women with formal employment opportunities

Low levels of secondary education completion amongst girls in several of the focal countries can make the formal employment of women in the WASH sector challenging. There is, however, positive experience emerging in South and South East Asia of women being successfully employed in WASH enterprises e.g. as sanitation sales agents, in meter reading and in rainwater harvesting tank construction.

Opportunities for promoting women's employment in low-skill positions in the WASH sector warrant exploration in the design of CI2 supported projects. This strategy is not well documented in available literature and further investigation into available experience would be beneficial. Understanding how employers successfully engaged these women and how men's concerns about women's independence, mobility and safety were allayed are important aspects to explore.

Challenges in employing women in the technical design and development of WASH facilities are particularly acute, with much smaller numbers of women going into tertiary education and even fewer focusing on STEM subjects such as engineering, although data on this is hard to find. These challenges will not be resolved quickly and will require medium to long-term support to encourage women to remain in education and to pursue careers in STEM. Possible support might involve scholarships for women to study STEM subjects in demand in WASH and work experience placements for women STEM students to gain practical experience in the workplace.

The literature review suggests that if cleaners or caretakers are not in post to maintain WASH facilities in schools and health centres, this work can often fall to female staff members. Any efforts to improve WASH facilities in schools and health centres should therefore include a resourced plan for the on-going cleaning and maintenance of the facilities.

As regards the protection and management of wetlands, coastal and marine ecosystems, relevant livelihoods initiatives for women have tended to accommodate, but failed to address, social norms which perpetuate gendered inequalities within industries and value chains. Initiatives to support sustainable practices in oceans-based industries should therefore not only take into account such restrictions – on women's time of mobility, for instance – but actively seek to tackle discriminatory norms which limit women's livelihoods opportunities.

The development of new climate-resilient ports offers opportunities to recruit women as employees. Attracting women recruits, and supporting their meaningful integration and progression means providing a combination of initiatives to build their skills and confidence, and provide a supportive work environment. These include training incentives, mentorship programmes to connect women to those in senior level roles, as well as family-friendly HR policies and safe (separate) sanitation facilities for women. Initiatives should include a focus on areas from which women tend to be most excluded, most notably port operations and management.

8.4 Opportunity Area 4

Understanding the gender equality benefits of gender-responsive facilities and initiatives

Our review has suggested that there is a limited body of experience of interventions working at the nexus between gender, climate change and each of WASH, protection and management of wetlands, coastal and marine ecosystems, and port infrastructure. There is then an enormous opportunity for CFM to contribute to building the evidence base on how gender equality can be promoted through, climate-smart WASH, ocean ecosystems protection initiatives and port infrastructure, using CI2 as the vehicle.

If CFM wishes to capitalise on this opportunity, it will need to operationalise robust monitoring, evaluation and learning frameworks for its new investments. This would involve:

- Having clear objectives for how gender equality is to be promoted through the investment and the intended outcomes.
- Specifying indicators with which to measure changes in gender equality and the achievement of intended outcomes.
- A methodology, which ideally makes use of quantitative and qualitative approaches to measuring change, and which collects data at different time points to understand the pace of change and offer opportunities to adapt interventions. Of particular interest would be conducting an impact assessment post project completion, when communities have had some time to enjoy the benefits of the investment.
- A skilled team to design and operationalise the monitoring, evaluation and learning, including team members with gender expertise.
- Financial resources to cover the cost of operationalising a robust monitoring, evaluation and learning framework.

Such an investment would contribute to CFM's internal learning, but if results could be published, they would be a public good, offering others working in the gender, climate change, WASH, ocean ecosystems and port infrastructure spaces the opportunity to learn from CFM's experience.

It is worth noting that data captured on initiatives to protect ocean ecosystems should include relevant informal economies (where women tend to be concentrated) as well as specific parts of value chains women in which they are represented disproportionately.

9 Understanding the results and impact of gender mainstreaming

Monitoring and reporting will be undertaken in accordance with the principles set out in CFM's Environmental and Social Management System (ESMS). CFM will monitor and evaluate the actions taken to promote gender equality, their results and impact in two ways. Firstly, by monitoring and evaluating key impact indicators at the Fund level. Secondly, by implementing monitoring plans for individual projects.

Fund Level Monitoring and Evaluation

The KPIs established for individual programmes will supplement the key impact indicators that have been adopted for the CI2 funds and against which CI2 is committed to monitor performance in all projects. These indicators are linked to selected UN Sustainable Development Goals (SDGs). The impact indicators that are relevant to gender are shown in Table 5 below.

Table 5 CI2 Impact Indicators

UN SDG	Gender-Relevant CI Two Impact Indicators
SDG 5 - Gender Equality	• Number of full-time, equivalent female employees • Number of females in management positions
SDG 6 - Clean Water and Sanitation	• Total number of people (gender disaggregated) with improved access to clean drinking water and sanitation
SDG 8 – Decent Work and Economic Growth	• Number of new direct jobs created; gender disaggregated • Catalytic effect (EUR million) (total DFI and commercial financing)

Project level monitoring and evaluation

Project companies will implement a monitoring and evaluation plan, which sets out:

- The indicators to be used to monitor how women and girls, as well as other priority social groups e.g., indigenous women, women from chronically poor households and women with disabilities, are being affected - both positively and negatively - by project interventions, and the impact these projects are having on their lives.
- The methodology for gathering and analysing monitoring data, as well as reporting and using the findings.

Each monitoring and evaluation plan will be bespoke to individual projects. However, they will contain comparable quantitative and qualitative indicators and data collection methodologies where appropriate. Depending on a project's life cycle, monitoring data will be collected where possible at regular intervals to understand progress during project implementation. The aim of project monitoring will be to provide timely information to make project adjustments and address any negative effects.

Relevant components of the monitoring and evaluation plan may be incorporated into contracts for EPCs or other sub-contractors, with reviews of the plan's implementation forming part of the contractor's normal review process.

There is a need for more evidence on effective approaches to promoting gender equality through WASH investments. In response to this, CFM may publish some of the results and learnings from its CI2 investments.

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Annex 3: Stakeholder mapping interview guide

Climate Investor 2 Gender Assessment Stakeholder Engagement Interview Guide

Introduction

Interviewer introduces her/himself and asks respondent to introduce her/himself. Note down:

- Name
- Organisation
- Role

We are working with Climate Fund Managers and the Green Climate Fund on a project proposal to leverage private sector investment into the development of climate resilient water and sanitation infrastructure. [Country X] is one of the countries where funding will be made available to leverage large scale private sector investment. As part of the proposal development process, we are seeking to understand how future private sector investments in WASH infrastructure development can help promote gender equality, for example, by improving access to quality water and sanitation services, enabling women's participation in the design and management of water and sanitation services and enabling women to benefit from the infrastructure development, including through formal and informal employment opportunities. We are keen to learn from the experience of experts in this field and would like to ask you a few questions. Your feedback will be anonymous. It will be used to inform the proposal to be submitted to the Green Climate Fund for financial support and the design of private sector investment leveraged through Climate Fund Managers. The interview will take approximately 45 minutes. Are you happy to proceed?

Interview Questions

1. In [Country X], what work is being done to promote gender equality in water and sanitation infrastructure development? Explore:
 - a. Expansion of WASH services to reduce domestic and health burden on women and girls;
 - b. Involvement of women in the design and management of water and sanitation services;
 - c. Employment of women in the development of water and sanitation infrastructure and services?
2. Which organisations are involved in this work, and what are their roles?
3. From your perspective, which organisations are most involved in promoting gender equality in water and sanitation infrastructure and services? What do they do to promote gender equality in water and sanitation infrastructure and service development? At what level(s) do they operate (national, sub-national, community)?
4. The assessment we are in the process of conducting identified four opportunities for promoting gender equality in water and sanitation infrastructure and services, which we shared with you in advance of this interview. We would like to discuss two of them in more detail with you.



- a. The first opportunity we'd like to discuss with you is giving women voice in the planning, design and management of WASH facilities. How do you think the project companies leading the infrastructure design and development can do this? Who do they need to work with and what do they need to do to ensure women have a voice in the planning, design and management of WASH facilities?
 - b. The second opportunity we'd like to discuss with you is providing women with formal employment opportunities in the WASH sector. What do you think private sector investors and project management companies can do to provide women with formal employment opportunities in the WASH sector?
5. Before we close, are there any recommendations you would like to give us to consider as we develop the proposal for private sector investment in WASH infrastructure?

Thank the respondent for their time and helpful feedback.

Annex 4: Stakeholders interviewed

Type of contact (Select)	Organisation
Uganda	
Government	MoW & E
Government	MoW & E
Government	MoW & E
Other	UNICEF -Uganda
Other	AfDB
Civil Society	UWASNET
Civil Society	KWDT
Civil Society	Katosi Women
Kenya	
Government	PMU Coordinator
Civil Society	KEWASNET
Civil Society	Centre for Social Planning and Administrative Development (CESPAD)
Other	Kenya Water for Health Organization (KWAHO)
Civil Society	SIMAVI
Civil Society	SNV
Government	Ministry of Health
Bangladesh	
Civil Society	WASH Alliance/SIMAVI
Civil Society	Independent
Civil Society	Care Bangladesh
Namibia	
Other	UNICEF - Namibia
Other	Independent
Nigeria	
Civil Society	Women Empowerment Initiative in Nigeria (WEIN)
Civil Society	Women Development Association for Self Sustenance
Civil Society	UNICEF-Nigeria
Djibouti	
Civil Society	Ecologie de Village Association (EVA)
Civil Society	Association for future and development of Obok(AADO)
Civil Society	
Civil Society	Union Nationale des Femmes Djiboutiennes (UNFD)
Ivory Coast	
Government	Ministry of Hydraulics
Civil Society	Génération Femme du 3ième Millénaire (GFM3)

Annex 5: Full Country Analyses

9.1 Bangladesh

Water, sanitation and hygiene in Bangladesh at a glance

Access to Drinking Water: 98.9% of the population has access to an improved source for drinking water, of which 58.5% has access to safely managed drinking water and 39.2% has access to basic drinking water.¹⁴³ Women are the main collectors of water¹⁴⁴ with the majority of women (76.6%) spending about 30 minutes to collect water.¹⁴⁵

Access to Sanitation: 78% of the population has access to an improved facility, of which 38.7% has access to safely managed sanitation services, 15.5% has access to basic sanitation services and 23.8% has access to limited sanitation services. None of the population is reported to practice open defecation. The remaining proportion of the population uses unimproved sanitation services.¹⁴⁶

Hygiene: 58.4% of the population has access to handwashing facilities with soap and water.¹⁴⁷

WASH in schools: Nationally, 82.2% of schools have access to basic service for drinking water, 56.3% of schools have access to basic service for sanitation and 50.7% of schools have access to basic service for hygiene.¹⁴⁸

WASH in health facilities: 64.4% of health care facilities have basic service for drinking water, 31.3% have basic service for sanitation and 37.6% have basic service for hygiene.¹⁴⁹

Policy context

Improving water, sanitation and hygiene is a key part of Bangladesh's national policy framework, as identified in the National Sustainable Development Strategy (2010). Recognising the increasing impacts of climate change, rapid urbanisation and water contamination, including salinity intrusion in fresh water and severe reduction on available groundwater, the National Sustainable Development Strategy addresses water, sanitation and hygiene under strategic priorities 7 on environment, natural resources and disaster management, and strategic priority 5 on the urban environment. To address the challenges faced, the policy aims to:

- Diversify water supply, with a shift to surface water and encourage rainwater harvesting;
- Encourage water recycling.
- Expand the use of technology to improve the efficient use of scarce resources, including water.
- Enhance investment in safe drinking water.

¹⁴³ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁴⁴ MICS (2019)

¹⁴⁵ MICS (2019)

¹⁴⁶ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁴⁷ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁴⁸ Joint Programme for Water Supply, Sanitation and Hygiene (2019)

¹⁴⁹ Joint Programme for Water Supply, Sanitation and Hygiene (2019)

Two other policies, the National Strategy for Water Supply and Sanitation (NSWSS) of 2014 and the National Hygiene Promotion Strategy for Water Supply and Sanitation Sector (NHPSWSS) of 2012, flesh out further policy intentions in the sector. The NSWSS identifies challenges around climate change adaption, rapid urbanisation, need for improved water resource and sanitation (both solid waste and faecal sludge) management and continued advancement up the sanitation ladder. It lays out 17 strategic objectives to combat these challenges, including ensuring safe drinking water, undertaking specific approaches for vulnerable people, gender mainstreaming, improving hygiene promotion, adapting to climate change and undertaking integrated and accountable development approaches.

The NHPSWSS seeks to promote the sustainable use of improved water supply and sanitation infrastructures, ensuring the creation of an environment for hygiene promotion and practice to reduce water and sanitation related diseases. It recognises challenges associated with increases in vulnerable water sources from weather extremities, including droughts and saline intrusion; continued open defecation in wetlands leading to a rise in water-borne illnesses; and continued, inequitable access to safe water and sanitation facilities, especially for women and people with disabilities. The policy aspires to overcome these challenges through 10 strategic objectives, including: exploring and replicating successful hygiene promotion models, gender mainstreaming, ensuring the potential use of the social organisations, ensuring hygiene promotion for hard-to-reach areas.

Strategy 10 seeks to include gender mainstreaming in all the service planning, implementation and monitoring by shifting organisational culture and ways of thinking and by developing procedures relating to gender responsibility, accountability, coordination and personnel policy and the consideration of women's role in hygiene promotion, both as a token role (such as gender advisor or officer) and also as an empowered advocate.

Improving hygiene awareness in schools is addressed in the National Education policy. This includes hand washing and installing modernized toilets for both girls and boys to decrease the spread of water-borne and vector-borne diseases.

Bangladesh's location makes it highly susceptible to the adverse effects of climate change, hindering its development, including access to safe and economic opportunities for women and other vulnerable groups. The Bangladesh Climate Change and Action Plan (BCCSAP) of 2008 looks further into the negative impacts caused by increased weather extremities, with climate-related impacts leading to a decrease in water availability and an increase in water contamination from rapid urbanisation, saline intrusion, poor sewerage management and riverbank erosion. It also looks at the impact this has on women and children, especially since women are responsible for water management in their families, bear the burden of collecting water from further distances due to degradation of water and are vulnerable to health risks associated with drinking saline water. It seeks to tackle these challenges through action under 6 pillars:

1. Food security, social protection and health;
2. Comprehensive disaster management;
3. Infrastructure;
4. Research and knowledge management;
5. Mitigation and low-carbon development;
6. Capacity building and institutional strengthening.

Pillar 1 places an importance on women and girls by increasing community involvement in planning and increasing resilience of vulnerable groups, including women and children, through development of community-level adaptation, livelihood diversification and better access to basic services and social protection. Programme T1P9 also seeks to complete a comprehensive study on the impact of climate change on women and gender relations and to develop recommendations to address these in all actions under the BCCSAP.

None of the policies reviewed explicitly considered advancing women's formal roles in the WASH sector, but many discussed the need for community involvement in the planning, implementation, operation and management of water and sanitation facilities. Alongside this, some policies also consider the advancement of women's involvement in entrepreneurial activities, with a focus on the non-farming sector in rural areas; training and use of technologies suitable for women of various socioeconomic status; creating initiatives for entrepreneurs for proper waste treatment before industrial waste disposal; and gender mainstreaming in all aspects of water, sanitation and hygiene supply chain, including WASH related committees and employment. Within the National Women Development Policy, there is a broader emphasis, within all sectors, on career development opportunities for women employees and equal participation of women in decision-making and governance at all levels. Additionally, due to only 21.1% of women completing upper secondary schooling¹⁵⁰ and 19.8% enrolment in tertiary education,¹⁵¹ women are limited in the careers prospects within the WASH sector, and a focus on policies is placed on opportunities in low-skilled jobs.

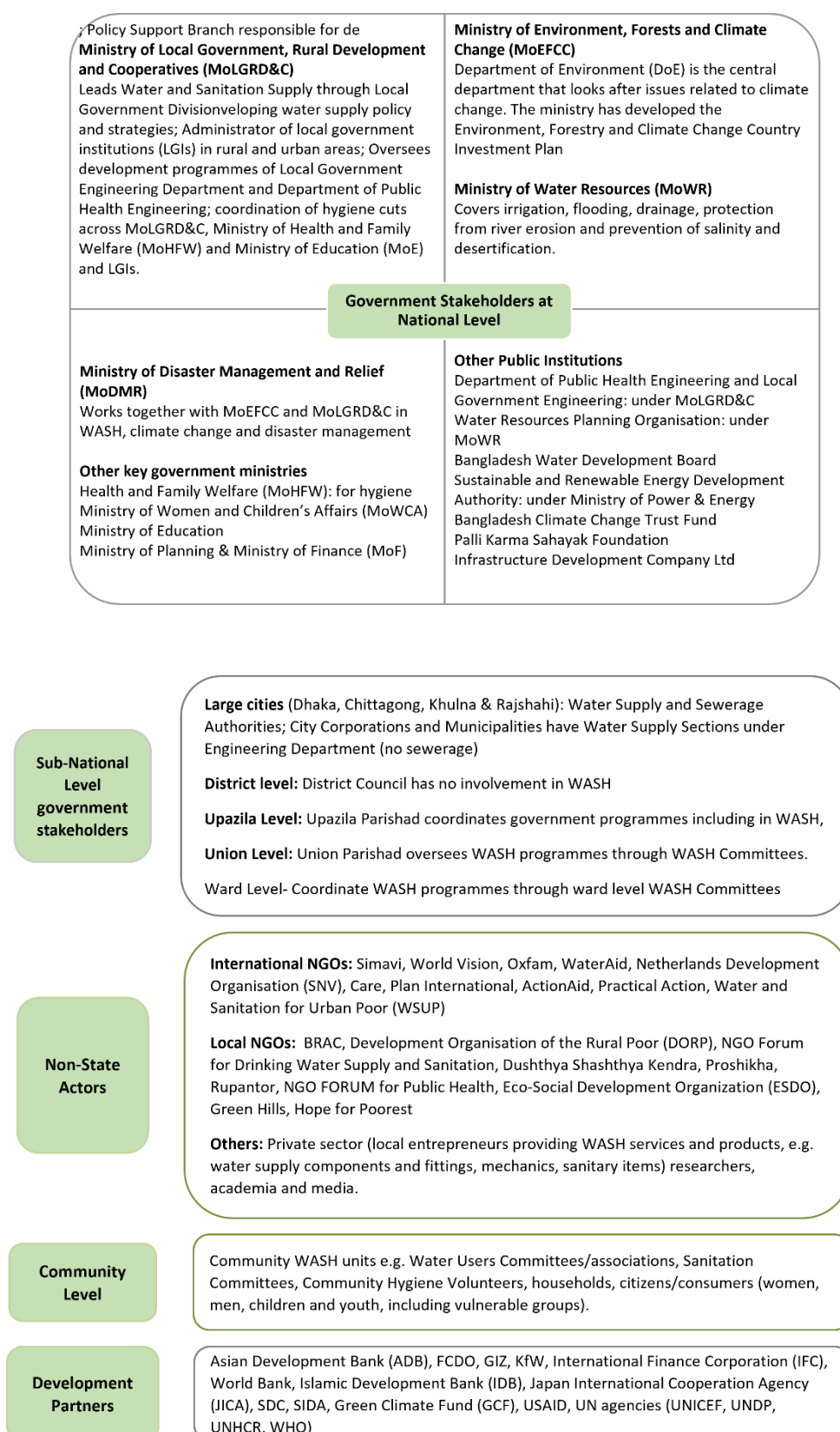
Stakeholder Mapping

In Bangladesh, the Policy Support branch of the Ministry of Local Government, Rural Development and Cooperatives has primary responsibility for Water and Sanitation Supply and Strategy. Other national Ministries with some WASH responsibility include the Ministry of Health and Family Welfare, for hygiene, and Ministry of Water Resources (MoWR), covering irrigation, flooding, and climate-related water protection. There are a range of agencies involved in the delivery of WASH services at the sub-national levels, including government agencies at the city, district, upazila, union, and ward level, as well as the private sector, national and international CSOs, and community-based committees. A wide range of development partners are involved in supporting government, private sector and CSO engagement in WASH. Figure A provides a visual mapping of the actors involved.

¹⁵⁰ UNIS (2019)

¹⁵¹ UNIS (2020)

Figure A: WASH stakeholders in Bangladesh



9.2 Botswana

Water, sanitation and hygiene in Botswana at a glance

Access to drinking water: 92.2% of the population has access to a basic level of drinking water and 4.7% of the population has access to a limited level of drinking water. 1.7% of the population relies on unimproved drinking water whilst 1.3% of the population has access only to surface drinking water. There is no data available for the proportion of the population with access to safely managed drinking water.¹⁵²

Access to sanitation: 80% of the population has access to basic sanitation services and 5.6% of the population has access to a limited level of sanitation services. 4.6% of the population relies on unimproved sanitation services whilst 9.8% of the population practices open defecation in sanitation services. There is no data available on the proportion of the population with access to safely managed sanitation services.¹⁵³

Data on hygiene, WASH in schools and WASH in health facilities is not available.

Policy context

Botswana's 11th National Development Plan covers the period April 2017 – March 2023. It was adopted in December 2016 and followed the country's celebration of the fifty years anniversary of Independence and the launching of Botswana's Vision 2036. The Vision 2036 provided an opportunity for the country to define its roadmap for the next twenty years. With an all-encompassing theme of "Achieving Prosperity for All", the Vision provides an inclusive development path to be achieved through the expansion of the domestic economy while empowering Botswanans to meaningfully participate in the development of their country. Achieving the goals and objectives of Vision 2036 will require a refocus of Botswana's current development model towards an export-oriented, labour-intensive, and private sector driven model.

Chapter two of the 11th National Development Plan looks at Botswana's population situation and development prospects, with sub-sections on the key development challenges facing Botswana's population. This includes a subsection on population and water resources and a subsection on migration, population distribution and urbanisation. In these subsections, some of Botswana's key WASH sector challenges are outlined. These challenges include water shortages and droughts in the water sector, and the inadequate capacity of sanitation infrastructure, and pressure on sanitation services because of increasing levels of urbanisation.

The plan outlines the negative health consequences of poor sanitation and waste management issues such as pollution, littering and illegal dumping. It also outlines the negative health consequences of water shortages and droughts, which cause food insecurity leading to malnutrition. The plan states

¹⁵² Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁵³ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

that the Government will continue with measures to address environmental health risks associated with poor sanitation, hygiene and waste management, and pollution throughout the value chain from the source of environmental contaminants to the affected populations. To combat water scarcity and its impact on the agricultural sector leading to food insecurity and malnutrition, the plan states that considerations will be made to adopt 'smart agriculture'. This approach will help Botswana to transform its agricultural systems towards more productive, efficient, resilient and sustainable systems.

The plan largely ignores specific gender considerations in the water and sanitation sectors. The plan does however state that the Government recognises the need for equal participation and representation of women and men in all development processes, and that NDP 11 provides an opportunity for all sectors to articulate gender issues that they will address in the execution of their respective mandates.

Although the policy largely ignores gender issues related to employment in the water and sanitation sectors, NDP11 does discuss the low participation of females in Science, Technology, Engineering and Mathematics (STEM) subjects as a key challenge facing the education sector, despite women having an upper secondary completion rate of 66.3% (compared to 62.6% for men) (UNIS), and a tertiary enrolment rate of 31.95% (compared to 26.09% for men) (UNIS). The policy fails to articulate how this challenge impacts women's employment in the water and sanitation sectors.

While the National Development Plan discusses the water and sanitation sectors, it does so in general terms within the broader context of Botswana's development as a whole. Botswana's National Water Policy however discusses the sectors in specific detail and provides a national framework that will facilitate access to water of suitable quality and standards for the citizenry and provide the foundations for sustainable development of water resources in support of economic growth, diversification and poverty eradication. The policy therefore aims to provide a strategic framework for the following goals:

- Protection, conservation and restoration of the nation's water resources
- Promotion of effective, sustainable management of water resources
- Promotion of the equitable and efficient use of water resources
- Reduction of the subsidies associated with water supply
- Assurance of access and affordability of water for all
- Protection and restoration of the environment
- Promotion of productive uses of water

Like the National Development Plan, the National Water Policy (2012) identifies water shortages and droughts and water supply for agriculture and industry as key challenges facing the sector. The policy also raises water use efficiency and the protection of water resources as important issues. The policy offers policy statements and objectives in a variety of thematic areas across the sector including Water for Growth, Water Demand Management and Conservation, Domestic Water Supply and Sanitation, Water for Environment and Tourism, Water for Agriculture, Water for Mining and Industry, Water for Energy, Information Management and Dissemination, Research and Development, International Cooperation and Monitoring & Evaluation.

However, also like the National Development Plan, women's roles in the water and sanitation sectors are largely ignored. The policy does state that gender and social equity in accessing water resources will be ensured and, in particular, women shall be empowered to participate fully in issues and

decisions relating to sustainable development and management of water resources but does not go into specifics on how this will be achieved.

As the National Water Policy offers a strategic policy framework for Botswana's water and sanitation sectors, the National Health Policy (2011) offers the same for the health sector. The National Health Policy offers valuable insight into the consequences of the challenges facing the water and sanitation sectors on public health. The National Health Policy discusses the impact of water shortages and their potential for causing malnutrition, and also identifies the negative public health consequences of a lack of access to adequate sanitation facilities and access to safe drinking water.

The National Health Policy also discusses the interplay between climate change, the water sector and public health. The policy discusses water scarcity and fluctuating rainfall patterns caused by climate change, asserting that the prevalence of malaria is expected to increase as temperatures rise and conditions are more conducive for mosquitoes to breed during rainy seasons. Cholera is also likely to increase during the same period. With a reduction in rainfall, food production is likely to decrease and may lead to malnutrition in the longer term. The policy does not however discuss the gender dynamics of these issues.

Botswana's draft Climate Change Response Policy (2016) and the National Climate Change Strategy for Botswana (2018) continue to look at the effects of climate change on the water and sanitation sectors. The draft Climate Change Response Policy is a statement of intent and vision of what Botswana aims to achieve on climate change, and is complemented by the National Climate Change Strategy, which aims to help operationalise the tenets of the draft policy and provide necessary guidance on how the policy objectives will be achieved.

The draft Climate Change Response Policy discusses several issues related to the water and sanitation sectors, including water availability, exacerbated water stress leading to food insecurity and water and sanitation management strategies. Like the National Health Policy, the draft Climate Change Response Policy also discusses how climate change is likely to negatively affect human health directly through increased temperatures, drought and floods, and also through the spread of water borne, water-related and vector borne diseases.

The policy recognises the need to mainstream gender considerations into development planning to ensure that climate change response measures are gender sensitive. It states that this shall be achieved through empowering women and youth to actively participate in the implementation of climate response measures, adopting strategies that are targeted at increasing women and youth's resilience to climate change, and including gender and climate change into academic curriculum at all levels.

The National Climate Change Strategy for Botswana helps to operationalise the draft Climate Change Response Policy's discussion of the water and sanitation sectors through the below policy goals to be realised for the sectors:

- Utilisation of shared water courses for the benefit of Botswana
- Integrating climate change response measures in the water planning processes across all economic sectors
- Consideration of defining potential water aquifers and adopting appropriate measures of protection for water security and sustainability

- Promotion of rainwater harvesting, water re-use, and recycling for domestic, agricultural, industrial, and commercial purposes
- Promotion of integrated watering systems for livestock, particularly in rural areas
- Employing accounting and valuation tools to support water management decision systems

These specific policy goals for the water and sanitation sectors feed into the broader policy objectives of the strategy, which are:

- To ensure climate change adaptation and mitigation action supports the achievement of Botswana's Vision 2030 of prosperity for all.
- To ensure climate change adaptation and mitigation action supports the achievement of current and future Botswana's National Development Plans (NDPs).
- To ensure climate change adaptation and mitigation action supports the achievement of the global Sustainable Development Goals (SDGs) in Botswana.
- To ensure climate change adaptation and mitigation action supports the achievement of Botswana's NDC, including the GHG emissions reduction target of 15% by 2030
- To ensure Botswana has a clear strategic agenda on climate change adaptation and mitigation actions, allowing for a spectrum of actors to be involved in implementation.

The National Climate Change Strategy devotes special attention to gender equality and ensuring that it serves the needs of women. There is a much more thorough review of gender as a cross-cutting issue than in the draft Climate Change Response Policy, and actionable Gender Based Strategies with strategic interventions including:

- Develop a national Climate Change Gender Action Plan (CC GAP), with an emphasis on women as drivers of climate resilience, and women's role in water and energy and healthcare provision in households.
- Promote equitable participation of women farmers and female-headed households in Climate Smart Agriculture (CSA) programmes, agritourism, and access to conservation agriculture technologies.
- Ensure equitable gender access to the proposed endowment fund providing low-cost finance to climate change adaptation projects, aligned to existing similar projects.
- Ensure that women's voices are included in natural resources management through their equitable participation in CBNRM processes.
- Ensure that gendered differences of climate change are mainstreamed into climate change education.
- Ensure the full participation of women and female-headed households in disaster management public gatherings, to address both the higher vulnerability of women and children, and to plan for the higher post-disaster burden placed on women due to their dual roles as producers and carers.

9.3 Brazil

Water, sanitation and hygiene in Ivory Coast at a glance

Access to drinking water: 99.3% of the population has access to an improved water source for drinking water, of which 85.8% has access to safely managed drinking water, 13.5% has access to

a basic level of drinking water, and 0.1% has access to a limited level of drinking water. The remainder of the population uses unimproved sources for drinking water¹.

Access to sanitation: 90.1% of the population has access to improved sanitation services, of which 48.7% has access to safely managed sanitation service, 41.4% has access to a basic level of sanitation service, whilst 0.1% has access to a limited level of sanitation service. 9.8% of the population uses unimproved sanitation services, including 0.3% who practice open defecation².

Data on hygiene is not available.

WASH in health facilities: Nationally, 80.2% had a basic service in drinking water, data on access to hygiene and sanitation is insufficient³.

WASH in schools: Nationally, 61% has a basic service in hygiene, data on access to drinking water and sanitation is insufficient⁴.

Policy context

The National Policy for Water Resources (2022) and the National Policy for Basic Sanitation (2019) both discuss in-depth the importance of WASH in daily life and lay out clear objectives to ensure the impact on human livelihoods and activities, such as business, is limited. The National Policy for Water Resources reviews the lack of subterranean water resource development, and water use in agriculture, tourism, industry, and domestic use. It examines the negative impacts that water scarcity creates in vulnerable communities, including increase in conflict, siltation of water, water pollution, and changes in natural characteristics from high degradation. It adds the impact of climate change leaders to further alterations to available water resources and impacts both the energy and agricultural sector. It seeks to increase environmental education and capacity building in technology and science for water resource management, establish a formal and sustainable water management system, utilize knowledge and experience of indigenous people and communities, promote post-graduate courses in water resources and environmentalism, support the increase in knowledge of ICT in climate change, and encourage the involvement of the population in identifying problems and finding solutions. It also seeks to incorporate gender equality in all actions and develop ongoing training that promotes female leadership and empowerment in the management of water resources but does not provide any detail on practical implications of this intention.

The National Policy of Basic Sanitation identifies issues with parts of the population continuing to lack access to basic sanitation facilities and/or proper disposal of solid waste as well as a burden on the current resources that are available, to not keep up the demand in solid waste disposal. Due to this there continues to be increases in health risks and the proliferation of parasites and infectious diseases caused by poor or inadequate sanitation. The policy lays out clear objectives and programmes to overcome these issues, such as increasing infrastructure, management, and monitoring of sanitation facilities, improve infrastructure to capture and store water resources, and create access to water in non-conventional ways such as the use of cisterns. It also sees a value in the utilization of the circular economy for solid waste disposal. It does not explicitly mention gender in the policy objectives.

The National Environmental Policy (1981, with several amendments, with the most recent being issued in 2013) and the National Policy for Climate Change (2010) elaborates on the importance of mitigating and adapting policies, programmes, and actions to prevent the impact on climate change for humans and economic activity but remains gender blind in both the challenges and objectives of the policy. The National Environmental Policy and its associated updates consider the degradation of sanitary conditions and the adverse effects that climate change has on the biota. It seeks to overcome

these challenges through rationing the use of water, expanding the use of environmental licensing for activities that utilise environmental resources, increase education on climate change and the environment, and incentivise studies and research related to technologies to use for the rationing and protection of the environment. The National Policy on Climate Change discusses the impacts extreme weather has on the health and livelihoods of people, which includes the effect it has on water reserves in the country. It reviews the idea of reusing water in urban planning and calls for action to protect water resources in the National Policy for Water Resources and National Policy for Basic Sanitation. It places civil responsibility on reducing the impact to climate change, as a key manner to achieve climate change actions, such as the reduction of green house gases (GHG).

The National Plan of Politics for Women (2013) explores the importance of including a gender lens in the sustainable use and handling of water, promotion of public policies to defend water as a public good, ensure access to water by building cisterns, ensuring agricultural security, and utilization of environmental technologies. It emphasises the importance of valuing women's contribution to farm work and equal participation in leadership and decision making. Finally, the Plan seeks to amplify access for women in the labour market, including within the agricultural sector and formal education in science and technology.

Though neither the National Education Plan (2014) nor the National Policy for Health (2017) discuss WASH specific issues or objectives, they both outline important factors related to gender equality and access to basic services. In Brazil 67.32% of the population has completed Upper Secondary school, of which 72.73% are female and 62.18% are male.⁵ While only 55.14% of the population is enrolled in tertiary education, a larger proportion are female, 64.35% compared to 46.26% males.⁶ The National Education Plan seeks to reduce discrimination and inequality, of all forms while also promoting socio-environmentalism in the education system. The National Policy for Health seeks to create equal access to healthcare and teach on the intersection of health, sustainable development, and the environment.

Several of the policies reviewed touch on employment opportunities e.g., the National Sanitation Policy promotion of employment within the sector, and the National Water Resource Policy promoting education in the ICT sector, but none of them explicitly consider employment opportunities for women in WASH related sectors. Brazil has a high percentage of entrepreneurship activity, with about 21% of the population in early stages of entrepreneurial activity, and 10% in established businesses⁷, and 8 to 10 ratio of female to male entrepreneurs⁸ in the country. The National Plan of Politics for Women (2013) wishes to promote economic and financial autonomy of women seeking entrepreneurial activities.

As a federal state, Brazil's lower tiers of government play an important role in the provision of sustainable WASH services, and most federal policies are created with broad objectives leaving each regional government responsible to expand on the objectives based on their needs. The first National Policy for Regional Development (2007) placed greater importance on climate resilience measures and gender equality, the current National Policy for Regional Development II (2019) emphasises achieving equality, within regions and between regions, strengthening science, technology, and innovation sectors, and climate action. The policy outlines 8 main principles to focus on in regional development, including sustainable development and the valorization of the environment. It also seeks to ensure equal access to economic opportunities within all the regions, both from a socio-economic and gender perspective. However, there is no specific discussion of WASH-related challenges and objectives in the policy, nor are there are gender specific strategies within outlined climate action measures.

9.4 Djibouti

Water, sanitation and hygiene in Djibouti at a glance

Access to drinking water: 90.8% of the population has access to an improved source of drinking water, although for the majority (76%), this is basic access.¹⁵⁴ None of the population has access to drinking water classified as safely managed. The remaining 9.2% of the population has access to either unimproved drinking water or surface water only.¹⁵⁵ Women are the main collectors of water.¹⁵⁶ 88.6% of the population has improved water available on the premises, while for 3%, it takes 30 minutes or less to get to an improved drinking source.¹⁵⁷

Access to sanitation: 73.8% of the population has access to improved sanitation services, of which 37.1% has access to safely managed sanitation services, 29.6% has access to basic sanitation services and 7.1% has access to limited sanitation services. 10% of the population uses unimproved sanitation services whilst 16.2% practices open defecation.¹⁵⁸

Data on hygiene, WASH in schools and WASH in health facilities is not available.

Policy context

After several decades of implementing various developmental policies that focused on the short and medium term, the Djibouti 2035 Vision (2010–2035) presents a long-term developmental vision for sustained growth. The vision has a strong focus on economic growth and private sector development, and water, sanitation and hygiene, as well as other social development factors, remain a secondary goal to support the achievement of this vision. Water, sanitation and hygiene remain a strong area of focus within the 2035 vision, specifically given the climate challenges Djibouti is starting to face. The vision sets out an overall strategy within which various ministries are tasked with setting up a more detailed action. The vision focuses on the following policy objectives for water, sanitation and hygiene:

- Providing good sanitation for the urban environments to have clean cities.
- Identifying more water sources and improving access to existing water sources.
- Modernization of infrastructure (sanitation, water access and port development) and increased quality of education.
- Simplifying the irrigation systems to ration water use.

The vision focuses primarily on water supply throughout the country and its implications for other areas of the primary sector, with a lesser focus on sanitation and hygiene. This is complemented by the climate section of the vision, which includes sustainable water management as an area of priority, amongst other strategy objectives unrelated to WASH. Furthermore, Vision 2035 identifies the link between climate change and WASH, specifically in terms of how they will experience increased instances of droughts and flooding which will impact food production and supplies throughout the country.

The 2010 Development Plan for the Primary Sector and the 2015 Strategy of Accelerated Growth and Promotion of Employment echo the focus on access to water for agriculture and livestock, as well as

¹⁵⁴ A basic level of access to drinking water means that the water is from an improved source and that the time for collection is not more than 30 mins for a roundtrip including queuing, Joint Programme for Water Supply, Sanitation and Hygiene .

¹⁵⁵ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁵⁶ MICS (2006)

¹⁵⁷ MICS (2006)

¹⁵⁸ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

the hygiene risks that a lack of sanitation infrastructure poses in cities. The main consequences identified in these policies are the spread of disease and further contamination of water sources and structural water distress for the population of Djibouti. Although the latter two policies touch on gender and the need to address gender inequity throughout the primary sector and for growth, they do not directly link WASH issues to gender. The problems identified, such as health consequences and hard to reach water points, are known to generally affect women more, but this is not specified in the policies. They do acknowledge that various stakeholders, including women and youth, should have a larger role in the policy-making process.

Beyond water and food shortages, Vision 2035 also identifies sanitation and hygiene issues related to water insecurity. Liquid and solid waste management is identified as one of the main WASH challenges, identifying that few households in Djibouti are connected through a collective waste management system. This contributes to the deteriorating health of the population. Although not directly linked, the strategy also identifies the disproportionate effects of poverty on women, including the provision of health care as evidenced by the growing impacts of the AIDS epidemic on women and children. Health consequences of open sewage and poor water management prove more of an obstacle for women, while water provision to health infrastructure (e.g., hospitals) also presents a challenge as women are the main users of primary health centres. The vision proposes to provide good sanitation facilities and modernise water and sanitation infrastructure, with no real specifics on how they will achieve this. They also emphasise the importance of identifying more water sources and increasing access to these, also by simplifying irrigation systems to ration water use. Lastly, they mention that, alongside infrastructure development, comes increased quality of education.

Education is a strong focus of Djibouti's 2011–2021 Gender Policy, the policy that most identifies how water and sanitation challenges are an extra burden on women. The policy covers accessibility to water and sanitation facilities in their communities, remoteness of water points and sanitation deficits in their immediate environments. It identifies the woman's role as mainly within the domestic sphere, meaning they carry the burden of walking to water points or suffer the consequences most severely when there are shortages within the household. Moreover, women disproportionately suffer the effects of desertification, caused by changes in the climate, in accessing water. The gender policy primarily advocates for education on hygiene practices, such as washing hands with soap, ways to preserve quality of drinking water at home and proper use of latrines in schools and in communities. The Gender Policy also most clearly outlines the informal role that women play in the household, and consequently, the informal role they might be able to play in reinforcing new sanitation practices in their household. This informal role can be supported through capacity building programmes for women and by relieving the time-poverty than women experience due to the burden of other housework.

The section on gender in Vision 2035 also acknowledges women's role in enforcing new behaviours and practices; however, it emphasises how 'in order to improve women's situations, men should be included in the project. It is important to work with men to push them into changing their approaches towards women and their approach with themselves' (Djibouti 2035 Vision, 2010, p. 98). Both policies further outline informal and formal roles to promote change, which can be seen as opportunities moving forward. Local authorities are mentioned as needing to play a decisive role in the planning, programming and budgeting of development actions to ensure gender mainstreaming. Tribal chiefs and traditional systems play an important role in community influence. Both policies discuss the inclusion of women in budgeting processes as an important starting point to establish their role in

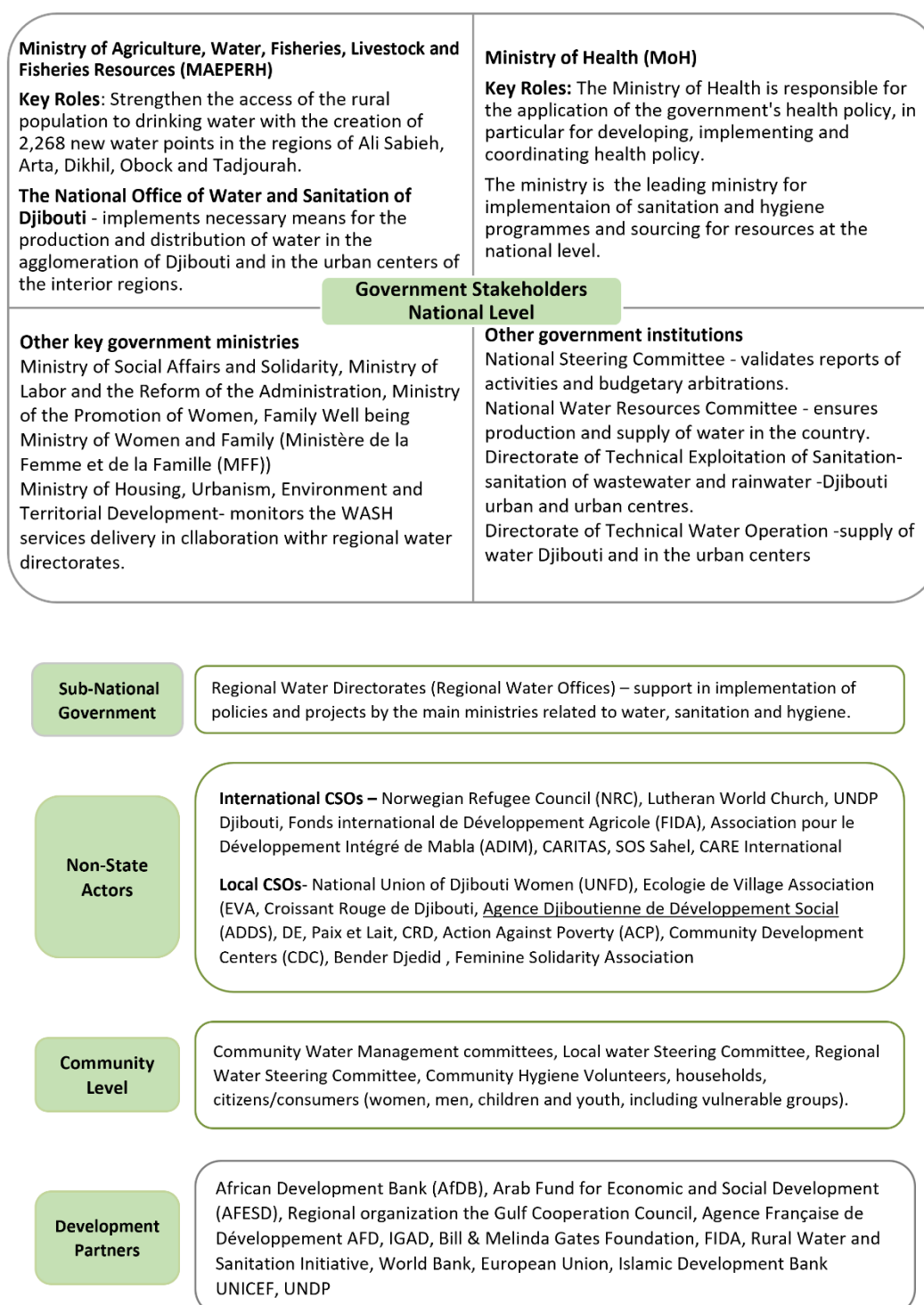
managing resources, both within households as well as at higher community or government levels. Lastly, the political, economic and social introduction of youth and teenagers to equality and equity concerns is highlighted in order to allow them to contribute to the country's development.

Currently, the completion rate for upper secondary is higher for women than for men; this presents a great opportunity to incorporate women into the workforce due to the higher levels of education they can take into the workforce. The 2018—2022 National Strategy for Social Protection identifies a role for women in the country's goal to increase primary sector employment, specifically in fishing and agriculture. The social protection strategy does not further specify jobs for women, but states that it aims to extend protection to more workers by (i) helping to strengthen the role of the private sector in creating jobs and (ii) seeking to integrate the maximum of informal sector assets into the formal sector, a women-dominated sector. Moreover, they suggest that the issue of water and food scarcity can be best managed in the short-term by immediate cash transfers to households. Overall, as with all other policies, they identify that the awareness of all the institutional, private and associative actors, the seriousness of the situation and how their contribution is most important. Djibouti's 2020—2024 Sanitation policy further echoes this and emphasizes that rural households should be taken care of specifically as their access issues are especially severe. The Sanitation policy does not further identify any issues or opportunities specific to women or men, making the policy gender blind.

Stakeholder Mapping

In Djibouti, the Ministry of Agriculture, Water, Fisheries, Livestock and Fisheries Resources and the National Office of Water and Sanitation have primary responsibility for WASH. Other national Ministries with some WASH responsibility include the Ministry of Health, the leading ministry in implementing sanitation and hygiene programmes at national level, and the Ministry of Housing, Urbanism, Environment and Territorial Development, who monitor WASH service delivery. There are a range of agencies involved in the delivery of WASH services at the sub-national levels, including regional water directorates, the private sector, national and international CSOs. A wide range of development partners are involved in supporting government, and international and national CSO engagement in WASH. Figure B provides a visual mapping of the actors involved.

Figure B: WASH stakeholders in Djibouti



9.5 Indonesia

Water, sanitation and hygiene in Ivory Coast at a glance

Access to drinking water: 93.3% of the population has access to an improved water source for drinking water, of which 92.4% has access to a basic level of drinking water, and 0.9% has access to a limited level of drinking water. The remainder of the population uses unimproved sources for drinking water¹⁵⁹.

Access to sanitation: 92.6% of the population has access to improved sanitation services, of which 86.5% has access to a basic level of sanitation service, whilst 6.1% has access to a limited level of sanitation service. 7.5% of the population uses unimproved sanitation services, including 6.1% who practice open defecation¹⁶⁰.

Access to hygiene: 94.1% of the population has access to basic hygiene while 5.9% has limited access¹⁶¹.

WASH in health facilities: Nationally, 80.2% has a basic service in drinking water, data on access to hygiene and sanitation is insufficient¹⁶².

WASH in schools: Nationally, 72.7% has a basic service in drinking water, 40.4% has a basic service in sanitation, and 58.9% has basic access to hygiene.¹⁶³

As a country with over 200 million spread out over 6,000 islands, and a major reliance on raw water supply to provide for their water resource, Indonesia has placed natural resource promotion and recovery as a priority in its Long-Term National Development Plan (2005). With the growing effects of climate change impacting the water cycle, rapid urbanization, and contamination of water supply it sees water supply and increasing sanitation facilities as a key component to ensuring self-reliance and development for all people. Whilst it recognises the need for gender mainstreaming in policies to reach the country's long-term goals, the specific gender dimensions of WASH are not explored.

National Medium-Term Development Plans serve to identify policy priorities in a given period, within the overarching framework of the Long-Term National Development Plan. The most recent National Medium-Term Development Plan (2020) elaborates further the importance of maintaining natural resources, improving sanitation, and mitigating the impacts of climate change to economic growth. The plan lays out 5 directives: human resource development, infrastructure development, regulatory simplification, bureaucratic simplification, and economic transformation. The Plan foresees the use of the 3R approach - Reduce, Reuse, Recycle - and the circular economy to increase effective waste disposal. It aims for 90% of the population to have access to safe and proper sanitation, including wastewater management. As for water resources, it seeks to improve infrastructure, including dam capacity and forest restoration, to meet the goal of 100% of the population having access to safe drinking water. The policy also sees community involvement as a main way to ensure implementation

¹⁵⁹ JMP 2020

¹⁶⁰ JMP 2020

¹⁶¹ JMP 2020

¹⁶² JMP 2019

¹⁶³ JMP 2019

and management of safe and adequate water and sanitation facilities and practices. This includes increasing women's role in development and decision-making, especially within health, education, labour market, economy, and public sectors.

While the Long-Term National Development Plan establishes the importance of improving water and sanitation access and infrastructure, the Long-Term Strategy for Low Carbon and Climate Resilience 2050 (2021) explores further the impacts that climate change has on the country's development, especially when it comes to WASH-related sectors. It notes that more extreme weather patterns are causing increased flooding and longer dry seasons, resulting in reduced availability of raw water supply, something which is exacerbated by rapid urbanisation and mass deforestation to increase economic activity. The Strategy identifies the limited access to channelled wastewater sewage for treatment of grey water as a high risk. It sees climate-related challenges as something that disproportionately impacts women due to limited access to water supplies for domestic and economic use, limited access to safe sanitation facilities, and an imbalance in capacity for understanding the importance of safe water and sanitation practices. It outlines objectives for both mitigation and adaptation to overcome these growing challenges while ensuring an equal balance to continue economic growth, including improving infrastructure and increasing community knowledge and capacity on water and sanitation management.

Along-side the Long-Term Strategy for Low Carbon and Climate Resilience, the National Strategy for Community-Based Total Sanitation (2008) touches on on-going sanitation challenges in Indonesia, with socio-cultural issues, continued health concerns due to poor handwashing and water contamination, and lack of progress by local governments being the main reasons given for the persistence of poor sanitation practices. The Strategy suggests Government will partner with NGOs and the private sector, subsidise communal sanitation facilities, and encourage community engagement to ensure every household has safe drinking water treatment and access to handwashing stations in public facilities. The Strategy does not consider gender-related issues in community-based sanitation and there is no consideration to the specific sanitation needs of women.

All the policies discuss increasing the capacity of communities and workers to lead to increased community engagement and awareness of these issues, but there is limited or no discussion about providing women specifically with career opportunities in the WASH sector. The Long-Term Strategy foresees supporting sanitation entrepreneurship in areas with potential, facilitating entrepreneurship to create products that comply with standards and the Long-Term Strategy for Low Carbon and Climate Resilience seeks to improve job activities with the consideration of gender equality. Based on available education data, women seem relatively well placed to take on roles in the WASH sector. Of the 62.7% of women complete upper secondary education, compared to 58.4% of men.¹⁶⁴ Whilst enrolment at tertiary levels of education remains relatively low at 36.3%, 38.9% of women enrol in tertiary education, compared to 33.7% of men.¹⁶⁵

The Presidential Instructions on Gender Mainstreaming (2000) emphasise the need to mainstream gender considerations in all policies and programmes. Most policies do recognise this but do not map out the implications or consistently differentiate how strategies will specifically target the different needs of women and men.

9.6 Ivory Coast

¹⁶⁴ UIS 2020

¹⁶⁵ UIS 2018

Water, sanitation and hygiene in Ivory Coast at a glance

Access to drinking water: 79.8% of the population has access to an improved water source, of which 35.2% has access to safely managed drinking water, 35.7% has access to basic drinking water and 8.9% has access to limited drinking water. The remaining 20.2% of the population access unimproved drinking water, including 6% who only access surface water.¹⁶⁶ Women are the main collectors of water.¹⁶⁷ Those with safely managed drinking water have water access on premises. Those with a basic level of access to drinking water spend less than 30 minutes collecting water whilst those with a limited level of access to drinking water spend more than 30 minutes collecting water.¹⁶⁸

Access to sanitation: 57.1% of the population has access to improved sanitation services, of whom 34.6% has access to basic sanitation services and 22.5% has access to limited sanitation services. 25.3% of the population practices open defecation whilst the remaining 17.6% of the population uses unimproved sanitation services.¹⁶⁹

Hygiene: 21.6% of the population has access to handwashing facilities with soap and water.¹⁷⁰

There is insufficient data on WASH in schools and health facilities available.

Policy context

In line with its 2040 Vision, the overall aim of Ivory Coast's 2016—2020 National Development Plan is to create a state in which the population as a whole is better off, translating into a significant reduction of poverty and a rise of the middle class to ensure economic growth. This development plan is more proactive than previous plans, and as such is a reflection of the country's relative stability and the increased levels of productivity the country has enjoyed in recent years. The 2016—2020 National Development Plan has five strategic areas that focus on:

1. Quality institutions and governance;
2. The preparedness and capacities needed for women and men to forge an emergent and prosperous Côte d'Ivoire;
3. Changes in production and consumption patterns so as to bring Ivory Coast onto the global stage;
4. Development of strategic infrastructure acting as a lever for Ivory Coast to move onto the global stage while abiding by environmental sustainability principles;
5. Beneficial integration into regional and global trade and other networks.

Although the plan has a strong focus on structural economic reforms, this is balanced with an understanding that these go hand in hand with the development of human capital. Water and sanitation is a cross-cutting issue that is identified under several of these five pillars of the plan, most

¹⁶⁶ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁶⁷ MICS (2016)

¹⁶⁸ Joint Programme for Water Supply, Sanitation and Hygiene (2016)

¹⁶⁹ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁷⁰ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

prominently, under pillars 1, 2 and 5. This highlights the challenges that Ivory Coast faces in water and sanitation, ranging from governance of resources, quality of health services and the population's health, and lastly, lacking infrastructure, especially in light of worsening climatic change. Beyond the general lack of access to water and sustainable sanitation systems, the national plan clearly acknowledges the heavier burden on populations in cities that are rapidly urbanizing and the lack of social safety nets to secure livelihoods for more vulnerable populations, including women amongst others. Consequences, such as the rapid spread of diseases like cholera, and food shortages due to water scarcity often impacts women more severely.

The 2015—2020 National Climate Change programme further discusses the health consequences of water scarcity and cleanliness. They have identified a positive correlation between variations in rainfall, temperature and the incidence rate of malaria, respiratory infections and diarrheal diseases. They have also demonstrated that urban populations are more vulnerable, particularly during the rainy season, due to inadequate wastewater drainage infrastructure. These challenges stem from the differences in rainfall between wet and dry seasons increasing, a drop in surface water availability and further effects of climate change on temperature and rainfall leading to extreme weather events. There is no mention of the differentiating impact between women and men, but the significant health consequences they identify present a clear challenge for the female population who are already at a disadvantage in this area. The programme promoted the introduction of climate change adaptation in all new legislation and policy instruments, identifying the importance of including diverse actors in their creation, including women's associations and NGOs who have localised knowledge of challenges and opportunities.

The National Development Plan broadly outlines that Ivory Coast should aim to increase accessibility to drinking water, build new water pumps and transfer the responsibility and ownership of adopting laws and regulations to local communities. Local communities should also include women's associations and further diversified representation. The wider goal in sanitation and drainage is to create a larger and more interconnected network through infrastructural development as well as improving the hygiene and sanitation conditions in the most vulnerable communities. Ivory Coast's Sector Policy Note on Sanitation (2012—2025) goes into more detail on improving sanitation in the country. Close to 5.5 million people do not have access to a latrine and practice defecation outdoors, mainly in rural areas, and many schools and health centres do not have improved sanitation facilities. Where there are improved sanitation facilities, this note identifies that these systems are not financed in a sustainable way, therefore upkeep is difficult. In terms of sanitation, there is a strong focus on eradicating open defecation, which has many negative impacts on women in terms of insecurity as well as hygiene and long-term, sustainable infrastructure improvements. The infrastructure development programme proposed raises the following considerations:

- Create demand for sanitation and create competition between communities' territories to promote on-site sanitation.
- Strengthen coordination with public and private actors.
- Strengthen the implementation and operating capacities of public and private actors and create a culture of social responsibility for these actors.
- Strengthen the sector's financial resources.

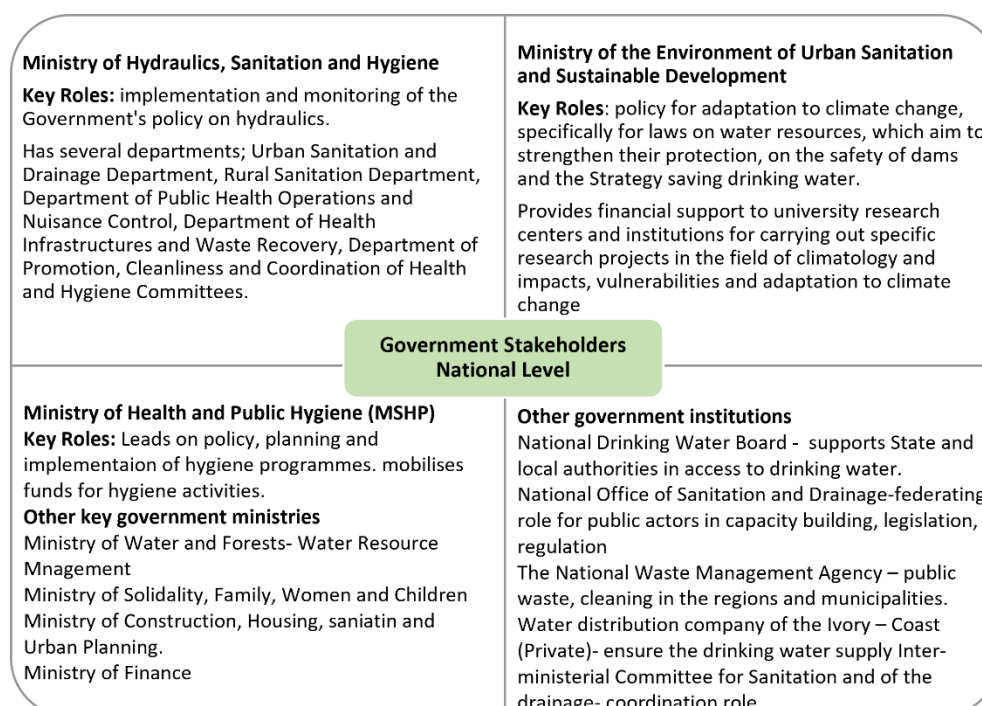
These suggested strategies focus on more than just the construction of new facilities and assure uptake of these new facilities to ensure their effectiveness. These strategies do not specifically

mention the inclusion of women or men in different ways. The National Development Plan does, however, identify how improved governance will result from better consideration of gender and the environment in public policies. With regard to gender, the strengthening of the institutional and regulatory framework will be necessary, as will the development of the skills of actors who apply it in plans, programs and projects. Furthermore, the plan proposes increased vocational training for women to include them in the employment sector, boosted by the improvements in water and sanitation infrastructure. Women and girl's education levels are low, however, with only 14% of girls completing upper secondary¹⁷¹ and 8.5% of women enrolling in tertiary education.¹⁷²

Stakeholder Mapping

In Ivory Coast, the Ministry of Hydraulics, Sanitation and Hygiene has primary responsibility for WASH. Other national Ministries with some WASH responsibility include the Ministry of the Environment of Urban Sanitation and Sustainable Development, who have a key role in the adaptation to climate change, and the Ministry of Health and Public Hygiene, leading on hygiene policy and programmes. There are a range of agencies involved in the delivery of WASH services at the sub-national levels, including regional and municipal government bodies and national and international CSOs. A wide range of development partners are involved in supporting government, private sector and CSO engagement in WASH. Figure C provides a visual mapping of the actors involved.

Figure C: WASH stakeholders in Ivory Coast



¹⁷¹ UNIS (2016)

¹⁷² UNIS (2019)



9.7 Kenya

Water, sanitation and hygiene in Kenya at a glance

Access to water: 71.1% of the population has access to an improved water source, of which 61.6% has access to basic drinking water and 9.5% has access to drinking water categorised as 'limited.' The remaining 28.8% of the population uses unimproved water sources for drinking water, of which 19% only have access to surface water.¹⁷³

Access to sanitation: 58.2% of the population has access to improved sanitation services, of which 32.7% has access to basic sanitation services and 25.5% has access to limited sanitation services. The remaining 41.8% of the population uses unimproved sanitation services, with 8.5% practicing open defecation.¹⁷⁴

Hygiene: 26.8% of the population has access to handwashing facilities with soap and water.¹⁷⁵

WASH in health facilities: 78% of health care facilities have basic service for drinking water. Data is not available on sanitation and hygiene facilities in health facilities.¹⁷⁶

Data is not available for WASH in schools.

¹⁷³ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁷⁴ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁷⁵ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁷⁶ Joint Programme for Water Supply, Sanitation and Hygiene (2019)

Policy context

The drive to advance livelihoods in Kenya is laid out through the four key areas, dubbed the Big Four Agenda, within the National Development plan, named the Kenya Vision 2030 (2008–2030). These include achieving food security, affordable housing, manufacturing and affordable healthcare. Water, sanitation and hygiene continues to be a key sub-focus area within the Social Pillar outlined in the Third Medium Term Plan (MTP III), 2018–2022, the midterm plan within the National Development Plan of 2030. Within the plan it sees environmental degradation, climate change, inadequate infrastructure and lack of water conservation strategies as hindrances to the development of a clean, safe environment and access to water and sanitation services. It explains that approaches to combat these challenges come from capacity building, public participation, adoption of innovative technical approaches, multi-sectoral collaboration and adequate water and sanitation facilities. Within this policy, there is no discussion on how inadequate water, sanitation and hygiene facilities and practices impact women, but it does touch on the under-valuation of women's contribution to the economy, under-employment and skills mismatch, and the implementation of the two-thirds gender rule that ensures that no more than 66% of one gender is represented in the national assembly and senate.

The Kenya Environmental Sanitation & Hygiene Policy (KESHP) 2016, launched two years before the MTP III, recognises the importance of water, sanitation and hygiene in achieving Kenya Vision 2030 and lays the foundation to include a people-centred approach to sanitation. The policy highlights key issues such as rapid urbanisation, the mortality and morbidity burden associated with unsafe water and sanitation and the risk of rape and domestic violence for women and girls due to inadequate sanitation facilities. To address these issues, the Policy intends to scale up access to improved rural and urban sanitation (Strategy 1) and build governance and leadership capacity for sanitation (Strategy 5). The policy discusses an increase in education, training and skill development for sanitation, including increasing the availability of skilled personnel in the planning and management of sanitation facilities, though it does not suggest these efforts are focused on any one gender/sex.

While the Policy discusses the importance of improved facilities, it does not discuss the negative impacts inflicted by climate change. However, the Ministry of Water and Sanitation (MOWS) Strategic Plan 2018 brings to light the impacts climate change has on the quality, quantity, timing, form and intensity of precipitation leading to an increase in natural disasters. It also highlights the decrease in available safe water supply. In addition, it touches on the importance of developing water infrastructure over focusing on water resources and management, and the impact that inadequate water and sanitation infrastructure and management have on the health of the population. In order to combat these challenges, the MOWS Strategic Plan lays out eight Key Result Areas (KRA), with Water and Sanitation Coverage (KRA 4 and 5), Human Resource Capacity (KRA 6), and Research and Technology (KRA 7) playing major roles in enhancing infrastructure, capacity building, employment opportunities and handling the adverse effects of climate change, though there is no clear discussion on a focus of women and girls.

The National Policy on Gender and Development (NPGD) elaborates on the burden women and girls face due to clean water deficits, with the need to travel long distances to collect water for domestic use, which affects girls' attendance at school and women's opportunities for income generating activities. The NPGD seeks to reduce these burdens by ensuring the availability of water for multiple

uses to women and to develop and review the environment management laws to ensure gender and equity compliance.

With climate change being a major discussion in most of the policies, the National Climate Change Policy examines how increases in weather extremities, including droughts, floods and water salinisation greatly impacts the water, sanitation and hygiene sector, with the biggest focus being on inadequate sanitation systems which lead to clogged drains and water contamination. The Ministry of Water and Sanitation Strategic Plan similarly addresses reversing degradation and climate change to ensure safe water and sanitation by strengthening capacity building in vulnerable communities.

Interwoven throughout these policies is the continual discussion on the inclusion of equal opportunities for stakeholder participation throughout the planning, implementation and management of water and sanitation projects/programmes and the push for participation of female leaders on community-led committees through gender mainstreaming. While most policies include opportunities for job creation, many do not discuss career opportunities for women specifically. With only 9.73% enrolment of women in tertiary education,¹⁷⁷ there continues to be a gap in highly skilled job opportunities within WASH. Policies focus more on low-skilled jobs, such as the KESHP, which includes training programmes for artisans and operators in sanitation and job creation in areas such as labour-intensive construction, sustainable livelihoods and long-term entrepreneurial activities. Finally, the NGDP seeks to encourage women's participation in the technologies and sciences and STEM based courses.

Whilst the KESHP and MOWS Strategic plan discuss the impact of inadequate water, sanitation and health in daily life, the National Education Sector Strategic Plan and the National Health Policy examine the importance of safe water, sanitation and hygiene within school and health facilities. Within schools, girls tend to be more affected by inadequate WASH facilities, and this impacts their attendance, especially when they are menstruating, and can expose them to increased risks of gender-based violence. To limit the impact on children's schooling, the National Education Sector Strategic Plan includes, among its objectives, the construction of facilities for safe and clean drinking water in schools and separate sanitation facilities for boys and girls.

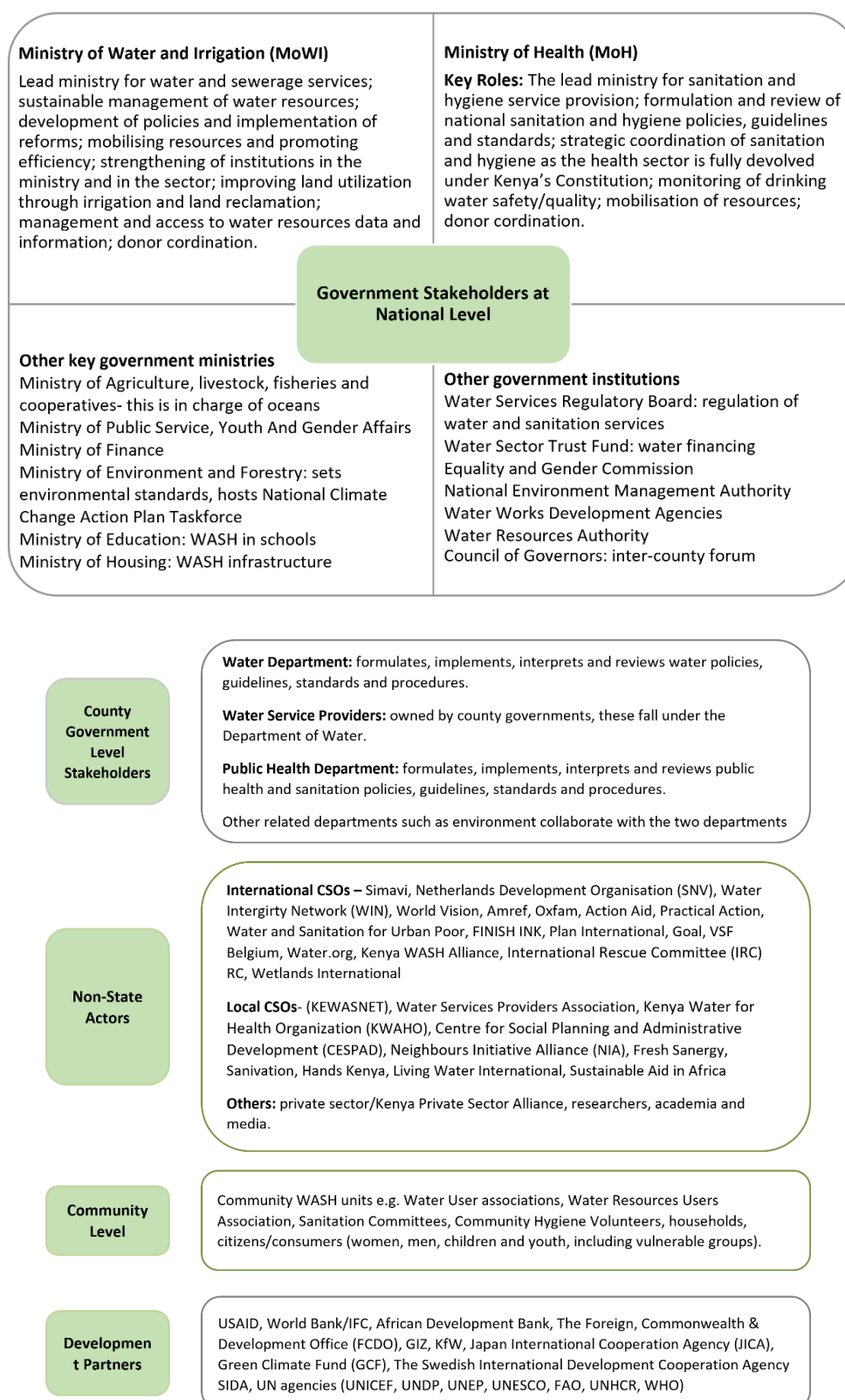
As the main users of primary health centres, poor WASH facilities in health centres disproportionately affects women and girls. The National Health Policy seeks to improve their WASH facilities to limit water-borne disease. The Policy also foresees mainstreaming gender issues in planning and implementation of programmes to minimise impacts on women and girls.

Stakeholder Mapping

In Kenya, the Ministry of Water and Irrigation has primary responsibility for WASH. Other national Ministries with some WASH responsibility include the Ministry of Health, responsible for sanitation and Hygiene provision and policy, and the Ministry of Environments and Forestry, who set environmental standards, and the Ministry of Housing, who handle WASH infrastructure. There are a range of agencies involved in the delivery of WASH services at the sub-national levels, including local water- and public health departments, the private sector, national and international CSOs. A wide range of development partners are involved in supporting government, private sector and CSO engagement in WASH. Figure D provides a visual mapping of the actors involved.

¹⁷⁷ UNIS (2017)

Figure D: WASH stakeholders in Kenya



9.8 Namibia

Water, sanitation and hygiene in Namibia at a glance

Access to drinking water: 91.4% of the population has access to an improved source of drinking water, of which 84.3% has access to basic drinking water and 7.1% has access to limited drinking water. The remainder of the population relies on unimproved water sources, with 4.9% of the population having access to surface water only.¹⁷⁸ Data on the proportion of the population with access to a safely managed water source is not available.

Access to sanitation: 48.1% of the population has access to improved sanitation services, of which 35.3% has access to basic sanitation services and 12.8% has access to limited sanitation services. 47.2% of the population continues to practice open defecation whilst 4.8% of the population uses unimproved sanitation services.¹⁷⁹ Data on the proportion of the population with access to safely managed sanitation services is not available.

Data on hygiene, WASH in schools and WASH in health facilities is not available.

Policy context

Namibia's 5th National Development Plan (NDP5) focuses on the period 2017/18–2021/22 and charts a course toward the cumulative targets outlined in Namibia's Vision 2030. The plan is organised around four key pillars for achieving sustainable development, namely Economic Progression, Social Transformation, Environmental Sustainability and Good Governance. NDP5 also identifies five 'game changers' that will move Namibia from a reactive, input-based economy towards a proactive, high-performing economy.

The plan looks at water and sanitation as two separate sectors, rather than looking at water and sanitation through a combined WASH lens. As such, analysis of the challenges faced in the water and sanitation sectors fall within two separate pillars of NDP5. Discussion of the water sector falls under the Economic Progression pillar of the plan, under the game changer subsection 'increase investment in infrastructure development.' The plan highlights key issues around safe water access in the water sector, including:

- Availability of water and reliable water supply.
- Increasing water demand for economic activities and urban consumption.
- Water demand from the agriculture sector for irrigation.
- Provision and financing of water infrastructure such as desalination plants in coastal areas.
- Improving management of existing water sources.

¹⁷⁸ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁷⁹ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

Discussion of issues related to sanitation and hygiene fall under the Social Transformation pillar of the policy, under the ‘social development’ subsection. The key issues in the sanitation sector that the 5th NDP highlights includes:

- Access to improved sanitation systems in rural areas.
- Promoting coordinated investment on sanitation infrastructure.
- Strengthening advocacy to tackle sanitation challenges.

While NDP5 summarises the key challenges of safe water supply and sanitation services access, particularly in rural areas, Namibia’s Water Supply and Sanitation Policy (2008) goes into much more detail on the challenges themselves, the context, the health consequences and Namibia’s policy objectives in response to these challenges.

The Water Supply and Sanitation Policy states that essential water supply and sanitation services should be available to all Namibians, that the improvement of water and sanitation services should be achieved equitably, that communities should have the right to determine which water and sanitation services are acceptable to them, and that utilisation of water resources should be environmentally sustainable. NDP5 then translates these broad policy objectives into actionable strategies as part of NDP5’s results framework.

As with the NDP5, the Water and Sanitation Policy deals with water supply and sanitation in isolation, concentrating on supply for domestic use, supply for economic activities and rural water supply in the water sector, as well as provision of rural and urban sanitation services in the sanitation sector.

Namibia has indicated its commitment to responding to climate change challenges through its National Policy on Climate Change (NPCC). The NPCC, adopted in 2011, is implemented through the National Climate Change Strategy Action Plan (NCCSAP) that was finalised in 2015 for the period 2013–2020.

Like the NDP5 and the Water and Sanitation Policy, Namibia’s 2011 National Policy on Climate Change also analyses the issues of water supply and provision of sanitation services but focuses much more heavily on the impact of climate change on both sectors, as well as the potential health consequences. The policy highlights the climate sensitivity of the two sectors, with Namibia’s already variable and unpredictable rainfall patterns being worsened by climate change. The policy argues that this in turn will have a significant impact on water supply and sanitation.

Namibia ratified the United Nations Framework Convention on Climate Change (UNFCCC) in 1995 and thus became obligated to prepare and submit national communications, completing national communications in 2002, 2011 and 2015. Namibia’s Fourth National Communication to the UNFCCC, submitted in 2020, covers the policy framework of the NPCC and summarises progress against the NCCSAP. The communication acknowledges the vulnerability of water supply to climate change, as Namibia relies on rainfall as its natural water source as well as dams, ephemeral rivers and aquifers in the absence of perennial rivers. The communication also highlights how high evaporation rates coupled with the country’s semi-arid climate means that Namibia is now at a water deficit.

While NDP5, the Water and Sanitation Policy, NPCC, the National Policy on Climate Change and the Fourth Communication to the UNFCCC all reference gender as a cross-cutting issue in the water and sanitation sectors, none explore in any great detail how women, youths and marginalised groups are impacted nor the roles that these groups can play in these sectors. In contrast, Namibia’s National

Gender Policy explicitly tackles the gender issues affecting the water and sanitation sectors. The policy argues that women continue to be under-represented in policy and decision-making forums on the utilisation of natural resources, environmental development and water management. The policy also explores the roles that women play within the water and sanitation sectors. For example, in most rural communities, women and girls constitute 75% of the workforce responsible for fetching water. Fetching water is primarily a female task with 36% of rural households having daughters fetch water compared to 15% which have sons fetch water.

In addition, the policy explores the health inequalities related to water and sanitation. As outlined in the other policies discussed, Namibia has a limited infrastructure with which to provide people, especially rural populations, with access to water and sanitation facilities. A lack of these basic amenities has a detrimental effect on the health of rural communities, with diseases linked to water and sanitation such as tuberculosis and malaria more prevalent in rural areas. Because more women than men live in rural areas and female-headed households tend to be poorer than male-headed households, a lack of rural water and sanitation infrastructure disproportionately affects women and female-headed households.

There is some discussion of women's education on water and sanitation sector issues in the National Policy on Climate Change for Namibia. The policy states that the government will include gender and climate change, including issues such as water supply, in the curriculum of education and training programmes. However, the policy stops short of discussing women's positioning for formal employment in the water and sanitation sectors. Currently, the completion rate for upper secondary education for women is relatively high, which implies that there is opportunity for WASH education policies that position women for employment in the sector to be successful.

All of the above policies are very limited in their consideration of women's employment levels in the water and sanitation sectors and in terms of opportunities for women to access employment in these sectors. However, the National Gender Policy (2010–2020) does put forward that, as a strategy for gender mainstreaming, women should be involved in interventions to achieve the objectives of environmental sustainability, including in the water and sanitation sectors. The policy stops short of discussing how this engagement could take place.

Stakeholder Mapping

In Namibia, the Ministry of Agriculture, Water and Land Reform has primary responsibility for WASH. Other national Ministries with some WASH responsibility include Ministry of Urban and Rural Development, ensuring implementation of national WASH strategy for urban settlements, and the Ministry of Health and Social Services, implementing and enforcing hygiene practices. There are a range of agencies involved in the delivery of WASH services at the sub-national levels, including district-level government agencies and traditional bodies, the private sector, national and international CSOs. A wide range of development partners are involved in supporting government, private sector and CSO engagement in WASH. Figure E provides a visual mapping of the actors involved.

Figure E: WASH stakeholders in Namibia

Ministry of Agriculture, Water and Land Reform (MAWLR)

Directorate of Water Supply and Sanitation Coordination (DWSSC): overall sector coordination, provision of rural sanitation services and hygiene and development of standards/guidelines.

Directorate of Water Resource Management (DWRM): enforcement of standards, compliance and water environment management.

Coordinates and chairs the National Water and Sanitation (WATSAN) committee.

Ministry of Urban and Rural Development (MURD)

Ensures implementation of national water, sanitation and hygiene strategy for urban informal settlement, provides technical support on water, sanitation and hygiene issues to Local Authorities.

Ministry of Health and Social Services (MoHSS)

Mandated for the implementation of sanitation and hygiene, enforcement and monitoring hygiene practices and health. Hygiene education materials adapted to both rural and urban areas

Government Stakeholders at National Level
Ministry of Works and Transportation

collaboration with MAWLR, provide standards and specifications and procedures to develop sanitation systems and on-site wastewater disposal and treatment, provided by line ministries, such as Ministry of Education, Arts and Culture (MoEAC) or MoHSS.

Ministry of Environment and Tourism (MET): Leads climate change programmes.

Other key government ministries: Ministry of Gender Equality, Poverty Eradication and Social Welfare (MGEPSW), Ministry of Education, Arts and Culture, Ministry of Finance

Other government institutions

Water Development Forum

Department of Water and Forestry (under MAWLR)
Namibia Water Corporation (NamWater): water and sewerage services in urban areas.

WATSAN Committee: body that coordinates WASH stakeholders at national, and sub-national levels;
Office of the Prime Minister: WASH related disaster risk management and response in the country.

National Planning Commission: coordinates WASH international development budget/funding.

Sub-National Government Stakeholders

Regional level MAWLR: coordinate with stakeholders the implementation of sanitation in selected communities and through setting up of community structures. Other regional level ministries MURD, MWT, MET, MoHSS, MoEAC, MET, MWT, MGEPSW

Local Authorities: Responsible for sanitation (hardware and software) in Local Authorities; gets support from MWT when required.

Constituency Development Committees (regional, local and constituency committees): Identify and prioritize areas/households to be considered for subsidised sanitation.

Traditional Authorities: Assist and cooperate with Government, Regional Councils and Local Authorities in the execution of their policies.

Non-State Actors

International CSOs: Red Cross Namibia

Local CSOs: Namibian Water Partnership, Development Workshop Namibia, Shack Dwellers Federation of Namibia which promotes construction of housing for female headed households in the informal urban areas, Project Hope Namibia.

Others: Namibian Chamber of Environment

Others; private sector, researchers, academia and media.

Community Level

Community WASH units e.g. Water User associations, Sanitation Committees, Community Hygiene Volunteers, households, citizens/consumers (women, men, children and youth, including vulnerable groups).

Development Partners

UN agencies (UNICEF, UNDP, UNEP, UNESCO, FAO, UNHCR, WHO)

9.9 Nigeria

Water, sanitation and hygiene in Nigeria at a glance

Access to drinking water: 78% of the population has access to an improved water source for drinking water, of which 21.7% has access to safely managed drinking water, 55.9% has access to a basic level of drinking water and 5% has access to a limited level of drinking water. The remainder of the population uses unimproved sources for drinking water, including 5.6% who only have access to surface water.¹⁸⁰ 40% of the population are women who are the main collectors of water. It takes 32.7% of the population, with improved drinking sources, 30 mins or less to collect water.¹⁸¹

Access to sanitation: 43% of the population has access to improved sanitation services, of which 30.5% has access to safely managed sanitation service, 12.2% has access to a basic level of sanitation service whilst 19.5% has access to a limited level of sanitation service. 19.1% of the population uses unimproved sanitation services, including 18.7% who practice open defecation.¹⁸²

Hygiene: 33.2% of the population has access to handwashing facilities with soap and water.¹⁸³

WASH in schools: Nationally, 36.4% of schools have a basic service for drinking water, 37.8% have a basic service for sanitation and 27.6% have a basic service in hygiene.¹⁸⁴

WASH in health facilities: 46.4% of health care facilities have a basic service for drinking water, 17.2% of health care facilities have basic service for sanitation and 65.8% of health care facilities have basic service for hygiene.¹⁸⁵

Policy context

As Nigeria continues to experience rapid urbanisation and growth, it seeks to improve infrastructure throughout the country to promote development opportunities, establish a strong foundation for a diversified economy and alleviate poverty. One of the main ways they aspire to achieve this is through social development, which covers water resources and sanitation, water and gender equity and employment and job creation, as outlined in the Mid Report of the National Development Plan (2021). The Mid Report points to the progress in improving water service delivery over the years, while also reviewing continued challenges. Challenges identified include ensuring economic diversification, private sector investment in WASH services, lack of coordination and communal disputes, negative impacts of climate change, contamination of water sources and its impacts on health and poor operation and maintenance culture. The report also touches on the importance of WASH in light of the COVID-19 pandemic. Some of the main ways it seeks to overcome these challenges is by continuing to improve sanitation services, extending water supply access and adopting an integrated planning and policy formulation approach, while also increasing training, developing the bioeconomy and circular economy, and increasing access to finance and technical support for businesses and projects in environmentally sustainable sectors.

¹⁸⁰ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁸¹ MICS (2017)

¹⁸² Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁸³ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁸⁴ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁸⁵ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

One recurring theme in the National Development Plan is the increase in negative impacts on Nigeria's development brought about by climate change. In order to address this increasing threat, the National Climate Change Policy for Nigeria was launched in 2021. The Climate Change Policy addresses these challenges through 10 strategic objectives, including implementing adaptation and mitigation measures that promote low-carbon development (Strategy 1); mainstreaming gender, youth and other vulnerable groups into all climate change interventions (Strategy 5); and promoting climate-proofing of construction and infrastructural development (Strategy 6). This policy examines the impact that extreme weather places on an already strained water, sanitation and hygiene sector due to an increase in droughts and floods. Nigeria seeks to overcome issues related to climate change and inadequate water and waste infrastructure through the implementation of a circular economy, as well as increasing infrastructure for wastewater management and reforms in the establishment, management, operation and maintenance of water supply and sanitation schemes and services. The policy recognises the importance of including women in the planning, implementation and management of related programmes, and seeks to integrate a gender perspective in the adoption of environmentally friendly and gender responsive technologies, and the development of gender-responsive, socially inclusive and resilient water and sanitation infrastructure.

One of the biggest challenges that Nigeria faces with implementing these policies and ensuring the adequate implementation of WASH facilities and infrastructure is the 3-tiered government structure which leads to a lack of coordination in policy and programme implementation as discussed in Nigeria Water and Sanitation Profile.¹⁸⁶

In all the policies reviewed, there was a clear theme around capacity building and job creation. Within the Climate Change Policy, there is discussion on collaboration with the Federal Ministry of Women Affairs to ensure equitable policies and programmes. This discussion seeks to encourage women's participation in the planning and management of climate change programmes. This includes investment to promote water security and climate change resilient growth and development to sustain jobs, employment, economic growth and social stability. With 66.8% of women completing upper secondary school¹⁸⁷ and no data available on those enrolled in tertiary education, there continues to be discrepancies in access to some highly skilled careers in WASH.

While most policies reviewed discussed the importance of WASH and ensuring water security in the country, the National Gender Policy of 2006 looks at the importance of including women in the work force and actively participating in decision making. The policy calls for equal inclusion in the design, implementation and management of all projects/programmes implemented. It also highlights the key role women play in water, sanitation and hygiene as women are usually the collectors of water and are in charge of maintenance of facilities, including water, within the family. However, they are usually the ones greatly impacted by the consequences of unsafe drinking water. This health risk is highlighted further in the National Health Policy which recognises that women and children face higher mortality and morbidity rates due to an increase in communicable diseases stemming from poor water and sanitation. Although not specifically related to WASH, both policies emphasize the importance of the equal participation of women and increased education and awareness. They also encourage community engagement in best practices.

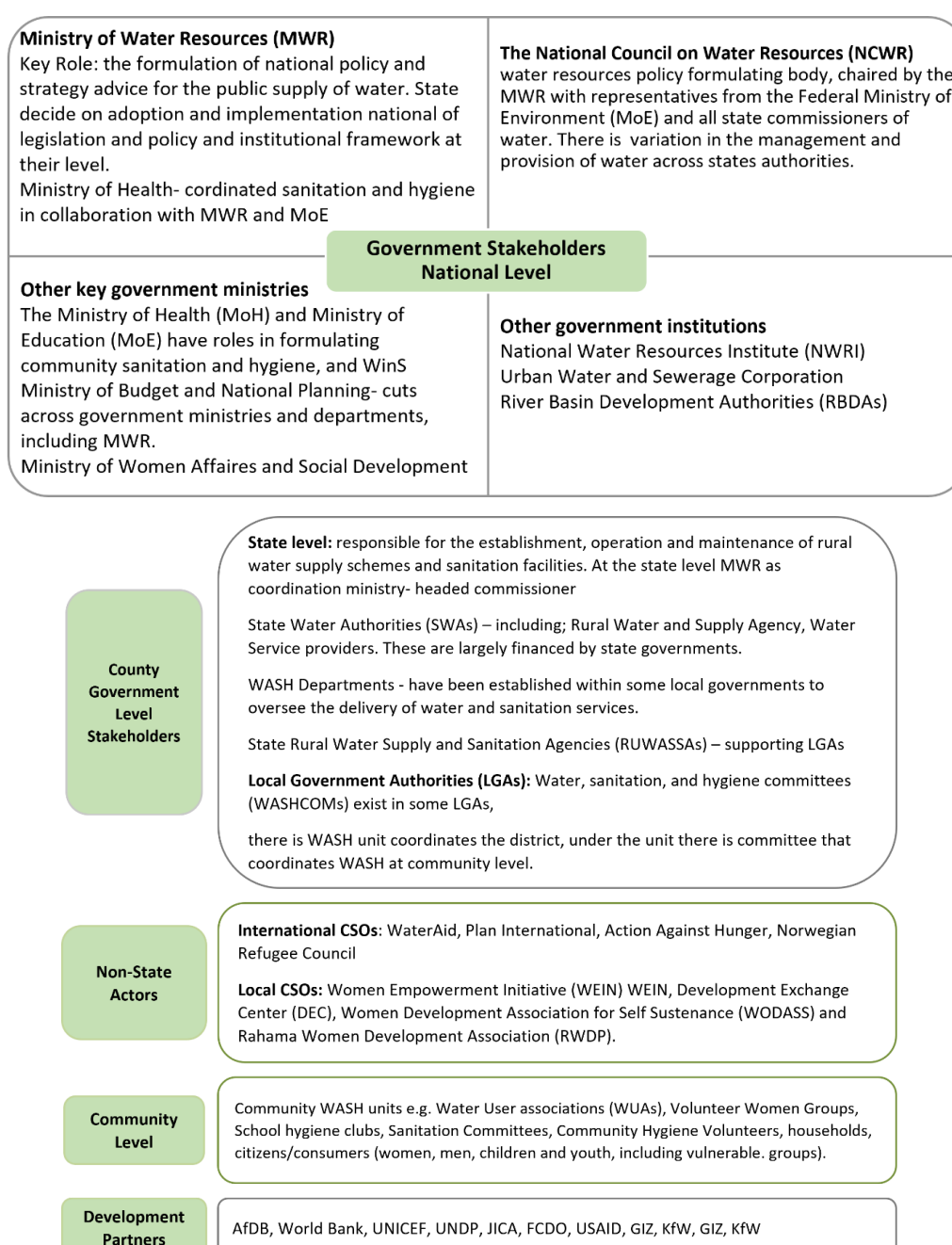
¹⁸⁶ Nigeria's National Water Supply and Sanitation Policy was unavailable to locate online.

¹⁸⁷ UNIS (2017)

Stakeholder Mapping

In Nigeria, the Ministry of Water Resources has primary responsibility for WASH. Other national Ministries with some WASH responsibility include The National Council on Water Resources, a water resources policy formulating body, and the Ministry of Health and Ministry of Education, who have roles formulating community sanitation and hygiene. There are a range of agencies involved in the delivery of WASH services at the sub-national levels, including state-level government agencies, national and international CSOs. A wide range of development partners are involved in supporting government, private sector and CSO engagement in WASH. Figure F provides a visual mapping of the actors involved.

Figure F: WASH stakeholders in Nigeria



9.10 Morocco

Water, sanitation and hygiene in Morocco at a glance

Access to drinking water: 90% of the population has access to an improved water source for drinking water, of which 79.9% has access to safely managed drinking water, 10.5% has access to a basic level of drinking water and 5.3% has access to a limited level of drinking water. The remainder of the population uses unimproved sources for drinking water, including 1.5% who only have access to surface water.¹⁸⁸

Access to sanitation: 87% of the population has access to improved sanitation services, of which 39.3% has access to safely managed sanitation service, 47.9% has access to a basic level of sanitation service whilst 1.5% has access to a limited level of sanitation service. 9.5% of the population uses unimproved sanitation services, including 1.7% who practice open defecation.¹⁸⁹

Data on hygiene and WASH in health facilities is not available.

WASH in schools: Nationally, 83.6% of schools have a basic service for drinking water, 69.9% has a basic service for sanitation and 89% has a basic service for hygiene.¹⁹⁰

Policy context

Due to continued threats to available water resources throughout the country, the National Sustainable Development Policy 2017 (NSDP) recognises the importance of water, sanitation and hygiene improvements for the development of the country, as outlined in Strategic Focus 6: ensure the conservation and rational management of water resources, Strategic Focus 14: promotion of integrated waste management to implement a circular economy, and Strategic Focus 16: securitize the needs (by ensuring needs are met) and strengthen water resources integrated management. Not only does it seek to improve facilities, access and programmes around water, sanitation and hygiene, it identifies the opportunities available for transitioning to a circular economy through recycling water usage. It also examines the need for human development and combat social inequalities within Stake 6: promote human development and reduce social and territorial inequalities. While the 6th Stake touches briefly on the importance of equality for all, it remains quite gender blind with regard to the incorporation of gender in policies. The NSDP further connects the importance of fighting climate change as a factor to ensure the advancement of water, sanitation and hygiene within the country, especially in regard to discrepancies between regions within the country.

Moroccan Climate Change Policy 2014 highlights the impact of climate change on health and water access in the country. The increase in temperatures and extreme weathers is leading to salinisation of water, desertification and an increase in water-borne diseases which impact women and girls more than men as women are typically responsible for managing water. It also points to the importance of

¹⁸⁸ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁸⁹ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁹⁰ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

increasing financing and programmes focused on combating climate change, including education and awareness on alternative solutions and the re-use of water resources.

While the Climate Change Policy touches on the importance of tackling climate change to ensure equitable access to water, sanitation and hygiene, the National Water Plan (2013) examines the current situation within Morocco, with the Moroccan climate being regulated by the Saharan depression and the anticyclones of the Azores leading to greater extremities when weather patterns change. With 20% of all water resources coming from ground water, a continued degradation of surface water, negative impacts from climate change and water contamination, Morocco has placed water as an important factor in its development. It outlines the importance of mobilising water resources, desalting ocean water, increasing the re-use of water for irrigation, enhancing water treatment and reusing wastewater. It also sets out the need to protect communities against floods, especially for sanitation facilities and agricultural drainage systems. Within the health policy that was reviewed, there was no mention of WASH specific strategies, though it does discuss the increase in water-borne diseases and the need to ensure adequate facilities.

Within all these policies, there is no clear examination of how to increase job opportunities for women. However, the National Sustainable Development Plan discusses increasing employment opportunities more generally. This includes strategic planning of the building usage as an economic opportunity with resource consumption (such as organizing the flow of water and waste) and adapting training to meet the transition to green-based trades as well as ensuring equitable representation of women and men within all sectors as outlined in the National Strategy for Gender Equality & Equity 2006. The Strategy sees the value of ensuring there is equal participation of women and men in communities, especially around decision-making, and identifies the impact that inadequate water and sanitation facilities place on women, such as leaving them more vulnerable to illness, it also discusses the importance of equal access to education. With only 29% of females completing upper secondary¹⁹¹ and 41.7% of women completing tertiary education,¹⁹² there continues to be a disproportion of women having the skills needed for certain WASH-related careers. The Climate Change Policy discusses the importance of capacity building and training for individuals, communities, interest groups and public and private enterprises to better adapt to and mitigate climate change. It seeks to do so by strengthening human and technical capacities of stakeholders, training public actors to take ownership of the vision, concepts and integrated approach, organizing workshops for capacity building on issues related to climate change and strengthening STEM and climate-related education and training. Additionally, the Water Plan discusses the strengthening of capacity building, including skill building for highly skilled positions, through building efficient water planning and management, and advancement of techniques, construction and management adapted to a national context.

Stakeholder Mapping

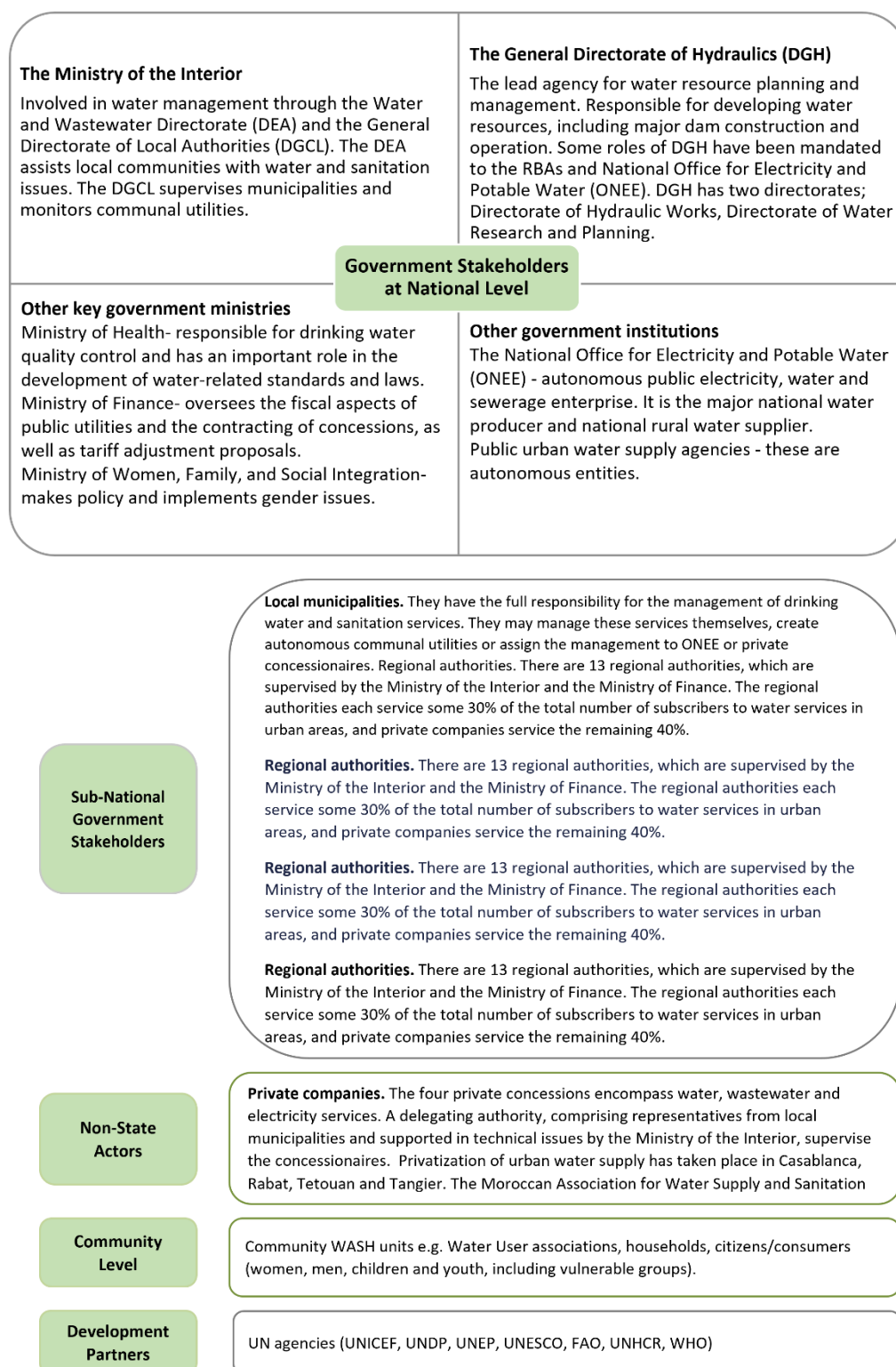
In Morocco, the Ministry of Interior has primary responsibility for WASH. Other national Ministries with some WASH responsibility include The General Directorate of Hydraulics, the lead agency for water resource planning, and The Ministry of Health, responsible for drinking water quality control and water-related laws. There are a range of agencies involved in the delivery of WASH services at the

¹⁹¹ UNIS (2020)

¹⁹² UNIS (2020)

sub-national levels, including district-level government agencies, the private sector, national and international CSOs. A wide range of development partners are involved in supporting government, private sector and CSO engagement in WASH. Figure G provides a visual mapping of the actors involved.

Figure G: WASH stakeholders in Morocco



9.11 Uganda

Water, sanitation and hygiene in Uganda at a glance

Access to drinking water: 83.1% of the population has access to an improved water source for drinking water, of which 16.6% has access to safely managed drinking water, 39.2% has access to a basic level of drinking water and 27.3% has access to a limited level of drinking water. The remainder of the population uses unimproved sources for drinking water, including 4.7% who only have access to surface water.¹⁹³

Access to sanitation: 37.4% of the population has access to improved sanitation services, of which 19.8% has access to a basic level of sanitation service whilst 17.6% has access to a limited level of sanitation service. 62.6% of the population uses unimproved sanitation services, including 4.7% who practice open defecation.¹⁹⁴

Hygiene: 22.6% of the population has access to handwashing facilities with soap and water.¹⁹⁵

WASH in schools: Nationally, 68.4% of schools have a basic service for drinking water, 79.6% has a basic service for sanitation and 30.1% has a basic service for hygiene.¹⁹⁶

WASH in health facilities: 44.3% of health care facilities have a basic service for drinking water. Data is not available on sanitation and hygiene facilities in health facilities.¹⁹⁷

Policy context

The shift away from a focus on poverty reduction to one of wealth creation through industrial development in Uganda's National Development Plan III (2020—2025) has resulted in decreased attention to water, sanitation and hygiene in national policy. Nevertheless, water, sanitation and hygiene remain a core area of Uganda's Human Development Programme, a sub-programme in the NDP III, and one which seeks to improve labour productivity for increased competitiveness and better quality of life for all, with the following targets having been set:

- Increased access to safe water supply from 70 to 85% (rural) and from 74 percent to 100% (urban).
- Increased access to basic sanitation from (improved toilet) 19 to 40% and hand washing from 34 to 50%.

In addition, the NDP III also foresees investing in the effective management of the entire WASH value chain segments, including containment, emptying, transportation, treatment, safe reuse and disposal.

The NDP III recognises that achievement of these, and other development targets, will be more challenging due to the effects of climate change, with more extreme weather patterns leading to

¹⁹³ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁹⁴ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁹⁵ Joint Programme for Water Supply, Sanitation and Hygiene (2020)

¹⁹⁶ Joint Programme for Water Supply, Sanitation and Hygiene (2019)

¹⁹⁷ Joint Programme for Water Supply, Sanitation and Hygiene (2019)

increased water stress, impacting water availability and quality for both domestic and commercial uses as well as the water and sanitation infrastructure already in place. The Uganda National Climate Change Policy (2015—2020) articulates the challenges further and highlights that, given their domestic role, women are likely to experience the effects of climate change on water and sanitation facilities most keenly. It commits to mainstream gender issues across climate change adaptation and mitigation actions including ensuring climate change related policies and activities are gender sensitive and developing a gender-sensitive approach in monitoring and evaluating systems.

The National Water and Sanitation Gender Strategy,¹⁹⁸ launched in 2018 (two years earlier than the National Development Plan III) retains a focus on poverty reduction and seeks to ‘empower men, women, boys, girls and vulnerable groups through ensuring equity in access to and control of resources in the water and sanitation sub-sector’. The Strategy highlights the burden and risks for women and girls that come with poor access to water, sanitation and hygiene. It reports that, on average, women and girls spend up to six hours a day fetching water, limiting their time for economic activities and attending school; that women and girls face shame and risk of attack as a result of inadequate sanitation facilities; and that women and girls are more at risk of health conditions such as diarrhoea, dysentery and cholera as a result of their domestic responsibilities. It does not, however, recognise the negative implications climate change is likely to have on the burden and risks women and girls face.

The Water and Sanitation Gender Strategy has five strategic objectives:

1. Integrate gender into all government policy, guidelines, plans and budgets;
2. Build the capacity of the Ministry of Water and Environment as well as other stakeholders to be aware of and to address gender differentials in water, sanitation and hygiene sectors and to promote a gender-sensitive work environment at the Ministry of Water and Environment;
3. Economically empower men, women, boys, girls and vulnerable groups, especially those with disabilities, through equitable access to and control of water resources, supply, sanitation and hygiene;
4. Collection and reporting of gender disaggregated data, with attention to gender issues in monitoring and reporting;
5. Building partnerships within the Sub-sector, Line Ministries, Civil Society, districts and Local Governments to move forward the gender agenda;
6. Gender coordination, partnership and networking.

The most direct benefits to women and girls are likely to come through strategic objective 3, which foresees the rehabilitation and expansion of water and sanitation facilities in rural and urban locations, as well as support so women and girls can take advantage of employment or income-generating opportunities associated with development of water and sanitation facilities.

This policy and other policies express a commitment to stakeholder participation in the planning, implementation and maintenance of water and sanitation projects, with the National Water Policy (1999) setting a minimum of at least half of Water and Sanitation Committees members being women. Whilst this commitment provides opportunities for women and girls to influence the development of water and sanitation infrastructure in their communities, they bring an additional time burden which

¹⁹⁸ Uganda’s Water and Sanitation Policy has not been accessible online and so has not formed part of our review. Interestingly, Uganda’s Gender Policy gives very little attention to gender issues in the areas of water and sanitation.

women and girls may struggle to take on. Social traditions which encourage women and girls to defer to men may inhibit women and girls' abilities to participate further.

Whilst the Water and Sanitation Gender Strategy does not identify inadequate water, sanitation and hygiene facilities within schools, health centres and hospitals as an issue, the Second National Health Policy and the National Education and Sports Strategic Plan do. As the main users of primary health centres, women and girls can be severely affected by inadequate water, sanitation and hygiene practices in these centres. Similarly, inadequate toilet facilities are often a barrier to girls' attendance at school, especially during menstruation. With this in mind, the Education and Sports Sector Plan includes, amongst its objectives, the expansion and improvement of safe water supply infrastructure to government supported primary schools as well as the construction of sanitation facilities primarily in primary schools. The National Health Policy also foresees improvements in health infrastructure and in waste disposal but does not provide specifics.

The Water and Sanitation Gender Strategy recognises the importance of building the capacity of Ministry of Water and Education staff on gender and of encouraging a gender-sensitive work environment, but stops short from promoting actions to increase women's participation in formal roles in the planning, development, management and maintenance of water and sanitation facilities. Interestingly, the National Development Plan III outlines clear plans for expanding education provision in Science, Technology, Engineering and Mathematics and increasing the number of graduates in these areas, but places no emphasis on encouraging women specifically into education and careers in these areas. Currently the proportion of women who have completed upper secondary school or enrolled in tertiary education remains low, with 20.2% and 4.33% respectively.

Stakeholder Mapping

In Uganda, the Ministry of Water and Environment has primary responsibility for WASH. Other national Ministries with some WASH responsibility include the Ministry of Health, responsible for sanitation and hygiene, as well as WASH in health care facilities, and the Ministry of Education and Sports, which has responsibility for WASH in schools. There are a range of agencies involved in the delivery of WASH services at the sub-national levels, including district-level government agencies, the private sector, national and international CSOs. A wide range of development partners are involved in supporting government, private sector and CSO engagement in WASH. Figure H provides a visual mapping of the actors involved.

Figure H: WASH stakeholders in Uganda

Ministry of Water and Environment (MWE)

Key Roles: Lead ministry for water and sewerage services. Sets national policies, standards, and priorities for their development. Departments include among others: the Water and Sanitation Development Facility; Water Utility and Regulation Department; Climate Change Department and Water and Environment Sector Liaison Department covering cross-cutting issues and with a mandate for gender and social safeguarding

Ministry of Health (MoH)

Key Roles: The lead ministry for sanitation and hygiene service provision; formulation and review of national sanitation and hygiene policies, guidelines and standards. Mandates of MoH, MWE & Ministry of Education and Sports (MoES) stipulated in a Memorandum of Understanding. MoH leads on sanitation and hygiene in households and health care facilities; MWE is responsible for sanitation and hygiene in public places and MoES is responsible in schools.

Government Stakeholders at National Level
Other key government ministries

Ministry of Local Government: supports local governments in developing district water plans; Ministry of Gender Labour and Social Development: has the national government Equity Strategy; Ministry of Education and Sports; Ministry of Finance, Planning and Economic Development: allocation of formulas for District WASH Conditional Grants; Ministry of Trade & Industry: development of policy regulation of the private sector

Other government institutions

National Water and Sewerage Corporation
National Environmental Management Authority
Water Utility and Regulation Department
Directorate of Water Development
Rural Water Supply and Sanitation Department
Urban Water Supply and Sewerage Department: supports water supply and sanitation services outside the NWSC's mandate.

District Level Government Stakeholders

Local Governments (districts, towns, sub-counties); District Water Office; Directorate of Health Services (sanitation and hygiene); Water and Sanitation Development Facilities: funds new water supply, sanitation schemes and major rehabilitations project in small towns and rural centres; Water Management Zones: monitors water quality at the catchment level; Umbrella Authorities: management of rural piped systems; Technical Support Units: links MWE with Local Government; Sub-county Water Supply and Sewage Board: works closely with Water for Production Regional Centres.

Non-State Actors

International CSOs: Caritas, Save the Children, Mercy Corps, World Vision Cooperation & Development, Amref, WaterAid, Netherlands Development Organisation (SNV), International Rescue Committee (IRC), Water for People

Local CSOs: Action for Rural Women's Empowerment, Action for Human Rights and Education Initiative Uganda, Advocates for Water & Environment Conservation

Others; Private Operators e.g association of private water drillers, entrepreneurs that manage small piped systems under a management contract with the Sub-County, researchers, academia and media.

Community Level

Community WASH units e.g. Water User Committees/associations, Catchment based Management Organisations, Sanitation Committees, Community Hygiene Volunteers, households, citizens/consumers (women, men, children and youth, including vulnerable groups).

Development Partners

World Bank/IFC, AfDB, USAID, The Foreign, Commonwealth & Development Office (FCDO), GIZ, KfW, Japan International Cooperation Agency (JICA), Green Climate Fund (GCF), The Swedish International Development Cooperation Agency (SIDA), GIZ, KfW, JICA, GCF, SIDA, Austrian Development Agency, French Government, UN agencies e.g. UNICEF, UNHCR, UNDP, UNEP, UNESCO, WHO)

Annex 6: Gender Action Plan

Introduction

This document sets out Climate Fund Managers (CFM) action plan for promoting gender equality in water, sanitation and hygiene (WASH), protection and management of wetlands, coastal and marine ecosystems, and climate resilient port infrastructure through investments by the Climate Investor Two (CI2) fund. By adopting a gender mainstreaming approach to CI2 and the projects it supports, CFM expects not only to mitigate the risks for women and men associated with funded projects, but also to benefit women and men equally, reducing gender differentials associated with vulnerabilities that are exacerbated by climate change, and promoting sustainable and equitable development.

CFM is committed to promoting gender equality and women's empowerment externally (in countries and communities where CFM's Managed Vehicles are invested) and internally, within CFM itself.¹⁹⁹ It does so through a twin-track approach; firstly, by mainstreaming gender considerations into its investments,²⁰⁰ and secondly, by putting in place organisational systems to promote gender equality within CFM. Procedures relating to the former are described in CFM's Environmental and Social Management System.

The development of this Gender Action Plan (GAP) has been guided by the Green Climate Fund (GCF) approach to gender mainstreaming (Figure 1). It has been informed by a Gender Assessment which comprised:

- i) A review of global evidence in the areas of WASH, the protection of wetlands, coastal and marine ecosystems and climate resilient port infrastructure;
- ii) A review of national policies, strategies, plans and contexts, relating to WASH, gender and climate change for 11 CI2 focal countries: Bangladesh, Botswana, Brazil, Djibouti, Indonesia, Ivory Coast, Kenya, Morocco, Namibia, Nigeria, Uganda;
- iii) Stakeholder analysis for eight out of the 11 countries;²⁰¹
- iv) A problem tree analysis for each of the three areas, which, following GCF guidance, drew on the Gender Assessment findings to unpack the causes and effects of the core problem identified; and
- v) Identification of opportunities for promoting gender equality through CI2 investments, based on the evidence reviewed.

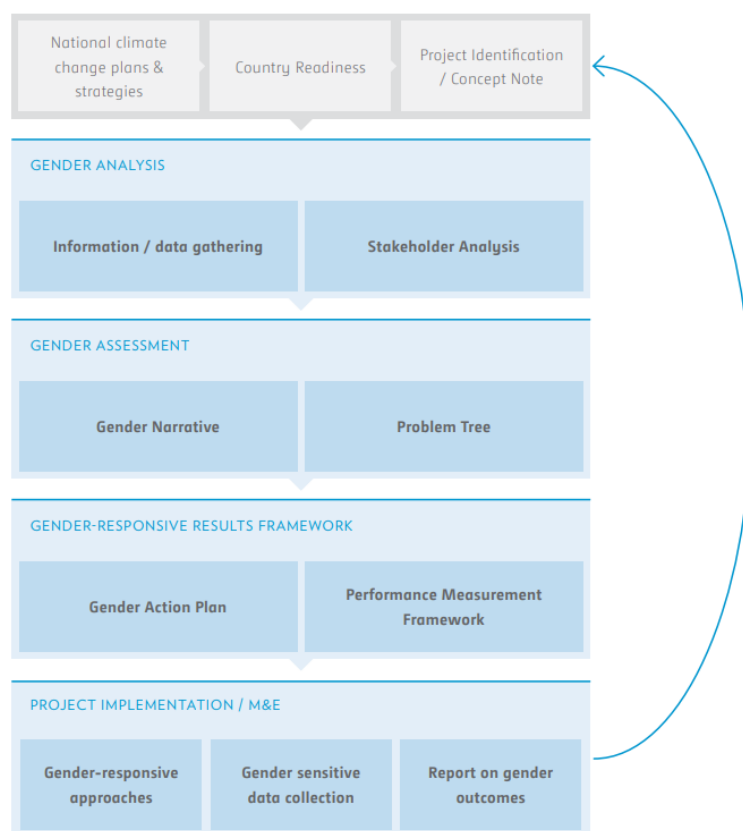
The GAP presents how CFM will use CI2 to take forward the opportunities for promoting gender equality identified in the Gender Assessment and provides a performance measurement framework which will be used to monitor CFM's implementation progress. As CI2 is expected to cover up to 25 countries, and financing investments which have yet to be identified, the actions included in the GAP are, by necessity, high level. Project specific GAPs, which will be developed during the project design phase, will allow this high-level framework to be tailored more directly.

¹⁹⁹ CFM Gender Policy 2021

²⁰⁰ Gender mainstreaming is the systematic process of ensuring that women and men have equal access and control over resources, development benefits and decision-making at all stages of the development process through ensuring that the respective needs, interests and priorities of women and men are integrated into all policies, programmes and projects. CFM Gender Policy 2021

²⁰¹ It was not possible to conduct stakeholder analysis for the remaining 3 countries as No Objection Letters were received too late in the process.

Figure 1: GCF's gender mainstreaming approach²⁰²



Overview of CI2 Gender Assessment

The CI2 Gender Assessment covered a broad set of issues relating to how men and women, girls and boys interact with WASH, wetland, coastal and marine ecosystems and port infrastructure (see Table 1).

Table 1: Issues covered in the Gender Assessment, by area

Sector	Issues covered
Water and Sanitation	• Women's and men's, girls' and boys' access to and use of WASH services, the barriers they encounter, including gender-based violence, and how climate change is affecting these barriers. • How women and men, girls and boys are affected by challenges in accessing quality WASH. • The roles that men and women, boys and girls play in the WASH sector, including formal employment opportunities. • Women and girls' participation and leadership in the design, planning and management of WASH infrastructure and services. • National policy in relation to gender, water and sanitation.

²⁰² Source: [guidelines-gcf-toolkit-mainstreaming-gender_0.pdf \(greenclimate.fund\)](#)

	Stakeholders involved in WASH.
Wetland, coastal and marine ecosystems	- Women's and men's use of and interaction with wetland, coastal and marine areas. - How climate change is affecting these different uses and interactions, and the impact on the populations concerned. - The different ways in which men and women are, are not, and could be involved in the protection and management of wetland, coastal and marine areas.
Port infrastructure	- The different ways in which men and women use and benefit from available port infrastructure. - The different ways in which men and women are, are not, and could be involved in and can benefit from the development of climate resilient port infrastructure.

In the WASH sector, the Gender Assessment found that women's and girls' domestic and care-giving roles mean that they are disproportionately affected by inadequate access to WASH infrastructure and services, a situation which is being exacerbated by volatilities associated with climate change. Whilst national policies often recognise this and make commitments to addressing gender issues across their policy implementation, there is very little detail as to how this will be done in practice. The causes and effects of this situation are multiple and are summarised in the Problem Tree Analysis below (Figure 2).

Figure 2: WASH Problem Tree Analysis

ROOT CAUSES	Governments have been slow to take action to mitigate effects of climate change and build resilience	A weak state-citizen social contract inhibits policy implementation and investment in essential public services	Gender norms relegate women and girls to the domestic space, undermine investment in girls' education and exclude women and girls from formal and informal decision-making fora		Women and girls' physiology	Weak public health systems
CAUSES	Climate change depletes available WASH infrastructure and resources	Insufficient investment into WASH infrastructure, maintenance and service management	Women and girls are responsible for household chores, including water collection, and caring for family members	Women and girls do not influence the design, planning and management of WASH services	Women and girls' have increased and specific WASH needs	Men and women, boys and girls have insufficient knowledge about the need to prioritise good hygiene practices
PURPOSE STATEMENT INADEQUATE WASH INFRASTRUCTURE AND SERVICES PLACE A HIGH BURDEN ON WOMEN AND GIRLS						

IMMEDIATE EFFECTS	Women and girls spend more time on household chores and family care giving Women and girls face increased risk of sexual harassment and gender-based violence Families, especially men, are unwilling to spend scarce resources on WASH services Girls, especially adolescent girls, do not attend school consistently Families, but especially women and girls, face an increased risk of ill-health and low levels of well-being, including mental well-being Available WASH infrastructure and services do not respond to women and girls' needs or priorities Women and girls are not able to attend to their physical needs
EFFECTS	Women and girls are not able to invest in empowering activities such as going to school, income generating activities, community leadership People, especially women and girls, have a poorer physical and mental health status Girls are not able to complete basic levels of education Few women have the education and skills to work in the WASH sector Gender norms and inequalities are sustained

As regards wetland, coastal and marine ecosystems, the Gender Assessment found that women are mostly excluded from the formal sector and most profitable parts of value chains in oceans-based economies, as well as from dialogue and decision making around community-based protection and management initiatives. At the same time, and in spite of gaps in the evidence base on climate-related impacts, they are likely to be disproportionately affected by damage to ecosystems. The causes and effects of this situation are summarized in the Problem Tree Analysis below (Figure 3).

Figure 3: Wetland, coastal, and marine ecosystem Problem Tree Analysis

ROOT CAUSES	Governments have been slow to take action to mitigate effects of climate change and build resilience	Gender norms relegate women to the domestic space: limiting the amount of time they can spend outside, and distance they can stray from, the home; and constraining women's access to credit and specialised equipment		Design of community protection and management initiatives does not take into account restrictions on women's participation
CAUSES	Climate change damages wetland, coastal and marine ecosystems	Women are mostly excluded from the formal sector in oceans-based industries, and are disproportionately represented in informal and unregulated economies	Women are concentrated in less profitable parts of value chains	Low female participation in community protection and management initiatives across relevant sectors
PURPOSE STATEMENT WOMEN ARE MARGINALISED IN THE USE, PROTECTION AND MANAGEMENT OF WETLAND, COASTAL AND MARINE ECOSYSTEMS				
IMMEDIATE EFFECTS	There is limited evidence on the gender-differentiated effects of damage to ecosystems At the same time, rising sea temperatures, levels and acidity are all expected to disproportionately affect women's livelihoods, with women facing disproportionate and additional barriers to adaptation Women working in oceans-based industries earn less than men for carrying out the same work Women working in informal or unregulated economies are vulnerable to exploitation, including unsanitary work environments			
EFFECTS	Policy- and decision-making are gender-blind, overlooking women's needs, interests and potential roles in protection and management Women and their families have poorer economic status and food security Gender norms and inequalities are sustained or exacerbated			

In spite of particularly limited evidence on the intersect between gender, climate and port infrastructure, the Gender Assessment likewise identified widespread exclusion of women from the

development and operation of port infrastructure, particularly from employment in operations and management. The Problem Tree Analysis below summarises the causes and effects of this situation (Figure 4).

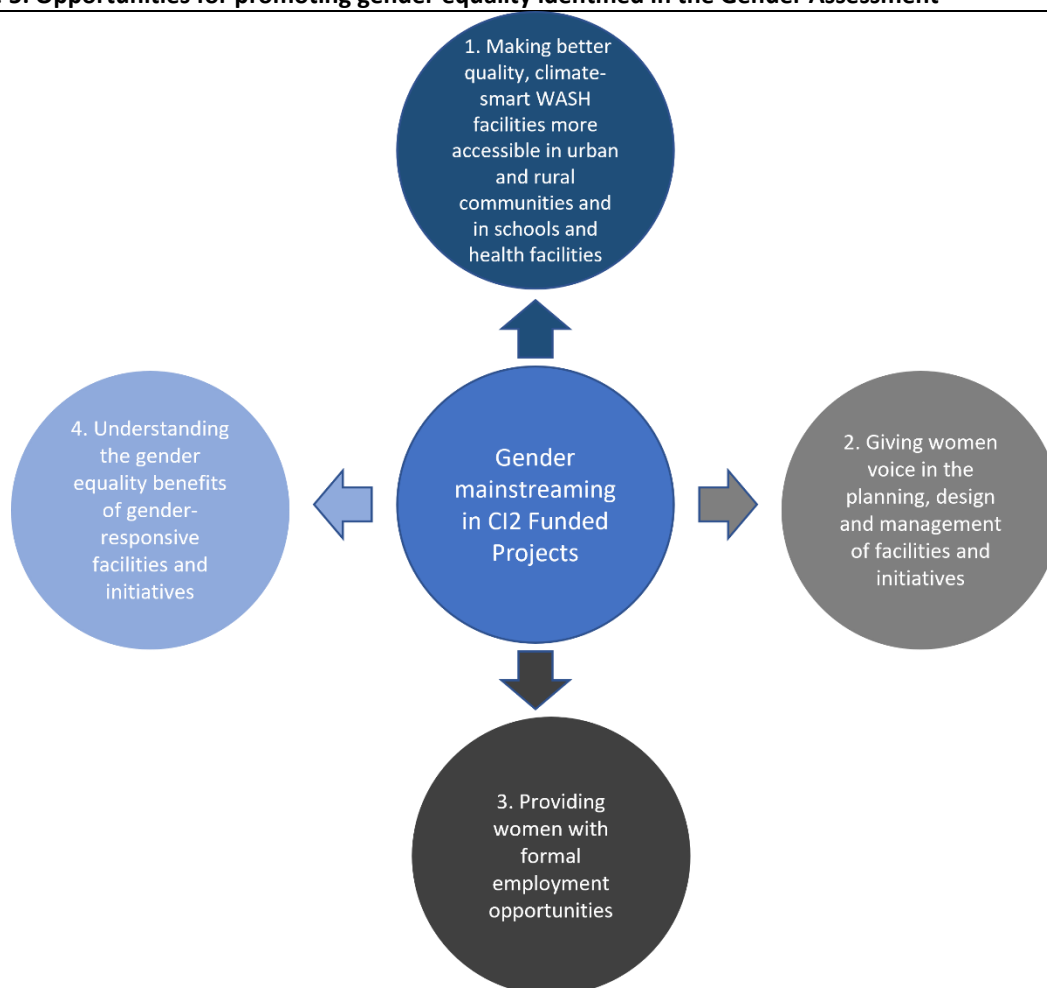
Figure 4: Resilient Port Infrastructure Problem Tree Analysis

ROOT CAUSES	Governments have been slow to take action to mitigate effects of climate change and build resilience		Gender norms dictate that port employment is overwhelmingly dominated by men		
	Operation of traditional port infrastructure damages ecosystems	Women comprise only a fraction of recruits and trainees to the port industry	Female employees are particularly underrepresented in port operations and management	There is a lack of support to the integration and progression of female employees (for example, through mentorship programmes)	Working environments are insufficient to enable retention of female recruitments (e.g. by providing separate sanitation facilities and family-friendly HR policies)
PURPOSE STATEMENT WOMEN ARE EXCLUDED FROM THE DEVELOPMENT AND OPERATION OF (CLIMATE-RESILIENT) PORT INFRASTRUCTURE					
IMMEDIATE EFFECTS	There is a lack of evidence on the gender-differentiated effects of ports' negative impacts on ecosystems, as well as of optimised infrastructure and energy efficiency Women do not benefit equally from the employment benefits of port infrastructure Women are excluded from discussions around new approaches such as no-impact port development, despite tending to be more willing to engage in sustainable or environmentally friendly practices				
EFFECTS	Policy- and decision-making are gender-blind, overlooking women's needs, interests and potential roles in the development of climate resilient port infrastructure Women have poorer economic status Gender norms and inequalities are sustained				

Overview of Gender Actions

Responding to the causes identified in the problem tree analyses, the Gender Assessment identified opportunities for promoting gender equality through CI2 investments in four main areas (Figure 3). These opportunities directly underpin the action areas presented below and in the action plan in Table 2, with a fifth action area being added to address the organisational environment needed within CFM Project Companies and contractors, to facilitate the Gender Action Plan's full implementation.

Figure 5: Opportunities for promoting gender equality identified in the Gender Assessment



Gender Action Areas for CI2

Improving women and girls' access to WASH

As far as possible, CFM and its Project Companies will use its investment projects and community development projects to address the physical, financial, attitudinal and safety barriers that may limit women's and girls' access to WASH.

Using normal project development processes e.g., the Environment and Social Impact Assessment (ESIA), the Community Needs Assessment and Local Gender Assessment etc., CFM will explicitly consider the barriers which affect women's and girls' access to WASH, and how the investment design can address these. It will incorporate appropriate actions into the design of investment projects, where possible, to address these barriers, and include relevant indicators in the results frameworks to monitor progress. As part of this, CFM will consider the barriers faced by different groups of women and girls, such as those from chronically poor households, from indigenous households and women and girls with disabilities. CFM will also consider the specific needs some of these groups might have in accessing WASH facilities.

CFM will seek to use community development projects to increase the direct benefits for women and girls resulting from investments, by addressing identified barriers which cannot be addressed through the main investment. Examples of potential projects include:

- Working with families and communities to build their understanding of and support for good WASH practices, including during menstruation, so that families are willing to invest scarce resources into WASH services, and communities are proactive in encouraging all members to observe good hygiene practices. Targeting men is critical in such interventions as they often control the use of financial resources in households.
- Modifying water points and toilet facilities so they can be used by women and girls with disabilities.
- Providing good lighting in places used as shared water points or sanitation facilities.

It is recognised that improving WASH facilities in schools and health centres often brings significant benefits to women and girls. Opportunities for doing this will be considered as part of the investment project and community development project design process.

Agreed objectives and actions to improve women's and girls' access to WASH, either through the investment project itself and/or through community development projects, will be set out in the project design, with relevant indicators to track progress and outcomes incorporated into the project's results framework (see Action 4 below). For the investment projects, these objectives will also be reflected in contracts with the Engineering Procurement Contractor (EPC) and will be reviewed as part of normal EPC performance review processes.

2. Giving women a voice in the planning, design and management of facilities and initiatives

As a priority beneficiary group for CI2 investments, it is essential that stakeholder engagement and consultation processes service the needs and experiences of women and girls. This is to ensure that investment projects and community development projects can be responsive, as far as possible, to the needs of women and girls.

Social norms in CI2 focal countries often devalue and silence women's voices, even when public consultation mechanisms are in place. To counter this and to ensure women's and girls' needs inform investment design, CFM and its Project Companies will adopt a stakeholder engagement approach that provides conducive spaces for women and girls to voice their needs and experiences. This process will involve:

- Identifying women and girls as a distinct stakeholder group with whom to engage directly, whilst recognising that, within this broad category, there are intersecting sub-groups with diverse needs and experiences who also need to be involved, e.g., indigenous women, women with disabilities and chronically poor women.
- At the community level, identifying women's leaders to work with to convene appropriate fora to hear a range of women's and girls' voices about their needs and challenges.
- Working with community and male leaders to help them understand the rationale for engaging women in the planning of facilities and initiatives and build their support for that to avoid domestic arguments or backlash.
- Holding consultation meetings at times and at venues which are convenient to the women and girls being targeted and considering how the direct and indirect costs of attending can be offset, particularly for the chronically poor, and for women and girls with disabilities.

- Using consultation sessions with women and girls as an opportunity to inform the design of the grievance mechanism so that it is accessible and so that women and girls have confidence in it as a system of redress and accountability. As part of this, a zero tolerance to all forms of gender-based violence and sexual harassment perpetrated by any staff or contractor from CFM, its Project Companies or contractors will be communicated.

In some projects, community-led management committees may play a role in the management of WASH facilities and ocean ecosystem protection initiatives. Where this is the case, CFM and its Project Companies will support committees to operate in a way that provides space for women's voices to be heard. The approach is likely to be context specific, but might involve:

- Identifying and recruiting to the committee local women who are perceived (by men and other women) to have some legitimacy in speaking out on behalf of the community. These women are likely to have other leadership roles in the community and have experience of publicly expressing their views.
- Building the skills of the women representatives on the committee so that they can consult with a diverse range of women and girls in the community to get their feedback on the WASH facilities, and to represent these views in committee meetings and decision-making.
- Building male committee members' understanding of the need for management to be informed by women's and girls' experiences, and to develop the skills to ensure women's voices inform committee decision-making.

Planned actions to provide a safe space for women's voices in the planning, design and management of facilities and initiatives will be set out in the investment's stakeholder engagement plan. Routine monitoring of the plan's implementation will give particular attention to actions taken to capture women's voices and to how project design and management is responding to the feedback received.

3. Providing women with formal employment opportunities

Low levels of secondary education completion amongst girls in several of the CI2 focal countries, combined with gender norms around acceptable productive roles for women and men, can make the formal employment of women in the WASH sector, as well as ocean-based industries and operation of port operations, challenging.

Project Companies will be proactive in hiring a gender-balanced team, making adjustments to working practices where this could level the playing field between male and female candidates. It is, however, recognised that in several focal countries there are systemic issues beyond CFM's control which are likely to limit the Project Companies' ability to recruit senior female project staff. In an effort to ameliorate this situation, CFM will scope out, in some focal countries, opportunities to encourage more women to enter the WASH sector. Options might include: providing scholarships for women embarking on tertiary education in relevant Science, Technology, Engineering and Mathematics (STEM) fields, providing work placements on projects for current female STEM students, or supporting women involved in the WASH sector to visit secondary and tertiary education facilities to encourage women to consider a career in the sector. Viable opportunities that can be accommodated within the overall investment will be pursued as pilot projects.

At the project level, Project Companies will identify employment opportunities for women in WASH enterprises and put in place appropriate mechanisms to recruit, train and support women in these roles. Opportunities might include roles as sanitation sale agents, in meter reading, water tank construction or cleaning, amongst others. It is recognised that there is limited experience of such approaches, and work done in this area through CI2 will be done on a pilot basis with the intention of

increasing knowledge about effective approaches. Figure 4 provides an example of how CFM has promoted women's employment opportunities in Sierra Leone. Where investments include improving WASH facilities in schools and health centres, Project Companies will seek to work with service managers to put in place a resourced plan for the on-going cleaning and maintenance of the facilities, and which does not default to traditional gender roles.

Box 1. Supporting women business leaders and women's employment in Sierra Leone

CFM's investment concerns the development of a waste to value portfolio in the city of Freetown, Sierra Leone. The assets comprise modular units which process organic waste and generate clean energy and digestate that can be used as fertiliser, thus contributing to the circular economy and providing a much-needed solution to the problem of waste management in the city. Currently most of the waste ends up in the city's waterways and in the marine environment.

The waste is collected by waste collection micro-enterprises ("youth groups"), which were established, trained and equipped by Freetown City Council to conduct door-to-door household waste collection in the city, and are jointly responsible for collecting up to 50% of the waste in city. The modules can be installed at a wide variety of premises and existing customers include a women's maternal health centre in Freetown.

The company in which CI2 is investing is a woman-led business as is its partner organisation. Female operators are actively sought, and the youth groups are approximately 30% female. With CI2's investment, a local gender action plan has been established, along with a programme of community development that includes a focus on women as well as youth groups.

As regards the protection and management of wetlands, coastal and marine ecosystems, projects supporting sustainable practices in oceans-based industries should not only take into account such restrictions – on women's time of mobility, for instance – but actively seek to tackle discriminatory norms which limit women's livelihoods opportunities.

Project Companies will seek to attract women recruits to climate-resilient port operations, and support their integration and progression, by providing a combination of initiatives to build their skills and confidence and provide a supportive work environment.

Understanding the results and impact of gender mainstreaming

Monitoring and reporting will be undertaken in accordance with the principles set out in CFM's Environmental and Social Management System (ESMS). CFM will monitor and evaluate the actions taken to promote gender equality, their results and impact, in two ways. Firstly, by monitoring and evaluating key impact indicators at the Fund level. Secondly, by implementing monitoring plans for individual projects.

Fund Level Monitoring and Evaluation

The KPIs established for individual programmes will supplement the key impact indicators that have been adopted for the CI2 funds, and against which CI2 is committed to monitor performance in all

projects. These indicators are linked to select UN Sustainable Development Goals (SDGs). The impact indicators that are relevant to gender are shown in Table 3 below.

Table 2 CI2 Impact Indicators

UN SDG	Gender-Relevant CI Two Impact Indicators
SDG 5 – Gender Equality	• Number of full-time, equivalent female employees. • Number of females in management positions.
SDG 6 – Clean Water and Sanitation	• Total number of people (gender disaggregated) with improved access to clean drinking water and sanitation.
SDG 8 – Decent Work and Economic Growth	• Number of new direct jobs created; gender disaggregated. • Catalytic effect (EUR million) (total DFI and commercial financing).

Project level monitoring and evaluation

Project companies will implement a monitoring and evaluation plan, which sets out:

- The indicators to be used to monitor how women and girls, as well as other priority social groups e.g., indigenous women, women from chronically poor households and women with disabilities, are being affected – both positively and negatively – by project interventions, and the impact these projects are having on their lives.
- The methodology for gathering and analysing monitoring data, as well as reporting and using the findings.

Each monitoring and evaluation plan will be bespoke to individual projects. However, they will contain comparable quantitative and qualitative indicators and data collection methodologies where appropriate. Depending on a project's life cycle, monitoring data will be collected where possible at regular intervals to understand progress made during project implementation. The aim of project monitoring will be to provide timely information to make project adjustments and address any negative effects.

Relevant components of the monitoring and evaluation plan may be incorporated into contracts for EPCs or other sub-contractors, with reviews of the plan's implementation forming part of the contractor's normal review process.

There is a need for more evidence on effective approaches to promoting gender equality through investments in WASH, protection of ocean ecosystems and climate-resilient port infrastructure. In response to this, CFM may publish some of the results and learnings from its CI2 investments.

Creating an enabling environment for gender mainstreaming in the CI2 Fund and the Project Companies

CFM recognises the need to draw on specialist skills and advice to guide the company's leadership on how to operationalise its gender commitments across the business and within the investments it manages. This is expected to be achieved through a network of gender and inclusion specialists who can be hired on an as-needed basis to provide both fund and project specific support. This network will benefit from contacts with local organisations with a strong track record in addressing gender issues in WASH, protection and management of wetlands, coastal and marine ecosystems, and climate resilient port infrastructure. It is anticipated that these specialists and local organisations could

potentially partner with Project Companies as a delivery agent e.g., in managing the stakeholder consultation with women or in community development activities.

CFM will support Project Companies to put in place essential internal systems and capacities that will facilitate gender mainstreaming through the project and its operations, thereby operationalising this CI2 Fund Gender Action Plan. Arrangements will include:

- Developing a local (project-specific) gender action plan based on an analysis of gender-related challenges and opportunities within the local context. This will integrate project-specific and organisational intentions for promoting gender equality.
- Agreeing and implementing a project-specific Gender Equality Policy which sets out the company's commitments to gender equality in their own internal operations and through the investment. This policy will set out the Project Company's position on gender mainstreaming and on specific topics such as Gender Based Violence and Harassment (GBVH) and the mechanisms in place to safeguard staff from it.
- Taking proactive measures to recruit a gender-balanced project team, seeking, where possible, to recruit qualified women into Project Company roles traditionally held by men.
- Building the understanding and support of company staff to promote gender equality, both within their organisational context and in the project itself.
- Integrating expectations of gender equal behaviours into staff job specifications and performance appraisal systems, with company leadership actively demonstrating a commitment to gender equality and setting Project Company expectations in this regard.
- Providing the project team responsible for health, safety, social and environmental management with the necessary training and skills to guide the Project Company in the effective implementation of its gender equality commitments.
- Implementing and managing a grievance mechanism for staff to report complaints, including cases of GBVH in the workplace.
- Monitoring and reporting progress against agreed indicators, tracking the impacts of the project on women and girls.

Project Companies will also take steps to require its principal contractors²⁰³ to give due attention to gender issues in their own operations. Requirements to be imposed on contractors are likely to include:

- An operational Gender Equality Policy.
- Proactive measures to recruit a gender-balanced team, seeking, where possible, to recruit qualified women into roles traditionally held by men.
- A grievance mechanism for the contractor (and sub-contractor) workforce to report complaints, including cases of GBVH.

These requirements would form part of the contract. Progress on delivering against these requirements would form part of the normal contractor performance review process.

²⁰³ In many cases the project is built by one principal contracting company – referred to as the Engineering, Procurement and Construction (EPC) contractor. In other cases a split contract may be awarded to different principal contracting parties. In such cases, each party would be required to follow the same requirements, with the actions coordinated by the Project Company.



Gender Action Plan (GAP)

The action plan on the following page summarises the actions to be taken by CFM, and at Fund level, in order to mainstream gender issues in the investment lifecycle of the CI2 fund.

CFM will monitor progress against this Gender Action Plan once a year using the indicators provided. Indicators may be updated over the life of CI2 to ensure they align well with CI2 investments and procedures.

Table 3 CI2 Gender Action Plan

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
Opportunity 1: Improve women and girls’ access to WASH								
1.1	Design and implement investment projects and community development projects which address physical, financial, attitudinal and safety barriers to women and girls’ WASH access.	X			Use ESIA and other associated assessments (including rapid gender analyses/assessments where applicable) to identify opportunities for addressing physical, financial, attitudinal and safety barriers for women and girls, and incorporate relevant actions into the project design and monitoring frameworks.	Investment projects and/or associated community development projects have identified and are taking forward actions to address at least two of the four types of barriers for women and girls. Number of beneficiaries, disaggregated by sex, with improved access to climate-resilient water. Number beneficiaries, disaggregated by sex, with improved access to climate-resilient sanitation services. Women and girls’ report reduced physical, financial, attitudinal and safety barriers to climate-resilient water and/or sanitation services.	External experts (consultants) to conduct ESIA, plus the community needs assessment and local gender analysis and action plan. Project Company and CFM staff to manage consultants and use the findings and inform investment design and/or associated community development projects. Project staff to manage community	USD\$ 150,000-200,000 per project (estimate) for the ESIA (which will include a gender expert). USD\$ 25,000 per project (estimate) for the needs assessment. USD\$ 25,000 per project (estimate) for the and gender analysis and action plan. Reimbursable costs (expenses). CFM staff time

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
							development projects. CFM staff to support investment implementation, ensuring it addresses identified barriers to women's and girls' access to WASH.	
Opportunity 2: Women's voice and agency in planning and operations								
2.1	Plan and operationalize stakeholder engagement so that women and girls' voices inform the design of investment projects and community development projects.	X	X	X	Stakeholder engagement includes consultations with women-only groups involving representatives of diverse groups of women. Stakeholder engagement informs men about the need to consult with women and girls.	Number of women/girls only consultation meetings held. Number of women or girls involved in women/girls only consultation meetings. ESIA report documents main issues raised by women and girls in consultation meetings. ESIA report documents how men were made aware of the need to	External experts (consultants) to conduct stakeholder engagement as part of studies described in Item 1.1.	Budget is included under Item 1.1. CFM and Project Staff time.

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
						consult directly with women and girls.		
2.2	Design and operationalise community grievance mechanisms that provide women and girls with confidence that the system provides redress and accountability on any form of GBV including SEAH.	X	X	X	Use stakeholder engagement with women and girls to understand how to operationalise a grievance mechanism that women and girls are ready to use. Put in place appropriate grievance mechanism and make women and girls aware of it, as well as the situations they would use it in.	ESIA report documents women and girls' recommendations for grievance mechanism. Number and proportion of grievances received from women and girls, with indication of type of grievance. Number of grievances received from women and girls that have been resolved, with an indication of the type of grievance.	External experts (consultants) to develop grievance mechanism as part of ESIA (Item 1.1). Project Company staff to implement and communicate the grievance mechanism. CFM staff to monitor Project compliance.	Budget is included under Item 1.1. CFM and Project staff time to .
2.3	Support community-led WASH management committees, as well as community governance mechanisms for	X	X	X	Provide training and support to male and female committee members so that female members are able to represent the views of women and	Percentage of committees taking proactive steps to ensure women's voices are heard, with examples of how women's voices inform committee decision-making.	External experts (consultants) to provide training if required. Project Company staff to monitor, and	Budget for external experts to be determined and allocated if required. CFM and Project staff time.

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
	initiatives relating to ocean ecosystems and port infrastructure, to operate in a way that allows women's experiences to influence decision-making.				men; provide space for women representatives to have a say in committee decision-making. Collect data from committees on how members are representing the views of women and how women's voices have informed committee decision-making.		collect and report data. CFM staff to monitor Project compliance.	
Opportunity 3: inclusive employment opportunities								
3.1	Proactively recruit women into WASH investments and/or in community development programmes.	X			Identify roles in WASH enterprises for which women can be proactively hired. Make training and support available for women and men hired. Conduct awareness-raising on	Percentage of positions in WASH enterprises held by women, disaggregated (where possible) by type of position e.g. management role, service provider role.	External experts (consultants) to develop recruitment strategy and provide training if required. Project Company staff to implement recruitment strategy	Budget for external experts to be determined and allocated if required. CFM and Project staff time.

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
					the rationale and benefits of engaging women in WASH enterprises, especially with men, to manage any potential backlash.		and collect and report data. CFM staff to monitor Project compliance.	Reimbursable costs(expenses).
3.2	Ensure that projects to support sustainable practices in oceans-based industries address structural barriers to women's livelihoods in related industries.		X		Account for gendered norms which restrict women's time use and mobility in project design to ensure accessibility and meaningful participation. Actively seek to tackle discriminatory norms which limit women's livelihoods opportunities in oceans-based industries to support their participation in the formal economy and in more	Percentage of women participating in oceans-related industries, disaggregated by formal vs. informal economy, and (where possible) by value chain portion.	External experts (consultants) to develop programme/strategy if required. Project Company staff to implement programme/strategy and collect and report data. CFM staff to monitor Project compliance.	Budget for external experts to be determined and allocated if required. CFM and Project staff time. Reimbursable costs (expenses).

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
					profitable parts of value chains.			
3.3	Proactively recruit women to the operation of climate-resilient ports and support their meaningful integration and progression.			X	Identify roles in port operations for which women can be proactively hired. Attract female recruits by providing training incentives and offer mentoring to female employees to support their progression into operational and management positions. Ensure a supportive working environment, which includes family-friendly HR facilities	Percentage of positions in climate-resilient ports held by women, disaggregated (where possible) by level of seniority and type of position.	External experts (consultants) to develop recruitment strategy and provide training if required. Project Company staff to implement recruitment strategy and collect and report data. CFM staff to monitor Project compliance.	Budget for external experts to be determined and allocated if required. CFM and Project staff time. Reimbursable costs

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
					and appropriate sanitation facilities.			
Opportunity 4: Monitoring and learning								
4.1	Monitor project level progress in benefitting women and girls.	X	X	X	Incorporate relevant indicators and methodologies for monitoring progress against project gender objectives into project monitoring plans. Review progress regularly with EPCs.	All projects include indicators²⁰⁴ in project monitoring plans to measure the number of women and girls benefitting, who these women and girls are (chronically poor, indigenous, women with disabilities etc.) and how they are benefitting. Qualitative evidence of how different types of women and girls (poor, chronically poor, indigenous, with disabilities etc.) have benefitted from improved infrastructure and associated employment	External experts (consultants) to define indicators as part of the local gender analysis and action plan. Project Company staff to implement system for the collection of data against the defined indicators.	Budget is included under Item 1.1. CFM and Project staff time.

²⁰⁴ Indicators will capture SDG-related benefits, as described in D3 of the proposal, and especially benefits accruing to women in relation to: employment, working conditions, financial resilience, livelihoods, time spent fetching water, incidence of WASH related disease and associated care burden.

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
						opportunities, as well as protection of oceans ecosystems. EPCs demonstrate reasonable progress against agreed indicators over the reporting timeframe and provide qualitative evidence of how women and girls have benefitted.	CFM staff to monitor Project compliance.	
4.2	Monitor Fund level progress in benefitting women and girls through CI2.	X	X	X	Incorporate gender-focused key performance indicators into the Fund's impact framework. Review progress against indicators regularly.	Fund impact framework includes indicators ²⁰⁵ to measure the number of women and girls benefitting, who these women and girls are (chronically poor, indigenous, women with disabilities etc.) and how they are benefitting. Data showing how the numbers of women and girls the Fund has benefitted and how they have benefitted available annually.	CFM Head of ESG supported by Regional ESG Managers to incorporate relevant indicators into Fund impact framework, prepare reports, and monitor the Project Companies.	CFM staff time
Cross-cutting issues to create an enabling environment								

²⁰⁵ Indicators will capture SDG-related benefits, as described in D3 of the proposal, and especially benefits accruing to women in relation to: employment, working conditions, financial resilience, livelihoods, time spent fetching water, incidence of WASH related disease and associated care burden.

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
5.1	Proactively hire a gender-balanced and diverse team into Project Companies.	N/A			Seek out suitably qualified women for Project Company roles, adjusting working practices where doing so alleviates critical barriers to employment.	Number and percentage of Project Company team positions held by women. Percentage of interviewees for Project Company positions who are women. Evidence of Project Companies making adjustments to 'normal' working practices to enable women to take on Project Company positions e.g., adjustment to normal working hours, to limit time spent in the office.	Project Company to develop and implement gender sensitive recruitment strategy. CFM staff to monitor compliance.	CFM and Project staff time.
5.2	Encourage more women into senior roles in Project Companies.	N/A			Scope out and implement measures that will encourage women to study relevant STEM subjects at tertiary level and gain practical experience in the WASH industry.	Project Companies have agreed a set of actions to encourage women to study relevant STEM subjects at tertiary level, with evidence of actions being implemented.	Project Company to develop and implement gender sensitive recruitment strategy. CFM staff to monitor compliance.	CFM and Project staff time.

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
5.3	Develop and implement project-specific Gender Action Plans.	N/A			Project level Gender Action Plans will set out actions to promote gender equality within the Project Company and on the project itself.	Project Company has approved Gender Action Plan, with evidence of its implementation.	External experts to conduct local gender analysis and preparation of action plan. Project staff to implement plan. CFM staff time to monitor compliance.	Budget is included under Item 1.1. CFM and Project staff time. Reimbursable costs (expenses).
5.4	Develop and implement a project-specific gender equality policy.	N/a			The project's gender equality policy will set out the company's commitments to gender equality in the Project Company's internal operations and through the investment. Communicate to staff the main points in	Project Company has an approved Gender Equality Policy. Evidence that Project Company staff are aware of the contents of the Gender Equality Policy and the Company's associated behavioural expectations.	CFM Head of ESG supported by Regional ESG Managers to develop policy; roll out to Project Companies, and conduct survey to assess awareness. CFM staff to monitor and compliance.	CFM and Project staff time.

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
					the policy, particularly in relation to gender discrimination, gender-based violence and harassment.		Project staff to implement policy and report relevant data.	
5.5	Build CFM and Project Company staff understanding of gender equality and women's empowerment.	N/A			Develop and deliver training to build gender awareness and skills in promoting gender equality within the Company and in the project investment.	Percentage of Project Company staff who have completed gender awareness and skills training. Pre- and post-training survey demonstrates that the majority of staff have an improved understanding of i) gender differences and their root causes, ii) how gender differences affect who benefits from WASH/oceans infrastructure development, and iii) how the Project Company can ensure women and girls benefit proportionately from WASH/oceans infrastructure development.	External experts (consultants) to develop and deliver training.	Budget to be defined and allocated based on Consultant proposal. CFM and Project staff time to attend gender awareness and skills building training.
5.6	Equip the Health, Safety, Social and Environmental teams	N/A			Provide gender focused training and	All ESIA processes designed and implemented to:	External experts (consultants) to	Budget to be defined and allocated based on Consultant proposal.

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
	within the Project Companies with the skills necessary to promote gender equality through infrastructure investments and community development projects.				mentoring support to the Team.	- Consult with a range of women and girls. - Identify ways in which the needs of women and girls can be met through the project investment or associated community development actions.	develop and deliver training. CFM staff to manage external experts (consultants).	CFM staff time. Project staff time to participate in training and mentoring.
5.7	Incorporate considerations of gender equality and the specific needs of women into the staff grievance mechanism in Project Companies.	N/A			Make staff aware of the staff grievance mechanism and the circumstances when they might call upon it.	Evidence that Project Company staff are aware of the staff grievance mechanism and how they can use it to express concerns relating to gender discrimination and/or sexual harassment. Number and percentage of staff grievances relating to gender discrimination and/or sexual harassment. Number of staff grievances relating to gender discrimination and/or sexual harassment resolved, with indication of any action taken.	CFM Head of ESG supported by Regional ESG Managers to support with design of mechanism. Project staff to implement, communicate and manage the grievance mechanism.	CFM and Project staff time.

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
5.8	Hold Project Company staff accountable for demonstrating gender equal behaviours.	N/A			Integrate expectations of gender equal behaviours into staff specifications and performance appraisal systems. Appraise staff performance against these in normal appraisal processes.	All Project Company job descriptions set out expectations of gender equal staff behaviour. All staff appraisals reflect on and document performance against expectations of gender equal staff behaviours.	CFM Head of ESG supported by Regional ESG Managers and Project Company leadership.	CFM and Project staff time.
5.9	Ensure principal contractors take steps to promote gender equality.	N/A			Obligations will be incorporated into contractor contracts and monitored through normal review processes.	Actions by which Principal contractors will promote gender equality are incorporated into all Project Company contracts with relevant indicators incorporated into project monitoring frameworks Principal contractors report reasonable progress against agreed gender related indicators over reporting timeframe and provide qualitative evidence of how women and girls have benefitted.	CFM Head of ESG supported by Regional ESG Managers and Project Company staff to identify, agree and incorporate into contracts relevant actions for Project Companies to promote gender equality.	CFM and Project staff time. Reimbursable costs (expenses).

	Components and Main Activities	Sector Relevance			Description	Indicators	Lead Responsibility	Budget Allocation
		WASH	Ocean ecosystems	Port infrastructure				
							Staff time to review Project Companies progress reports against gender equality related actions..	



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